ESC European Society of Cardiology

European Heart Journal (2023) **44**, 3720–3826
https://doi.org/10.1093/eurheartj/ehad191

ESC GUIDELINES

2023 ESC Guidelines for the management of acute coronary syndromes

Developed by the task force on the management of acute coronary syndromes of the European Society of Cardiology (ESC)

Authors/Task Force Members: Robert A. Byrne *†, (Chairperson) (Ireland), Xavier Rossello ‡, (Task Force Co-ordinator) (Spain), J.J. Coughlan ‡, (Task Force Co-ordinator) (Ireland), Emanuele Barbato (Italy), Colin Berry (United Kingdom), Alaide Chieffo (Italy), Marc J. Claeys (Belgium), Gheorghe-Andrei Dan (Romania), Marc R. Dweck (United Kingdom), Mary Galbraith (United Kingdom), Martine Gilard (France), Lynne Hinterbuchner (Austria), Ewa A. Jankowska (Poland), Peter Jüni (United Kingdom), Takeshi Kimura (Japan), Vijay Kunadian (United Kingdom), Margret Leosdottir (Sweden), Roberto Lorusso (Netherlands), Roberto F.E. Pedretti (Italy), Angelos G. Rigopoulos (Greece), Maria Rubini Gimenez (Germany), Holger Thiele (Germany), Pascal Vranckx (Belgium), Sven Wassmann (Germany), Nanette Kass Wenger (United States of America), Borja Ibanez *†, (Chairperson) (Spain), and ESC Scientific Document Group

* Corresponding authors: Robert A. Byrne, Department of Cardiology and Cardiovascular Research Institute (CVRI) Dublin, Mater Private Network, Dublin, Ireland, and School of Pharmacy and Biomolecular Sciences, RCSI University of Medicine and Health Sciences, Dublin, Ireland. Tel: +353-1-2483190, E-mail: robertabyrne@rcsi.ie; and Borja Ibanez, Clinical Research Department, Centro Nacional de Investigaciones Cardiovasculares Carlos III (CNIC), Madrid, Spain, and Cardiology Department, IIS-Fundación Jiménez Díaz University Hospital, Madrid, Spain, CIBERCV, ISCIII, Madrid, Spain. Tel: +3491 4531200, E-mail: bibanez@cnic.es

† The two Chairpersons contributed equally to the document and are joint corresponding authors.

‡ The two Task Force Co-ordinators contributed equally to the document.

Author/Task Force Member affiliations are listed in author information.

ESC Clinical Practice Guidelines (CPG) Committee: listed in the Appendix.

ESC subspecialty communities having participated in the development of this document:

Associations: Association of Cardiovascular Nursing & Allied Professions (ACNAP), Association for Acute CardioVascular Care (ACVC), European Association of Cardiovascular Imaging (EACVI), European Association of Preventive Cardiology (EAPC), European Association of Percutaneous Cardiovascular Interventions (EAPCI), European Heart Rhythm Association (EHRA), and Heart Failure Association (HFA).

Working Groups: Cardiovascular Pharmacotherapy, Cardiovascular Surgery, E-Cardiology, Myocardial and Pericardial Diseases, Thrombosis.

Patient Forum

The content of these European Society of Cardiology (ESC) Guidelines has been published for personal and educational use only. No commercial use is authorized. No part of the ESC Guidelines may be translated or reproduced in any form without written permission from the ESC. Permission can be obtained upon submission of a written request to Oxford University Press, the publisher of the *European Heart Journal*, and the party authorized to handle such permissions on behalf of the ESC (journals.permissions@oup.com).

Disclaimer. The ESC Guidelines represent the views of the ESC and were produced after careful consideration of the scientific and medical knowledge and the evidence available at the time of their publication. The ESC is not responsible in the event of any contradiction, discrepancy, and/or ambiguity between the ESC Guidelines and any other official recommendations or guidelines issued by the relevant public health authorities, in particular in relation to good use of healthcare or therapeutic strategies. Health professionals are encouraged to take the ESC Guidelines fully into account when exercising their clinical judgment, as well as in the determination and the implementation of preventive, diagnostic or therapeutic medical strategies; however, the ESC Guidelines do not override, in any way whatsoever, the individual responsibility of health professionals to make appropriate and accurate decisions in consideration of each patient's health condition and in consultation with that patient and, where appropriate and/or necessary, the patient's caregiver. Nor do the ESC Guidelines exempt health professionals from taking into full and careful consideration the relevant official updated recommendations or guidelines issued by the competent public health authorities, in order to manage each patient's case in light of the scientifically accepted data pursuant to their respective ethical and professional obligations. It is also the health professional's responsibility to verify the applicable rules and regulations relating to drugs and medical devices at the time of prescription.

This article is co-published with permission in European Heart Journal and European Heart Journal - Acute Cardiovascular Care. All rights reserved. © The European Society of Cardiology 2023. The articles are identical except for stylistic differences in keeping with each journal's style. Either citation can be used when citing this article. For permissions, please e-mail: journals.permissions@oup.com

No part of this publication may be reproduced, stored in a retrieval system, transmitted, used for text and data mining, or used for training artificial intelligence, in any form or by any means, without the prior permission in writing of Oxford University Press, or as expressly permitted by law, by license or under terms agreed with the appropriate reprographics rights organization. Inquiries concerning reproduction outside the scope of the above should be sent to the Rights Department, Oxford University Press, at the address above.

ESC Guidelines are protected by copyright. As per opt out option under Article 4 §3 of EU Directive 2019/790, the European Society of Cardiology (ESC) reserves all rights to license uses of ESC Guidelines to train or develop generative artificial intelligence (AI) models, large language models or other deep learning or machine learning models.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Document Reviewers: Sigrun Halvorsen, (Clinical Practice Guidelines Review Co-ordinator) (Norway), Stefan James, (Clinical Practice Guidelines Review Co-ordinator) (Sweden), Magdy Abdelhamid (Egypt), Victor Aboyans (France), Nina Ajmone Marsan (Netherlands), Sotiris Antoniou (United Kingdom), Riccardo Asteggiano (Italy), Maria Bäck (Sweden), Davide Capodanno (Italy), Ruben Casado-Arroyo (Belgium), Salvatore Cassese (Germany), Jelena Čelutkienė (Lithuania), Maja Cikes (Croatia), Jean-Philippe Collet (France), Gregory Ducrocq (France), Volkmar Falk (Germany), Laurent Fauchier (France), Tobias Geisler (Germany), Diana A. Gorog (United Kingdom), Lene Holmvang (Denmark), Tiny Jaarsma (Sweden), Hywel Wynne Jones (United Kingdom), Lars Køber (Denmark), Konstantinos C. Koskinas (Switzerland), Dipak Kotecha (United Kingdom), Konstantin A. Krychtiuk (Austria), Ulf Landmesser (Germany), George Lazaros (Greece), Basil S. Lewis (Israel), Bertil Lindahl (Sweden), Ales Linhart (Czech Republic), Maja-Lisa Løchen (Norway), Mamas A. Mamas (United Kingdom), John William McEvoy (Ireland), Borislava Mihaylova (United Kingdom), Richard Mindham (United Kingdom), Christian Mueller (Switzerland), Lis Neubeck (United Kingdom), Josef Niebauer (Austria), Jens Cosedis Nielsen (Denmark), Alexander Niessner (Austria), Valeria Paradies (Netherlands), Agnes A. Pasquet (Belgium), Steffen E. Petersen (United Kingdom), Eva Prescott (Denmark), Amina Rakisheva (Kazakhstan), Bianca Rocca (Italy), Giuseppe M.C. Rosano (Italy), Leyla Elif Sade (United States of America / Türkiye), François Schiele (France), Jolanta M. Siller-Matula (Austria), Christian Sticherling (Switzerland), Robert F. Storey (United Kingdom), Matthias Thielmann (Germany), Christiaan Vrints (Belgium), Stephan Windecker (Switzerland), Rune Wiseth (Norway), and Adam Witkowski (Poland)

All experts involved in the development of these guidelines have submitted declarations of interest. These have been compiled in a report and simultaneously published in a supplementary document to the guidelines. The report is also available on the ESC website www.escardio.org/Guidelines

See the *European Heart Journal* online for supplementary documents that include background information and evidence tables.

Keywords Guidelines • Acute cardiac care • Acute coronary syndrome • Antithrombotic therapy • Fibrinolysis • High-sensitivity troponin • Invasive strategy • MINOCA • Myocardial infarction • Non-ST-elevation myocardial infarction • Patient-centred care • Percutaneous coronary intervention • Recommendations • Reperfusion therapy • Revascularization • Secondary prevention • ST-segment elevation myocardial infarction • Unstable angina

Table of contents

1. Preamble 3727
2. Introduction 3728
2.1. Definitions | Acute coronary syndromes and myocardial infarction 3730
2.2. Epidemiology of acute coronary syndromes 3732
2.3. Number and breakdown of classes of recommendations 3732
2.4. What is new 3733
3. Triage and diagnosis 3735
3.1. Clinical presentation and physical examination 3735
3.1.1. Clinical presentation 3735
3.1.2. History taking and physical examination 3737
3.2. Diagnostic tools | Electrocardiogram 3737
3.2.1. Acute coronary syndrome with persistent ST-segment elevation (suspected ST-elevation myocardial infarction) 3738
3.2.2. Acute coronary syndrome without persistent ST-segment elevation (non-ST elevation acute coronary syndrome) 3738
3.3. Diagnostic tools | Biomarkers 3739
3.3.1. High-sensitivity cardiac troponins 3739
3.3.2. Central laboratory vs. point of care 3739
3.3.3. Confounders of cardiac troponin concentration 3739
3.3.4. Rapid 'rule-in' and 'rule-out' algorithms 3739
3.3.4.1. European Society of Cardiology 0 h/1 h and 0 h/2 h algorithms 3740
3.3.4.1.1. Rule-out 3740
3.3.4.1.2. Rule-in 3740
3.3.4.1.3. Observe 3740
3.3.4.2. Practical guidance on how to implement the European Society of Cardiology 0 h/1 h algorithm 3742
3.3.5. Other biomarkers 3742
3.4. Diagnostic tools | Non-invasive imaging 3742
3.4.1. Echocardiography 3742
3.4.2. Computed tomography 3742
3.4.3. Cardiac magnetic resonance imaging with or without stress testing 3742
3.5. Differential diagnosis for acute chest pain 3743
4. Initial measures for patients presenting with suspected acute coronary syndrome | Initial treatment 3743
4.1. Pre-hospital logistics of care 3743
4.1.1. Time to treatment 3743

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

4.1.2. Healthcare systems and system delays 3743
4.1.3. Emergency medical services 3743
4.1.4. General practitioners 3744
4.1.5. Organization of ST-elevation myocardial infarction treatment in networks 3744
4.2. Emergency care 3744
4.2.1. Initial diagnosis and monitoring 3744
4.2.2. Acute pharmacotherapy 3744
4.2.2.1. Oxygen 3744
4.2.2.2. Nitrates 3744
4.2.2.3. Pain relief 3744
4.2.2.4. Intravenous beta-blockers 3744
5. Acute-phase management of patients with acute coronary syndrome 3745
5.1. Selection of invasive strategy and reperfusion therapy 3745
5.2. Acute coronary syndrome managed with invasive strategy 3745
5.2.1. Primary percutaneous coronary intervention strategy for ST-elevation myocardial infarction 3745
5.2.1.1. Invasive strategy in ST-elevation myocardial infarction late presenters 3747
5.2.2. Immediate invasive strategy for non-ST elevation acute coronary syndrome 3747
5.2.3. Routine vs. selective invasive strategy 3747
5.2.3.1. Early vs. delayed invasive strategy for non-ST elevation acute coronary syndrome 3747
5.2.4. Summary of invasive strategies for patients with non-ST elevation acute coronary syndrome 3747
5.3. Fibrinolysis and pharmaco-invasive strategy in patients with ST-elevation myocardial infarction 3749
5.3.1. Benefit and indication of fibrinolysis 3749
5.3.1.1. Pre-hospital fibrinolysis 3749
5.3.1.2. Angiography and percutaneous coronary intervention after fibrinolysis (pharmaco-invasive strategy) 3749
5.3.1.2.1. Comparison of fibrinolytic agents 3749
5.3.1.2.2. Hazards of fibrinolysis and contraindications 3749
5.4. Patients not undergoing reperfusion 3749
5.4.1. Patients who are not candidates for invasive coronary angiography 3749
5.4.2. Patients with coronary artery disease not amenable to revascularization 3749
6. Antithrombotic therapy 3750
6.1. Antiplatelet therapy in the acute phase 3752
6.1.1. Oral antiplatelet therapy 3752
6.1.2. Timing of loading dose of oral antiplatelet therapy 3753
6.1.2.1. Pre-treatment in patients with suspected ST-elevation myocardial infarction 3753
6.1.2.2. Pre-treatment in patients with non-ST-elevation acute coronary syndrome 3753
6.1.2.3. Summary of pre-treatment strategies 3753
6.1.3. Intravenous antiplatelet drugs 3753
6.2. Anticoagulant treatment in the acute phase 3754
6.2.1. Anticoagulation in patients with ST-elevation myocardial infarction undergoing primary percutaneous coronary intervention 3754
6.2.2. Anticoagulation in patients with non-ST-elevation acute coronary syndrome undergoing angiography and percutaneous coronary intervention if indicated 3754
6.3. Maintenance antithrombotic therapy after revascularization 3755
6.3.1. Shortening dual antiplatelet therapy 3756
6.3.2. De-escalation from potent $P2Y_{12}$ inhibitor to clopidogrel 3756
6.3.3. Summary of alternative antiplatelet strategies to reduce bleeding risk in the first 12 months after acute coronary syndrome 3757
6.4. Long-term treatment 3759
6.4.1. Prolonging antithrombotic therapy beyond 12 months 3759
6.5. Antiplatelet therapy in patients requiring oral anticoagulation 3760
6.5.1. Acute coronary syndrome patients requiring anticoagulation 3760
6.5.2. Patients requiring vitamin K antagonists or undergoing coronary artery bypass surgery 3762
6.6. Antithrombotic therapy as an adjunct to fibrinolysis 3762
6.7. Antithrombotic therapy in patients not undergoing reperfusion 3762
7. Acute coronary syndrome with unstable presentation 3762
7.1. Out-of-hospital cardiac arrest in acute coronary syndrome 3763
7.1.1. Systems of care 3763
7.2. Cardiogenic shock complicating acute coronary syndrome 3763
8. Management of acute coronary syndrome during hospitalization 3764
8.1. Coronary care unit/intensive cardiac care unit 3764
8.1.1. Monitoring 3764
8.1.2. Ambulation 3764
8.1.3. Length of stay in the intensive cardiac care unit 3765
8.2. In-hospital care 3765
8.2.1. Length of hospital stay 3765
8.2.2. Risk assessment 3765
8.2.2.1. Clinical risk assessment 3765
8.2.2.2. Imaging risk assessment 3765
8.2.2.3. Biomarkers for risk assessment 3765
8.2.2.4. Bleeding risk assessment 3765
8.2.2.5. Integrating ischaemic and bleeding risks 3765
9. Technical aspects of invasive strategies 3766
9.1. Percutaneous coronary intervention 3766
9.1.1. Vascular access 3766
9.1.2. Intravascular imaging/physiology of the infarct-related artery 3766
9.1.2.1. Intravascular imaging 3766
9.1.2.2. Intravascular physiology 3767
9.1.3. Timing of revascularization with percutaneous coronary intervention 3767
9.1.4. Balloons and stents 3768
9.1.5. Embolic protection and microvascular salvage strategies 3768
9.1.5.1. Thrombus aspiration 3768
9.1.5.2. Interventions to protect the microcirculation 3768
9.2. Coronary artery bypass grafting 3768
9.2.1. Indication and timing of coronary artery bypass grafting in acute coronary syndrome patients 3768
9.2.2. Technical considerations specific to acute coronary syndrome patients 3768
9.3. Spontaneous coronary artery dissection 3768
9.3.1. Intravascular imaging 3769
9.3.2. Revascularization 3769
10. Management of patients with multivessel disease 3769
10.1. Management of multivessel disease in acute coronary syndrome complicated by cardiogenic shock 3769

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

10.2. Patients with multivessel coronary artery disease undergoing primary percutaneous coronary intervention 3770
10.3. Timing of non-infarct-related artery revascularization in acute coronary syndrome 3771
10.3.1. Patients presenting with ST-elevation myocardial infarction and multivessel coronary artery disease 3771
10.3.2. Patients presenting with non-ST-elevation acute coronary syndrome and multivessel coronary artery disease . 3771
10.4. Evaluation of non-infarct-related artery stenosis severity (angiography vs. physiology) 3771
10.5. Hybrid revascularization 3772
11. Myocardial infarction with non-obstructive coronary arteries . 3772
12. Special situations 3775
12.1. Type 2 myocardial infarction and acute myocardial injury 3775
12.2. Complications 3775
12.2.1. Heart failure 3775
12.2.2. Mechanical complications 3776
12.2.3. Left ventricular thrombus 3776
12.2.4. Post-acute coronary syndrome pericarditis 3776
12.2.5. Arrhythmias 3776
12.2.5.1. Atrial fibrillation 3776
12.2.5.2. Ventricular arrhythmias 3777
12.2.6. Bleeding 3777
12.2.6.1. Management of bleeding 3777
12.3. Comorbid conditions 3778
12.3.1. Patients at high bleeding risk and with blood disorders (anaemia and thrombocytopaenia) 3778
12.3.2. Chronic kidney disease 3779
12.3.3. Diabetes mellitus 3779
12.3.4. Older adults with frailty and multimorbidity 3779
12.3.4.1. The older person 3779
12.3.4.2. Frailty and multimorbidity 3779
12.3.5. Pregnancy 3780
12.3.6. Drug abuse 3780
12.3.7. Patients with cancer 3780
12.3.8. Coronavirus disease (COVID-19) 3780
13. Long-term treatment 3781
13.1. Cardiac rehabilitation 3783
13.1.1. Comprehensive cardiac rehabilitation 3783
13.1.2. Digital health 3783
13.1.3. Adherence and persistence 3783
13.2. Lifestyle management 3783
13.2.1. Tobacco 3783
13.2.2. Nutrition and alcohol 3783
13.2.3. Physical activity and exercise 3784
13.2.4. Psychological considerations 3784
13.2.5. Resumption of activities 3784
13.3. Pharmacological treatment 3784
13.3.1. Antithrombotic therapy 3784
13.3.2. Lipid-lowering therapy 3784
13.3.3. Beta-blockers 3785
13.3.4. Nitrates and calcium channel blockers 3786
13.3.5. Renin–angiotensin–aldosterone system inhibitors 3786
13.3.6. Medications for diabetes 3786
13.3.6.1. Sodium–glucose co-transporter 2 inhibitors 3786
13.3.6.2. Glucagon-like peptide-1 receptor agonists 3787
13.3.7. Proton pump inhibitors 3787
13.3.8. Vaccination 3787
13.3.9. Anti-inflammatory drugs 3787
13.3.10. Hormone replacement therapy 3787
14. Patient perspectives 3788
14.1. Patient-centred care 3788
14.2. Shared decision-making 3789
14.3. Informed consent 3789
14.4. Research participation and consent in the acute setting 3790
14.5. Patient satisfaction and expectations 3790
14.6. Patient-reported outcome measures and patient-reported experience measures 3791
14.7. Preparation for discharge 3791
15. Key messages 3791
16. Gaps in evidence 3793
17. Sex differences 3795
18. 'What to do' and 'What not to do' messages from the Guidelines 3796
19. Quality indicators 3801
20. Supplementary data 3801
21. Data availability statement 3801
22. Author information 3801
23. Appendix 3801
24. References 3802

Tables of Recommendations

Recommendation Table 1 — Recommendations for clinical and diagnostic tools for patients with suspected acute coronary syndrome 3738
Recommendation Table 2 — Recommendations for non-invasive imaging in the initial assessment of patients with suspected acute coronary syndrome 3742
Recommendation Table 3 — Recommendations for the initial management of patients with acute coronary syndrome 3745
Recommendation Table 4 — Recommendations for reperfusion therapy and timing of invasive strategy 3750
Recommendation Table 5 — Recommendations for antiplatelet and anticoagulant therapy in acute coronary syndrome 3758
Recommendation Table 6 — Recommendations for alternative antithrombotic therapy regimens 3759
Recommendation Table 7 — Recommendations for fibrinolytic therapy 3762
Recommendation Table 8 — Recommendations for cardiac arrest and out-of-hospital cardiac arrest 3763
Recommendation Table 9 — Recommendations for cardiogenic shock 3764
Recommendation Table 10 — Recommendations for in-hospital management 3766
Recommendation Table 11 — Recommendations for technical aspects of invasive strategies 3769
Recommendation Table 12 — Recommendations for management of patients with multivessel disease 3772
Recommendation Table 13 — Recommendations for myocardial infarction with non-obstructive coronary arteries 3775
Recommendation Table 14 — Recommendations for acute coronary syndrome complications 3777
Recommendation Table 15 — Recommendations for acute coronary syndrome comorbid conditions 3781

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Recommendation Table 16 — Recommendations for long-term management 3787
Recommendation Table 17 — Recommendations for patient perspectives in acute coronary syndrome care 3791

List of tables

Table 1 Classes of recommendations 3728
Table 2 Levels of evidence 3728
Table 3 Definitions of terms related to invasive strategy and reperfusion therapy commonly used in this document 3731
Table 4 New recommendations 3733
Table 5 Revised recommendations 3734
Table 6 Dose regimen of antiplatelet and anticoagulant drugs in acute coronary syndrome patients 3751
Table 7 Suggested strategies to reduce bleeding risk related to percutaneous coronary intervention 3760
Table 8 Gaps in evidence 3793
Table 9 'What to do' and 'What not to do' 3796

List of figures

Figure 1 Central illustration 3729
Figure 2 The spectrum of clinical presentations, electrocardiographic findings, and high-sensitivity cardiac troponin levels in patients with acute coronary syndrome 3730
Figure 3 Classification of patients presenting with suspected acute coronary syndrome: from a working to a final diagnosis 3732
Figure 4 An overview of the initial triage, management and investigation of patients who present with signs and symptoms potentially consistent with acute coronary syndrome 3736
Figure 5 The A.C.S. assessment for the initial evaluation of patients with suspected acute coronary syndrome 3737
Figure 6 The 0 h/1 h or 0 h/2 h rule-out and rule-in algorithms using high-sensitivity cardiac troponin assays in patients presenting to the emergency department with suspected NSTEMI and without an indication for immediate invasive angiography 3741
Figure 7 Modes of presentation and pathways to invasive management and myocardial revascularization in patients presenting with STEMI 3746
Figure 8 Selection of invasive strategy and reperfusion therapy in patients presenting with NSTE-ACS 3748
Figure 9 Antithrombotic treatments in acute coronary syndrome: pharmacological targets 3752
Figure 10 Recommended default antithrombotic therapy regimens in acute coronary syndrome patients without an indication for oral anticoagulation 3755
Figure 11 Alternative antiplatelet strategies to reduce bleeding risk in the first 12 months after an ACS 3757
Figure 12 Antithrombotic regimens in patients with acute coronary syndrome and an indication for oral anticoagulation 3761
Figure 13 A practical algorithm to guide intravascular imaging in acute coronary syndrome patients 3767
Figure 14 Algorithm for the management of acute coronary syndrome patients with multivessel coronary artery disease 3770
Figure 15 Underlying causes for patients with a working diagnosis of myocardial infarction with non-obstructive coronary arteries 3773
Figure 16 Evaluation of patients with a working diagnosis of MINOCA 3774
Figure 17 Long-term management after acute coronary syndrome 3782
Figure 18 Lipid-lowering therapy in ACS patients 3785
Figure 19 A person-centred approach to the ACS journey 3789
Figure 20 Acute coronary syndrome patient expectations 3790

Abbreviations and acronyms

AβYSS	Beta Blocker Interruption After Uncomplicated Myocardial Infarction
ACCOAST	A Comparison of Prasugrel at the Time of Percutaneous Coronary Intervention or as Pretreatment at the Time of Diagnosis in Patients with Non-ST Elevation Myocardial Infarction
ACE	Angiotensin-converting enzyme
ACS	Acute coronary syndrome
AF	Atrial fibrillation
AFIRE	Atrial Fibrillation and Ischemic Events With Rivaroxaban in Patients With Stable Coronary Artery Disease
AMI	Acute myocardial infarction
ARB	Angiotensin receptor blocker
ARC-HBR	Academic Research Consortium for High Bleeding Risk
ARNI	Angiotensin receptor/neprilysin inhibitor
ASCVD	Atherosclerotic cardiovascular disease
ASSENT 3	ASsessment of the Safety and Efficacy of a New Thrombolytic 3
ATLANTIC	Administration of Ticagrelor in the Cath Lab or in the Ambulance for New ST Elevation Myocardial Infarction to Open the Coronary Artery
AUGUSTUS	An Open-Label, 2 × 2 Factorial, Randomized Controlled, Clinical Trial to Evaluate the Safety of Apixaban Versus Vitamin K Antagonist and Aspirin Versus Aspirin Placebo in Patients With Atrial Fibrillation and Acute Coronary Syndrome or Percutaneous Coronary Intervention
AV	Atrioventricular
BARC	Bleeding Academic Research Consortium
b.i.d.	*Bis in die* (twice a day)
BBB	Bundle branch block
BEACON	Better Evaluation of Acute Chest Pain with Coronary Computed Tomography Angiography
BETAMI	BEtablocker Treatment After Acute Myocardial Infarction in Patients Without Reduced Left Ventricular Systolic Function
BMS	Bare metal stent
BNP	Brain natriuretic peptide
CABG	Coronary artery bypass grafting
CAD	Coronary artery disease
CAPITAL-RCT	Carvedilol Post-Intervention Long-Term Administration in Large-scale Randomized Controlled Trial
CAPRICORN	CArvedilol Post-infaRct survIval COntRolled evaluatioN
CCS	Chronic coronary syndrome

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

CCTA	Coronary computed tomography angiography
CCU	Coronary care unit
CHA_2DS_2-VASc	Congestive heart failure, Hypertension, Age, Diabetes, Stroke or TIA-Vascular disease
CHAMPION PCI	Cangrelor versus Standard Therapy to Achieve Optimal Management of Platelet Inhibition
CHAMPION PHOENIX	A Clinical Trial Comparing Cangrelor to Clopidogrel Standard Therapy in Subjects Who Require Percutaneous Coronary Intervention
CHAMPION PLATFORM	Cangrelor Versus Standard Therapy to Achieve Optimal Management of Platelet Inhibition
CKD	Chronic kidney disease
CMR	Cardiac magnetic resonance
CI	Confidence interval
COACT	Coronary Angiography after Cardiac Arrest
COLCOT	Colchicine Cardiovascular Outcomes Trial
COMFORTABLE-AMI	Comparison of Biolimus Eluted From an Erodible Stent Coating With Bare Metal Stents in Acute ST-Elevation Myocardial Infarction
COMPARE-ACUTE	Comparison Between FFR Guided Revascularization Versus Conventional Strategy in Acute STEMI Patients With MVD
COMPASS	Cardiovascular Outcomes for People Using Anticoagulation Strategies
COMPLETE	Complete vs. Culprit-only Revascularization to Treat Multivessel Disease After Early PCI for STEMI
COVID-19	Coronavirus disease 2019
CR	Cardiac rehabilitation
CRT	Cardiac resynchronization therapy—defibrillator/pacemaker
CS	Cardiogenic shock
CT	Computed tomography
CV	Cardiovascular
CVD	Cardiovascular disease
CvLPRIT	Complete versus Lesion-only Primary PCI Trial
cTn	Cardiac troponin
CULPRIT-SHOCK	Culprit Lesion Only PCI versus Multivessel PCI in Cardiogenic Shock
DANAMI-3–PRIMULTI	Third Danish Study of Optimal Acute Treatment of Patients with ST-Segment Elevation Myocardial Infarction—Primary PCI in Multivessel Disease
DANBLOCK	Danish Trial of Beta Blocker Treatment After Myocardial Infarction Without Reduced Ejection Fraction
DAPT	Dual antiplatelet therapy
DAT	Dual antithrombotic therapy
DCB	Drug-coated balloon
DES	Drug-eluting stent(s)
DM	Diabetes mellitus
ECG	Electrocardiography/gram
ECMO	Extracorporeal membrane oxygenation
eGFR	Estimated glomerular filtration rate
ED	Emergency department
EMS	Emergency medical service(s)
EPHESUS	Eplerenone Post-AMI Heart failure Efficacy and SUrvival Study
ESC	European Society of Cardiology
EXAMINATION	Everolimus-Eluting Stents Versus Bare-Metal Stents in ST Segment Elevation Myocardial Infarction
ExTRACT-TIMI 25	Enoxaparin and Thrombolysis Reperfusion for Acute myocardial infarction Treatment Thrombolysis In Myocardial Infarction—Study 25
FAME	Fractional Flow Reserve versus Angiography for Multivessel Evaluation
FAMOUS-NSTEMI	Fractional flow reserve (FFR) versus angiography in guiding management to optimise outcomes in non-ST segment elevation myocardial infarction
FAST-MI	French Registry of Acute ST-elevation and non-ST-elevation Myocardial Infarction
FFR	Fractional flow reserve
FLOWER-MI	Flow Evaluation to Guide Revascularization in Multivessel ST-Elevation Myocardial Infarction
FMC	First medical contact
GLP-1RA	Glucagon-like peptide-1 receptor agonist
GP	Glycoprotein
GRACE	Global Registry of Acute Coronary Events
HBR	High bleeding risk
HCR	Hybrid coronary revascularization
HF	Heart failure
HFrEF	Heart failure with reduced ejection fraction
HOST-REDUCE-P-OLYTECH-ACS	Harmonizing Optimal Strategy for Treatment of Coronary Artery Diseases Trial—Comparison of REDUCTION of PrasugrEl Dose & POLYmer TECHnology in ACS Patients
HR	Hazard ratio
HR-QoL	Health-related quality of life
hs-cTn	High-sensitivity cardiac troponin
IABP	Intra-aortic balloon counter pulsation/pumping
IABP-SHOCK II	Intraaortic Balloon Pump in Cardiogenic Shock II
ICA	Invasive coronary angiography
ICCU	Intensive cardiac care unit
ICD	Implantable cardioverter defibrillator
ICU	Intensive care unit
IMPROVE-IT	Improved Reduction of Outcomes: Vytorin Efficacy International Trial
INR	International normalized ratio
IRA	Infarct-related artery
ISAR-REACT 5	Intracoronary stenting and Antithrombotic regimen Rapid Early Action for Coronary Treatment
ISIS-4	Fourth International Study of Infarct Survival
i.v.	Intravenous
IVUS	Intravascular ultrasound
LAD	Left anterior descending
LBBB	Left bundle branch block
LD	Loading dose
LDL-C	Low-density lipoprotein-cholesterol
LIMA	Left internal mammary artery
LMWH	Low-molecular-weight heparin
LoDoCo2	Low-dose Colchicine trial-2
LV	Left ventricular(cle)
LVAD	Left ventricular assist device
LVEF	Left ventricular ejection fraction
MACE	Major adverse cardiovascular events

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

MASTER DAPT	Management of High Bleeding Risk Patients Post Bioresorbable Polymer Coated Stent Implantation With an Abbreviated Versus Prolonged DAPT Regimen
MATRIX	Minimizing Adverse Haemorrhagic Events by Transradial Access Site and Systemic Implementation of angioX
MCS	Mechanical circulatory support
MD	Maintenance dose
MI	Myocardial infarction
MINOCA	Myocardial infarction with non-obstructive coronary arteries
MRA	Mineralocorticoid receptor antagonist
MVD	Multivessel disease
MVO	Microvascular obstruction
NOAC	Non-vitamin K antagonist oral anticoagulant
NORSTENT	Norwegian Coronary Stent Trial
NPV	Negative predictive value
NRT	Nicotine replacement therapy
NSTE	Non-ST elevation
NSTE-ACS	Non-ST elevation acute coronary syndrome
NSTEMI	Non-ST-elevation myocardial infarction
NT-pro BNP	N-terminal pro B-type natriuretic peptide
NYHA	New York Heart Association
o.d.	Once a day
OAC	Oral anticoagulant/ation
OASIS-5	Fifth Organization to Assess Strategies in Acute Ischemic Syndromes
OASIS-6	The Safety and Efficacy of Fondaparinux Versus Control Therapy in Patients With ST Segment Elevation Acute Myocardial Infarction
OAT	Occluded Artery Trial
OCT	Optical coherence tomography
ODYSSEY OUTCOMES	Evaluation of Cardiovascular Outcomes After an Acute Coronary Syndrome During Treatment With Alirocumab
OHCA	Out-of-hospital cardiac arrest
OR	Odds ratio
PARADISE-MI	Prospective ARNI vs ACE Inhibitor Trial to Determine Superiority in Reducing Heart Failure Events After MI
PCI	Percutaneous coronary intervention
PCSK9	Proprotein convertase subtilisin/kexin type 9
PE	Pulmonary embolism
PEGASUS-TIMI 54	PrEvention with TicaGrelor of SecondAry Thrombotic Events in High-RiSk Patients with Prior AcUte Coronary Syndrome—Thrombolysis In Myocardial Infarction
PEPCAD NSTEMI	Bare Metal Stent Versus Drug Coated Balloon With Provisional Stenting in Non-ST-Elevation Myocardial Infarction
PLATO	PLATelet inhibition and patient Outcomes
POC	Point of care
POPular Genetics	Cost-effectiveness of CYP2C19 Genotype Guided Treatment With Antiplatelet Drugs in Patients With ST-segment-elevation Myocardial Infarction Undergoing Immediate PCI With Stent Implantation: Optimization of Treatment
PPCI	Primary percutaneous coronary intervention
PPI	Proton pump inhibitor
PPV	Positive predictive value
PRAMI	Preventive Angioplasty in Myocardial Infarction
PREM	Patient-reported experience measure
PROM	Patient-reported outcome measure
QI	Quality indicator
RAAS	Renin–angiotensin–aldosterone system
RAPID-CTCA	Rapid Assessment of Potential Ischaemic heart Disease with CTCA
RCT	Randomized controlled trial
REALITY	Restrictive and Liberal Transfusion Strategies in Patients With Acute Myocardial Infarction
REBOOT-CNIC	TREatment With Beta-blockers After myOcardial Infarction withOut Reduced Ejection fracTion
REDUCE-SWEDE-HEART	Evaluation of Decreased Usage of Betablockers After Myocardial Infarction in the SWEDEHEART Registry
REMINDER	Double-Blind, Randomized, Placebo-Controlled Trial Evaluating The Safety And Efficacy Of Early Treatment With Eplerenone In Patients With Acute Myocardial Infarction
REVELATION	REVascularization With PaclitaxEL-Coated Balloon Angioplasty Versus Drug-Eluting Stenting in Acute Myocardial InfarcTION
RIVAL	Radlal Vs femorAL access for coronary intervention
ROMICAT II	Multicenter Study to Rule Out Myocardial Infarction by Cardiac Computed Tomography
ROSC	Return of spontaneous circulation
RR	Relative risk
RV	Right ventricular
SAPT	Single antiplatelet therapy
SBP	Systolic blood pressure
s.c.	Subcutaneous
SCAD	Spontaneous coronary artery dissection
SHOCK	Should We Emergently Revascularize Occluded Coronaries for Cardiogenic Shock
SGLT2	Sodium–glucose co-transporter 2
SMART-DECISION	Long-term Beta-blocker Therapy After Acute Myocardial Infarction
SPECT	Single-photon emission computerized tomography
STE	ST elevation
STEMI	ST-elevation myocardial infarction
STOPDAPT-2-ACS	ShorT and OPtimal Duration of Dual AntiPlatelet Therapy-2 Study for the Patients With ACS
STREAM	Strategic Reperfusion Early After Myocardial Infarction
SWEDEHEART	Swedish Web-System for Enhancement and Development of Evidence-Based Care in Heart Disease Evaluated According to Recommended Therapies
TALOS-AMI	TicAgrelor Versus CLOpidogrel in Stabilized Patients With Acute Myocardial Infarction
TAT	Triple antithrombotic therapy

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

TICO	Ticagrelor Monotherapy After 3 Months in the Patients Treated With New Generation Sirolimus Stent for Acute Coronary Syndrome
TIMI	Thrombolysis In Myocardial Infarction
TLR	Target lesion revascularization
TOMAHAWK	Immediate Unselected Coronary Angiography Versus Delayed Triage in Survivors of Out-of-hospital Cardiac Arrest Without ST-segment Elevation
TOPIC	Timing of Platelet Inhibition After Acute Coronary Syndrome
TOTAL	Trial of routine aspiration ThrOmbecTomy with PCI vs. PCI ALone in patients with STEMI
TRITON-TIMI 38	TRial to Assess Improvement in Therapeutic Outcomes by Optimizing Platelet InhibitioN with Prasugrel Thrombolysis In Myocardial Infarction 38
TROPICAL-ACS	Testing Responsiveness to Platelet Inhibition on Chronic Antiplatelet Treatment For Acute Coronary Syndromes
TTE	Transthoracic echocardiography
TWILIGHT	Ticagrelor With Aspirin or Alone in High-Risk Patients After Coronary Intervention
UA	Unstable angina
UFH	Unfractionated heparin
VA-ECMO	Veno-arterial extracorporeal membrane oxygenation
VALIANT	VALsartan In Acute myocardial iNfarcTion
VF	Ventricular fibrillation
VKA	Vitamin K antagonist
VT	Ventricular tachycardia

1. Preamble

Guidelines evaluate and summarize available evidence with the aim of assisting health professionals in proposing the best diagnostic or therapeutic approach for an individual patient with a given condition. Guidelines are intended for use by health professionals and the European Society of Cardiology (ESC) makes its Guidelines freely available.

ESC Guidelines do not override the individual responsibility of health professionals to make appropriate and accurate decisions in consideration of each patient's health condition and in consultation with that patient or the patient's caregiver where appropriate and/or necessary. It is also the health professional's responsibility to verify the rules and regulations applicable in each country to drugs and devices at the time of prescription, and, where appropriate, to respect the ethical rules of their profession.

ESC Guidelines represent the official position of the ESC on a given topic and are regularly updated. ESC Policies and Procedures for formulating and issuing ESC Guidelines can be found on the ESC website (https://www.escardio.org/Guidelines).

The Members of this Task Force were selected by the ESC to represent professionals involved with the medical care of patients with this pathology. The selection procedure aimed to include members from across the whole of the ESC region and from relevant ESC Subspecialty Communities. Consideration was given to diversity and inclusion, notably with respect to gender and country of origin. The Task Force performed a critical evaluation of diagnostic and therapeutic approaches, including assessment of the risk-benefit ratio. The strength of every recommendation and the level of evidence supporting them were weighed and scored according to predefined scales as outlined below. The Task Force followed ESC voting procedures, and all approved recommendations were subject to a vote and achieved at least 75% agreement among voting members.

The experts of the writing and reviewing panels provided declaration of interest forms for all relationships that might be perceived as real or potential sources of conflicts of interest. Their declarations of interest were reviewed according to the ESC declaration of interest rules and can be found on the ESC website (http://www.escardio.org/Guidelines) and have been compiled in a report published in a supplementary document with the guidelines. The Task Force received its entire financial support from the ESC without any involvement from the healthcare industry.

The ESC Clinical Practice Guidelines (CPG) Committee supervises and co-ordinates the preparation of new guidelines and is responsible for the approval process. ESC Guidelines undergo extensive review by the CPG Committee and external experts, including members from across the whole of the ESC region and from relevant ESC Subspecialty Communities and National Cardiac Societies. After appropriate revisions, the guidelines are signed off by all the experts involved in the Task Force. The finalized document is signed off by the CPG Committee for publication in the *European Heart Journal*. The guidelines were developed after careful consideration of the scientific and medical knowledge and the evidence available at the time of their writing. Tables of evidence summarizing the findings of studies informing development of the guidelines are included. The ESC warns readers that the technical language may be misinterpreted and declines any responsibility in this respect.

Off-label use of medication may be presented in this guideline if a sufficient level of evidence shows that it can be considered medically appropriate for a given condition. However, the final decisions concerning an individual patient must be made by the responsible health professional giving special consideration to:

- The specific situation of the patient. Unless otherwise provided for by national regulations, off-label use of medication should be limited to situations where it is in the patient's interest with regard to the quality, safety, and efficacy of care, and only after the patient has been informed and has provided consent.
- Country-specific health regulations, indications by governmental drug regulatory agencies, and the ethical rules to which health professionals are subject, where applicable.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Table 1 **Classes of recommendations**

Classes of recommendations	Definition	Wording to use
Class I	Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective.	Is recommended or is indicated
Class II	Conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure.	
Class IIa	Weight of evidence/opinion is in favour of usefulness/efficacy.	Should be considered
Class IIb	Usefulness/efficacy is less well established by evidence/opinion.	May be considered
Class III	Evidence or general agreement that the given treatment or procedure is not useful/effective, and in some cases may be harmful.	Is not recommended

©ESC 2023

Table 2 **Levels of evidence**

Level of evidence A	Data derived from multiple randomized clinical trials or meta-analyses.
Level of evidence B	Data derived from a single randomized clinical trial or large non-randomized studies.
Level of evidence C	Consensus of opinion of the experts and/or small studies, retrospective studies, registries.

©ESC 2023

2. Introduction

The major aspects of the management of patients with acute coronary syndromes described in this European Society of Cardiology (ESC) Guideline are summarized in *Figure 1*.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

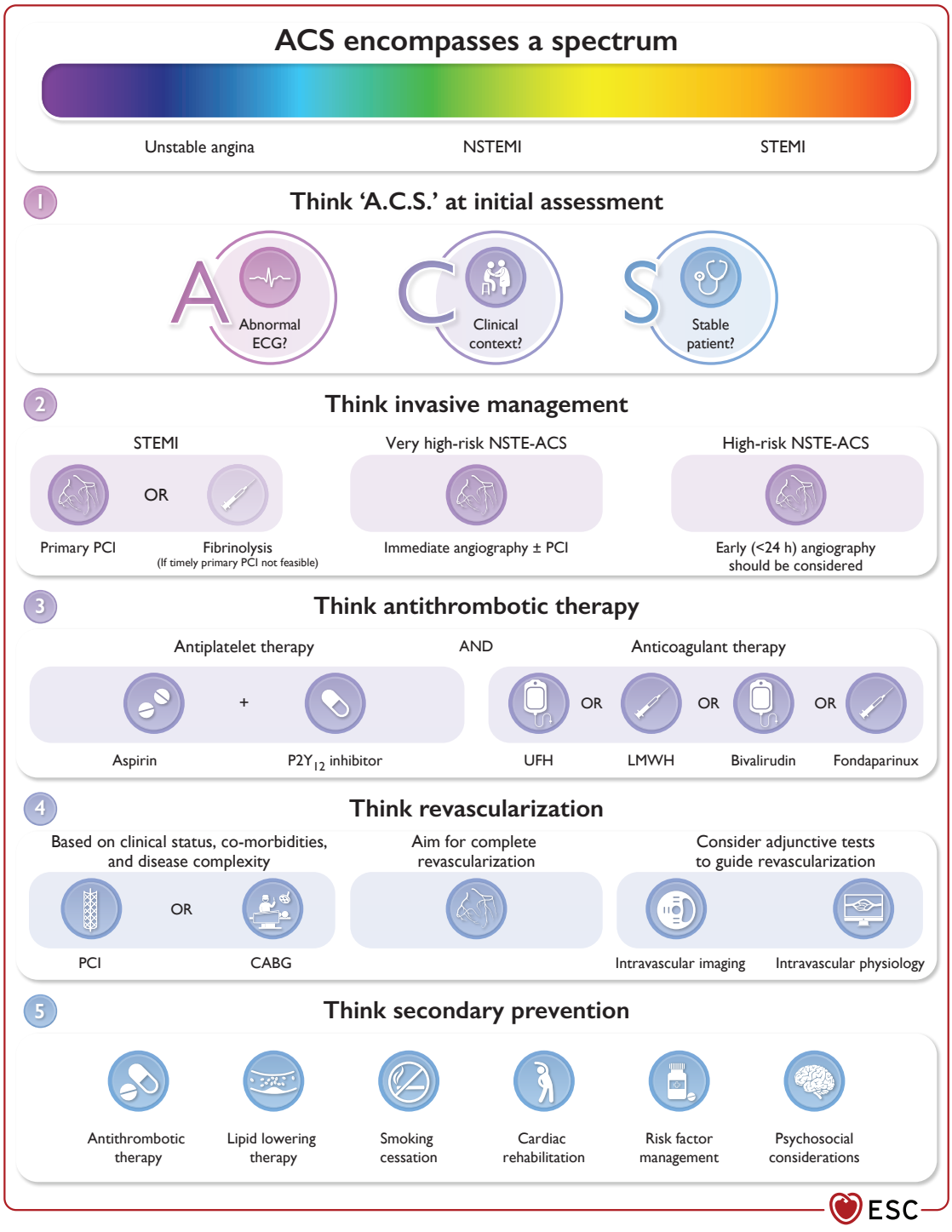


Figure 1 Central illustration. ACS, acute coronary syndrome; CABG, coronary artery bypass grafting; ECG, electrocardiogram; LMWH, low molecular-weight heparin; NSTE-ACS, non-ST-elevation acute coronary syndrome; PCI, percutaneous coronary intervention; PPCI, primary percutaneous coronary intervention; STEMI, ST-elevation myocardial infarction; UFH, unfractionated heparin. Patients with acute coronary syndrome (ACS) can initially present with a wide variety of clinical signs and symptoms and it is important that there is a high degree of awareness of this amongst both the general public and healthcare providers. If ACS is suspected, think 'A.C.S.' for the initial triage and assessment. This involves performing an electrocardiogram (ECG) to assess for **A**bnormalities or evidence of ischaemia, taking a targeted clinical history to assess the clinical **C**ontext of the presentation, and carrying out a targeted clinical examination to assess for clinical and haemodynamic **S**tability. Based on the initial assessment, the healthcare provider can decide whether immediate invasive management is required. Patients with ST-elevation myocardial infarction (STEMI) require primary percutaneous coronary intervention (PPCI) (or fibrinolysis if PPCI within 120 min is not feasible); patients with non-ST-elevation ACS (NSTE-ACS) with very high-risk features require immediate angiography ± PCI if indicated; patients with NSTE-ACS and high-risk features should undergo inpatient angiography (angiography within 24 h should be considered). A combination of antiplatelet and anticoagulant therapy is indicated acutely for patients with ACS. The majority of patients with ACS will eventually undergo revascularization, most commonly with PCI. Once the final diagnosis of ACS has been established, it is important to implement measures to prevent recurrent events and to optimize cardiovascular risk. This consists of medical therapy, lifestyle changes and cardiac rehabilitation, as well as consideration of psychosocial factors.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

2.1. Definitions | Acute coronary syndromes and myocardial infarction

Acute coronary syndromes (ACS) encompass a spectrum of conditions that include patients presenting with recent changes in clinical symptoms or signs, with or without changes on 12-lead electrocardiogram (ECG) and with or without acute elevations in cardiac troponin (cTn) concentrations (*Figure 2*). Patients presenting with suspected ACS may eventually receive a diagnosis of acute myocardial infarction (AMI) or unstable angina (UA). The diagnosis of myocardial infarction (MI) is associated with cTn release and is made based on the fourth universal definition of MI.[1] UA is defined as myocardial ischaemia at rest or on minimal exertion in the absence of acute cardiomyocyte injury/necrosis. It is characterized by specific clinical findings of prolonged (>20 min) angina at rest; new onset of severe angina; angina that is increasing in frequency, longer in duration, or lower in threshold; or angina that occurs after a recent episode of MI. ACS are associated with a broad range of clinical presentations, from patients who are symptom free at presentation to patients with ongoing chest discomfort/symptoms and patients

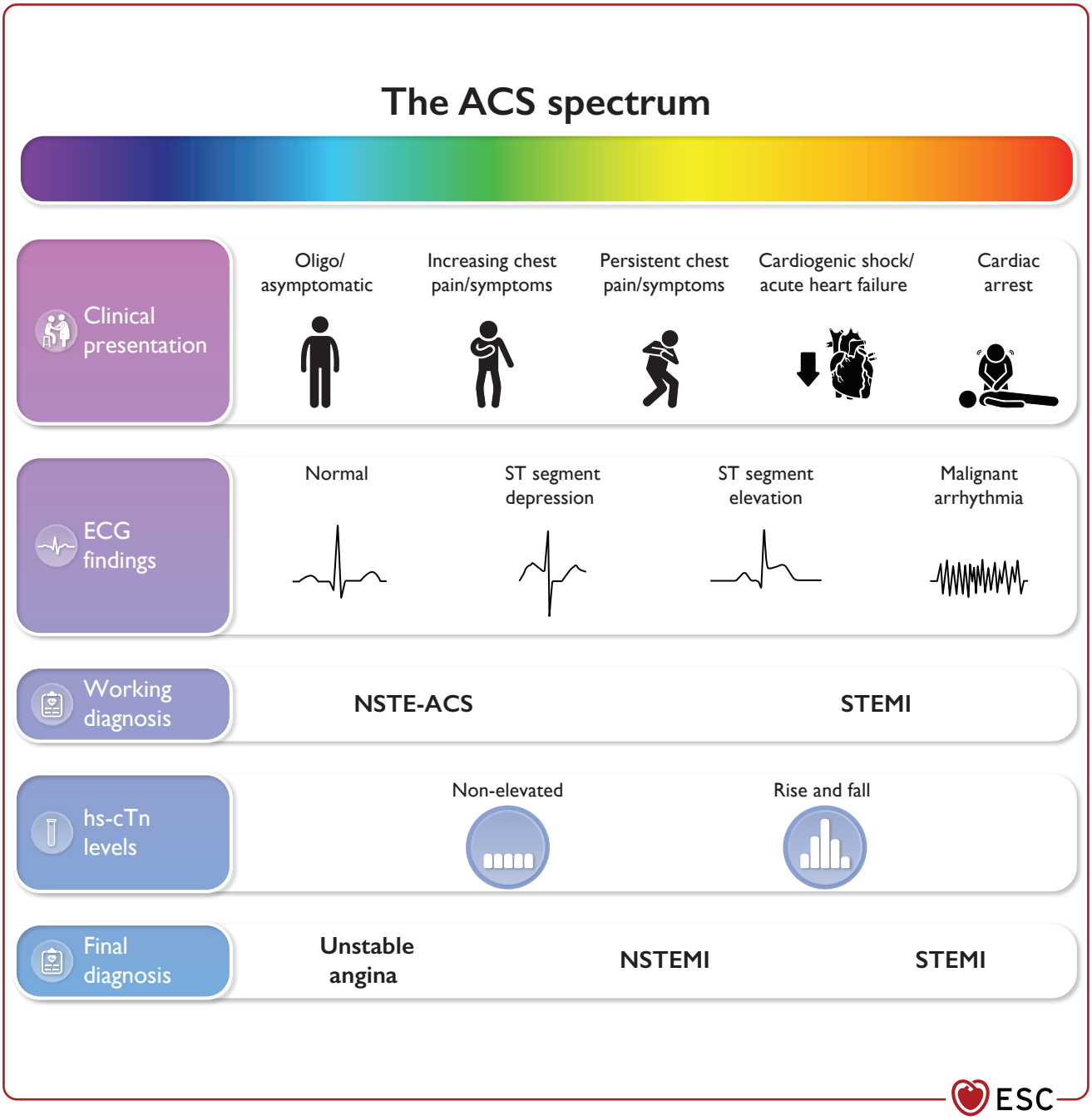


Figure 2 The spectrum of clinical presentations, electrocardiographic findings, and high-sensitivity cardiac troponin levels in patients with acute coronary syndrome. ACS, acute coronary syndrome; ECG, electrocardiogram; hs-cTn, high-sensitivity cardiac troponin; NSTE-ACS, non-ST-elevation acute coronary syndrome; NSTEMI, non-ST-elevation myocardial infarction; STEMI, ST-elevation myocardial infarction.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

with cardiac arrest, electrical/haemodynamic instability, or cardiogenic shock (CS) (*Figure 2*).

Patients presenting with suspected ACS are typically classified based on ECG at presentation for the purposes of initial management. After this, patients can be further classified based on the presence or absence of cardiac troponin elevation (once these results are available), as demonstrated in *Figures 2* and *3*. These features (ECG changes and cardiac troponin elevation) are important in the initial triage and diagnosis of patients with ACS, helping to risk stratify patients and guide the initial management strategy. However, after the acute management and stabilization phase, most aspects of the subsequent management strategy are common to all patients with ACS (regardless of the initial ECG pattern or the presence/absence of cardiac troponin elevation at presentation) and can therefore be considered under a common pathway. A glossary of the terms related to invasive strategies and reperfusion therapy commonly used in this document, and their associated definitions, is provided in *Table 3*.

While they are closely related, it is important to recognize that ACS is not the same as MI.[1] AMI is defined as cardiomyocyte necrosis in the clinical setting of acute myocardial ischaemia. This includes MI due to atherothrombotic events (Type 1 MI) and also other potential causes of myocardial ischaemia and myocyte necrosis (Type 2–5 MI) (Supplementary data online, *Table S1*). Myocardial injury is another distinct entity, used to describe troponin release due to mechanisms other than myocardial ischaemia and not meeting the criteria for MI outlined in Supplementary data online, *Table S1*. Myocardial injury can be acute or chronic depending on whether there is evidence of dynamic change in the elevated troponins on serial testing. Some causes of myocardial injury include myocarditis, sepsis, takotsubo cardiomyopathy, heart valve disease, cardiac arrhythmias, and heart failure (HF).

The focus of this guideline is largely centred on the management of patients who will eventually receive a diagnosis of Type 1 MI. However, at every stage of the management of patients presenting with ACS, physicians must carefully consider other differential diagnoses in their clinical assessment because they are common, associated with different underlying pathological mechanisms, have different prognoses, and frequently require different treatment approaches. More information is provided in the Supplementary data online. In general, detailed information regarding the results of individual trials will not be provided in the main guideline. However, where appropriate, this information is provided in the Supplementary data online evidence tables.

Table 3 Definitions of terms related to invasive strategy and reperfusion therapy commonly used in this document

Term	Definition
First medical contact (FMC)	The time point when the patient is initially assessed by a physician, paramedic, nurse, or other trained emergency medical services worker who can obtain and interpret the ECG and deliver initial interventions (e.g. defibrillation). FMC can be either in the pre-hospital setting or upon patient arrival at the hospital (e.g. the emergency department)
STEMI diagnosis	The time at which a patient with ischaemic symptoms is interpreted as presenting with ACS and ST-segment elevation (or ST-segment elevation equivalent)
Primary PCI[a]	Emergent PCI with balloon, stent, or other approved device, performed on the IRA without previous fibrinolytic treatment
Primary PCI strategy[a]	Emergency coronary angiography and PCI of the IRA if indicated
Rescue PCI[a]	Emergency PCI performed as soon as possible in cases of failed fibrinolytic treatment
Routine early PCI strategy after fibrinolysis[a]	Coronary angiography, with PCI of the IRA if indicated, performed between 2 h and 24 h after successful fibrinolysis
Pharmaco-invasive strategy[a]	Fibrinolysis combined with rescue PCI (in cases of failed fibrinolysis) or routine early PCI strategy (in cases of successful fibrinolysis)
Immediate invasive strategy	Emergency coronary angiography (i.e. as soon as possible) and PCI/CABG of the IRA if indicated
Early invasive strategy	Early coronary angiography (<24 h from diagnosis of ACS) and PCI/CABG of the IRA if indicated
Selective invasive strategy	Coronary angiography ± PCI/CABG based on clinical assessment and/or non-invasive testing

ACS, acute coronary syndrome; CABG, coronary artery bypass grafting; ECG, electrocardiogram; IRA, infarct-related artery; PCI, percutaneous coronary intervention; STE-ACS, ST-segment-elevation acute coronary syndrome.
[a]CABG may also be indicated instead of PCI in certain circumstances.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026
© ESC 2023

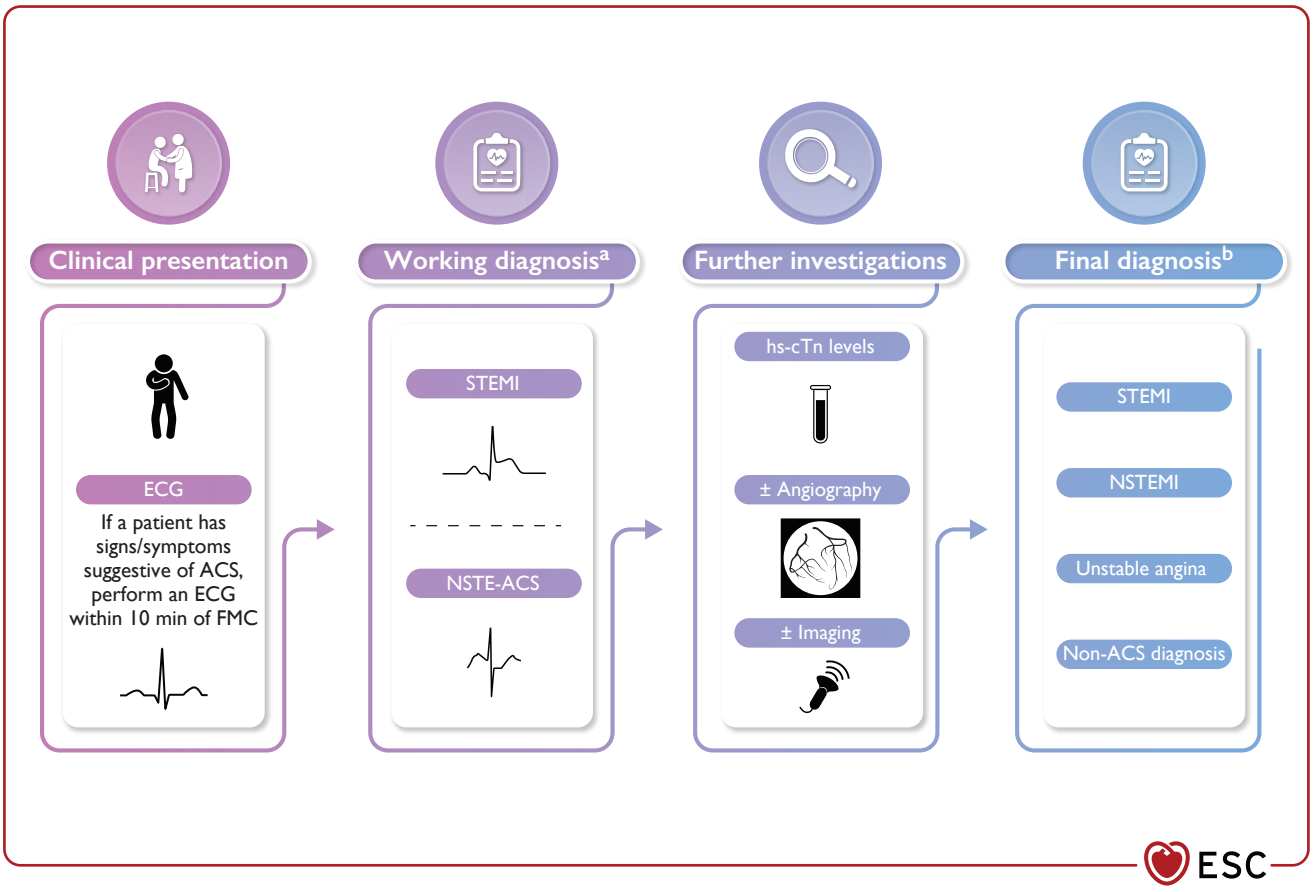


Figure 3 Classification of patients presenting with suspected acute coronary syndrome: from a working to a final diagnosis. ACS, acute coronary syndrome; ECG, electrocardiogram; FMC, first medical contact; hs-cTn, high-sensitivity cardiac troponin; MI, myocardial infarction; NSTE-ACS, non-ST-elevation acute coronary syndrome; NSTEMI, non-ST-elevation myocardial infarction: STEMI, ST-elevation myocardial infarction. [a]The working ACS diagnosis can be classified as STEMI or NSTE-ACS on the basis of available clinical information and ECG findings. This allows for initial triage and assessment. [b]The final diagnosis is based on symptoms, ECG and troponin for the diagnosis of MI as well as the results of other tests (i.e. imaging and/or angiography) to facilitate understanding of the mechanism and subclassification of the type of MI. Patients initially assigned a working diagnosis of STEMI or NSTE-ACS may eventually receive a final non-ACS diagnosis.

2.2. Epidemiology of acute coronary syndromes

Cardiovascular disease (CVD) is the most common cause of mortality and morbidity worldwide, with a substantial portion of this burden borne by low- and middle-income countries.[2,3] ACS is often the first clinical manifestation of CVD. In 2019, there were an estimated 5.8 million new cases of ischaemic heart disease in the 57 ESC member countries.[3] The median age-standardized incidence estimate per 100 000 people was 293.3 (interquartile ratio 195.8–529.5). CVD remains the most common cause of death within ESC member countries, accounting for just under 2.2 million deaths in females and just over 1.9 million deaths in males in the most recent year of available data. Ischaemic heart disease is the most common cause of CVD death, accounting for 38% of all CVD deaths in females and 44% in males.[3]

2.3. Number and breakdown of classes of recommendations

The total number of recommendations in this guideline is 193. A summary of the recommendations according to Class of Recommendation and Level of Evidence (LoE) is also provided. As per Class of Recommendation, there were 106 Class I, 70 Class II, and 17 Class III recommendations. As per LoE, there were 56 LoE A, 64 LoE B, and 73 LoE C recommendations.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

2.4. What is new

Table 4 New recommendations

Recommendations	Class[a]	Level[b]
Recommendations for antiplatelet and anticoagulant therapy in acute coronary syndrome		
If patients presenting with ACS stop DAPT to undergo coronary artery bypass grafting, it is recommended they resume DAPT after surgery for at least 12 months.	I	C
In older ACS patients, especially if HBR, clopidogrel as the $P2Y_{12}$ receptor inhibitor may be considered.	IIb	B
Recommendations for alternative antithrombotic therapy regimens		
In patients who are event-free after 3–6 months of DAPT and who are not high ischaemic risk, single antiplatelet therapy (preferably with a $P2Y_{12}$ receptor inhibitor) should be considered.	IIa	A
$P2Y_{12}$ inhibitor monotherapy may be considered as an alternative to aspirin monotherapy for long-term treatment.	IIb	A
In HBR patients, aspirin or $P2Y_{12}$ receptor inhibitor monotherapy after 1 month of DAPT may be considered.	IIb	B
In patients requiring OAC, withdrawing antiplatelet therapy at 6 months while continuing OAC may be considered.	IIb	B
De-escalation of antiplatelet therapy in the first 30 days after an ACS event is not recommended.	III	B
Recommendations for cardiac arrest and out-of-hospital cardiac arrest		
Evaluation of neurological prognosis (no earlier than 72 h after admission) is recommended in all comatose survivors after cardiac arrest.	I	C
Transport of patients with out-of-hospital cardiac arrest to a cardiac arrest centre according to local protocol should be considered.	IIa	C
Recommendations for technical aspects of invasive strategies		
In patients with spontaneous coronary artery dissection, PCI is recommended only for patients with symptoms and signs of ongoing myocardial ischaemia, a large area of myocardium in jeopardy, and reduced antegrade flow.	I	C
Intravascular imaging should be considered to guide PCI.	IIa	A
Intravascular imaging (preferably optical coherence tomography) may be considered in patients with ambiguous culprit lesions.	IIb	C
Recommendations for multivessel disease in ACS patients presenting in cardiogenic shock		
Staged PCI of non-IRA should be considered.	IIa	C
Recommendations for multivessel disease in haemodynamically stable STEMI patients undergoing primary PCI		
It is recommended that PCI of the non-IRA is based on angiographic severity.	I	B
Invasive epicardial functional assessment of non-culprit segments of the IRA is not recommended during the index procedure.	III	C
Recommendations for acute coronary syndrome complications		
Implantation of a permanent pacemaker is recommended when high-degree AV block does not resolve within a waiting period of at least 5 days after MI.	I	C
Cardiac magnetic resonance imaging should be considered in patients with equivocal echocardiographic images or in cases of high clinical suspicion of LV thrombus.	IIa	C
Following an acute anterior MI, a contrast echocardiogram may be considered for the detection of LV thrombus if the apex is not well visualized on echocardiography.	IIb	C
In selected patients with high-degree AV block in the context of an anterior wall MI and acute heart failure, early device implantation (cardiac resynchronization therapy—defibrillator/pacemaker) may be considered.	IIb	C
In patients with recurrent life-threatening ventricular arrhythmias, sedation or general anaesthesia to reduce sympathetic drive may be considered.	IIb	C
Recommendations for acute coronary syndrome comorbid conditions		
It is recommended to base the choice of long-term glucose-lowering treatment on the presence of comorbidities, including heart failure, chronic kidney disease, and obesity.	I	A
For frail older patients with comorbidities, a holistic approach is recommended to individualize interventional and pharmacological treatments after careful evaluation of the risks and benefits.	I	B
An invasive strategy is recommended in cancer patients presenting with high-risk ACS with expected survival ≥6 months.	I	B
A temporary interruption of cancer therapy is recommended in patients in whom the cancer therapy is suspected to be a contributing cause of ACS.	I	C
A conservative non-invasive strategy should be considered in ACS patients with poor cancer prognosis (i.e. with expected life survival <6 months) and/or very high bleeding risk.	IIa	C
Aspirin is not recommended in cancer patients with a platelet count <10 000/μL.	III	C

Continued

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Clopidogrel is not recommended in cancer patients with a platelet count <30 000/μL.	III	C
In ACS patients with cancer and <50 000/μL platelet count, prasugrel or ticagrelor are not recommended.	III	C
Recommendations for long-term management		
It is recommended to intensify lipid-lowering therapy during the index ACS hospitalization for patients who were on lipid-lowering therapy before admission.	I	C
Low-dose colchicine (0.5 mg once a day) may be considered, particularly if other risk factors are insufficiently controlled or if recurrent cardiovascular disease events occur under optimal therapy.	IIb	A
Combination therapy with a high-dose statin plus ezetimibe may be considered during index hospitalization.	IIb	B
Recommendations for patient perspectives in acute coronary syndrome care		
Patient-centred care is recommended by assessing and adhering to individual patient preferences, needs and beliefs, ensuring that patient values are used to inform all clinical decisions.	I	B
It is recommended to include ACS patients in decision-making (as much as their condition allows) and to inform them about the risk of adverse events, radiation exposure, and alternative options. Decision aids should be used to facilitate the discussion.	I	B
It is recommended to assess symptoms using methods that help patients to describe their experience.	I	C
Use of the 'teach back' technique for decision support during the securing of informed consent should be considered.	IIa	B
Patient discharge information should be provided in both written and verbal formats prior to discharge. Adequate preparation and education for patient discharge using the teach back technique and/or motivational interviewing, giving information in chunks, and checking for understanding, should be considered.	IIa	B
Assessment of mental well-being using a validated tool and onward psychological referral when appropriate should be considered.	IIa	B

ACS, acute coronary syndrome; AV, atrioventricular; DAPT, dual antiplatelet therapy; HBR, high bleeding risk; IRA, infarct-related artery; LV, left ventricular(cle); MI, myocardial infarction; OAC, oral anticoagulant/ation; PCI, percutaneous coronary intervention; STEMI, ST-elevation myocardial infarction.
[a]Class of recommendation.
[b]Level of evidence.

Table 5 Revised recommendations

Recommendations in 2017 and 2020 versions	Class[a]	LoE[b]	Recommendations in 2023 version	Class[a]	LoE[b]
Recommendations for imaging for patients with suspected NSTE-ACS					
In patients with no recurrence of chest pain, normal ECG findings, and normal levels of cardiac troponin (preferably high sensitivity), but still with suspected ACS, a non-invasive stress test (preferably with imaging) for inducible ischaemia or CCTA is recommended before deciding on an invasive approach.	I	B	In patients with suspected ACS, non-elevated (or uncertain) hs-cTn, no ECG changes and no recurrence of pain, incorporating CCTA or a non-invasive stress imaging test as part of the initial workup should be considered.	IIa	A
Recommendations for timing of invasive strategy in NSTE-ACS					
An early invasive strategy within 24 h is recommended in patients with any of the following high-risk criteria: • Diagnosis of NSTEMI suggested by the diagnostic algorithm recommended in *Section 3* • Dynamic or presumably new contiguous ST/T-segment changes suggesting ongoing ischaemia • Transient ST-segment elevation • GRACE risk score >140.	I	A	An early invasive strategy within 24 h should be considered in patients with at least one of the following high-risk criteria: • Confirmed diagnosis of NSTEMI based on current recommended ESC hs-cTn algorithms • Dynamic ST-segment or T wave changes • Transient ST-segment elevation • GRACE risk score >140.	IIa	A
Recommendations for antiplatelet and anticoagulant therapy in STEMI					
A potent $P2Y_{12}$ inhibitor (prasugrel or ticagrelor), or clopidogrel if these are not available or are contraindicated, is recommended before (or at latest at the time of) PCI, and maintained over 12 months, unless there are contraindications such as excessive risk of bleeding.	I	A	Pre-treatment with a $P2Y_{12}$ receptor inhibitor may be considered in patients undergoing a primary PCI strategy.	IIb	B
Recommendations for long-term antithrombotic therapy					
After stent implantation in patients undergoing a strategy of DAPT, stopping aspirin after 3–6 months should be considered, depending on the balance between the ischaemic and bleeding risks.	IIa	A	In patients who are event-free after 3–6 months of DAPT and who are not high ischaemic risk, SAPT (preferably with a $P2Y_{12}$ receptor inhibitor) should be considered.	IIa	A

Continued

© ESC 2023

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Recommendations for cardiac arrest and out-of-hospital cardiac arrest					
Delayed as opposed to immediate angiography should be considered among haemodynamically stable patients without ST-segment elevation successfully resuscitated after out-of-hospital cardiac arrest.	**IIa**	**B**	Routine immediate angiography after resuscitated cardiac arrest is not recommended in haemodynamically stable patients without persistent ST-segment elevation (or equivalents).	**III**	**A**
Targeted temperature management (also called therapeutic hypothermia), aiming for a constant temperature between 32 and 36 C for at least 24 h, is indicated in patients who remain unconscious after resuscitation from cardiac arrest (of presumed cardiac cause).	**I**	**B**	Temperature control (i.e. continuous monitoring of core temperature and active prevention of fever [i.e. >37.7°C]) is recommended after either out-of-hospital or in-hospital cardiac arrest for adults who remain unresponsive after return of spontaneous circulation.	**I**	**B**
Recommendations for in-hospital management					
When echocardiography is suboptimal/inconclusive, an alternative imaging method (CMR preferably) should be considered.	**IIa**	**C**	When echocardiography is suboptimal/inconclusive, CMR imaging may be considered.	**IIb**	**C**
Recommendations for management of multivessel disease in haemodynamically stable STEMI patients undergoing primary PCI					
Routine revascularization of non-IRA lesions should be considered in STEMI patients with multivessel disease before hospital discharge.	**IIa**	**A**	Complete revascularization is recommended either during the index PCI procedure or within 45 days.	**I**	**A**
Recommendations for acute coronary syndrome comorbid conditions					
Glucose-lowering therapy should be considered in ACS patients with blood glucose >10 mmol/L (>180 mg/dL), with the target adapted to comorbidities, while episodes of hypoglycaemia should be avoided.	**IIa**	**B**	Glucose-lowering therapy should be considered in patients with ACS with persistent hyperglycaemia, while episodes of hypoglycaemia should be avoided.	**IIa**	**C**

ACS, acute coronary syndrome; CCTA, coronary computed tomography angiography; CMR, cardiac magnetic resonance; DAPT, dual antiplatelet therapy; ECG, electrocardiography/gram; ESC European Society of Cardiology; GRACE, Global Registry of Acute Coronary Events; hs-cTn, high-sensitivity cardiac troponin; IRA, infarct-related artery; NSTE-ACS, non-ST-elevation acute coronary syndrome; NSTEMI, non-ST-elevation myocardial infarction; PCI, percutaneous coronary intervention; SAPT, single antiplatelet therapy; STEMI, ST-elevation myocardial infarction.
[a]Class of recommendation.
[b]Level of evidence.

© ESC 2023

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

New/revised concepts

- ACS should be considered a spectrum, which encompasses both non-ST-elevation (NSTE)-ACS and ST-elevation MI (STEMI).
- A section on the management of ACS in patients with cancer is provided.
- A section on patient perspectives is provided.

3. Triage and diagnosis

3.1. Clinical presentation and physical examination

3.1.1. Clinical presentation

Acute chest discomfort—which may be described as pain, pressure, tightness, heaviness, or burning—is the leading presenting symptom prompting consideration of the clinical diagnosis of ACS and the initiation of testing aligned with specific diagnostic algorithms (*Figure 4*).

Chest pain descriptors should be classified as cardiac, possibly cardiac, and likely non-cardiac. Further information on the suggested use of these terms is provided in the Supplementary data online. The use of the descriptor 'atypical' should be avoided. Chest pain-equivalent symptoms include dyspnoea, epigastric pain, and pain in the left or right arm or neck/jaw.

Misdiagnosis or delayed diagnosis is sometimes due to an incomplete history or difficulty in eliciting symptoms from the patient. In order to understand the complexity of ACS-related symptomatology, careful history taking and comprehensive interaction with the patient are crucial and may help to facilitate an early and accurate diagnosis. Further information is provided in the Supplementary data online, including *Figure S1*, which outlines some of the most common symptoms of ACS in women and men.

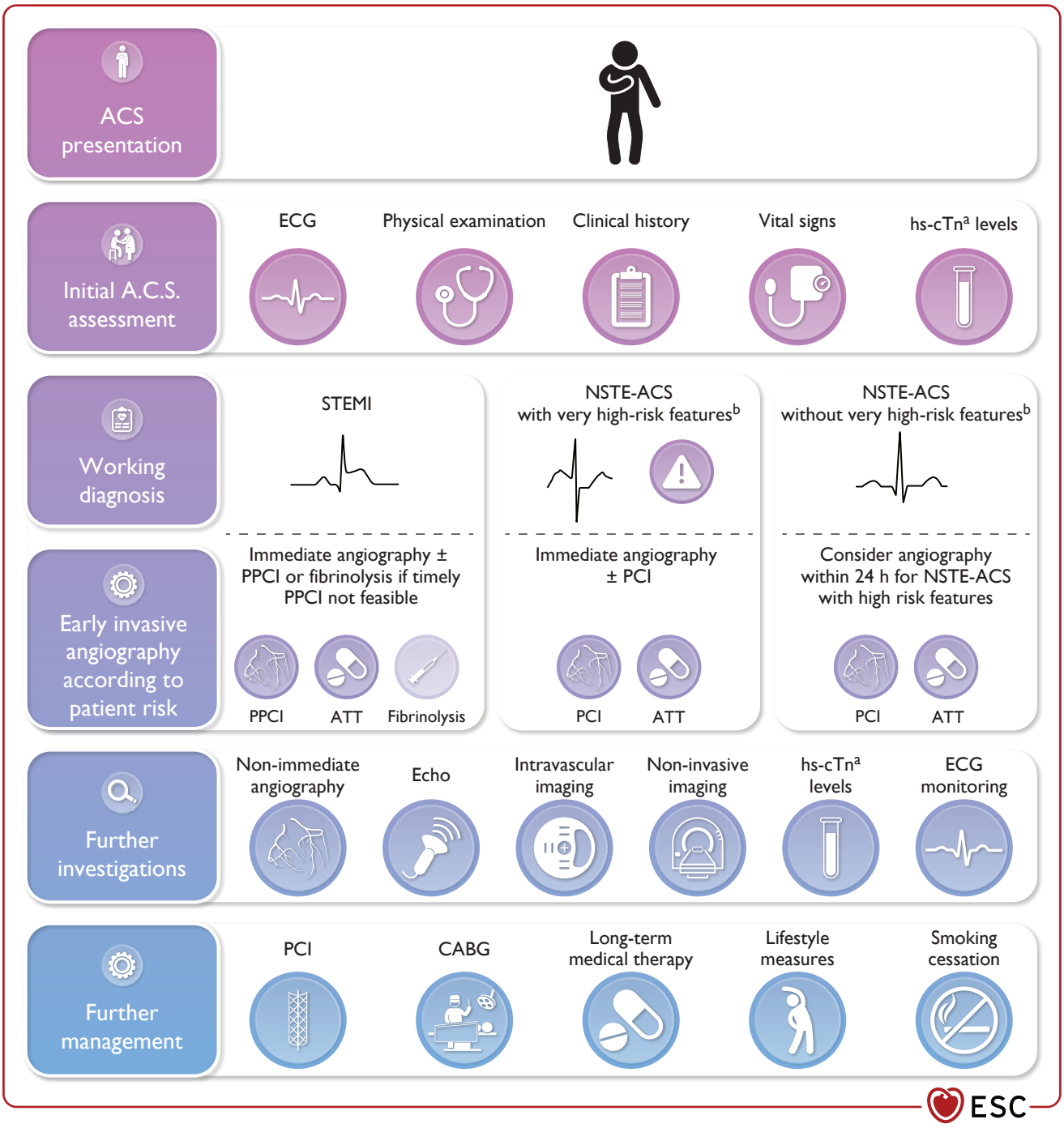


Figure 4 An overview of the initial triage, management and investigation of patients who present with signs and symptoms potentially consistent with acute coronary syndrome. ACS, acute coronary syndrome; ATT, antithrombotic therapy; CABG, coronary artery bypass grafting; ECG, electrocardiogram; hs-cTn, high-sensitivity cardiac troponin; NSTE-ACS, non-ST-elevation acute coronary syndrome; PPCI, primary percutaneous coronary intervention; STEMI, ST-elevation myocardial infarction. The 'A.C.S.' assessment is detailed in *Figure 5*. [a]Results of hs-cTn measurements are not required for the initial stratification of ACS and the initial emergency management (i.e. for patients with a working diagnosis of STEMI or very high-risk NSTE-ACS) should not be delayed based on this. [b]For patients with NSTE-ACS with very high-risk features, immediate angiography is recommended. For patients with NSTE-ACS with high-risk features, early invasive angiography (i.e. <24 h) should be considered and inpatient invasive angiography is recommended. See Recommendation Table 4 for details.

It is important that awareness of the symptoms associated with ACS is high among the general population, in particular red flag symptoms such as prolonged chest pain (>15 min) and/or recurrent pain within 1 h, which should prompt patients or other members of the public to seek urgent medical help. Continuous education, promotion, and advocacy efforts are important to make sure that this information is as widely available as possible to the general population.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

3.1.2. History taking and physical examination

Patients with suspected ACS present in a broad range of clinical scenarios, including in the community, at the emergency department (ED), or in the inpatient setting. It is crucial to take a focused medical history and accurately characterize the presenting symptoms in order to manage the patient via the appropriate care pathway as soon as possible.

Prompt assessment of vital signs is recommended at first medical contact (FMC), at the same time as acquisition of an initial ECG (*Figure 5*). In patients presenting with suspected ACS, physical examination is recommended and is useful both to eliminate differential diagnoses and to identify very high-risk and high-risk ACS features. This may be particularly relevant for patients presenting with cardiac arrest, signs of CS, and/or haemodynamic or electrical instability.[4] Focused physical examination should include checking for the presence of all major pulses, measurement of blood pressure in both arms, auscultation of the heart and lungs, and assessing for signs of HF or circulatory compromise.

3.2. Diagnostic tools | Electrocardiogram

The resting 12-lead ECG is the first-line diagnostic tool in the assessment of patients with suspected ACS. It is recommended that an ECG is obtained immediately upon FMC and interpreted by a qualified emergency medical technician or physician within 10 min.[4,5] It should be repeated as necessary, especially if symptoms have waned at FMC. Based on the initial ECG, patients with suspected ACS can be differentiated into two working diagnoses:

- **Patients with acute chest pain (or chest pain-equivalent signs/symptoms) and persistent ST-segment elevation (or ST-segment elevation equivalents) on ECG (working diagnosis: ST-segment elevation MI: STEMI)**. The vast majority of these patients will sustain myocardial necrosis and troponin elevation, fulfilling the criteria for an MI, but MI will not be the final diagnosis in all patients with a working diagnosis of STEMI.
- **Patients with acute chest pain (or chest pain-equivalent signs/symptoms) but without persistent ST-segment**

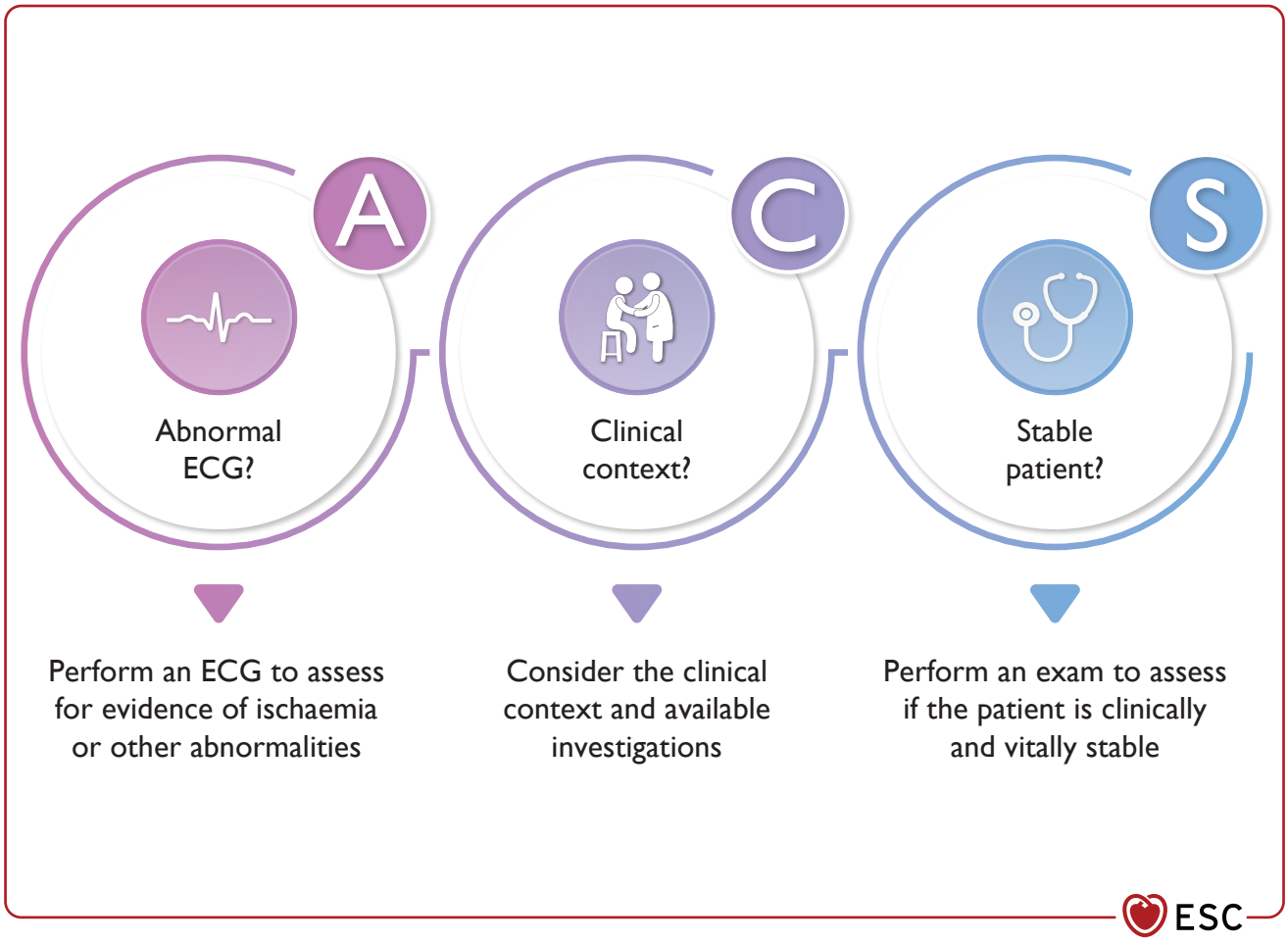


Figure 5 The A.C.S. assessment for the initial evaluation of patients with suspected acute coronary syndrome. ECG, electrocardiogram. This figure summarizes the initial 'A.C.S. assessment' that can be performed for a patient presenting with suspected ACS. 'A' stands for 'Abnormal ECG?': an ECG should be performed within 10 min of FMC and assessed for evidence of abnormalities or ischaemia. 'C' stands for 'Clinical Context?': it is important to consider the clinical context of the patient's presentation and the results of any investigations that are available. This should also include a targeted history with the aim of determining the patient's symptoms and elucidating any other relevant background information. 'S' stands for 'Stable Patient?': the patient should be quickly assessed to determine if they are clinically stable—this should include assessment of the clinical vital signs, including heart rate, blood pressure, and oxygen saturations, if possible, as well as checking for potential signs of CS.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

elevation (or ST-segment elevation equivalents) on ECG (working diagnosis: non-ST-elevation [NSTE]-ACS). These patients may exhibit other ECG alterations, including transient ST-segment elevation, persistent or transient ST-segment depression, and T wave abnormalities, including hyperacute T waves, T wave inversion, biphasic T waves, flat T waves, and pseudo-normalization of T waves. Alternatively, the ECG may be normal. The majority of patients in this category who subsequently display a typical rise and fall in cardiac troponin levels (i.e. fulfilling MI criteria as per the fourth universal definition of MI) will receive a final diagnosis of non-ST-elevation MI (NSTEMI). In other patients, the troponin level will remain below the 99th centile and they will receive a final diagnosis of UA, although with high-sensitivity troponin assays this diagnosis has become less common. It is also important to recognize that NSTEMI or UA will not be the final diagnosis in all patients with an initial working diagnosis of NSTE-ACS.

3.2.1. Acute coronary syndrome with persistent ST-segment elevation (suspected ST-elevation myocardial infarction)

The priority for these patients is the implementation of reperfusion therapy as soon as possible (see *Section 5*). In the appropriate clinical context, ST-segment elevation (measured at the J-point) is considered suggestive of ongoing coronary artery acute occlusion in the following cases:

New ST elevation at the J-point in at least two contiguous leads:

- ≥2.5 mm in men <40 years, ≥2 mm in men ≥40 years, or ≥1.5 mm in women regardless of age in leads V2–V3
- and/or ≥1 mm in the other leads (in the absence of left ventricular [LV] hypertrophy or left bundle branch block [LBBB]).

In patients with suspected inferior STEMI, it is recommended to record right precordial leads (V3R and V4R) in order to assess for ST-segment elevation.[6] Posterior leads (V7–V9) can also be recorded to investigate for posterior STEMI, particularly in patients with ongoing symptoms and an inconclusive standard 12-lead ECG.

The diagnosis of ongoing acute coronary artery occlusion on ECG can sometimes be challenging, and some cases may warrant prompt management and triage for immediate reperfusion therapy despite the absence of ST-segment elevation. It is also important to recognize that while the most sensitive sign for ongoing acute coronary artery occlusion is ST-segment elevation, there are other ECG findings that can be suggestive of ongoing coronary artery occlusion (or severe ischaemia). If these findings are present, prompt triage for immediate reperfusion therapy is indicated (see Supplementary data online, *Figure S2*).

ST-segment depression in leads V1–V3 (especially when the terminal T wave is positive) and/or ST-segment elevation in V7–V9 are highly suggestive of posterior coronary artery occlusion (often the left circumflex artery).[1,7] ST-segment elevation in V3R and V4R is highly suggestive of ongoing RV ischaemia.[8] ST depression ≥1 mm in ≥6 surface leads (inferolateral ST depression), coupled with ST-segment elevation in aVR and/or V1, suggests multivessel ischaemia or left main coronary artery obstruction, particularly if the patient presents with haemodynamic compromise.[9–11]

Bundle branch block (BBB). In patients with a high clinical suspicion of ongoing myocardial ischaemia, the presence of LBBB, right bundle branch block (RBBB), or a paced rhythm precludes an accurate assessment of the presence or absence of ST-segment elevation. Therefore, patients presenting with these ECG patterns in combination with signs/symptoms that are highly suspicious for ongoing myocardial ischaemia should be managed similarly to those with clear ST-segment elevation, regardless of whether the BBB is previously known (see Supplementary data online).[4]

3.2.2. Acute coronary syndrome without persistent ST-segment elevation (non-ST elevation acute coronary syndrome)

While the ECG in the setting of NSTE-ACS may be normal in more than one-third of patients, characteristic ECG abnormalities are frequently present and increase the diagnostic probability of ACS.[12–16] These ECG abnormalities include ST depression and T wave changes (especially biphasic T waves or prominent negative T waves [Wellens' sign, related to severe proximal left anterior descending artery stenosis]), (see Supplementary data online, *Figure S3*).

Recommendation Table 1 — Recommendations for clinical and diagnostic tools for patients with suspected acute coronary syndrome

Recommendations	Class[a]	Level[b]
It is recommended to base the diagnosis and initial short-term risk stratification of ACS on a combination of clinical history, symptoms, vital signs, other physical findings, ECG, and hs-cTn.[1,17,18]	I	B
ECG		
Twelve-lead ECG recording and interpretation is recommended as soon as possible at the point of FMC, with a target of <10 min.[5,19]	I	B
Continuous ECG monitoring and the availability of defibrillator capacity is recommended as soon as possible in all patients with suspected STEMI, in suspected ACS with other ECG changes or ongoing chest pain, and once the diagnosis of MI is made.[20,21]	I	B
The use of additional ECG leads (V3R, V4R, and V7–V9) is recommended in cases of inferior STEMI or if total vessel occlusion is suspected and standard leads are inconclusive.[22–24]	I	B
An additional 12-lead ECG is recommended in cases with recurrent symptoms or diagnostic uncertainty.	I	C

Continued

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Blood sampling		
It is recommended to measure cardiac troponins with high-sensitivity assays immediately after presentation and to obtain the results within 60 min of blood sampling.[15,25–27]	I	B
It is recommended to use an ESC algorithmic approach with serial hs-cTn measurements (0 h/1 h or 0 h/2 h) to rule in and rule out NSTEMI.[28–44]	I	B
Additional testing after 3 h is recommended if the first two hs-cTn measurements of the 0 h/1 h algorithm are inconclusive and no alternative diagnoses explaining the condition have been made.[45,46]	I	B
The use of established risk scores (e.g. GRACE risk score) for prognosis estimation should be considered.[47–49]	IIa	B
Triage for emergency reperfusion strategy		
It is recommended that patients with suspected STEMI are immediately triaged for an emergency reperfusion strategy.[50–52]	I	A

© ESC 2023

ACS, acute coronary syndrome; ECG, electrocardiogram; ESC, European Society of Cardiology; FMC, first medical contact; GRACE, Global Registry of Acute Coronary Events; hs-cTn, high-sensitivity cardiac troponin; MI, myocardial infarction; NSTEMI, non-ST-elevation myocardial infarction; STEMI, ST-elevation myocardial infarction.
[a]Class of recommendation.
[b]Level of evidence.

3.3. Diagnostic tools | Biomarkers

3.3.1. High-sensitivity cardiac troponins

After excluding clinical and ECG signs suggestive of STEMI or very high-risk NSTE-ACS, biomarkers play a complementary role in the diagnosis, risk stratification, and management of patients with suspected ACS. Measurement of a biomarker of cardiomyocyte injury, preferably high-sensitivity cardiac troponin (hs-cTn), is recommended in all patients with suspected ACS.[15,17,25–27,53,54] If the clinical presentation is compatible with myocardial ischaemia, then a rise and/or fall in cTn above the 99th percentile of healthy individuals points to a diagnosis of MI as per the criteria in the fourth universal definition of MI.[1] In patients with MI, levels of cTn rise rapidly (i.e. usually within 1 h if using high-sensitivity assays) after symptom onset and remain elevated for a variable period of time (usually several days).[1,15,26,53,55–58]

Advances in technology have led to a refinement in cTn assays and have improved their accuracy in detecting and quantifying cardiomyocyte injury.[1,12–15,18,26,34,35,53,55–60] Data from large multicentre studies have consistently shown that hs-cTn assays increase diagnostic accuracy for MI at the time of presentation in comparison to conventional assays, especially in patients presenting early after chest pain onset, enabling more rapid 'rule-in' and 'rule-out' of MI.[1,12–15,26,34,35,53,55–58] Overall, hs-cTn T and hs-cTn I subunit assays appear to provide comparable diagnostic accuracy in the early diagnosis of MI.[28,32,61,62] The use of the terms 'normal' and 'abnormal' to describe hs-cTn levels should be avoided; instead, the terms *'non-elevated'* and *'elevated'* should be used to refer to hs-cTn levels below and above the 99th percentile. Some of the clinical implications of hs-cTn assays are detailed in Supplementary data online, *Table S2*.

It is also important to consider that there are other clinical conditions apart from Type 1 MI in which elevations in cTn can be observed (see Supplementary data online, *Section 3.3.1* and *Table S3*).

3.3.2. Central laboratory vs. point of care

The vast majority of cTn assays that run on automated platforms in the central laboratory are sensitive (i.e. allow for the detection of cTn in ~20–50% of healthy individuals) or high-sensitivity (i.e. allow for the detection of cTn in ~50–95% of healthy individuals) assays. High-sensitivity assays are recommended over lower-sensitivity assays, as they provide higher diagnostic accuracy at an identical low cost.[1,12,15,25–27,57,63]

The majority of currently used point-of-care (POC) tests cannot be considered high-sensitivity assays.[64] The advantage of POC tests is a shorter turnaround time. However, this is counterbalanced by lower sensitivity, lower diagnostic accuracy, and lower negative predictive value (NPV). A randomized trial in low-risk chest pain patients with suspected NSTE-ACS and onset of symptoms ≥2 h before ambulance presentation reported that the use of a pre-hospital rule-out strategy (with a single POC conventional troponin T test) resulted in a significant reduction of 30-day healthcare costs and a comparable major adverse cardiovascular event (MACE) rate in comparison to an ED rule-out strategy (with evaluation as per standard local practice).[65]

Overall, automated assays have been more thoroughly evaluated than POC tests and are currently preferred.[1,12–15,26,34,35,53,55–58] However, this is a rapidly developing field and it will be important to re-evaluate this preference when more extensively validated high-sensitivity POC tests are clinically available.[66–68]

3.3.3. Confounders of cardiac troponin concentration

In patients presenting with suspected NSTE-ACS, four clinical variables affect hs-cTn concentrations beyond the presence or absence of MI. These variables are: age (concentrations in healthy very young vs. 'healthy' very old individuals differ by up to 300%); renal dysfunction (differences between otherwise healthy patients with very high vs. very low estimated glomerular filtration rate [eGFR] of up to 300%); time from chest pain onset (>300%); and, to a lesser extent, sex (≈40%).[28,34,35,69–76] Despite the potential baseline differences in hs-cTn values based on these four variables, absolute changes in hs-cTn levels are still of diagnostic and prognostic value. Current data on the use of sex-specific hs-cTn values in the diagnosis of MI have been controversial and failed to demonstrate a clear clinical benefit.[74,75,77–80] Therefore, until automated tools (i.e. risk assessment calculators) incorporating the effect of all four clinical variables (age, eGFR, time from chest pain onset, and sex) are available, the use of uniform cut-off concentrations should remain the standard of care for the early diagnosis of MI.[28,30,31,34,35,73,81,82]

3.3.4. Rapid 'rule-in' and 'rule-out' algorithms

Due to their higher sensitivity and diagnostic accuracy for the detection of MI at presentation, the time interval to the second cTn assessment can be shortened with the use of hs-cTn assays. This substantially reduces the delay to diagnosis, translating into shorter stays in the ED, lower costs, and less diagnostic uncertainty for patients.[15,83–88] It is recommended to use the 0 h/1 h algorithm (best option) or the 0 h/2 h

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

algorithm (second-best option) (*Figure 6*). These algorithms have been derived and validated in large multicentre diagnostic studies using central adjudication of the final diagnosis for all currently available hs-cTn assays.[27–39,62,70,73,82,89–93] Optimal thresholds for rule-out were selected to allow a sensitivity and NPV of at least 99%. Optimal thresholds for rule-in were selected to allow a positive predictive value (PPV) of at least 70%. These algorithms were developed from large derivation cohorts and then validated in large independent validation cohorts. The previous ESC 0 h/3 h algorithm was considered as an alternative,[40,56] but three recent large diagnostic studies suggested that the ESC 0 h/3 h algorithm appears to balance efficacy and safety less well than more rapid protocols using lower rule-out concentrations, including the ESC 0 h/1 h algorithm.[41–43] The very high safety and high efficacy of applying the ESC 0 h/1 h algorithm was recently confirmed in three real-life implementation studies, including one randomized controlled trial (RCT).[44,94,95] Therefore, the ESC 0 h/3 h algorithm is an alternative for cases where the ESC 0 h/1 h or 0 h/2 h algorithms are not available. Of note, patients assigned to the 'rule-out' pathway using the ESC 0 h/1 h or 0 h/2 h algorithms have a very low rate of clinical events through to 30 days.[95,96]

3.3.4.1. European Society of Cardiology 0 h/1 h and 0 h/2 h algorithms

The ESC 0 h/1 h and 0 h/2 h algorithms are based on two underlying concepts: firstly, hs-cTn is a continuous variable and the probability of MI increases with increasing hs-cTn values.[28,30,31,34,35,73,82] Secondly, early absolute changes in the levels within 1 h or 2 h can be used as surrogates for absolute changes over 3 h or 6 h and provide incremental diagnostic value to the single cTn assessment at presentation.[27,28,30,31,34,35,73,82,97] The cut-off concentrations within the 0 h/1 h and 0 h/2 h algorithms are assay specific (Supplementary data online, *Table S4*).[27,28,30,31,34,35,73,82]

3.3.4.1.1. Rule-out. The NPV for MI in patients assigned to the 'rule-out' pathway has exceeded 99% in several large validation cohorts.[28–30,34,35,73] Assignment to the rule-out pathway does not always equal outpatient management. However, when used in conjunction with clinical and ECG findings, the 0 h/1 h and 0 h/2 h algorithms will enable the identification of appropriate candidates for early discharge and outpatient management. Even after the ruling out of MI, elective non-invasive or invasive imaging may be appropriate according to clinical and risk assessment, and an alternative diagnosis to MI should be identified.

3.3.4.1.2. Rule-in. The PPV for MI in patients meeting the 'rule-in' pathway criteria in several studies has been ~70–75%. Most of the 'rule-in' pathway patients with diagnoses other than MI still have conditions that require specialist cardiology input and either coronary angiography or non-invasive imaging in order to establish an accurate final diagnosis.[28,30,31,34,35,73,82] Therefore, the vast majority of patients triaged towards the 'rule-in' pathway by these algorithms will require hospital admission and invasive coronary angiography (ICA).

3.3.4.1.3. Observe. Patients who do not qualify for the 'rule-out' or 'rule-in' pathways are assigned to the 'observe' pathway. These patients represent a heterogeneous group and have been shown to have a mortality rate that is comparable to rule-in patients.[98] Therefore, an individual assessment based on the particular risk profile of the patient (i.e. risk scores) is of paramount importance for patients in this group. Additionally, a third measurement of cTn at 3 h (± echocardiography) is recommended as the next step in order to guide further management.[45,46]

Most patients in the observe zone with a high degree of clinical suspicion of ACS (e.g. relevant increase in cTn from presentation to 3 h) are candidates for ICA. Conversely, most patients with a low to intermediate likelihood for ACS according to clinical judgment are candidates for non-invasive imaging after transfer from the ED to the ward. Computed tomography (CT) angiography can be used to aid diagnosis and, in particular, to identify patients with non-obstructed coronary arteries who can be discharged if other relevant diseases have been excluded. CT angiography can also identify patients with obstructive coronary disease in whom revascularization may be considered. In the appropriate clinical context, if alternative conditions have been identified that explain the cTn values (i.e. rapid ventricular rate response to atrial fibrillation [AF], marked anaemia, or a hypertensive emergency), further diagnostic testing (i.e. ICA) may not be required.

The same concepts apply to the 0 h/2 h algorithm. Cut-off levels for both the 0 h/1 h and 0 h/2 h algorithms are also assay specific, and these cut-off levels are shown in Supplementary data online, *Table S4*.[99]

The ESC 0 h/1 h and 0 h/2 h algorithms should always be integrated with a detailed clinical assessment and a 12-lead ECG. Repeat blood sampling is mandatory in cases where there is ongoing or recurrent chest pain. Recently, artificial intelligence models that include serial hs-cTn measurements in conjunction with individual risk profiles have been proposed to be useful to facilitate a personalized diagnostic evaluation of patients with suspected MI. Similarly, risk-assessment models combining hs-cTn values at presentation and after early or late resampling have been developed to predict MI events during the first 30 days. These models may facilitate alternative hs-cTn cut-offs based on the balance between NPV and PPV best suited to individual clinical sites.[27] A diagnostic approach to the use of the ESC 0 h/1 h and 0 h/2 h algorithms is shown in *Figure 6*.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

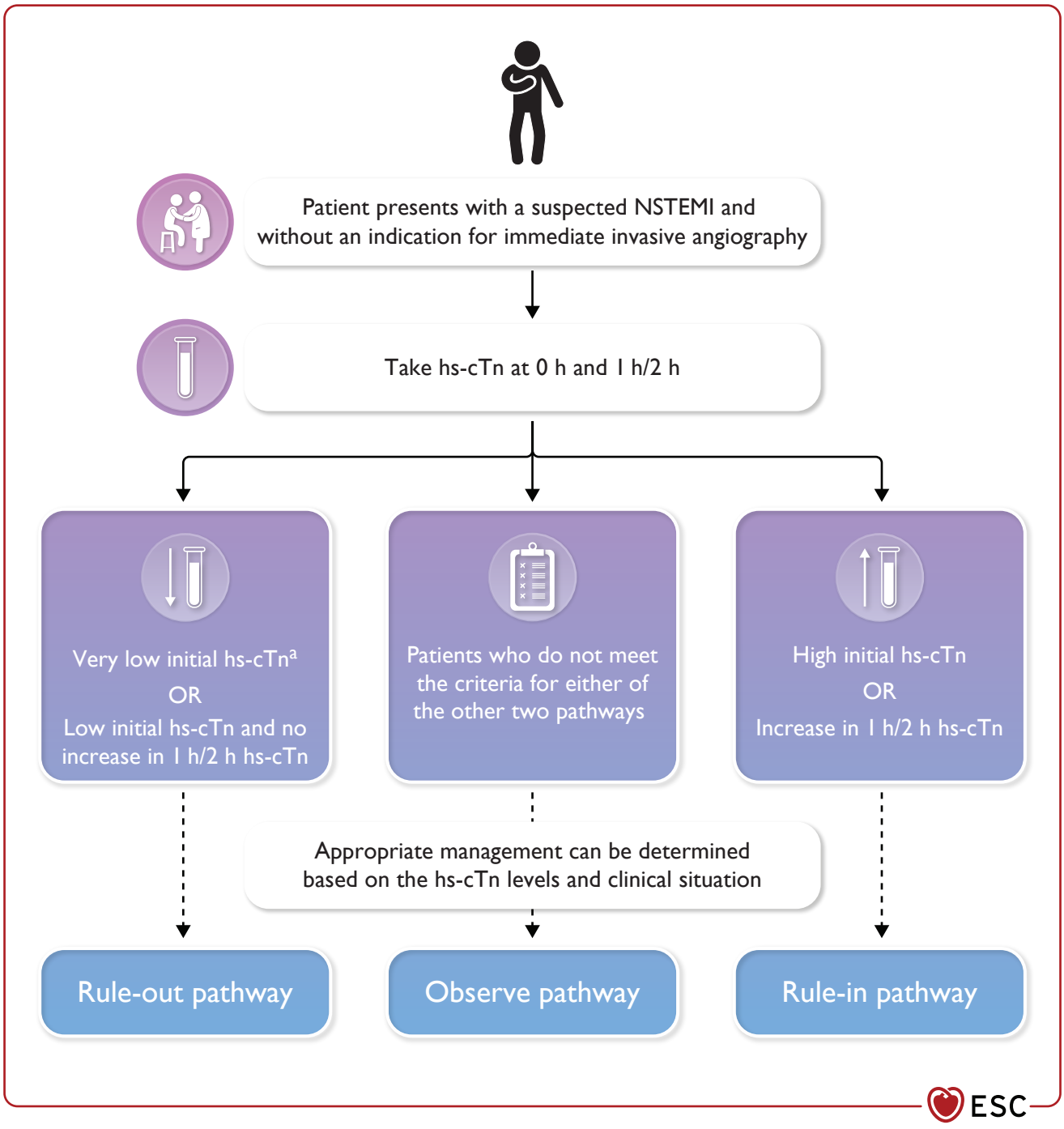


Figure 6 The 0 h/1 h or 0 h/2 h rule-out and rule-in algorithms using high-sensitivity cardiac troponin assays in patients presenting to the emergency department with suspected NSTEMI and without an indication for immediate invasive angiography. hs-cTn, high-sensitivity cardiac troponin; NSTEMI, non-ST-elevation myocardial infarction. Patients are classified into one of three pathways as per the results of their hs-cTn values at 0 h (time of initial blood test) and 1 h or 2 h later. Patients with a very low initial hs-cTn value or patients with a low initial value and no 1 h/2 h change in hs-cTn are assigned to the 'rule-out' pathway. Patients with a high initial hs-cTn value or a 1 h/2 h change in hs-cTn are assigned to the 'rule-in' pathway. Patients who do not meet the criteria for the rule-out or rule-in strategies are assigned to the 'observe' pathway, and these patients should have hs-cTn levels checked at 3 h ± echocardiography in order to decide on further management. Cut-offs are assay specific (see Supplementary material online, *Table S4*) and derived to meet pre-defined criteria for sensitivity and specificity for NSTEMI. Potential management and testing options for each of the three strategies are provided in the relevant sections of the main text.[12–15,26,27,53,55–58,100,101] [a]Only applicable if the chest pain onset was >3 h prior to the 0 h hs-cTn measurement.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

3.3.4.2. Practical guidance on how to implement the European Society of Cardiology 0 h/1 h algorithm

In order to maximize the safety and feasibility of implementing the 0 h/1 h algorithm, blood samples for hs-cTn at 0 h and 1 h should be obtained irrespective of other clinical details and pending results (see caveats of using rapid algorithms in Supplementary data online, *Section 3.3.2.2*). This may result in unnecessary cTn measurements in the ~10–15% of patients with very low 0 h concentrations and chest pain onset >3 h, but substantially facilitates the process and thereby further increases patient safety. Similarly, the 0 h blood sample should be obtained immediately after admission to the ED.

3.3.5. Other biomarkers

The use of biomarkers other than cTn for the diagnosis of ACS is not recommended (unless cTn is not available). Among the multitude of additional biomarkers evaluated for the diagnosis of NSTEMI, only creatine kinase myocardial band isoenzyme, myosin-binding protein C, and copeptin may have clinical relevance when used in combination with (standard) cTn T/I, although in most clinical situations their incremental value above and beyond cTn is limited.[45,46,83,102–114]

3.4. Diagnostic tools | Non-invasive imaging

3.4.1. Echocardiography

In emergency rooms and chest pain units, transthoracic echocardiography (TTE) performed or interpreted by trained healthcare professionals should be routinely available. In cases of suspected ACS with diagnostic uncertainty, TTE can be useful to identify signs suggestive of ongoing ischaemia or prior MI. However, this should not result in relevant delays in transfer to the cardiac catheterization laboratory if there is suspicion of an acute coronary artery occlusion. TTE can also be useful to suggest alternative aetiologies associated with chest pain (i.e. acute aortic disease, RV signs in pulmonary embolism [PE]). All patients presenting with CS or haemodynamic instability should undergo emergency TTE to try to identify the underlying cause—in particular, to assess LV and RV function and look for evidence of mechanical complications.

3.4.2. Computed tomography

Upon clinical presentation, CT is often the diagnostic tool of choice for ruling out alternative potentially life-threatening differential diagnoses of ACS, like PE or aortic dissection (this should be an ECG-gated contrast CT angiogram with full coverage of the thoracic aorta and the proximal head and neck vessels). Generally, CT does not have a role in patients presenting with suspicion of ongoing acute coronary occlusion, for whom emergency ICA is the priority.

Coronary CT angiography (CCTA) has been investigated in many trials for the assessment of patients presenting to the ED with suspected NSTE-ACS. However, trials investigating CCTA in the era of hs-cTn assays may be of greater relevance for contemporary practice. The BEACON (Better Evaluation of Acute Chest Pain with Coronary Computed Tomography Angiography) study showed no reduction of in-hospital duration of stay or hospital admission in the CCTA arm compared with patients investigated with hs-cTn, with similar results to those observed in the ROMICAT II (Rule Out Myocardial Ischemia/Infarction by Computer Assisted Tomography) and RAPID-CTCA (Rapid Assessment of Potential Ischaemic Heart Disease with CTCA) trials.[115–117] In the latter study, a default approach using early non-invasive CCTA in patients with suspected NSTE-ACS did not improve clinical outcomes at 1 year and was associated with a modest increase in the duration and cost of the hospital stay. A default approach using CCTA as the first-line imaging investigation in patients with suspected NSTE-ACS is therefore not recommended. However, CCTA may provide added value in certain clinical settings (i.e. for patients in the observe zone in whom cTn and ECG results remain inconclusive). A normal CCTA (ruling out both obstructive and non-obstructive plaque) has a high NPV to exclude ACS and is associated with excellent clinical outcomes.

The systematic use of CCTA in rule-out patients after hospital discharge may identify the presence of obstructive or non-obstructive plaque and guide preventative medical therapies.[118] CCTA can also be used to risk stratify selected low-risk NSTEMI patients. Such patients, who are found to have normal coronary arteries, non-obstructive coronary disease, or distal obstructive disease, may then not require ICA.[119–121] Of note, the utility of CCTA may be limited in patients with tachycardia, established coronary artery disease (CAD), previous stents, or extensive coronary calcification.

3.4.3. Cardiac magnetic resonance imaging with or without stress testing

Cardiac magnetic resonance (CMR) imaging delineates cardiac structure and function, and also has the ability to provide assessments of myocardial perfusion and the pattern of myocardial injury. CMR is the imaging test of choice when poor echocardiographic windows preclude diagnostic echocardiographic evaluation. CMR allows direct visualization of infarcted regions, providing information on scarring and viability that can be differentiated from other forms of myocardial injury (e.g. myocarditis). CMR is therefore of particular clinical value in establishing a diagnosis of AMI where there is diagnostic uncertainty. CMR can also be useful in identifying the culprit vascular territory and in confirming a diagnosis of myocarditis or takotsubo cardiomyopathy, amongst other differentials. CMR is of particular value in establishing a diagnosis in patients presenting with a working diagnosis of myocardial infarction with non-obstructive coronary arteries (MINOCA) following invasive angiography and is the gold standard for the assessment of LV thrombus.

Recommendation Table 2 — Recommendations for non-invasive imaging in the initial assessment of patients with suspected acute coronary syndrome

Recommendations	Class[a]	Level[b]
Emergency TTE is recommended in patients with suspected ACS presenting with cardiogenic shock or suspected mechanical complications.	I	C
In patients with suspected ACS, non-elevated (or uncertain) hs-cTn levels, no ECG changes and no recurrence of pain, incorporating CCTA or a non-invasive stress imaging test as part of the initial workup should be considered.[116,122–127]	IIa	A
Emergency TTE should be considered at triage in cases of diagnostic uncertainty but this should not result in delays in transfer to the cardiac catheterization laboratory if there is suspicion of an acute coronary artery occlusion.	IIa	C
Routine, early CCTA in patients with suspected ACS is not recommended.[117]	III	B

ACS, acute coronary syndrome; CCTA, coronary computed tomography angiography; ECG, electrocardiogram; hs-cTn, high-sensitivity cardiac troponin; TTE, transthoracic echocardiography.
[a]Class of recommendation.
[b]Level of evidence.

© ESC 2023

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Cardiac magnetic resonance can also assess myocardial perfusion with pharmacological stress. This can be used as an alternative to CCTA in the assessment of patients in the observe zone following ECG and hs-cTn assessments, particularly in those with advanced, established CAD, in whom assessments of myocardial perfusion and viability may provide more useful information than CCTA. Some additional information on CMR, single-photon emission computerized tomography (SPECT) perfusion imaging and stress echocardiography is provided in the Supplementary data online.

Depending on local expertise and availability, other forms of stress imaging (e.g. SPECT, nuclear, stress echo) can be used to assess patients in the observe zone.

3.5. Differential diagnosis for acute chest pain

Several cardiac and non-cardiac conditions that may mimic ACS should be considered in the differential diagnosis of acute chest pain as part of the clinical assessment. More information about the differential diagnosis of acute chest pain is provided in the sections on MINOCA and Type 2 MI and in the Supplementary data online, *Table S5*.

4. Initial measures for patients presenting with suspected acute coronary syndrome | Initial treatment

4.1. Pre-hospital logistics of care

Individuals experiencing acute chest pain in the community represent an undifferentiated population, often presenting *ad hoc* to first medical responders in the pre-hospital setting. These patients should undergo immediate risk assessment and triage following local protocols established within the emergency medical service (EMS) (*Figures 7* and *8*).

If the first responding medical professional suspects ACS, a 12-lead ECG should be acquired and analysed as soon as possible. It is recommended that all medical and paramedical personnel caring for ACS patients within the EMS setting have access to defibrillation equipment and are trained in basic cardiac life support. Patients with suspected ACS are initially categorized on the basis of the 12-lead ECG and triaged into two initial treatment pathways: (i) one for patients with an ECG consistent with STEMI (persistent ST-segment elevation or equivalent ECG patterns) (*Figure 7*); and (ii) one for patients without ST-segment elevation or equivalent ECG patterns (suspected NSTE-ACS) (*Figure 8*). The initial ECG-guided risk stratification should also trigger treatment decisions in the pre-hospital setting, including the choice of target hospital, and serve to determine the sequence of initial investigations and interventions (including pharmacological), in particular, the timing of ICA.

An initial diagnosis of suspected STEMI portends a higher risk of immediate, life-threatening complications (e.g. ventricular fibrillation [VF]). Accordingly, there is an indication for initiating an emergency reperfusion strategy and direct transfer to a centre with 24/7 PCI capabilities. Patients who present with an ECG without ST-segment elevation (or equivalent ECG patterns) but have ongoing ischaemic symptoms should undergo pre-hospital triage in accordance with protocols for patients in the STEMI pathway, since they also face immediate risks, including ventricular arrhythmias.

4.1.1. Time to treatment

Time to treatment is vital for the care of patients triaged to the STEMI pathway. Components of the total ischaemic time, contributors to delays in initial management, and the selection of reperfusion strategy for STEMI patients are shown in *Figure 7*. Treatment times reflect the efficiency and quality of care of a system taking care of patients with suspected STEMI. The multidisciplinary STEMI treatment pathway should be subject to continuous clinical audit in order to assess the treatment times for individual patients and identify opportunities for healthcare improvement through quality indicators (QIs). If projected QIs are not met, interventions are needed to improve the performance of the system.

Recognition of ischaemic symptoms by individuals in the community has pivotal importance in activation of the out-of-hospital pathway, and this is especially relevant to first responders without healthcare training. The recommended action should be to contact the EMS rather than to self-present to an ED or primary care clinician.

The time from symptom onset to 'first call for help' is associated with socioeconomic factors and sex.[128] In order to avoid delays through failure to recognize and act on symptoms of a 'heart attack', community education initiatives should target less well-served groups (i.e. those from deprived communities, ethnic minority groups) and use targeted public health messaging (i.e. avoiding stereotyped messaging that underpins a negative bias based on sex, ethnicity, or social background, and using language and images that will resonate with those groups). System delays are representative of the quality of care and it is recommended to measure these as QIs.

4.1.2. Healthcare systems and system delays

For patients with suspected STEMI, the system delay (the time from when the patient contacts the healthcare system to reperfusion) is amenable to improvement by organizational measures, whereas patient delay is multifactorial. System delay is a predictor of mortality in STEMI patients treated with primary PCI (PPCI).[129–131] When a working diagnosis of STEMI is made in the pre-hospital setting (EMS), immediate activation of the catheterization laboratory team reduces treatment delays and mortality.[132–136]

When a STEMI working diagnosis is made by the EMS in the pre-hospital setting and the patient is triaged for emergency invasive management, they should bypass the ED and go straight to the catheterization laboratory. Bypassing the ED is associated with a significant saving in the time from FMC to wire crossing and may be associated with improved survival.[137–139] For patients presenting to a non-PCI centre with a suspected STEMI, the 'door-in to door-out time'—defined as the duration between arrival of the patient at the hospital to discharge of the patient in an ambulance 'en route' to the PCI centre—is also an important clinical performance measure, and a door-in to door-out time of ≤30 min is recommended to expedite reperfusion therapy.[140]

4.1.3. Emergency medical services

At a national level, an EMS with an easily recalled and well-publicised unique medical dispatching number (112 for most European Union countries) is important to speed-up system activation. Parallel circuits for the referral and transport of patients with suspected STEMI that bypass the EMS should be avoided. The ambulance system plays a critical role in the early management of patients with suspected STEMI, including immediately establishing the initial diagnosis, triage, and treatment.[129,141]

Ambulances in the EMS must be equipped with ECG recorders, defibrillators, telemetry devices, and at least one person trained in advanced life support. The quality of the care provided depends on the

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

training of the staff involved. Ambulance personnel must be trained to recognize ischaemic symptoms, administer oxygen when appropriate, secure intravenous (i.v.) access, effectively relieve pain, administer fibrinolysis when indicated, and provide basic life support.[142] Ambulance staff should record an ECG as soon as possible for diagnostic purposes and either interpret the ECG or transmit it so that it can be reviewed by experienced staff to establish or refute a working diagnosis of STEMI. Regular and structured training of ambulance staff is mandatory for a high-quality pre-hospital service.

4.1.4. General practitioners

In some countries, primary care clinicians (general practitioners) play an important role in the early care of patients with suspected ACS and may provide the FMC. Education and training of general practitioners in emergency, pre-hospital care is essential for the delivery of optimal pre-hospital care in this setting. The responsibilities of the primary care clinicians may include diagnosis, activation of the EMS, risk stratification, and initiation of pre-hospital treatment. However, in most settings, consultation with a general practitioner instead of a direct call to the EMS will increase the pre-hospital delay. Therefore, the public should be educated to call the EMS directly rather than a primary care physician for symptoms suggestive of ACS.

4.1.5. Organization of ST-elevation myocardial infarction treatment in networks

It is recommended that a regional reperfusion strategy is established to maximize the efficiency of care for patients with a working diagnosis of STEMI.[143] The optimal treatment of patients with a working diagnosis of STEMI should be based on the implementation of networks between hospitals with various levels of clinical service provision (the 'hub and spoke' model), linked by a prioritized and efficient ambulance service. A PCI centre is a multidisciplinary acute care centre that provides emergency invasive management 24/7 for patients presenting with suspected STEMI. This centre should also provide intensive care facilities, and more advanced centres should offer cardio-thoracic services, advanced haemodynamic support, and surgery.

The goal of STEMI networks is to provide optimal care while minimizing delays, thereby improving clinical outcomes. Cardiologists should actively collaborate with all stakeholders, particularly emergency physicians, in establishing such networks. The main features of such a network are detailed in Supplementary data online, *Table S6*. It is recommended that the EMS should transport patients with a working diagnosis of STEMI to hospitals with a 24/7 service for PCI, bypassing non-PCI-capable hospitals.[144] Further information on this topic is provided in the Supplementary data online.

Geographic areas where the expected transfer time to the primary PCI centre makes it impossible to routinely achieve the maximal allowable delays indicated in the recommendations should develop protocols for rapid fibrinolysis at the place of STEMI diagnosis, with the aim of treatment within 10 min of FMC, followed by immediate transfer to a centre with 24/7 service for PCI. Such networks increase the proportion of patients receiving reperfusion with the shortest possible treatment delay.[145–147] The quality of care, time delays, and patient outcomes should be measured and reported to the healthcare professionals contributing to the EMS.

4.2. Emergency care

4.2.1. Initial diagnosis and monitoring

Management of ACS starts from the point of FMC, when a working diagnosis of ACS is established. The working diagnosis of ACS is usually based on symptoms consistent with myocardial ischaemia and the signs on a 12-lead ECG (see *Section 3.2*). It is recommended to initiate ECG monitoring as soon as possible in all patients with suspected ACS in order to detect life-threatening arrhythmias and to allow prompt defibrillation if indicated.

4.2.2. Acute pharmacotherapy

4.2.2.1. Oxygen

Oxygen supplementation is recommended in ACS patients with hypoxaemia (oxygen saturations <90%). Oxygen supplementation in patients who are not hypoxic (oxygen saturations >90%) is not associated with clinical benefits and is therefore not recommended.[148,149]

4.2.2.2. Nitrates

Sublingual nitrate may be helpful to relieve ischaemic symptoms. However, a reduction in chest pain after nitroglycerine administration can be misleading and is not recommended as a diagnostic manoeuvre.[150] In patients with an ECG compatible with ongoing STEMI and symptom relief after nitroglycerine administration, it is recommended to obtain another 12-lead ECG. Complete normalization of ST-segment elevation, along with relief of symptoms, after nitroglycerine administration is suggestive of coronary spasm, with or without associated MI. Nitrates should not be given to patients with hypotension, marked bradycardia or tachycardia, RV infarction, known severe aortic stenosis, or phosphodiesterase 5 inhibitor use within the previous 24–48 h.

4.2.2.3. Pain relief

Intravenous opioids (e.g. morphine 5–10 mg) should be considered for the relief of severe chest pain. Other forms of pain relief (e.g. nitrous oxide/oxygen plus i.v. acetaminophen/paracetamol) have been reported to be inferior to morphine.[151] However, morphine may enhance nausea and vomiting and slow the gastrointestinal absorption of oral medicines, which may delay the onset of action of orally administered antiplatelet therapy.[152,153] Evidence from small-scale trials suggests that i.v. morphine may also reduce myocardial and microvascular damage when given to patients with ongoing acute coronary artery occlusion, though co-administration with metoclopramide appears to negate this effect. Conversely, morphine has also been reported to reduce antiplatelet activity after administration of ticagrelor, though this effect was rescued by metoclopramide administration.[154,155] The positive effects of morphine on myocardial damage may potentially be related to reduced oxygen consumption as a result of decreased preload and negative inotropy and chronotropy.

Platelet inhibition induced by oral $P2Y_{12}$ receptor antagonists may be delayed in patients with ongoing MI. Morphine may also further reduce absorption, delay the onset of action, and decrease the antiplatelet effect of oral $P2Y_{12}$ receptor inhibitors in MI patients, although this effect may vary between the different $P2Y_{12}$ inhibitors.[153,156–158] Further research is ongoing in this area, but at present it should be noted that currently available clinical data have not demonstrated any increase in the risk of adverse clinical outcomes as a result of any interaction between morphine and antiplatelet agents in the setting of ACS.[159–161]

4.2.2.4. Intravenous beta-blockers

Few RCTs testing early i.v. beta-blockers have been performed in the era of invasive management for patients with a working diagnosis of STEMI. Not all beta-blockers appear to exert the same

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

cardio-protective effect in the context of ongoing acute coronary occlusion, with metoprolol demonstrating the greatest protective effect in experimental studies.[162] Intravenous metoprolol is also the most widely tested beta-blocker in trials enrolling patients undergoing PPCI.[163,164] While the long-term clinical benefits associated with early i.v. metoprolol administration are not clear, it is safe when used in patients without signs of acute HF and has been consistently associated with a reduction in the incidence of VF and microvascular obstruction (MVO).[163–171] Based on these data, i.v. beta-blockers (preferably metoprolol) should be considered at the time of presentation in patients with a working diagnosis of STEMI undergoing PPCI with no signs of acute HF, a systolic blood pressure (SBP) >120 mmHg, and without other contraindications.[163–166,169] Administration of i.v. beta-blockers in patients with suspected NSTE-ACS has not been tested.

Recommendation Table 3 — Recommendations for the initial management of patients with acute coronary syndrome

Recommendations	Class[a]	Level[b]
Hypoxia		
Oxygen is recommended in patients with hypoxaemia (SaO_2 <90%).	I	C
Routine oxygen is not recommended in patients without hypoxaemia (SaO_2 >90%).[148,172]	III	A
Symptoms		
Intravenous opioids should be considered to relieve pain.	IIa	C
A mild tranquilizer should be considered in very anxious patients.	IIa	C
Intravenous beta-blockers		
Intravenous beta-blockers (preferably metoprolol) should be considered at the time of presentation in patients undergoing PPCI with no signs of acute heart failure, an SBP >120 mmHg, and no other contraindications.[163–167,169]	IIa	A
Pre-hospital logistics of care		
It is recommended that the pre-hospital management of patients with a working diagnosis of STEMI is based on regional networks designed to deliver reperfusion therapy expeditiously and effectively, with efforts made to make PPCI available to as many patients as possible.[145]	I	B
It is recommended that PPCI-capable centres deliver a 24/7 service and are able to perform PPCI without delay.[173,174]	I	B
It is recommended that patients transferred for PPCI bypass the emergency department and CCU/ICU and are transferred directly to the catheterization laboratory.[137,175–178]	I	B
It is recommended that EMS transfer patients with suspected STEMI to a PCI-capable centre, bypassing non-PCI centres.	I	C
It is recommended that ambulance teams are trained and equipped to identify ECG patterns suggestive of acute coronary occlusion and to administer initial therapy, including defibrillation, and fibrinolysis when applicable.[142]	I	C
It is recommended that all hospitals and EMS participating in the care of patients with suspected STEMI record and audit delay times and work together to achieve and maintain quality targets.	I	C

CCU, cardiac care unit; ECG, electrocardiogram; EMS, emergency medical services; ICU, intensive care unit; i.v., intravenous; PPCI, primary percutaneous coronary intervention; SaO_2, saturation of oxygen; SBP, systolic blood pressure; STEMI, ST-elevation myocardial infarction.
[a]Class of recommendation.
[b]Level of evidence.

© ESC 2023

5. Acute-phase management of patients with acute coronary syndrome

5.1. Selection of invasive strategy and reperfusion therapy

The definitions of the terms related to invasive strategy and reperfusion therapy are presented in *Table 3*.

Depending on the initial assessment of the ECG, the clinical context and haemodynamic stability, patients with suspected ACS should be classified as either:

(i) Patients with a working diagnosis of STEMI. These patients should be triaged for immediate reperfusion therapy (i.e. a PPCI strategy or fibrinolysis if PPCI is not possible within 120 min of diagnosis) (*Figure 7*).
Or
(ii) Patients with a working diagnosis of NSTE-ACS. For these patients:
- An inpatient invasive strategy is recommended.
- An immediate invasive strategy is recommended when any very high-risk feature is present (*Figure 8*).
- An early (i.e. within 24 h) invasive strategy should be considered when any high-risk features are present (*Figure 8*).

5.2. Acute coronary syndrome managed with invasive strategy

Invasive management strategies are time sensitive. It is recommended that patients triaged to an immediate invasive strategy (those with high suspicion of ongoing acute coronary artery occlusion [i.e. persistent ST-segment elevation or equivalents] or NSTE-ACS with any very high-risk characteristics) receive emergency angiography as soon as possible. High-risk NSTE-ACS patients (e.g. ruled in as NSTEMI as per the 0 h/1 h or 0 h/2 h ESC algorithms, with dynamic ST-segment or T wave changes, with transient ST-segment elevation, or with a Global Registry of Acute Coronary Events [GRACE] risk score >140) should be considered for an early invasive strategy (i.e. undergoing angiography within 24 h).

5.2.1. Primary percutaneous coronary intervention strategy for ST-elevation myocardial infarction

In patients with a working diagnosis of STEMI, a PPCI strategy (i.e. immediate angiography and PCI as needed) is the preferred reperfusion strategy, provided it can be performed in a timely manner (i.e. within

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

120 min of the ECG-based diagnosis, *Figure 7*). RCTs have shown that if the delay to treatment is similar, PPCI is superior to fibrinolysis in reducing mortality, non-fatal reinfarction, and stroke.[52,179] However, in some circumstances, PPCI is not an immediate option and fibrinolysis should be initiated expeditiously as part of a pharmaco-invasive strategy, provided the patient has presented within 12 h of symptom onset (see *Section 5.3*).

There is a lack of contemporaneous data to inform the treatment delay limit at which the advantage of PCI over fibrinolysis is lost. For simplicity, an absolute time of 120 min from STEMI diagnosis to PCI-mediated reperfusion (i.e. wire crossing of the infarct-related artery [IRA]) rather than a relative PCI-related delay over fibrinolysis has been chosen. Given the recommended time interval of 10 min from STEMI diagnosis to administration of a bolus of fibrinolytics (see below), the 120 min absolute time delay would correspond to a relative PCI-related delay in the range of 110–120 min. This is within the range of the times identified as the limit of delay below which PCI should be chosen in older studies and registries.[176,180–184]

For patients who undergo fibrinolysis, rescue PCI is indicated if fibrinolysis fails (i.e. ST-segment resolution <50% within 60–90 min of fibrinolytic administration) or in the presence of haemodynamic or electrical instability, worsening ischaemia, or persistent chest pain.[184,185] Patients with successful fibrinolysis should undergo early invasive angiography (i.e. within 2–24 h from the time of the lytic bolus injection) (see *Section 5.3*).[186]

Patients with a working diagnosis of STEMI who present to a non-PCI centre should be immediately transferred to a PCI-capable centre (*Figure 7*) for a timely PPCI strategy. If PPCI is not feasible within 120 min, patients should undergo immediate fibrinolysis followed by transfer to a PCI centre without waiting for signs of reperfusion. For patients presenting after 12 h from symptom onset, a PPCI strategy is preferred over fibrinolysis in all cases.

Emergency coronary artery bypass grafting (CABG) surgery should be considered for patients with a patent IRA but with unsuitable anatomy for PCI, and either a large myocardial area at jeopardy or with CS. In patients with MI-related mechanical complications who require coronary revascularization, CABG is recommended at the time of surgical repair. In STEMI patients with failed PCI or with an acute coronary occlusion not amenable to PCI, emergency CABG is infrequently performed because the benefits of surgical revascularization in this setting are less certain.[185,187,188] Because there will be a delay to reperfusion with CABG in this situation, the probability of myocardial salvage to a degree sufficient to impact on prognosis is considered low. In addition, the surgical risks of CABG in this setting may be elevated.

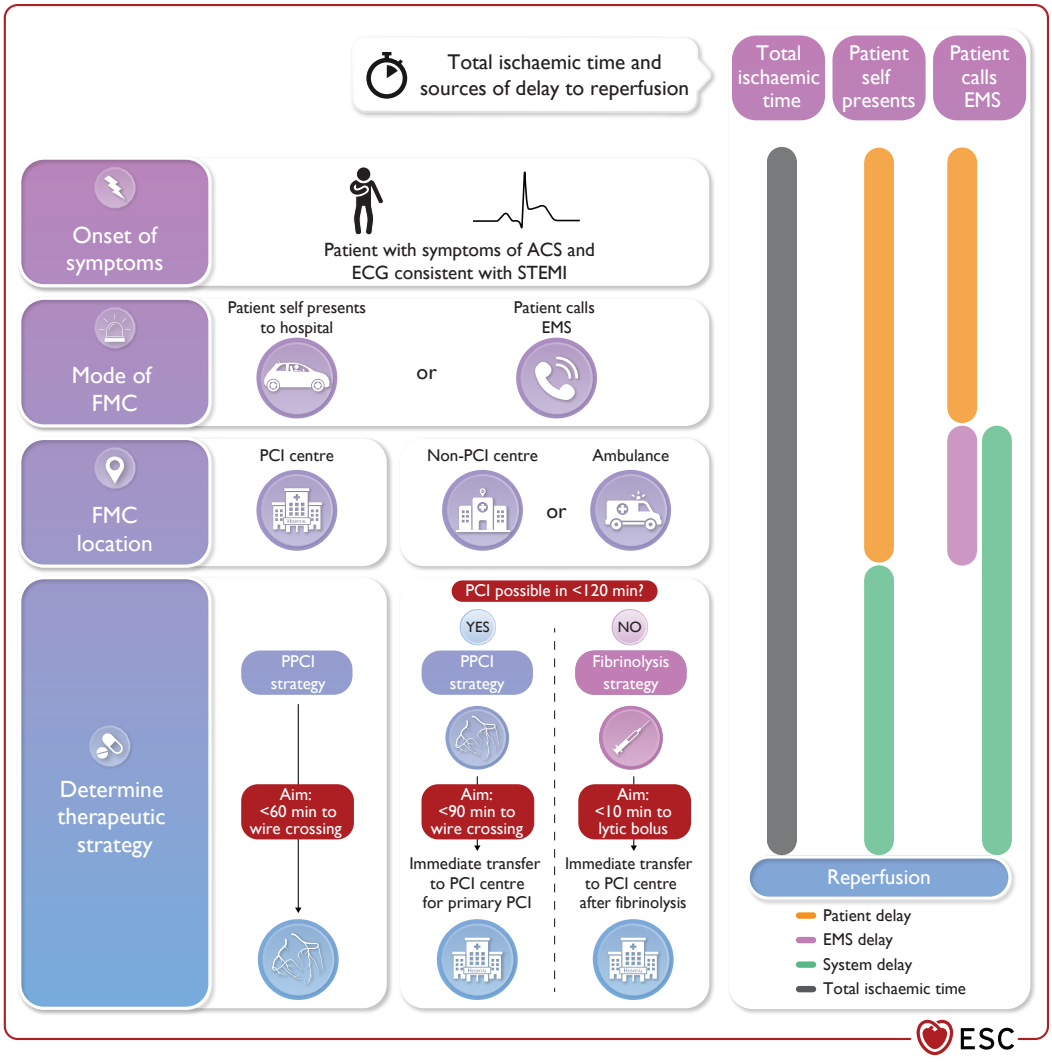


Figure 7 Modes of presentation and pathways to invasive management and myocardial revascularization in patients presenting with STEMI. ACS, acute coronary syndrome; ECG, electrocardiogram; EMS, emergency medical services; FMC, first medical contact; PCI, percutaneous coronary intervention; PPCI, primary percutaneous coronary intervention; STEMI, ST-segment elevation myocardial infarction.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

5.2.1.1. Invasive strategy in ST-elevation myocardial infarction late presenters

While routine immediate angiography and PCI (if indicated) are clearly associated with clinical benefit in patients presenting within 12 h of symptom onset, the value of a routine PPCI strategy in STEMI patients presenting later than 12 h after symptom onset is less well established.

A small RCT in 347 STEMI patients presenting 12–48 h after symptom onset and without persistent symptoms reported that a routine PPCI strategy improved myocardial salvage and long-term survival compared with conservative treatment.[189,190] This observation is supported by a recent analysis of data from three nationwide observational studies from the FAST-MI (French Registry of Acute ST-elevation and non-ST-elevation Myocardial Infarction) programme, which showed a significant lower rate of all-cause death at 1 month (2.1% vs. 7.2%) and after a median follow-up of 58 months (30.4% vs. 78.7%) with an invasive strategy in comparison to conservative treatment.[191] However, in stable patients with persistent occlusion of the IRA 3–28 days after MI, the large ($n = 2166$) Occluded Artery Trial (OAT) reported no clinical benefit from routine coronary intervention with medical management in comparison to medical management alone.[192,193] A meta-analysis of trials testing whether late recanalization of an occluded IRA is beneficial also showed no benefit of reperfusion.[194] Therefore, routine PCI of an occluded IRA in STEMI patients presenting >48 h after onset of symptoms and without persistent symptoms is not indicated.[192,193] These patients should be managed in the same way as patients with chronic total occlusion according to the ESC Guidelines for the diagnosis and management of chronic coronary syndromes (CCS).[195]

5.2.2. Immediate invasive strategy for non-ST elevation acute coronary syndrome

An immediate invasive strategy refers to emergency (i.e. as soon as possible) angiography and PCI if indicated. This is recommended for patients with a working diagnosis of NSTE-ACS and any of the following very high-risk criteria:

- Haemodynamic instability or CS.
- Recurrent or ongoing chest pain refractory to medical treatment.
- Acute HF presumed secondary to ongoing myocardial ischaemia.
- Life-threatening arrhythmias or cardiac arrest after presentation.
- Mechanical complications.
- Recurrent dynamic ECG changes suggestive of ischaemia (particularly with intermittent ST-segment elevation).

5.2.3. Routine vs. selective invasive strategy

A routine invasive strategy with inpatient coronary angiography is recommended for patients with a confirmed diagnosis of NSTEMI or a working diagnosis of NSTE-ACS and a high index of suspicion for UA. In patients with a working diagnosis of NSTE-ACS, multiple RCTs comparing routine vs. selective invasive strategies have been conducted and their results have been pooled in several meta-analyses.[196–200] The available evidence indicates that a routine invasive strategy does not reduce all-cause mortality risk in the overall population of NSTE-ACS patients, but reduces the risk of composite ischaemic endpoints, particularly in high-risk patients. A routine invasive strategy can increase the risk of peri-procedural complications and bleeding. However, most of the available evidence is based on old RCTs that were conducted before the implementation of several important developments in PCI, including radial access, modern drug-eluting stents (DES), complete functional revascularization for multivessel disease (MVD), improved co-adjuvant pharmacological therapies, and contemporary biomarker assays.

5.2.3.1. Early vs. delayed invasive strategy for non-ST elevation acute coronary syndrome

An early invasive strategy refers to routine invasive angiography (and PCI if needed) within 24 h of presentation. This should be considered in patients with a working diagnosis of NSTE-ACS and any of the following high-risk criteria:

- A confirmed diagnosis of NSTEMI based on current recommended ESC hs-cTn algorithms.
- Dynamic ST-segment or T wave changes.
- Transient ST-segment elevation.
- A GRACE risk score >140.

Several meta-analyses have pooled data from multiple RCTs assessing different timing intervals of invasive angiography among NSTE-ACS patients. None of these studies observed superiority of early invasive strategies compared with routine invasive strategies for death or non-fatal MI, although early invasive strategies were associated with a lower risk of recurrent/refractory ischaemia and a shorter duration of hospital stay.[201–203] A collaborative meta-analysis comparing an early vs. a delayed invasive strategy using a modified individual patient data approach observed no difference in mortality overall but a survival benefit in high-risk patients, including those with a GRACE risk score >140 and those with positive troponin, although tests for interaction were inconclusive.[202] The largest meta-analysis to date (17 RCTs >10 000 patients) reported that, in all-comers with NSTE-ACS, early ICA only significantly reduced the risk of recurrent ischaemia and duration of stay, with no significant reductions in all-cause mortality, MI, admission for HF, or repeat revascularization.[203] The main limitation of the interpretation of meta-analyses of these RCTs is the variability of the time to invasive angiography in the individual trials: while invasive angiography was virtually always performed within 24 h of randomization in the early invasive strategy groups, the time from randomization to angiography was heterogeneous in the delayed invasive groups. In many trials, delayed angiography was performed within 24 h of randomization (albeit later than in the early angiography arm of the respective trial). Additionally, the diagnosis of NSTEMI was not based on the current recommended ESC hs-cTn algorithms. Moreover, studies assessing the value of a GRACE risk score >140 to guide the timing of ICA and revascularization in the era of hs-cTn for the diagnosis of NSTEMI are lacking. Further detail on the interaction between treatment effect according to GRACE score and its components in individual trials is provided in the Supplementary data online. Data from observational studies is concordant with trial data, without a strong signal of benefit with early versus delayed coronary angiography.[204]

A selective invasive approach after appropriate ischaemia testing or detection of obstructive CAD by CCTA is recommended in patients without very high- or high-risk features and a low index of suspicion for NSTE-ACS. These patients should be managed as per the ESC Guidelines for the management of CCS.[195] A selective invasive approach is also appropriate for patients with NSTEMI or UA who are not deemed good candidates for coronary angiography.

5.2.4. Summary of invasive strategies for patients with non-ST elevation acute coronary syndrome

In summary, very high-risk NSTE-ACS patients are recommended to undergo an immediate invasive strategy with emergency angiography

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

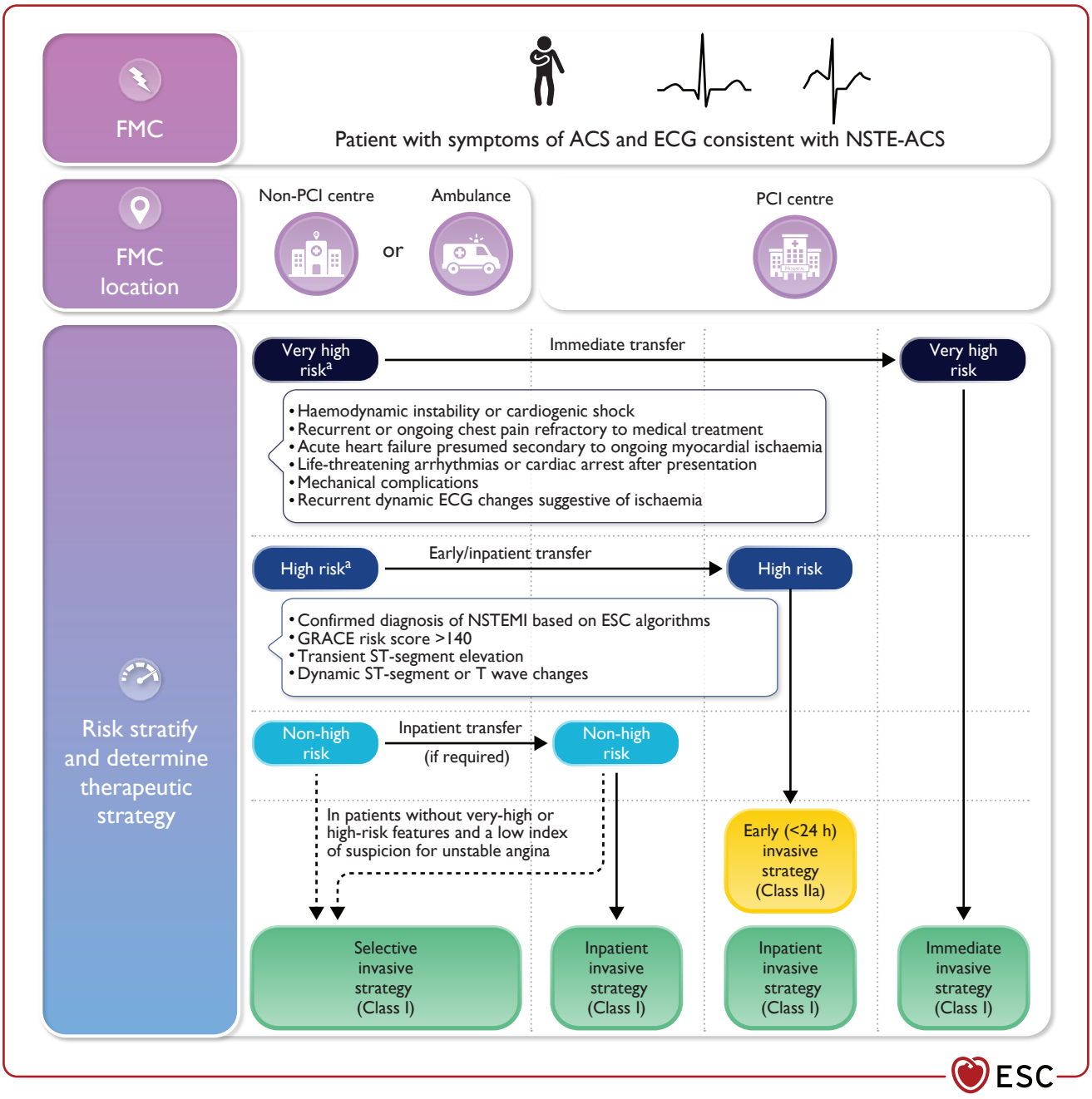


Figure 8 Selection of invasive strategy and reperfusion therapy in patients presenting with NSTE-ACS. ACS, acute coronary syndrome; CS, cardiogenic shock; ECG, electrocardiogram; FMC, first medical contact; GRACE, Global Registry of Acute Coronary Events; hs-cTn, high-sensitivity cardiac troponin; NSTE-ACS, non-ST-elevation acute coronary syndrome; NSTEMI, non-ST-elevation myocardial infarction; PCI, percutaneous coronary intervention; UA, unstable angina. This figure summarizes the selection of invasive strategy and reperfusion therapy in patients presenting with ACS. [a]Risk criteria: Patients who meet any one of the 'very high-risk' NSTE-ACS criteria should undergo an immediate invasive strategy; these very high-risk criteria include haemodynamic instability or CS, recurrent or refractory chest pain despite medical treatment, life-threatening arrhythmias, mechanical complications of MI, HF clearly related to ACS, and recurrent dynamic ST-segment or T wave changes, particularly with intermittent ST-segment elevation. Patients with NSTE-ACS who meet any of the 'high-risk' criteria (confirmed NSTEMI as per the hs-cTn-based ESC algorithm, NSTE-ACS with GRACE score >140, dynamic ST-segment or T wave changes, or transient ST-segment elevation) should be considered for early invasive angiography (i.e. within 24 h) and should undergo an inpatient invasive strategy. An invasive strategy during hospital admission is recommended in NSTE-ACS patients with high-risk criteria or with a high index of suspicion for UA. In selected patients a selective invasive strategy can also be an option. See Recommendation Table 4 for full details.

and PCI if required. High-risk NSTE-ACS patients are recommended to undergo an inpatient invasive strategy and should be considered for an early invasive strategy (i.e. within 24 h). For patients who do not meet any of the very high-risk or high-risk criteria (generally patients with clinical suspicion for NSTE-ACS and non-elevated troponins or patients with elevated troponins not meeting the criteria for MI), the strategy can be tailored based on the degree of clinical suspicion. For patients with a high index of suspicion for UA, an

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

inpatient invasive strategy is recommended. Conversely, for patients with a low index of suspicion, a selective invasive approach is recommended.

5.3. Fibrinolysis and pharmaco-invasive strategy in patients with ST-elevation myocardial infarction

5.3.1. Benefit and indication of fibrinolysis

Fibrinolytic therapy is an important reperfusion strategy for STEMI patients presenting within 12 h of symptom onset when PPCI cannot be performed in a timely manner; it prevents 30 early deaths per 1000 patients treated within 6 h of symptom onset.[205] The largest absolute treatment benefit is seen among those patients at the highest risk, including the elderly. Successful reperfusion is generally associated with significant improvement in ischaemic symptoms, ≥50% ST-segment resolution, and haemodynamic stability. The doses of fibrinolytic agents and concomitant antithrombotic therapies are given in the Fibrinolysis and Pharmaco-invasive Strategy provided in the Supplementary data online, *Section 6.3*.

5.3.1.1. Pre-hospital fibrinolysis

If trained medical or allied health staff can interpret the ECG on site, or transmit the ECG for remote interpretation, it is recommended to initiate fibrinolytic therapy in the pre-hospital setting. A fibrin-specific agent (i.e. tenecteplase, alteplase, or reteplase) is the preferred agent. The goal is to start fibrinolytic therapy within 10 min of the STEMI diagnosis. Fibrinolytic therapy initiation should not be delayed by waiting for the results of cardiac biomarker testing. In a meta-analysis of six randomized trials ($n = 6434$), pre-hospital fibrinolysis compared with in-hospital fibrinolysis reduced early mortality by 17%, particularly when administered in the first 2 h after symptom onset.[51,206] These, and more recent, data support the pre-hospital initiation of fibrinolytic treatment when a reperfusion strategy is indicated.[145,207–209] The STREAM (Strategic Reperfusion Early After Myocardial Infarction) trial demonstrated that pre-hospital fibrinolysis followed by an early PCI strategy was associated with a similar outcome to transfer for PPCI in STEMI patients presenting within 3 h of symptom onset who could not undergo PPCI within 1 h of FMC, although a slight excess of intracranial bleeding was observed with the investigational strategy.[184,210] This excess in intracranial bleeding was blunted by halving the dose of tenecteplase in patients >75 years of age.

5.3.1.2. Angiography and percutaneous coronary intervention after fibrinolysis (pharmaco-invasive strategy)

It is recommended that patients should be transferred to a PCI centre immediately after initiation of lytic therapy (*Figure 7*). In cases of failed fibrinolysis or evidence of re-occlusion or re-infarction with recurrence of ST-segment elevation, immediate angiography and rescue PCI are indicated.[185,211] In this setting, re-administration of fibrinolysis is not beneficial and is discouraged.[185] Even if it is likely that fibrinolysis is successful (e.g. ST-segment resolution >50% at 60–90 min; typical reperfusion arrhythmia; and disappearance of chest pain), routine early angiography (i.e. within 2–24 h) is recommended. Several randomized trials have shown that routine early angiography with subsequent PCI (if required) after fibrinolysis reduced the rates of re-infarction and recurrent ischaemia in comparison to a 'watchful waiting' strategy (i.e. a strategy in which angiography and revascularization were performed only in patients with spontaneous or induced severe ischaemia or LV dysfunction, or in patients with a positive outpatient ischaemia test).[186,209,212–215] A network meta-analysis including 15 357 STEMI patients treated with fibrinolytic therapy ($n = 4212$), PPCI ($n = 6139$), or fibrinolysis followed by routine immediate or early PCI ($n = 5006$) investigated whether STEMI patients should be transferred to a PCI-capable facility immediately (defined as a facilitated PCI approach) or within a day (e.g. <24 h, defined as a pharmaco-invasive approach).[209] After PPCI, the pharmaco-invasive strategy was the second most favourable approach, with an odds ratio (OR) for death of 0.79 (95% confidence interval [CI], 0.59–1.08) compared with conventional fibrinolytic therapy. This supports the safety of transferring STEMI patients to a PCI-capable centre for angiography within 2–24 h. The benefit of routine early PCI after fibrinolysis was demonstrated without an increased risk of adverse events (stroke or major bleeding), and across the spectrum of the investigated patient subgroups.[209,216] Therefore, early angiography with subsequent PCI if required is the recommended standard of care after successful fibrinolysis (*Figure 7*). Observational analysis of registry data has also provided some further support for the use of a pharmaco-invasive strategy.[130]

The optimal time delay between successful fibrinolysis and PCI has not been clearly defined; there has been a wide variation in this time delay in trials, ranging from a median of 1.3 to 17 h.[184,185,206,215,217] Based on these data, a time window for PCI of 2–24 h after successful lysis is recommended.

5.3.1.2.1. Comparison of fibrinolytic agents. Some information on comparisons of fibrinolytic agents is provided in the Supplementary data online, *Section 6.3.1*.

5.3.1.2.2. Hazards of fibrinolysis and contraindications. Some information regarding the hazards of, and contraindications to, fibrinolysis is provided in the Supplementary data online, *Section 6.3.2*.

5.4. Patients not undergoing reperfusion

The management of ACS patients not undergoing reperfusion is discussed in the Supplementary data online, *Section 5.2*.

5.4.1. Patients who are not candidates for invasive coronary angiography

Information regarding the management of NSTE-ACS patients who are not candidates for invasive angiography is provided in the Supplementary data online, *Section 5.2.1*.

5.4.2. Patients with coronary artery disease not amenable to revascularization

Information regarding the management of ACS patients with CAD that is not amenable to revascularization is provided in the Supplementary data online, *Section 5.2.2*.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Recommendation Table 4 — Recommendations for reperfusion therapy and timing of invasive strategy

Recommendations	Class[a]	Level[b]
Recommendations for reperfusion therapy for patients with STEMI		
Reperfusion therapy is recommended in all patients with a working diagnosis of STEMI (persistent ST-segment elevation or equivalents[c]) and symptoms of ischaemia of ≤12 h duration.[51,182]	I	A
A PPCI strategy is recommended over fibrinolysis if the anticipated time from diagnosis to PCI is <120 min.[52,218,219]	I	A
If timely PPCI (<120 min) cannot be performed in patients with a working diagnosis of STEMI, fibrinolytic therapy is recommended within 12 h of symptom onset in patients without contraindications.[176,183]	I	A
Rescue PCI is recommended for failed fibrinolysis (i.e. ST-segment resolution <50% within 60–90 min of fibrinolytic administration) or in the presence of haemodynamic or electrical instability, worsening ischaemia, or persistent chest pain.[184,185]	I	A
In patients with a working diagnosis of STEMI and a time from symptom onset >12 h, a PPCI strategy is recommended in the presence of ongoing symptoms suggestive of ischaemia, haemodynamic instability, or life-threatening arrhythmias.[220]	I	C
A routine PPCI strategy should be considered in STEMI patients presenting late (12–48 h) after symptom onset.[189–191,221]	IIa	B
Routine PCI of an occluded IRA is not recommended in STEMI patients presenting >48 h after symptom onset and without persistent symptoms.[189,192,193]	III	A
Transfer/interventions after fibrinolysis		
Transfer to a PCI-capable centre is recommended in all patients immediately after fibrinolysis.[184–186,212,213,222–224]	I	A
Emergency angiography and PCI of the IRA, if indicated are recommended in patients with new-onset or persistent heart failure/shock after fibrinolysis.[185,225]	I	A
Angiography and PCI of the IRA, if indicated, is recommended between 2 and 24 h after successful fibrinolysis.[186,212,213,217,224]	I	A
Invasive strategy in NSTE-ACS		
An invasive strategy during hospital admission is recommended in NSTE-ACS patients with high-risk criteria or a high index of suspicion for unstable angina.[196–200]	I	A
A selective invasive approach is recommended in patients without very high- or high-risk NSTE-ACS criteria and with a low index of suspicion for NSTE-ACS.[196–200]	I	A
An immediate invasive strategy is recommended in patients with a working diagnosis of NSTE-ACS and with at least one of the following very high-risk criteria: • Haemodynamic instability or cardiogenic shock • Recurrent or refractory chest pain despite medical treatment • In-hospital life-threatening arrhythmias • Mechanical complications of MI • Acute heart failure presumed secondary to ongoing myocardial ischaemia • *Recurrent* dynamic ST-segment or T wave changes, particularly intermittent ST-segment elevation.	I	C
An early invasive strategy within 24 h should be considered in patients with at least one of the following high-risk criteria: • Confirmed diagnosis of NSTEMI based on current recommended ESC hs-cTn algorithms • Dynamic ST-segment or T wave changes • Transient ST-segment elevation • GRACE risk score >140[202,226–230]	IIa	A

Continued

ACS, acute coronary syndrome; ECG, electrocardiogram; ESC, European Society of Cardiology; GRACE, Global Registry of Acute Coronary Events; hs-cTn, high-sensitivity cardiac troponin; IRA, infarct-related artery; MI, myocardial infarction; NSTE-ACS, non-ST-elevation acute coronary syndrome; NSTEMI, non-ST-elevation myocardial infarction; PPCI, primary percutaneous coronary intervention; STEMI, ST-elevation myocardial infarction.
[a]Class of recommendation.
[b]Level of evidence.
[c]ST-segment elevation equivalents are presented in Supplementary data online, *Figure S2*.

6. Antithrombotic therapy

Antithrombotic treatment is an important component of the management of all patients presenting with ACS. The specific choice and combination of therapy, the time of its initiation, and the treatment duration depend on various patient and procedural factors. Treatment decisions must be made weighing the benefits of antithrombotic therapy against the risk of bleeding, including severe, life-threatening bleeding.[231,232] Recommended anticoagulant and antiplatelet drugs and their dosing (for use during and after ACS) are summarized in *Table 6* and illustrated in *Figure 9*.

© ESC 2023

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Table 6 Dose regimen of antiplatelet and anticoagulant drugs in acute coronary syndrome patients

I. Antiplatelet drugs	
Aspirin	LD of 150–300 mg orally or 75–250 mg i.v. if oral ingestion is not possible, followed by oral MD of 75–100 mg o.d.; no specific dose adjustment in CKD patients.
$P2Y_{12}$ receptor inhibitors (oral or i.v.)	
Clopidogrel	LD of 300–600 mg orally, followed by an MD of 75 mg o.d.; no specific dose adjustment in CKD patients. Fibrinolysis: at the time of fibrinolysis an initial dose of 300 mg (75 mg for patients older than 75 years of age).
Prasugrel	LD of 60 mg orally, followed by an MD of 10 mg o.d. In patients with body weight <60 kg, an MD of 5 mg o.d. is recommended. In patients aged ≥75 years, prasugrel should be used with caution, but a MD of 5 mg o.d. should be used if treatment is deemed necessary. No specific dose adjustment in CKD patients. Prior stroke is a contraindication for prasugrel.
Ticagrelor	LD of 180 mg orally, followed by an MD of 90 mg b.i.d.; no specific dose adjustment in CKD patients.
Cangrelor	Bolus of 30 mcg/kg i.v. followed by 4 mcg/kg/min infusion for at least 2 h or the duration of the procedure (whichever is longer). In the transition from cangrelor to a thienopyridine, the thienopyridine should be administered immediately after discontinuation of cangrelor with an LD (clopidogrel 600 mg or prasugrel 60 mg); to avoid a potential DDI, prasugrel may also be administered 30 min before the cangrelor infusion is stopped. Ticagrelor (LD 180 mg) should be administered at the time of PCI to minimize the potential gap in platelet inhibition during the transition phase.
GP IIb/IIIa receptor inhibitors (i.v.)	
Eptifibatide	Double bolus of 180 mcg/kg i.v. (given at a 10-min interval) followed by an infusion of 2.0 mcg/kg/min for up to 18 h. For CrCl 30–50 mL/min: first LD, 180 mcg/kg i.v. bolus (max 22.6 mg); maintenance infusion, 1 mcg/kg/min (max 7.5 mg/h). Second LD (if PCI), 180 mcg/kg i.v. bolus (max 22.6 mg) should be administered 10 min after the first bolus. Contraindicated in patients with end-stage renal disease and with prior ICH, ischaemic stroke within 30 days, fibrinolysis, or platelet count <100 000/mm^3.
Tirofiban	Bolus of 25 mcg/kg i.v. over 3 min, followed by an infusion of 0.15 mcg/kg/min for up to 18 h. For CrCl ≤60 mL/min: LD, 25 mcg/kg i.v. over 5 min followed by a maintenance infusion of 0.075 mcg/kg/min continued for up to 18 h. Contraindicated in patients with prior ICH, ischaemic stroke within 30 days, fibrinolysis, or platelet count <100 000/mm^3.
II. Anticoagulant drugs	
UFH	Initial treatment: i.v. bolus 70–100 U/kg followed by i.v. infusion titrated to achieve an aPTT of 60–80 s. During PCI: 70–100 U/kg i.v. bolus or according to ACT in case of UFH pre-treatment.
Enoxaparin	Initial treatment: for treatment of ACS 1 mg/kg b.i.d. subcutaneously for a minimum of 2 days and continued until clinical stabilization. In patients whose CrCl is below 30 mL per minute (by Cockcroft–Gault equation), the enoxaparin dosage should be reduced to 1 mg per kg o.d. During PCI: for patients managed with PCI, if the last dose of enoxaparin was given less than 8 h before balloon inflation, no additional dosing is needed. If the last s.c. administration was given more than 8 h before balloon inflation, an i.v. bolus of 0.3 mg/kg enoxaparin sodium should be administered.
Bivalirudin	During PPCI: 0.75 mg/kg i.v. bolus followed by i.v. infusion of 1.75 mg/kg/h for 4 h after the procedure. In patients whose CrCl is below 30 mL/min (by Cockcroft–Gault equation), maintenance infusion should be reduced to 1 mg/kg/h.
Fondaparinux	Initial treatment: 2.5 mg/d subcutaneously. During PCI: A single bolus of UFH is recommended. Avoid if CrCl <20 mL/min.

ACS, acute coronary syndrome; ACT, activated clotting time; aPPT, activated partial thromboplastin time; b.i.d., bis in die (twice a day); CKD, chronic kidney disease; CrCl, creatinine clearance; DDI, drug–drug interactions; ICH, intracranial haemorrhage; i.v. intravenous; LD, loading dose; MD, maintenance dose; o.d., once a day; PPCI, primary percutaneous coronary intervention; s.c. subcutaneous; UFH, unfractionated heparin.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

© ESC 2023

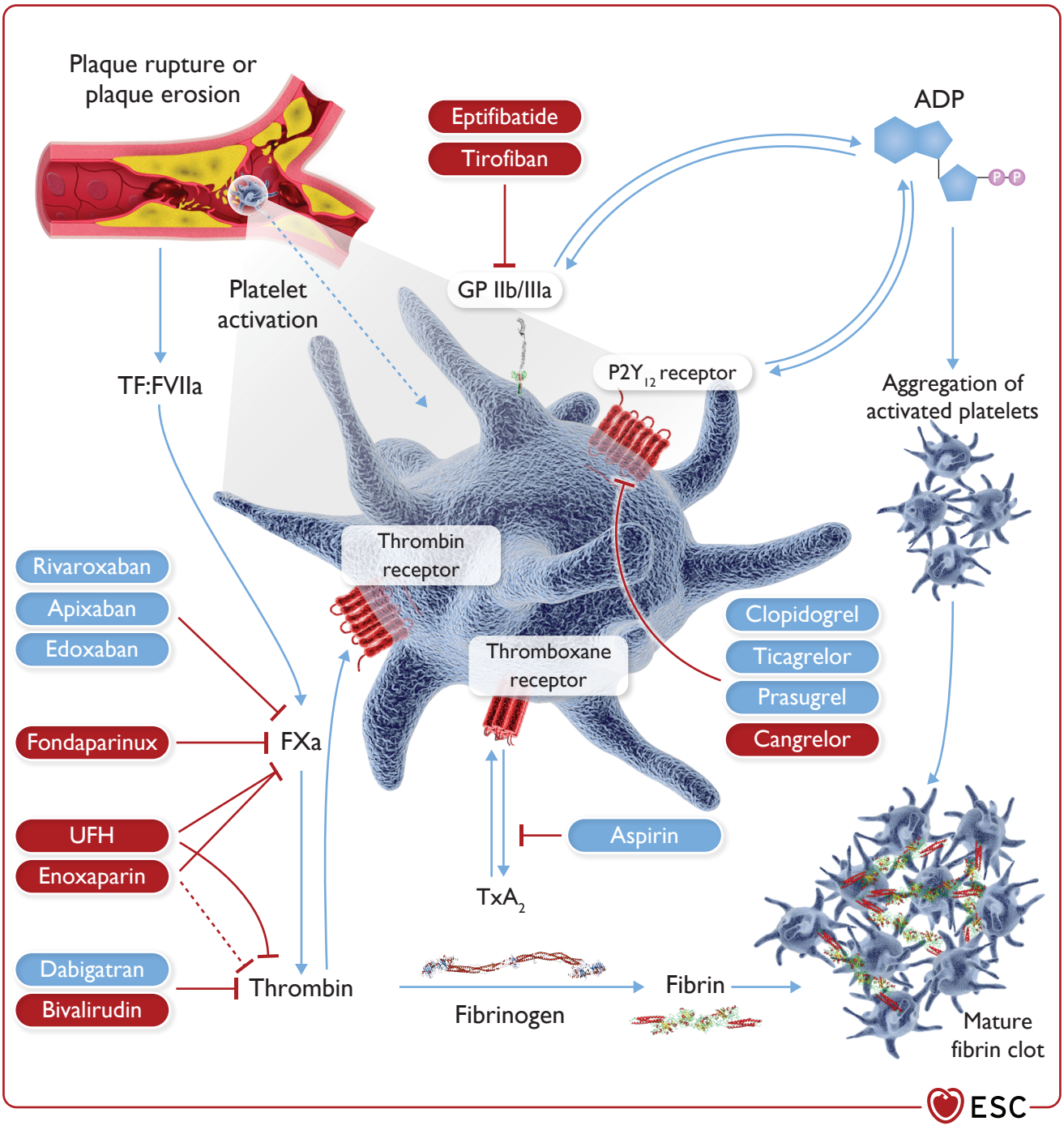


Figure 9 Antithrombotic treatments in acute coronary syndrome: pharmacological targets. ADP, adenosine diphosphate; FVIIa, Factor VIIa; FXa, Factor Xa; GP, glycoprotein; TF, tissue factor; TxA_2, thromboxane A_2; UFH, unfractionated heparin. Drugs with oral administration are shown in blue and drugs with preferred parenteral administration in red.

6.1. Antiplatelet therapy in the acute phase

6.1.1. Oral antiplatelet therapy

Antiplatelet drugs play a key role in the acute phase of treatment for ACS. *Table 6* summarizes the dosing regimens of the available oral and i.v. antiplatelet drugs. The choice of antiplatelet regimen should take the bleeding risk of the patient into account. Factors associated with an elevated bleeding risk have been detailed by the Academic Research Consortium on High Bleeding Risk (ARC-HBR).[233] The presence of one major or two minor ARC-HBR risk factors indicates high bleeding risk (HBR). Of note, the presence of multiple major risk factors is associated with a progressive increase in the bleeding risk.[234]

Aspirin treatment is started with a loading dose (LD) as soon as possible, followed by maintenance treatment (*Table 6*).[235] Current evidence supports an aspirin maintenance dose (MD) of 75–100 mg once a day (o.d.).[236,237]

Based on the results of the phase III PLATelet inhibition and patient Outcomes (PLATO) and TRial to Assess Improvement in Therapeutic Outcomes by Optimizing Platelet InhibitioN with Prasugrel

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Thrombolysis In Myocardial Infarction 38 (TRITON-TIMI 38) studies, dual antiplatelet therapy (DAPT) including aspirin and a potent $P2Y_{12}$ receptor inhibitor (prasugrel or ticagrelor) is recommended as the default DAPT strategy for ACS patients.[238,239] Clopidogrel, which is characterized by less effective and more variable platelet inhibition, should only be used when prasugrel or ticagrelor are contraindicated/not available, or in some patients considered otherwise at HBR (e.g. ≥1 major or ≥2 minor ARC-HBR criteria).[233,240–242] In addition, the use of clopidogrel may be considered in older patients (e.g. ≥70 years).[242,243]

Prasugrel should be considered in preference to ticagrelor for ACS patients who proceed to PCI. The Intracoronary Stenting and Antithrombotic Regimen Rapid Early Action for Coronary Treatment (ISAR-REACT) 5 RCT is the largest head-to-head comparison of 1-year DAPT with prasugrel vs. DAPT with ticagrelor in patients with ACS planned for invasive evaluation, >80% of whom underwent PCI.[244] A treatment strategy with prasugrel (LD given as soon as possible after randomization for patients undergoing PPCI and after delineation of coronary anatomy for patients presenting with NSTE-ACS) vs. ticagrelor (LD given as soon as possible after randomization in all cases) significantly reduced the composite endpoint of death, MI, or stroke (6.9% vs. 9.3%, $P = 0.006$) without any increase in bleeding complications (4.8% vs. 5.4%, $P = 0.46$). Limitations of this study include an open-label design and limited data on medically managed or CABG-treated patients.

6.1.2. Timing of loading dose of oral antiplatelet therapy

Both aspirin and oral $P2Y_{12}$ inhibitors achieve platelet inhibition more rapidly following an oral LD. Pre-treatment refers to a strategy in which an antiplatelet drug, usually a $P2Y_{12}$ receptor inhibitor, is given before coronary angiography and, therefore, before the coronary anatomy is known. Although a potential benefit with pre-treatment in the setting of ACS has been hypothesized, large-scale randomized trials supporting a routine pre-treatment strategy with $P2Y_{12}$ receptor inhibitors are lacking. Caution in relation to pre-treatment may be of particular relevance in patients at HBR (e.g. those receiving an oral anticoagulant [OAC]).

6.1.2.1. Pre-treatment in patients with suspected ST-elevation myocardial infarction

The Administration of Ticagrelor in the Cath Lab or in the Ambulance for New ST Elevation Myocardial Infarction to Open the Coronary Artery (ATLANTIC) trial is the only randomized study testing the safety and efficacy of different timings of $P2Y_{12}$ receptor inhibitor initiation in patients with a working diagnosis of STEMI undergoing PPCI.[245] In this trial, patients were randomized to receive a ticagrelor LD either during transfer to a PPCI centre or immediately before angiography.[245] The median difference between the timing of $P2Y_{12}$ receptor inhibitor loading with the two treatment strategies was 31 min. In this study, the pre-treatment strategy failed to meet the pre-specified primary endpoint of improved ST-segment elevation resolution or Thrombolysis In Myocardial Infarction (TIMI) flow before intervention. Rates of major and minor bleeding events were identical in both treatment arms. These results were supported by real-world data obtained from the SWEDEHEART (Swedish Web-System for Enhancement and Development of Evidence-Based Care in Heart Disease Evaluated According to Recommended Therapies) registry in STEMI patients.[246] Prasugrel pre-treatment has not been directly investigated in patients with STEMI.

6.1.2.2. Pre-treatment in patients with non-ST-elevation acute coronary syndrome

The randomized A Comparison of Prasugrel at the Time of Percutaneous Coronary Intervention or as Pretreatment at the Time of Diagnosis in Patients with Non-ST Elevation Myocardial Infarction (ACCOAST) trial not only demonstrated a lack of benefit with respect to ischaemic outcomes with prasugrel pre-treatment, but also a substantially higher bleeding risk.[247] In this study, the median time from first LD to the start of coronary angiography in the pre-treatment group was 4.4 h. With respect to pre-treatment data for ticagrelor, the ISAR-REACT 5 trial showed that a ticagrelor-based strategy with routine pre-treatment was inferior to a prasugrel-based strategy with a deferred LD in NSTE-ACS patients.[244] The DUBIUS (Downstream Versus Upstream Strategy for the Administration of P2Y12 Receptor Blockers) trial also attempted to address this question but was stopped early for futility as there was no difference between upstream vs. downstream oral $P2Y_{12}$ administration in patients with NSTE-ACS (both NSTEMI and UA) scheduled for coronary angiography within 72 h of hospital admission.[248]

6.1.2.3. Summary of pre-treatment strategies

In patients with a working diagnosis of STEMI undergoing PPCI, pre-treatment with a $P2Y_{12}$ receptor inhibitor may be considered.[245] In patients with a working diagnosis of NSTE-ACS, routine pre-treatment with a $P2Y_{12}$ receptor inhibitor before knowing the coronary anatomy in patients anticipated to undergo an early invasive strategy (i.e. <24 h) is not recommended.[244,245,247] For patients with a working diagnosis of NSTE-ACS, where there is an anticipated delay to invasive angiography (i.e. >24 h), pre-treatment with a $P2Y_{12}$ receptor inhibitor may be considered according to the bleeding risk of the patient. In all ACS patients proceeding to PCI who did not receive $P2Y_{12}$ receptor inhibitor pre-treatment, an LD is recommended at the time of PCI.

6.1.3. Intravenous antiplatelet drugs

Peri-interventional i.v. antiplatelet drugs include $P2Y_{12}$ receptor inhibitors (cangrelor) and glycoprotein (GP) IIb/IIIa inhibitors (eptifibatide and tirofiban). Most of the trials evaluating GP IIb/IIIa inhibitors in PCI-treated ACS patients pre-date the era of routine DAPT, in particular, early initiation of DAPT including an LD of a potent $P2Y_{12}$ receptor inhibitor.[249,250] There is no strong evidence for any additional benefit with the routine use of GP IIb/IIIa inhibitors in ACS patients scheduled for coronary angiography. Nevertheless, their use should be considered for bailout if there is evidence of no-reflow or a thrombotic complication during PCI. Another potential use for GP IIb/IIIa inhibitors is in the setting of high-risk PCI in patients who have not been pre-treated with $P2Y_{12}$ receptor inhibitors.

Cangrelor is a direct reversible, short-acting $P2Y_{12}$ receptor inhibitor that has been evaluated during PCI for CCS and ACS in clinical trials against clopidogrel, both with administration before (Cangrelor versus Standard Therapy to Achieve Optimal Management of Platelet Inhibition [CHAMPION PCI]) and after (CHAMPION PLATFORM and CHAMPION PHOENIX [A Clinical Trial Comparing Cangrelor to Clopidogrel Standard Therapy in Subjects Who Require Percutaneous Coronary Intervention]) PCI.[251–253] A meta-analysis of these trials showed that the benefit of cangrelor with respect to major ischaemic endpoints was counterbalanced by an increase in minor bleeding complications.[254] It is also important to note that the benefit of cangrelor with respect to ischaemic endpoints was attenuated in CHAMPION PCI with upfront administration of clopidogrel, and

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

data for its use in conjunction with ticagrelor or prasugrel treatment are limited. Due to its proven efficacy in preventing intra-procedural and post-procedural stent thrombosis in $P2Y_{12}$ receptor inhibitor-naïve patients, cangrelor may be considered on a case-by-case basis in $P2Y_{12}$ receptor inhibitor-naïve ACS patients undergoing PCI, including in patients for whom it may not be feasible to give oral drugs in the setting of emergent PCI (e.g. CS patients and/or patients on mechanical ventilation).

6.2. Anticoagulant treatment in the acute phase

Anticoagulation is an important component of the initial treatment of ACS and of the peri-procedural treatment for ACS patients managed with an invasive strategy. Therefore, parenteral anticoagulation is recommended for all ACS patients at the time of diagnosis.[255] *Table 6* provides an overview of the relevant anticoagulant drugs and their dosing in ACS patients.

In general, a crossover between anticoagulants should be avoided in patients with ACS (especially between unfractionated heparin [UFH] and low-molecular-weight heparin [LMWH]), with the exception of adding UFH to fondaparinux when a patient presenting with NSTE-ACS proceeds to PCI after a period of fondaparinux treatment (see below for further detail).[256,257] Anticoagulants should generally be discontinued immediately after PCI, except in specific clinical settings such as the confirmed presence of LV aneurysm with thrombus formation or AF requiring anticoagulation. In addition, for bivalirudin in patients with STEMI undergoing PCI, a full dose post-PCI infusion is recommended.

In this section of the guideline, we summarize the recommendations for anticoagulant treatment in the acute phase for patients with STEMI undergoing PPCI and for patients with NSTE-ACS undergoing angiography (and PCI if indicated).

6.2.1. Anticoagulation in patients with ST-elevation myocardial infarction undergoing primary percutaneous coronary intervention

Unfractionated heparin has been established as the standard of care in patients with STEMI undergoing PPCI due to its favourable risk/benefit profile. In these patients, anticoagulation should be given during the invasive procedure. High-quality evidence with respect to the benefit of administering anticoagulation at an earlier time point in patients undergoing a PPCI strategy is lacking.

Alternatives to UFH that should be considered in patients with STEMI undergoing PPCI include enoxaparin (a LMWH) and bivalirudin (a direct thrombin inhibitor). The ATOLL (STEMI Treated With Primary Angioplasty and Intravenous Lovenox or Unfractionated Heparin) trial reported a reduction in the primary endpoint at 30 days (incidence of death, complication of MI, procedure failure, or major bleeding) with enoxaparin in comparison to UFH in patients with STEMI undergoing PPCI.[258]

In the BivaliRudin with prolonged full-dose Infusion durinG primary PCI versus Heparin Trial 4 (BRIGHT-4), 6016 patients with STEMI undergoing PPCI were randomized to either bivalirudin (with a full dose post-PCI infusion) or UFH.[259] The use of GP IIb/IIIa inhibitors was restricted to patients who experienced thrombotic complications. The primary endpoint (a composite of all-cause mortality or Bleeding Academic Research Consortium [BARC] type 3–5 bleeding at 30 days), the individual components of the primary endpoint, and definite or probable stent thrombosis were all significantly reduced in the bivalirudin group.[259] Based on the totality of the available data, bivalirudin with a full-dose post-PCI infusion should be considered as an alternative to UFH, although further studies to confirm these findings in non-East Asian populations are required. Bivalirudin is also the recommended alternative to UFH in patients presenting with ACS who have a history of heparin-induced thrombocytopaenia. Additional information about bivalirudin, including evidence tables summarizing the relevant clinical trials, is provided in the Supplementary data online.

Based on the results of the OASIS-6 (The Safety and Efficacy of Fondaparinux Versus Control Therapy in Patients With ST Segment Elevation Acute Myocardial Infarction) trial, fondaparinux is not recommended in patients with STEMI undergoing PPCI.[260]

To summarize, parenteral anticoagulation is recommended for patients with STEMI undergoing PPCI and UFH is the default choice of anticoagulant at present. Enoxaparin and bivalirudin should be considered as alternatives to UFH in these patients but fondaparinux is not recommended.

6.2.2. Anticoagulation in patients with non-ST-elevation acute coronary syndrome undergoing angiography and percutaneous coronary intervention if indicated

Patients with NSTE-ACS are also recommended to receive parenteral anticoagulation. In patients with NSTE-ACS who are anticipated to undergo immediate or early (i.e. <24 h from the time of diagnosis) invasive angiography and PCI if indicated, parenteral anticoagulation at the time of diagnosis is recommended, and UFH has been historically established as the anticoagulant of choice. However, in a meta-analysis of trials comparing UFH with enoxaparin, mortality and major bleeding was not different between both agents in patients with NSTE-ACS or stable patients scheduled for PCI.[261] Therefore, enoxaparin should be considered as an alternative to UFH in these patients (especially in cases where monitoring of clotting times is complex).

NSTE-ACS patients who do not undergo early invasive angiography (i.e. within 24 h of diagnosis) will have an extended initial treatment phase consisting of only pharmacological treatment. In these patients, fondaparinux therapy is recommended in preference to enoxaparin while awaiting invasive angiography, based on the favourable outcomes demonstrated with fondaparinux in comparison to enoxaparin in the Fifth Organization to Assess Strategies in Acute Ischemic Syndromes (OASIS-5) trial.[262] Of note, guiding catheter thrombus formation was of concern with fondaparinux and, therefore, a full-dose bolus of UFH should be given if the patient proceeds to PCI. The potential impact of contemporary changes in clinical practice (including radial access, early catheterization, and infrequent GP IIb/IIIa inhibitor therapy) on the treatment effect observed in OASIS-5 should also be considered. If fondaparinux is not available, enoxaparin should be considered for these patients.

Intravenous enoxaparin should also be considered as an anticoagulant for PCI in patients with NSTE-ACS in whom subcutaneous (s.c.) enoxaparin was used while awaiting coronary angiography.[261]

In summary, parenteral anticoagulation is recommended for patients with NSTE-ACS. For patients with NSTE-ACS undergoing immediate or early angiography (± PCI if indicated), UFH is recommended but enoxaparin should be considered as an alternative to UFH. For patients with NSTE-ACS who are not anticipated to undergo early angiography, fondaparinux (with a UFH bolus at time of PCI) is recommended in preference to enoxaparin, although enoxaparin should be considered if fondaparinux is not available.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

6.3. Maintenance antithrombotic therapy after revascularization

While continuation of anticoagulation after PCI is not necessary in the vast majority of patients (i.e. those without an indication for long-term OAC), post-interventional antiplatelet treatment is mandatory in ACS patients. Following PCI, a default DAPT regimen consisting of a potent $P2Y_{12}$ receptor inhibitor (prasugrel or ticagrelor) and aspirin is generally recommended for 12 months, irrespective of the stent type, unless there are contraindications.[236,238,239,244,263] In specific clinical scenarios, the default DAPT duration can be shortened (<12 months), extended (>12 months), or modified (switching DAPT, DAPT de-escalation). The recommended default antithrombotic treatment options for ACS patients without an indication for OAC are shown in *Figure 10*.

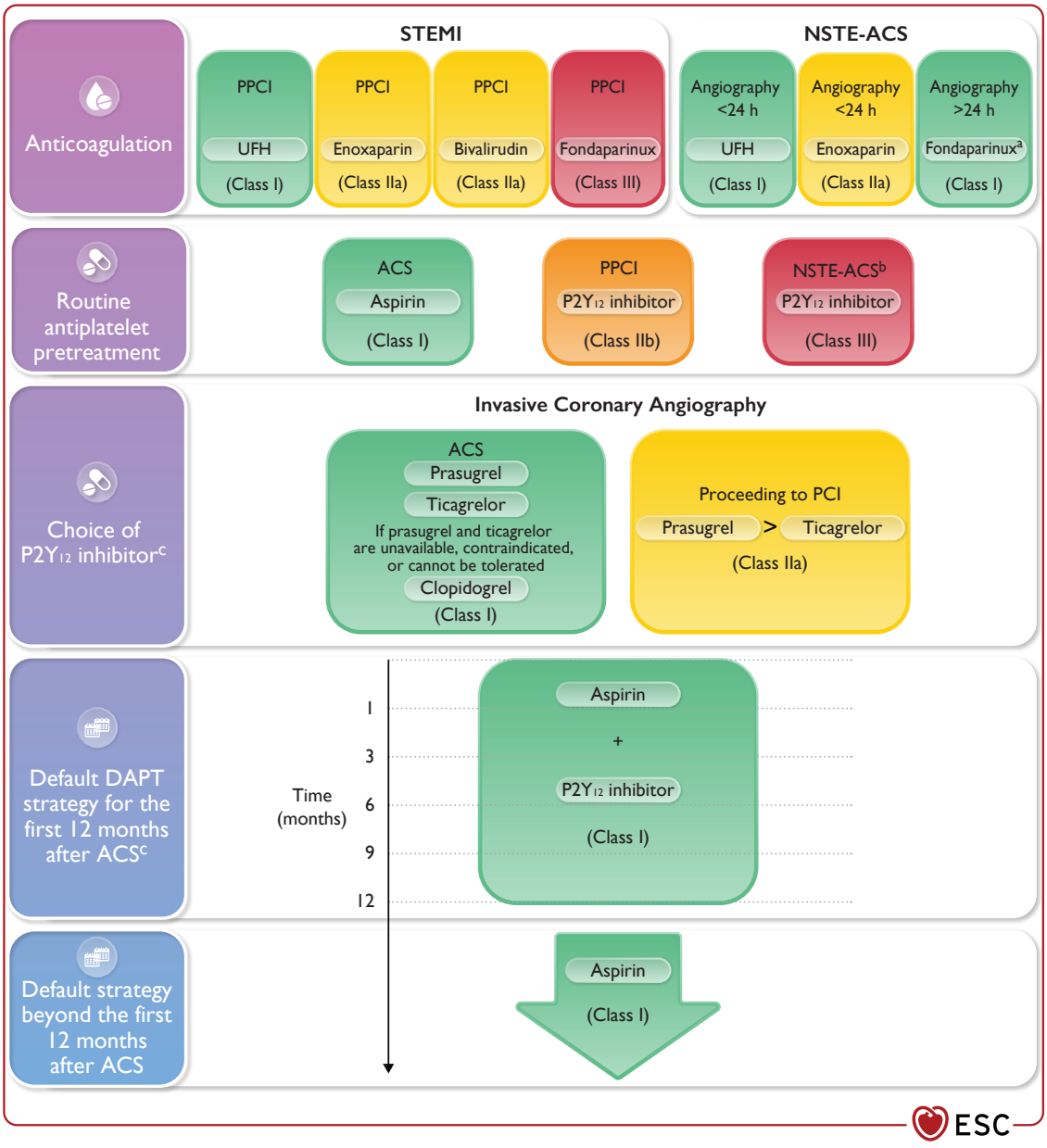


Figure 10 Recommended default antithrombotic therapy regimens in acute coronary syndrome patients without an indication for oral anticoagulation. ACS, acute coronary syndrome; DAPT, dual antiplatelet therapy; HBR, high bleeding risk; NSTE-ACS, non-ST-elevation acute coronary syndrome; PCI, percutaneous coronary intervention; PPCI, primary percutaneous coronary intervention; UFH, unfractionated heparin. Algorithm for antithrombotic therapy in ACS patients without an indication for oral anticoagulation undergoing invasive evaluation. [a]Fondaparinux (plus a single bolus of UFH at the time of PCI) is recommended in preference to enoxaparin for NSTE-ACS patients in cases of medical treatment or logistical constraints for transferring the NSTE-ACS patient to PCI within 24 h of symptom onset. [b]Routine pre-treatment with a $P2Y_{12}$ receptor inhibitor in NSTE-ACS patients in whom coronary anatomy is not known and early invasive management (<24 h) is planned is not recommended, but pre-treatment with a $P2Y_{12}$ receptor inhibitor may be considered in NSTE-ACS patients who are not expected to undergo an early invasive strategy (<24 h) and do not have HBR. [c]Clopidogrel is recommended for 12 months DAPT if prasugrel and ticagrelor are not available, cannot be tolerated, or are contraindicated, and may be considered in older ACS patients (typically defined as older than 70–80 years of age).

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

6.3.1. Shortening dual antiplatelet therapy

Several RCTs and meta-analyses have compared standard 12-month DAPT with ≤6 months DAPT followed by aspirin monotherapy in ACS patients.[264–267] In some of these trials, the reduction in bleeding events associated with abbreviated DAPT regimens came at the cost of an increase in the rates of ischaemic complications. In a large-scale network meta-analysis, 3-month DAPT but not 6-month DAPT was associated with higher rates of MI or stent thrombosis in ACS patients.[264]

A number of large RCTs have investigated DAPT duration further shortened to 1–3 months followed by $P2Y_{12}$ receptor inhibitor monotherapy in patients with and without ACS.[268–271] In general, low to intermediate ischaemic risk patients were included, and early monotherapy with clopidogrel or ticagrelor was used. Some trials included a comparison with more prolonged DAPT than usual in the control arm. Patients with STEMI tended to be excluded or under-represented.

The TWILIGHT (Ticagrelor With Aspirin or Alone in High-Risk Patients After Coronary Intervention) trial examined the effect of ticagrelor monotherapy vs. ticagrelor plus aspirin for 1 year after 3 months of DAPT (ticagrelor and aspirin) on clinically relevant bleeding. This study enrolled 'high-risk' patients as per the trial inclusion criteria, which meant that the enrolled patients had at least one clinical feature and one angiographic feature associated with a high risk of ischaemic or bleeding events. However, in order to be randomized the patients were also required to have not experienced a major bleeding or ischaemic event in the 3 months following hospital discharge.[271] STEMI patients were excluded from this study. Bleeding events (BARC type 2, 3, or 5 bleeding) were significantly reduced by omitting aspirin after 3 months, without a signal of increased ischaemic risk. A dedicated subgroup analysis suggested these findings were consistent in 4614 patients with NSTEMI/UA.[272] In the TICO (Ticagrelor Monotherapy After 3 Months in the Patients Treated With New Generation Sirolimus Stent for Acute Coronary Syndrome) trial, ticagrelor monotherapy vs. ticagrelor plus aspirin for up to 1 year after 3 months of DAPT (ticagrelor and aspirin) was tested in 3056 ACS patients (36% STEMI).[273] Net adverse clinical events and major bleeding events were significantly reduced with ticagrelor monotherapy, and major adverse cardiac and cerebrovascular events were not significantly different. Limitations of this study included the selected population assessed and the lower than expected event rates. A study-level meta-analysis of outcomes in a population of patients (with both ACS and CCS) fitted with a DES also reported a beneficial effect of shortened DAPT for 1–3 months on major bleeding events, as well as a neutral effect on death, MI, and stroke.[274]

The STOPDAPT-2-ACS (ShorT and OPtimal Duration of Dual AntiPlatelet Therapy-2 Study for the Patients With ACS) trial investigated a short DAPT strategy in ACS patients.[275] At 1–2 months, patients were randomized to either clopidogrel monotherapy or continued DAPT for 12 months. Non-inferiority of the investigational strategy for the composite endpoint of cardiovascular (CV) or bleeding events was not proven, suggesting that systematic very short duration DAPT (i.e. <3 months) followed by clopidogrel monotherapy is not a useful strategy in ACS patients.

The MASTER DAPT (Management of High Bleeding Risk Patients Post Bioresorbable Polymer Coated Stent Implantation With an Abbreviated Versus Prolonged DAPT Regimen) trial examined a strategy of abbreviated DAPT (1 month) followed by either aspirin or $P2Y_{12}$ inhibitor monotherapy vs. DAPT ≥3 months (standard therapy) in a cohort of 4579 HBR patients (49% ACS, 12% STEMI) undergoing PCI with a bioabsorbable polymer-coated stent.[276] Net adverse clinical events and major adverse cardiac or cerebral events were comparable between the groups, whereas major or clinically relevant non-major bleeding events were significantly reduced in the abbreviated therapy group.

6.3.2. De-escalation from potent $P2Y_{12}$ inhibitor to clopidogrel

The need to switch between oral $P2Y_{12}$ receptor inhibitors is not uncommon as a consequence of bleeding complications (or concern regarding bleeding), non-bleeding side effects (e.g. dyspnoea on ticagrelor, allergic reactions), and socioeconomic factors.[277,278] As such, switching between oral $P2Y_{12}$ receptor inhibitors may be considered in selected cases.

$P2Y_{12}$ receptor inhibitor de-escalation (i.e. switching from prasugrel/ticagrelor to clopidogrel) in ACS patients may be considered as an alternative strategy to the default treatment regimen in order to reduce the risk of bleeding events. However, it is important to note that there is a potential risk of increased ischaemic events with de-escalation and this strategy is not recommended in the first 30 days after the index ACS event. In the TROPICAL-ACS (Testing Responsiveness to Platelet Inhibition on Chronic Antiplatelet Treatment For Acute Coronary Syndromes) trial (44% NSTE-ACS, 56% STEMI), an approach of DAPT de-escalation from prasugrel to clopidogrel (at 2 weeks after ACS) was guided by platelet function testing and was non-inferior to standard treatment with prasugrel at 1 year after PCI in terms of net clinical benefit.[279] In the Cost-effectiveness of CYP2C19 Genotype Guided Treatment With Antiplatelet Drugs in Patients With ST-segment-elevation Myocardial Infarction Undergoing Immediate PCI With Stent Implantation: Optimization of Treatment (POPular Genetics) trial, DAPT de-escalation from ticagrelor/prasugrel to clopidogrel guided by CYP2C19 genotyping in ACS patients undergoing PPCI within the previous 48 h was non-inferior to standard treatment with ticagrelor or prasugrel at 12 months with respect to thrombotic events and resulted in a lower incidence of bleeding.[280]

The single-centre TOPIC (Timing of Platelet Inhibition After Acute Coronary Syndrome) trial used an unguided de-escalation approach in 645 ACS patients (60% NSTE-ACS, 40% STEMI) from ticagrelor/prasugrel to clopidogrel after 1 month of DAPT with ticagrelor/prasugrel and aspirin. Net adverse clinical events and bleeding events were reduced, whereas the rate of ischaemic endpoints was unchanged.[281] The TALOS-AMI (TicAgrelor versus CLOpidogrel in Stabilised Patients with Acute Myocardial Infarction) trial investigated unguided de-escalation in 2697 ACS patients (46% NSTEMI/UA, 54% STEMI) from ticagrelor to clopidogrel after 1 month of DAPT with ticagrelor and aspirin.[282] This uniform unguided de-escalation strategy led to significant 12-month reductions in net adverse clinical events and bleeding events. The HOST-REDUCE-POLYTECH-ACS (Harmonizing Optimal Strategy for Treatment of Coronary Artery Diseases Trial—Comparison of REDUCTION of PrasugrEl Dose & POLYmer TECHnology in ACS Patients) trial tested a different method of de-escalation—a reduction in prasugrel dose rather than switching to clopidogrel. In this trial, 2338 low-risk ACS patients <75 years of age (14% STEMI, 25% NSTEMI, and 61% UA) were randomized to low-dose prasugrel (5 mg daily) or standard-dose prasugrel (10 mg daily) after 1 month of DAPT with standard-dose prasugrel.[283] Prasugrel dose de-escalation was associated with fewer net adverse clinical events and bleeding events, mainly by reducing bleeding events without an increase in ischaemic events. It should be noted that the TALOS-AMI and HOST-REDUCE-POLYTECH-ACS trials only included East Asian populations.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

6.3.3. Summary of alternative antiplatelet strategies to reduce bleeding risk in the first 12 months after acute coronary syndrome

Considering the totality of evidence from the scientific literature, alternatives to the default strategy of 12 months DAPT in patients with ACS include shortening the DAPT duration to 1 or 3–6 months (depending on the balance of bleeding and ischaemic risks) and de-escalating DAPT from prasugrel/ticagrelor-based DAPT to clopidogrel-based DAPT. However, it should be noted that much of the evidence on these strategies in ACS patients is derived from trials powered primarily for bleeding outcomes, many of which had a non-inferiority design and were, therefore, not powered to detect potentially relevant differences in ischaemic outcomes. The patient populations enrolled in these studies were also often relatively selected, often excluding or under-representing the highest risk ACS patients. As such, it is important to reflect that even meta-analyses of the available randomized evidence cannot overcome the potential selection bias at the point of entry in the relevant randomized trials.

These important limitations explain why these strategies should at present remain considered as alternative strategies to the default of 12 months DAPT. From a practical perspective, this means that these strategies should not be employed as a default strategy in the wider ACS population but can be considered when there is a specific motivation for their use (i.e. aiming to reduce the risk of bleeding events in HBR patients or if there are other specific concerns regarding a 12-month potent $P2Y_{12}$ inhibitor-based DAPT regimen). De-escalation of antiplatelet therapy in the first 30 days is not recommended, but de-escalation of $P2Y_{12}$ receptor inhibitor therapy may be considered as an alternative strategy beyond 30 days after an ACS, in order to reduce the risk of bleeding events. DAPT abbreviation strategies (followed preferably by $P2Y_{12}$ inhibitor monotherapy within the first 12 months post-ACS) should be considered in patients who are event-free after 3–6 months of DAPT and who are not high ischaemic risk, with the duration of DAPT guided by the ischaemic and bleeding risks of the patient. For HBR patients, aspirin or $P2Y_{12}$ receptor inhibitor monotherapy after 1 month of DAPT may be considered. Please see Recommendation Table 6 for full details. These alternative antiplatelet strategies to reduce bleeding risk in the first 12 months after ACS are also summarized in *Figure 11*.

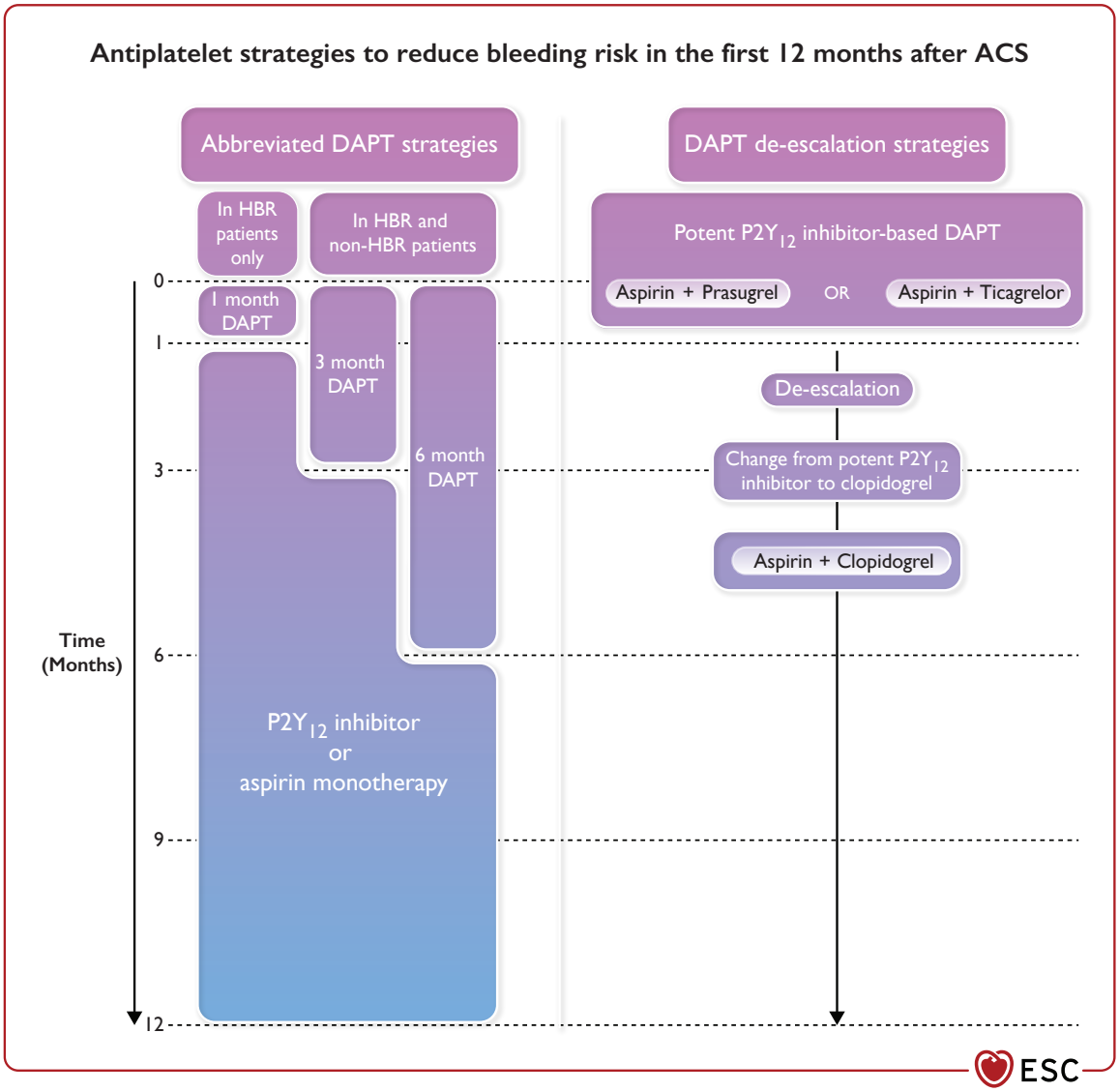


Figure 11 Alternative antiplatelet strategies to reduce bleeding risk in the first 12 months after an ACS. ACS, acute coronary syndrome; DAPT, dual antiplatelet therapy; HBR, high bleeding risk; PFT, platelet function test.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

To summarise, antiplatelet strategies to reduce bleeding risk in the first 12 months after an ACS can be divided into abbreviated DAPT strategies and DAPT de-escalation strategies. Twelve-month DAPT (preferably with prasugrel or ticagrelor) remains the default strategy for patients with ACS (*Figure 10*) and these strategies should only be used as alternatives to this strategy, in general driven by a motivation to reduce the risk of bleeding events (i.e. if the patient is HBR or if there are other specific concerns regarding 12-month potent $P2Y_{12}$ inhibitor-based DAPT).

The specific alternative antiplatelet strategies employed (i.e. choice of $P2Y_{12}$ inhibitor, duration of DAPT, choice of SAPT agent) to reduce bleeding risk should be chosen based on the bleeding risk of the patient (HBR or not) and these recommendations are summarized in Recommendation Table 6.

Recommendation Table 5 — Recommendations for antiplatelet and anticoagulant therapy in acute coronary syndrome

Recommendations	Class[a]	Level[b]
Antiplatelet therapy		
Aspirin is recommended for all patients without contraindications at an initial oral LD of 150–300 mg (or 75–250 mg i.v.) and an MD of 75–100 mg o.d. for long-term treatment.[284,285]	I	A
In all ACS patients, a $P2Y_{12}$ receptor inhibitor is recommended in addition to aspirin, given as an initial oral LD followed by an MD for 12 months unless there is HBR[c].[238,239,263,286]	I	A
A proton pump inhibitor in combination with DAPT is recommended in patients at high risk of gastrointestinal bleeding.[287,288]	I	A
Prasugrel is recommended in $P2Y_{12}$ receptor inhibitor-naïve patients proceeding to PCI (60 mg LD, 10 mg o.d. MD, 5 mg o.d. MD for patients aged ≥75 years or with a body weight <60 kg).[239]	I	B
Ticagrelor is recommended irrespective of the treatment strategy (invasive or conservative) (180 mg LD, 90 mg b.i.d. MD).[238]	I	B
Clopidogrel (300–600 mg LD, 75 mg o.d. MD) is recommended when prasugrel or ticagrelor are not available, cannot be tolerated, or are contraindicated.[263,289]	I	C
If patients presenting with ACS stop DAPT to undergo CABG, it is recommended they resume DAPT after surgery for at least 12 months.	I	C
Prasugrel should be considered in preference to ticagrelor for ACS patients who proceed to PCI.[244,290]	IIa	B
GP IIb/IIIa receptor antagonists should be considered if there is evidence of no-reflow or a thrombotic complication during PCI.	IIa	C
In $P2Y_{12}$ receptor inhibitor-naïve patients undergoing PCI, cangrelor may be considered.[251–254]	IIb	A
In older ACS patients,[d] especially if HBR,[c] clopidogrel as the $P2Y_{12}$ receptor inhibitor may be considered.[242,243,291]	IIb	B
Pre-treatment with a $P2Y_{12}$ receptor inhibitor may be considered in patients undergoing a primary PCI strategy.[244,245]	IIb	B
Pre-treatment with a $P2Y_{12}$ receptor inhibitor may be considered in NSTE-ACS patients who are not expected to undergo an early invasive strategy (<24 h) and do not have HBR[c].[263]	IIb	C
Pre-treatment with a GP IIb/IIIa receptor antagonist is not recommended.[292]	III	A
Routine pre-treatment with a $P2Y_{12}$ receptor inhibitor in NSTE-ACS patients in whom coronary anatomy is not known and early invasive management (<24 h) is planned is not recommended.[244,247,248,293–295]	III	A
Anticoagulant therapy		
Parenteral anticoagulation is recommended for all patients with ACS at the time of diagnosis.[255,296]	I	A
Routine use of a UFH bolus (weight-adjusted i.v. bolus during PCI of 70–100 IU/kg) is recommended in patients undergoing PCI.	I	C
Intravenous enoxaparin at the time of PCI should be considered in patients pre-treated with subcutaneous enoxaparin.[256,261,297]	IIa	B
Discontinuation of parenteral anticoagulation should be considered immediately after an invasive procedure.	IIa	C
Patients with STEMI		
Enoxaparin should be considered as an alternative to UFH in patients with STEMI undergoing PPCI.[258,261,298]	IIa	A
Bivalirudin with a full-dose post PCI infusion should be considered as an alternative to UFH in patients with STEMI undergoing PPCI.[259,299,300–303]	IIa	A
Fondaparinux is not recommended in patients with STEMI undergoing PPCI.[260]	III	B
Patients with NSTE-ACS		
For patients with NSTE-ACS in whom early invasive angiography (i.e. within 24 h) is not anticipated, fondaparinux is recommended.[262,304]	I	B
For patients with NSTE-ACS in whom early invasive angiography (i.e. within 24 h) is anticipated, enoxaparin should be considered as an alternative to UFH.[256]	IIa	B
Combining antiplatelets and OAC		
As the default strategy for patients with atrial fibrillation and CHA_2DS_2-VASc score ≥1 in men and ≥2 in women, after up to 1 week of triple antithrombotic therapy following the ACS event, dual antithrombotic therapy using a NOAC at the recommended dose for stroke prevention and a single oral antiplatelet agent (preferably clopidogrel) for up to 12 months is recommended.[305–310]	I	A
During PCI, a UFH bolus is recommended in any of the following circumstances: • if the patient is on a NOAC • if the INR is <2.5 in VKA-treated patients.	I	C

Continued

Continued

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

In patients with an indication for OAC with VKA in combination with aspirin and/or clopidogrel, careful regulation of the dose intensity of VKA with a target INR of 2.0–2.5 and a time in the therapeutic range >70% should be considered.[305–308,311]	IIa	B
When rivaroxaban is used and concerns about HBR prevail over ischaemic stroke, rivaroxaban 15 mg o.d. should be considered in preference to rivaroxaban 20 mg o.d. for the duration of concomitant SAPT or DAPT.[307]	IIa	B
In patients at HBR,[c] dabigatran 110 mg b.i.d. should be considered in preference to dabigatran 150 mg b.i.d. for the duration of concomitant SAPT or DAPT, to mitigate bleeding risk.[305]	IIa	B
In patients requiring anticoagulation and treated medically, a single antiplatelet agent in addition to an OAC should be considered for up to 1 year.[308,312]	IIa	B
In patients treated with an OAC, aspirin plus clopidogrel for longer than 1 week and up to 1 month should be considered in those with high ischaemic risk or with other anatomical/procedural characteristics that are judged to outweigh the bleeding risk.[e]	IIa	C
In patients requiring OAC, withdrawing antiplatelet therapy at 6 months while continuing OAC may be considered.[313]	IIb	B
The use of ticagrelor or prasugrel as part of triple antithrombotic therapy is not recommended.	III	C

© ESC 2023

ACS, acute coronary syndrome; b.i.d., *bis in die* (twice a day); CHA_2DS_2-VASc, Congestive heart failure, Hypertension, Age ≥75 years, Diabetes mellitus, Stroke or transient ischaemic attack, Vascular disease; DAPT, dual antiplatelet therapy; GP, glycoprotein; HBR, high bleeding risk; INR, international normalized ratio; i.v., intravenous; LD, loading dose; MD, maintenance dose; NOAC, non-vitamin K antagonist oral anticoagulant; NSTE-ACS, non-ST-elevation acute coronary syndrome; OAC, oral anticoagulant; PPCI, primary percutaneous coronary intervention; SAPT, single antiplatelet therapy; STEMI, ST-elevation myocardial infarction; UFH, unfractionated heparin; VKA, vitamin K antagonist.

[a]Class of recommendation.

[b]Level of evidence.

[c]HBR should be assessed in a structured manner, e.g. presence of a single major or two minor characteristics as defined by ARC-HBR (see section 8.2.2.3 in Supplementary data online).

[d]The definition of older patients varies across trials, ranging from 70 to 80 years of age. Frailty and comorbidities should also be taken in consideration.

[e]See Antiplatelet therapy in patients requiring oral anticoagulation *Section 6.2* in Supplementary data online for more information on high-risk features of stent-driven recurrent events.

6.4. Long-term treatment

By default, DAPT consisting of a potent $P2Y_{12}$ receptor inhibitor in addition to aspirin is recommended for a minimum of 12 months after an ACS event; exceptions include patients for whom surgery is urgently needed, patients in whom OAC is indicated, and patients in whom the risk of bleeding is too high for other reasons.[238,239,263] After PCI for ACS, ischaemic and bleeding events both markedly decrease over time.

Further information regarding long-term antithrombotic strategies (i.e. beyond 12 months) is provided in the Supplementary data online.

6.4.1. Prolonging antithrombotic therapy beyond 12 months

Prolonged antithrombotic therapy options: See Supplementary data online, *Figure S4; Tables S7* and *S8* for additional information.[314–319]

Recommendation Table 6 — Recommendations for alternative antithrombotic therapy regimens

Recommendations	Class[a]	Level[b]
Shortening/de-escalation of antithrombotic therapy		
In patients who are event-free after 3–6 months of DAPT and who are not high ischaemic risk, single antiplatelet therapy (preferably with a $P2Y_{12}$ receptor inhibitor) should be considered.[264,268–271,273,274,276,313,320]	IIa	A
De-escalation of $P2Y_{12}$ receptor inhibitor treatment (e.g. with a switch from prasugrel/ticagrelor to clopidogrel) may be considered as an alternative DAPT strategy to reduce bleeding risk.[279–282,321,322]	IIb	A
In HBR patients, aspirin or $P2Y_{12}$ receptor inhibitor monotherapy after 1 month of DAPT may be considered.[276,313]	IIb	B
De-escalation of antiplatelet therapy in the first 30 days after an ACS event is not recommended.[238,323]	III	B
Prolonging antithrombotic therapy		
Discontinuation of antiplatelet treatment in patients treated with an OAC is recommended after 12 months.[324,325]	I	B
Adding a second antithrombotic agent to aspirin for extended long-term secondary prevention should be considered in patients with high ischaemic risk and without HBR[c].[314–318]	IIa	A
Adding a second antithrombotic agent to aspirin for extended long-term secondary prevention may be considered in patients with moderate ischaemic risk and without HBR[c].[314–318]	IIb	A
$P2Y_{12}$ inhibitor monotherapy may be considered as an alternative to aspirin monotherapy for long-term treatment.[326,327]	IIb	A

© ESC 2023

ACS, acute coronary syndrome; DAPT, dual antiplatelet therapy; HBR, high bleeding risk; OAC, oral anticoagulant.

[a]Class of recommendation.

[b]Level of evidence.

[c]The evidence supporting this approach (prolonged treatment with a second antithrombotic agent) is based on trials in which the duration of prolonged treatment was as follows: mean of 23 months (COMPASS), mean of 18 months (DAPT trial), and median of 33 months (PEGASUS-TIMI 54). Therefore, the benefits and risks associated with continuation of these respective treatments beyond these time points is at present unclear.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

6.5. Antiplatelet therapy in patients requiring oral anticoagulation

6.5.1. Acute coronary syndrome patients requiring anticoagulation

In 6–8% of patients undergoing PCI, long-term OAC is indicated and should also be continued during the invasive procedure. Interruption of the long-term OAC and bridging with parenteral anticoagulants may lead to an increase in thrombo-embolic episodes and bleeds.[328–330] In patients undergoing PCI, it is unknown whether it is safer to bridge non-vitamin K antagonist (VKA) OACs (NOACs) with parenteral anticoagulants or to continue NOACs without additional parenteral anticoagulation. In VKA-treated patients, no parenteral anticoagulation is needed if the international normalized ratio (INR) is >2.5.[311,331,332] Strategies to minimize PCI-related complications in patients on OAC are listed in *Table 7*.

Evidence on the management of ACS patients with an indication for long-term OAC undergoing PCI is derived from subgroups of RCTs.[305–309,333] Patients with STEMI (who generally carry a higher atherothrombotic risk) were under-represented (~10% of the study populations) in the major RCTs.[305,307–309] Pivotal trials testing the benefit of NOACs as part of the antithrombotic regimen in patients with an indication for long-term anticoagulation undergoing PCI are discussed in the Supplementary data online.

All of these trials were individually powered to address the safety of the tested strategy with regard to bleeding events, but not to reliably assess differences in individual ischaemic endpoints. In a meta-analysis of all four NOAC-based RCTs comparing dual antithrombotic therapy (DAT) with triple antithrombotic therapy (TAT) in AF patients undergoing PCI (encompassing 10 234 patients), the primary safety endpoint (International Society on Thrombosis and Haemostasis major or clinically relevant non-major bleeding) was significantly lower with DAT vs. TAT (relative risk [RR] 0.66, 95% CI, 0.56–0.78; P <0.001).[310] There were no significant differences in all-cause and CV death, stroke, or trial-defined major adverse cardiovascular events (MACE). However, DAT was associated with a borderline increased risk of MI (RR 1.22, 95% CI, 0.99–1.52; $P = 0.07$) and a significant increase in stent thrombosis (RR 1.59, 95% CI, 1.01–2.50; $P = 0.04$). This translates into an absolute reduction in major bleeding events of 2.3% compared with an absolute increase in stent thrombosis of 0.4%, without an effect on overall MACE. When interpreting the results of these studies, an important general point is that the treatment effect is confounded by the use of NOACs in the DAT treatment arms and VKAs in the TAT arms.

Secondary analyses from the AUGUSTUS (An Open-Label, 2 × 2 Factorial, Randomized Controlled, Clinical Trial to Evaluate the Safety of Apixaban Versus Vitamin K Antagonist and Aspirin Versus Aspirin Placebo in Patients With Atrial Fibrillation and Acute Coronary Syndrome or Percutaneous Coronary Intervention) trial indicate that the stent thrombosis rate was highest within the first 30 days after randomization, with higher rates in the non-aspirin group.[334] Aspirin treatment reduced ischaemic events (CV death, MI, stroke, stent thrombosis) but also increased major bleeding events in the first 30 days. Aspirin treatment did not impact on ischaemic event rates after 30 days and for up to 6 months, but did increase the bleeding risk during this time period.[334,335] In the MASTER DAPT trial, 4579 HBR patients were allocated to 1 month vs. 6 months of DAPT after implantation of a biodegradable-polymer sirolimus-eluting stent; half of the patients presented with ACS and a third were on OAC treatment.[276] A sub-analysis of this study reported that stopping DAPT after 1 month and stopping single antiplatelet therapy (SAPT) after 6 months while maintaining OAC was safe with respect to ischaemic events in patients taking clinically indicated long-term OAC therapy.[313]

In patients with ACS, the indication for OAC should be re-assessed and treatment continued only if a compelling indication exists (e.g. paroxysmal, persistent, or permanent AF with a CHA_2DS_2-VASc [Congestive heart failure, Hypertension, Age ≥75 years, Diabetes mellitus, Stroke or transient ischaemic attack, Vascular disease] score ≥1 in men and ≥2 in women; mechanical heart valve; or recent/a history of recurrent or unprovoked deep vein thrombosis or PE). Although they have been tested in a minority of patients in the major RCTs, in the absence of robust safety and efficacy data, the use of prasugrel or ticagrelor as part of TAT is not recommended. The intensity of OAC should be carefully monitored, with a target INR of 2.0–2.5 in patients treated with VKA (with the exception of individuals with a mechanical prosthetic valve in the mitral position).

Overall, in patients with AF without mechanical prosthetic valves or moderate to severe mitral stenosis, the evidence supports the use of NOACs over VKAs as they reduce bleeding risk. DAT with a NOAC at the recommended dose for stroke prevention and SAPT (preferably clopidogrel, which was used in >90% of patients in the major RCTs) is recommended as the default strategy for up to 12 months after up to 1 week of TAT (with NOAC and DAPT consisting of aspirin and clopidogrel) (*Figure 12*)—the up to 1 week duration of TAT is based on the median treatment duration in the investigational arm of the AUGUSTUS trial.[308] Although none of the available RCTs were designed to detect differences in ischaemic events, the numerically higher risk of stent thrombosis and MI is offset by the lower risk of bleeding, with a resultant neutral effect on total mortality.[310,336–338]

Table 7 Suggested strategies to reduce bleeding risk related to percutaneous coronary intervention

- Anticoagulant doses adjusted to body weight and renal function, especially in women and older patients
- Radial artery approach as default vascular access
- Proton pump inhibitors in patients on dual antiplatelet therapy at higher-than-average risk of gastrointestinal bleeds (i.e. history of gastrointestinal ulcer/haemorrhage, anticoagulant therapy, chronic non-steroidal anti-inflammatory drug/corticosteroid use), or two or more of:
  (a) Age ≥65 years
  (b) Dyspepsia
  (c) Gastro-oesophageal reflux disease
  (d) *Helicobacter pylori* infection
  (e) Chronic alcohol use
- In patients on OAC:
  (a) PCI performed without interruption of VKAs or NOACs
  (b) In patients on VKAs, do not administer UFH if INR >2.5
  (c) In patients on NOACs, regardless of the timing of the last administration of NOACs, add low-dose parenteral anticoagulation (e.g. enoxaparin 0.5 mg/kg i.v. or UFH 60 IU/kg)
- Aspirin is indicated but avoid pre-treatment with $P2Y_{12}$ receptor inhibitors
- GP IIb/IIIa receptor inhibitors only for bailout or peri-procedural complications

GP, glycoprotein; INR, international normalized ratio; i.v., intravenous; NOAC, non-vitamin K antagonist oral anticoagulant; OAC, oral anticoagulation/anticoagulant; PCI, percutaneous coronary intervention; UFH, unfractionated heparin; VKA, vitamin K antagonist.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

© ESC 2023

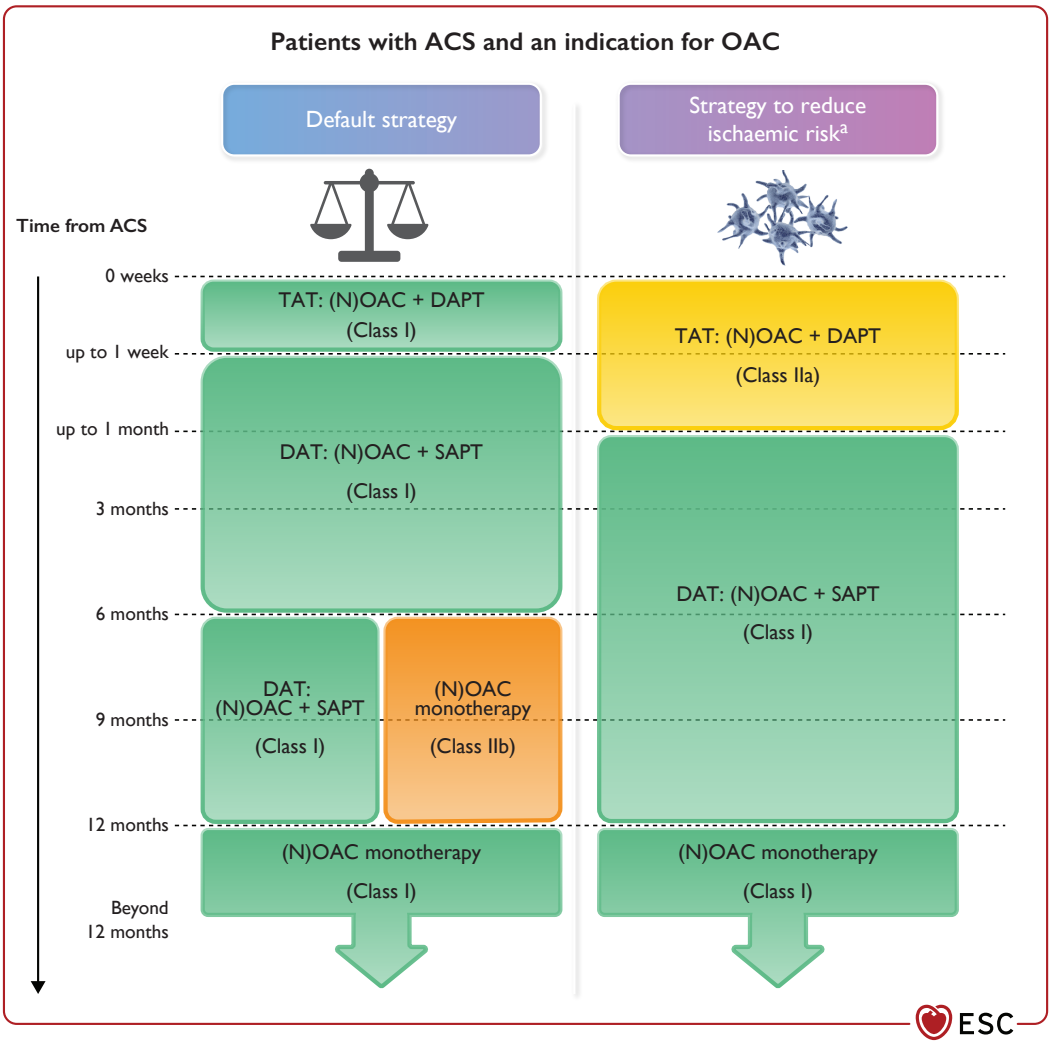


Figure 12 Antithrombotic regimens in patients with acute coronary syndrome and an indication for oral anticoagulation. ACS, acute coronary syndrome; ARC-HBR, Academic Research Consortium for High Bleeding Risk; DAPT, dual antiplatelet therapy; DAT, dual antithrombotic therapy; NOAC, non-vitamin K antagonist oral anticoagulant; OAC, oral anticoagulation/anticoagulant; SAPT, single antiplatelet therapy; TAT, triple antithrombotic therapy; VKA, vitamin K antagonist. OAC: preference for a NOAC over VKA for the default strategy and in all other scenarios if no contraindications. For both TAT and DAT regimens, the recommended doses for the NOACs are as follows: Apixaban 5 mg b.i.d., Dabigatran 110 mg or 150 mg b.i.d., Edoxaban 60 mg o.d., Rivaroxaban 15 mg or 20 mg o.d. NOAC dose reductions are recommended in patients based on certain criteria for each of the NOACs (including renal function, body weight, concomitant medications and age). SAPT: preference for a $P2Y_{12}$ receptor inhibitor (usually clopidogrel) over aspirin. See Bleeding risk assessment in Supplementary data online, *Section 8.2.2.3* for details on the ARC-HBR criteria. In addition, patients with a PRECISE-DAPT score of ≥25 are regarded as high bleeding risk. [a]See Supplementary material online, *Table S9* for examples of high-risk features of stent-driven recurrent events.

At variance with the default strategy, DAT may be shortened to 6 months by withdrawing the antiplatelet therapy in certain patients; for example, in patients with multiple HBR factors. In patients with high ischaemic risk or other anatomical/procedural characteristics that outweigh the bleeding risk, TAT should be prolonged for up to 1 month, followed by DAT for up to 12 months.

There is currently limited evidence to support the use of OAC with ticagrelor or prasugrel as DAT after ACS and/or PCI as an alternative to TAT; ticagrelor was used in 5–12% and prasugrel in 1–2% of patients, respectively, in the four pivotal RCTs.[305,307–309,339]

In medically managed ACS patients, current data support DAT over TAT, with a single antiplatelet agent (most commonly clopidogrel) for at least 6 months.[308] In the AUGUSTUS trial, ~24% of enrolled patients presented with medically managed ACS.[308] In these patients, apixaban significantly reduced bleeding events compared with a VKA, while no significant differences were observed in death or ischaemic events. The use of aspirin, in comparison to placebo, led to more bleeding events but no significant differences in death, hospitalization, or ischaemic events were observed.[308]

Regarding the need to continue with any antiplatelet agent beyond 12 months after ACS and/or PCI in patients with an indication for OAC, the AFIRE (Atrial Fibrillation and Ischemic Events With Rivaroxaban in Patients With Stable Coronary Artery Disease) trial randomized 2236 AF patients treated with PCI or CABG more than 1 year earlier or with documented CAD to receive either rivaroxaban monotherapy or combination therapy with rivaroxaban plus a single antiplatelet agent.[324] Rivaroxaban monotherapy was non-inferior to combination therapy for the primary efficacy composite endpoint of stroke, systemic embolism, MI, UA requiring revascularization, or overall death, and superior with regard to the primary safety endpoint of major bleeding. This trial and another prematurely terminated trial support the recommendation to stop antiplatelet therapy after 12 months and continue with OAC monotherapy in most patients.[325]

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

6.5.2. Patients requiring vitamin K antagonists or undergoing coronary artery bypass surgery

In patients for whom a VKA is mandated (e.g. patients with mechanical prosthetic valves), DAT with a VKA and SAPT (preferably clopidogrel) is indicated after an up to 1-week period of TAT (with aspirin and clopidogrel).[306] A network meta-analysis has reported that compared with TAT (consisting of VKA plus aspirin and clopidogrel), DAT (VKA plus clopidogrel) was associated with a trend towards a reduction in TIMI major bleeding, with no significant difference observed in MACE.[336]

In ACS patients undergoing CABG with an established indication for OAC, anticoagulation in combination with SAPT should be resumed after CABG as soon as possible and TAT should be avoided.

6.6. Antithrombotic therapy as an adjunct to fibrinolysis

ISIS-2 (Second International Study Of Infarct Survival) demonstrated that the benefits of aspirin and fibrinolytics (i.e. streptokinase) were additive.[340] The first dose of aspirin (162–325 mg) should be chewed or given i.v. and a low dose (75–100 mg) given orally daily from the next day thereafter. Clopidogrel added to aspirin reduces the risk of CV events and overall mortality in patients treated with fibrinolysis and should be added to aspirin following lytic therapy.[341,342] Based on the available RCTs, there is insufficient evidence to support or refute improved outcomes with ticagrelor or prasugrel in patients with STEMI treated with thrombolytics.[343–345] There is no evidence that administration of GP IIb/IIIa receptor inhibitors improves myocardial perfusion or outcomes in patients treated with fibrinolysis, and it may increase the risk of bleeding events.[346]

Parenteral anticoagulation is recommended until revascularization, if performed. Despite an increased risk of major bleeding, the net clinical benefit favoured enoxaparin over UFH in the ASsessment of the Safety and Efficacy of a New Thrombolytic 3 (ASSENT 3) trial ($n = 6095$).[347] In the large Enoxaparin and Thrombolysis Reperfusion for Acute myocardial infarction Treatment–Thrombolysis In Myocardial Infarction 25 (ExTRACT–TIMI 25) trial ($n = 20\,506$), a lower dose of enoxaparin was given to patients ≥75 years old and to those with impaired renal function (estimated creatinine clearance <30 mL/min). Enoxaparin was associated with a reduction in the risk of death and re-infarction at 30 days when compared with a weight-adjusted UFH dose, but at the cost of a significant increase in non-cerebral bleeding complications. The net clinical benefit (i.e. absence of death, non-fatal infarction, and intracranial haemorrhage) favoured enoxaparin.[348,349] In the large OASIS-6 trial, fondaparinux was superior to placebo or UFH in preventing death and re-infarction, especially in patients who received streptokinase.[260,350] In a large trial with streptokinase, significantly fewer re-infarctions were seen with bivalirudin given for 48 h compared with UFH, although at the cost of a modest non-significant increase in non-cerebral bleeding complications.[351] Bivalirudin has not been studied with fibrin-specific agents, and there is no evidence to support direct thrombin inhibitors as an adjunct to fibrinolysis.[260,350]

Weight-adjusted i.v. tenecteplase, low-dose aspirin, clopidogrel given orally, and enoxaparin i.v. followed by s.c. administration until the time of PCI (revascularization) represents the most extensively studied antithrombotic regimen as part of a pharmaco-invasive strategy.[184,186,213,346,352] Further information on fibrinolytic therapy, including antithrombotic co-therapies and contraindications is provided in Supplementary data online, *Tables S10* and *S11*.

Recommendation Table 7 — Recommendations for fibrinolytic therapy

Recommendations	Class[a]	Level[b]
Fibrinolytic therapy		
When fibrinolysis is the reperfusion strategy, it is recommended to initiate this treatment as soon as possible after diagnosis in the pre-hospital setting (aim for target of <10 min to lytic bolus).[206,353–355]	I	A
A fibrin-specific agent (i.e. tenecteplase, alteplase, or reteplase) is recommended.[356,357]	I	B
A half-dose of tenecteplase should be considered in patients >75 years of age.[184]	IIa	B
Antiplatelet co-therapy with fibrinolysis		
Aspirin and clopidogrel are recommended.[340–342]	I	A
Anticoagulation co-therapy with fibrinolysis		
Anticoagulation is recommended in patients treated with fibrinolysis until revascularization (if performed) or for the duration of hospital stay (up to 8 days).[260,347,348,350,357–360]	I	A
Enoxaparin i.v. followed by s.c. is recommended as the preferred anticoagulant.[347,348,357–360]	I	A
When enoxaparin is not available, UFH is recommended as a weight-adjusted i.v. bolus, followed by infusion.[357]	I	B
In patients treated with streptokinase, an i.v. bolus of fondaparinux followed by an s.c. dose 24 h later should be considered.[260]	IIa	B

i.v., intravenous; s.c, subcutaneous; UFH, unfractionated heparin.
[a]Class of recommendation.
[b]Level of evidence.

6.7. Antithrombotic therapy in patients not undergoing reperfusion

Patients with a final diagnosis of ACS who do not undergo reperfusion should receive a $P2Y_{12}$ receptor inhibitor in addition to aspirin, maintained over 12 months unless there is HBR. Among ACS patients who are medically managed without revascularization, the combination of aspirin and ticagrelor for up to 12 months has demonstrated a benefit in comparison to aspirin and clopidogrel.[238,361] The combination of aspirin and prasugrel can also be justified in preference to aspirin and clopidogrel if coronary angiography has been performed and CAD is confirmed.[239,362] As such, potent $P2Y_{12}$ inhibitor-based DAPT is a reasonable option for patients with a final diagnosis of ACS not undergoing reperfusion, unless concerns over the bleeding risk prevail (e.g. based on ARC-HBR criteria).[238,361] A DAPT regimen based on clopidogrel and aspirin may provide a good net clinical benefit among older ACS patients.[242,363] Further information regarding antithrombotic therapy in ACS patients who do not undergo reperfusion is provided in the Supplementary data online.

7. Acute coronary syndrome with unstable presentation

In some cases, ACS patients can present with haemodynamic compromise (i.e. out-of-hospital cardiac arrest [OHCA] and/or CS).

© ESC 2023

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

7.1. Out-of-hospital cardiac arrest in acute coronary syndrome

While a small minority of all patients with ACS present as OHCA, ACS is the most common cause of OHCA.[364–366] In patients with OHCA, resuscitation efforts should follow the European Resuscitation Council Guidelines.[367] The majority of adult cardiac arrest cases are associated with obstructive CAD and ACS should be included in the differential diagnosis.[365,368] Therefore, ICA can be part of the post-resuscitation management for patients who are estimated to have a high probability of acute coronary occlusion (e.g. persistent ST-segment elevation or equivalents and/or haemodynamic and/or electrical instability).[367,369] Neurological status (e.g. comatose vs. non-comatose) and survival probability (i.e. favourable benefit/risk ratio vs. futility) should also be included in the decision-making algorithm.

Despite the lack of dedicated trials, patients with return of spontaneous circulation (ROSC) and persistent ST-segment elevation should, in general, undergo a PPCI strategy (immediate ICA and PCI if indicated), based on the overall clinical situation and a reasonable benefit/risk ratio. Based on registry reports, emergent ICA and PCI are associated with good outcomes in this setting, particularly in patients who are non-comatose at initial assessment.[368,370,371]

The management of patients with ROSC without evidence of ST-segment elevation should be individualized according to haemodynamic and neurological status. In OHCA with an initial shockable rhythm and without ST-segment elevation or equivalents and without CS, routine immediate ICA is not superior to a delayed invasive strategy based on data from the COACT (Coronary Angiography after Cardiac Arrest) and TOMAHAWK (Immediate Unselected Coronary Angiography Versus Delayed Triage in Survivors of Out-of-hospital Cardiac Arrest Without ST-segment Elevation) RCTs.[372,373] Smaller, underpowered trials (EMERGE [EMERGEncy versus delayed coronary angiogram in survivors of out-of-hospital cardiac arrest with no obvious non-cardiac cause of arrest], PEARL [A Pilot Randomized Clinical Trial of Early Coronary Angiography Versus No Early Coronary Angiography for Post-Cardiac Arrest Patients Without ECG ST Segment Elevation], and COUPE [Coronariography in OUt of hosPital Cardiac arrEst]) have also pointed to the same conclusion.[372–377] Further detail on these trials is provided in the Supplementary data online, *Evidence Tables*.

Based on data from the COACT and TOMAHAWK trials, it appears reasonable to delay ICA in haemodynamically stable patients with resuscitated OHCA without ST-segment elevation or equivalents. Initial evaluation in the ED or intensive cardiac care unit (ICCU) should focus on excluding non-coronary causes (cerebrovascular events, respiratory failure, non-cardiogenic shock, PE, or intoxication). Echocardiography is also useful in the evaluation of these patients. The decision to perform selective coronary angiography (and PCI if indicated) should also consider factors associated with poor neurological outcome and the likelihood of ACS.

In patients who remain unresponsive after ROSC, monitoring of core temperature and actively preventing fever (defined as a temperature >37.7°C) is recommended to improve neurological outcome.[367,378–385] A recent study compared device-based temperature control of 36°C for 24 h followed by targeting of 37°C for either 12 or 48 h (for total intervention times of 36 and 72 h, respectively) or until the patient regained consciousness in 789 patients with OHCA of a presumed cardiac cause (~45% with ST segment elevation on ECG; immediate coronary angiography performed in 92% and PCI in 43%). This study reported comparable outcomes with both strategies with respect to the primary endpoint (death, severe disability, or coma) at 90 days.[384] In all comatose survivors, evaluation of neurological prognosis no earlier than 72 h after admission is recommended.[367,378–383,386]

7.1.1. Systems of care

There is increasing evidence suggesting that specialized hospitals for patients following OHCA (referred to as cardiac arrest centres) may be associated with clinical benefits.[367] See Supplementary data online, *Section 7.1.1* for expanded information on this topic.

Recommendation Table 8 — Recommendations for cardiac arrest and out-of-hospital cardiac arrest

Recommendations	Class[a]	Level[b]
Cardiac arrest and OHCA		
A PPCI strategy is recommended in patients with resuscitated cardiac arrest and an ECG with persistent ST-segment elevation (or equivalents).[368,387,388]	I	B
Routine immediate angiography after resuscitated cardiac arrest is not recommended in haemodynamically stable patients without persistent ST-segment elevation (or equivalents).[373–377]	III	A
Temperature control		
Temperature control (i.e. continuous monitoring of core temperature and active prevention of fever [i.e. >37.7°C]) is recommended after either out-of-hospital or in-hospital cardiac arrest for adults who remain unresponsive after return of spontaneous circulation.[378–385,389]	I	B
Systems of care		
It is recommended that healthcare systems implement strategies to facilitate transfer of all patients in whom ACS is suspected after resuscitated cardiac arrest directly to a hospital offering 24/7 PPCI via one specialized EMS.[390–392]	I	C
Transport of patients with OHCA to a cardiac arrest centre according to local protocols should be considered.[391,393]	IIa	C
Evaluation of neurological prognosis		
Evaluation of neurological prognosis (no earlier than 72 h after admission) is recommended in all comatose survivors after cardiac arrest.[386]	I	C

ACS, acute coronary syndrome; ECG, electrocardiogram; EMS, emergency medical services; OHCA, out-of-hospital cardiac arrest; PPCI, primary percutaneous coronary intervention.
[a]Class of recommendation.
[b]Level of evidence.

7.2. Cardiogenic shock complicating acute coronary syndrome

Early revascularization with either PCI or CABG is recommended for patients with AMI complicated by CS, based on the results of the SHOCK (Should We Emergently Revascularize Occluded Coronaries

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026
© ESC 2023

for Cardiogenic Shock) trial.[394–396] While most patients will proceed to PCI at the time of diagnostic angiography if myocardial revascularization is indicated, surgical revascularization represents a valuable treatment option in patients in whom attempted PCI of the IRA has failed or if the coronary anatomy is not amenable to PCI.[395,397,398] In the presence of CS due to AMI-related mechanical complications, surgical or percutaneous treatment may also be indicated and the strategy should be decided based on discussion between members of the Heart Team.

In the IABP-SHOCK II (Intraaortic Balloon Pump in Cardiogenic Shock II) trial, intra-aortic balloon pump (IABP) use was not associated with lower 30-day mortality.[399] Therefore, in the absence of mechanical complications, the routine use of an IABP is not recommended for CS complicating AMI. The role of mechanical circulatory devices (veno-arterial extracorporeal membrane oxygenation [VA-ECMO], micro-axial pump) in the AMI setting is not well established and large-scale randomized trials are warranted.[400,401] The Extracorporeal Membrane Oxygenation in the Therapy of Cardiogenic Shock trial randomized 122 patients (51% with STEMI) with rapidly deteriorating or severe CS to either immediate implementation of VA-ECMO or an initially conservative strategy (which allowed for downstream use of VA-ECMO).[402] The immediate implementation of VA-ECMO did not result in improved clinical outcomes.[402] However, the interpretation of this trial is challenging because of the ~40% crossover rate to VA-ECMO in the conservative arm, the inclusion of heterogenous phenotypes of CS, and inclusion of crossover in the combined primary endpoint. As a result of these limitations, this trial cannot adequately answer if mechanical circulatory support (MCS) is able to reduce mortality in this setting.

It is important to note that while there is still a lack of high-quality randomized data supporting the use of MCS in ACS patients presenting with CS, some recent observational analyses have reported that the use of intravascular LV assist devices may be associated with an increased risk of adverse events in comparison to IABP in this setting, including mortality and bleeding.[401,403] Therefore, while MCS may be considered in selected patients with ACS and severe/refractory CS, caution should be exercised in this regard until further randomized data are available. The management of patients with CS complicating AMI and MVD is presented in *Section 10*.

Recommendation Table 9 — Recommendations for cardiogenic shock

Recommendations	Class[a]	Level[b]
Immediate coronary angiography and PCI of the IRA (if indicated) is recommended in patients with CS complicating ACS.[394,396,404]	I	B
Emergency CABG is recommended for ACS-related CS if PCI of the IRA is not feasible/unsuccessful.[394,395]	I	B
In cases of haemodynamic instability, emergency surgical/catheter-based repair of mechanical complications of ACS is recommended, based on Heart Team discussion.	I	C
Fibrinolysis should be considered in STEMI patients presenting with CS if a PPCI strategy is not available within 120 min from the time of STEMI diagnosis and mechanical complications have been ruled out.[184,354]	IIa	C
In patients with ACS and severe/refractory CS, short-term mechanical circulatory support may be considered.[402]	IIb	C
The routine use of an IABP in ACS patients with CS and without mechanical complications is not recommended.[399,405–407]	III	B

© ESC 2023

ACS, acute coronary syndrome; CABG, coronary artery bypass grafting; CS, cardiogenic shock; IABP, intra-aortic balloon pump; IRA, infarct-related artery; PPCI, primary percutaneous coronary intervention; STEMI, ST-elevation myocardial infarction.
[a]Class of recommendation.
[b]Level of evidence.

8. Management of acute coronary syndrome during hospitalization

8.1. Coronary care unit/intensive cardiac care unit

Following reperfusion, it is recommended to admit high-risk ACS patients (including all STEMI patients) to a coronary care unit (CCU) or ICCU. Conditions in patients with ACS that act as acute risk modifiers include ongoing myocardial ischaemia (e.g. failed reperfusion), acute HF and/or hypoperfusion, CS, cardiac arrest with coma, malignant (life-threatening) cardiac arrhythmias, high-degree atrioventricular block, and acute renal failure (with oliguria). All ICCUs must have appropriate diagnostic facilities to guide the delivery of pharmacological and invasive treatment. The staff should be thoroughly familiar with the management of all aspects of ACS, including: arrhythmias, HF, mechanical circulatory support, invasive and non-invasive haemodynamic monitoring (arterial and pulmonary artery pressures), respiratory monitoring, mechanical ventilation, and temperature control.[408] The CCU/ICCU should also be able to manage patients with renal and pulmonary disease. The desirable organization, structure, and criteria of CCU/ICCUs have been detailed in an ESC–Acute CardioVascular Care Association position paper.[408]

8.1.1. Monitoring

It is recommended to initiate ECG monitoring as soon as possible in all patients with ACS in order to detect life-threatening arrhythmias and allow prompt defibrillation if indicated. ECG monitoring for arrhythmias and new ST-segment elevation/depression is recommended for at least 24 h after symptom onset in all high-risk patients with ACS, including all STEMI patients.[409] Longer monitoring could be considered in patients at intermediate to high risk of cardiac arrhythmias (i.e. those with more than one of the following criteria: haemodynamically unstable, presenting with major arrhythmias, left ventricular ejection fraction [LVEF] <40%, failed reperfusion, additional critical coronary stenoses of major vessels, or complications related to PCI). Further monitoring for arrhythmias will be dependent on the estimated risk. When a patient leaves the ICCU or equivalent, monitoring may be continued by telemetry. It is recommended that personnel adequately equipped and trained to manage life-threatening arrhythmias and cardiac arrest accompany patients who are transferred between facilities during the time window in which they require continuous rhythm monitoring.[409]

8.1.2. Ambulation

Early ambulation (i.e. out of bed on day 1) is recommended in the majority of patients with ACS. This is facilitated by using radial access for

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

invasive management. Patients with extensive myocardial damage, HF, hypotension, or arrhythmias may initially rest in bed before assessment of myocardial function and clinical stabilization. Prolongation of bed rest and limitation of physical activity may occasionally be required in patients with large infarcts or severe complications.

8.1.3. Length of stay in the intensive cardiac care unit

The optimal length of stay in the ICCU and hospital should be individualized according to the patient's clinical situation, taking into account their baseline cardiac risk and comorbidities, baseline mental/functional status, and social support.[410,411] Of note, the majority of adverse in-hospital events occur early after admission and the initiation of treatment.

8.2. In-hospital care

8.2.1. Length of hospital stay

The impact of both successful reperfusion and knowledge of the coronary anatomy (due to increasing rates of ICA) has resulted in progressive reductions in the length of stay after ACS, alongside significant reductions in 30-day mortality, suggesting that discharge within 72 h is not associated with late mortality.[411–417] Candidates for early discharge after PCI can be identified using simple criteria.[413,414] In one study, patients meeting the following criteria were considered to be 'low risk' and suitable for early discharge: age <70 years, LVEF >45%, one- or two-vessel disease, successful PCI, and no persistent arrhythmias.[413] A recently published consensus document also presents a template and flow chart to support reasonable decision-making regarding post-procedural length of stay for a broad spectrum of patients undergoing PCI.[418]

Early (i.e. same day) transfer to a local hospital following successful PPCI is routine practice. This can be done safely under adequate monitoring and supervision in selected patients (i.e. patients without signs or symptoms consistent with ongoing myocardial ischaemia, without arrhythmias, who are haemodynamically stable, who are not requiring vasoactive or mechanical support, and who are not scheduled for further revascularization).[419]

8.2.2. Risk assessment

Early and late risk stratification soon after presentation is useful to aid decision-making in patients presenting with ACS.

8.2.2.1. Clinical risk assessment

All patients with ACS (in particular, patients with STEMI) should have an early assessment of short-term risk, including an evaluation of the extent of myocardial damage, the achievement of successful reperfusion, and the presence of clinical markers of high risk of further events (i.e. older age, tachycardia, hypotension, Killip class >I, anterior MI, previous MI, elevated initial serum creatinine, history of HF, peripheral arterial disease or anaemia). Several risk scores have been developed based on readily identifiable parameters in the acute phase before reperfusion.[420,421] A number of prognostic models that aim to estimate the longer-term risk of all-cause mortality, or the combined risk of all-cause mortality or MI, have also been developed. These models have been formulated into clinical risk scores and, among these, the GRACE risk score offers the best discriminative performance and is therefore recommended for risk assessment.[48,421–425] Additional information regarding the GRACE score is provided in the Supplementary data online.

8.2.2.2. Imaging risk assessment

LV dysfunction is a key prognostic factor for patients with ACS.[426] It is recommended that the LVEF is determined before hospital discharge in all patients with ACS. Routine echocardiography after PPCI is recommended to assess resting LV, RV, and valvular function. In addition, echocardiography can be used to exclude early post-infarction mechanical complications and LV thrombus. In the limited number of cases in which echocardiography is suboptimal or inconclusive, CMR may be a valuable alternative.[427–431]

In patients presenting days after an acute ACS event with a completed MI, the presence of recurrent angina or documented ischaemia and proven viability in a large myocardial territory may help to guide the strategy of planned revascularization of an occluded IRA.[192,432,433]

In patients with a pre-discharge LVEF of ≤40%, re-evaluation of the LVEF 6–12 weeks after complete revascularization and optimal medical therapy is recommended to assess the potential need for primary prevention implantable cardioverter defibrillator (ICD) implantation.[434] Additional parameters that are measured by imaging in these patients and that have been used as endpoints in clinical trials include: (i) infarct size (CMR, SPECT, and positron emission tomography); (ii) myocardium at risk (SPECT, CMR); (iii) MVO (CMR); and (iv) intra-myocardial haemorrhage (CMR). Infarct size, MVO and intra-myocardial haemorrhage are predictors of both long-term mortality and HF in STEMI survivors.[435–438]

8.2.2.3. Biomarkers for risk assessment

Beyond diagnostic utility, initial cTn levels add prognostic information in addition to clinical and ECG variables in terms of predicting the risk of short- and long-term mortality. While hs-cTn T and I have comparable diagnostic accuracy, hs-cTn T has slightly greater prognostic accuracy regarding mortality.[61,439–441] Serial measurements are useful to identify peak levels of cTn for risk stratification purposes in patients with established MI. The higher the hs-cTn levels, the greater the risk of death.[31,55,442] However, evidence is limited regarding the optimal time points of serial hs-cTn measurement. Serum creatinine and eGFR should also be determined in all patients with ACS because they affect prognosis and are key elements of the GRACE risk score.[443] Similarly, natriuretic peptides (brain natriuretic peptide [BNP] and N-terminal pro-BNP [NT-pro BNP]) provide prognostic information in addition to cTn regarding the risk of death and acute HF, and the development of AF.[444] Additional information on the use of biomarkers for this purpose is presented in the Supplementary data online.

8.2.2.4. Bleeding risk assessment

Major bleeding events are associated with increased mortality in patients with ACS.[231] Further detail on scores that may be considered for estimation of bleeding risk is provided in the Supplementary data online, including *Table S12*.

8.2.2.5. Integrating ischaemic and bleeding risks

Major bleeding events affect prognosis in a similar way to spontaneous ischaemic complications.[445,446] Given the trade-off between ischaemic and bleeding risks for any antithrombotic regimen, risk scores may be useful to tailor antithrombotic duration and intensity, in order to maximize ischaemic protection and minimize bleeding risk in the individual patient. Specific risk scores have been developed for patients on DAPT following PCI, in the settings of both CCS and ACS. Further detail on available scores is provided in the Supplementary data online.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Recommendation Table 10 — Recommendations for in-hospital management

Recommendations	Class[a]	Level[b]
Logistical issues for hospital stay		
It is recommended that all hospitals participating in the care of high-risk patients have an ICCU/CCU equipped to provide all required aspects of care, including treatment of ischaemia, severe heart failure, arrhythmias, and common comorbidities.	I	C
It is recommended that high-risk patients (including all STEMI patients and very high-risk NSTE-ACS patients) have ECG monitoring for a minimum of 24 h.	I	C
It is recommended that high-risk patients with successful reperfusion therapy and an uncomplicated clinical course (including all STEMI patients and very high-risk NSTE-ACS patients) are kept in the CCU/ICCU for a minimum of 24 h whenever possible, after which they may be moved to a step-down monitored bed for an additional 24–48 h.	I	C
Discharge of selected high-risk patients within 48–72 h should be considered if early rehabilitation and adequate follow-up are arranged.[411,413,415,447]	IIa	A
Same-day transfer in selected stable patients after successful and uneventful PCI should be considered.[419]	IIa	C
Imaging		
Routine echocardiography is recommended during hospitalization to assess regional and global LV function, detect mechanical complications, and exclude LV thrombus.	I	C
When echocardiography is suboptimal/inconclusive, CMR imaging may be considered.	IIb	C

© ESC 2023

ACS, acute coronary syndrome; CCU, cardiac care unit; CMR, cardiac magnetic resonance; ECG, electrocardiogram; ICCU, intensive cardiac care unit; LV, left ventricular; NSTE-ACS, non-ST-segment elevation acute coronary syndrome; PCI, percutaneous coronary intervention; STEMI, ST-elevation myocardial infarction.
[a]Class of recommendation.
[b]Level of evidence.

9. Technical aspects of invasive strategies

9.1. Percutaneous coronary intervention

9.1.1. Vascular access

Timely PCI with concomitant antithrombotic drugs has reduced the ischaemic risk in patients with ACS. However, this strategy is also associated with an increased bleeding risk, which affects prognosis at least as much as ischaemic complications and is associated with impaired survival.[448,449] Among patients undergoing PCI, access-related bleeding accounts for 30–70% of total bleeding events.[450] There is strong evidence demonstrating that reducing access-site bleeding events with the use of radial access translates into significant clinical benefits.[448,449] The largest randomized trials on this topic in patients with ACS are the RadIal Vs femorAL access for coronary intervention (RIVAL) trial with 7021 ACS patients and the Minimizing Adverse Haemorrhagic Events by TRansradial Access Site and Systemic Implementation of angioX (MATRIX) trial with 8404 ACS patients (47.6% with STEMI).[451,452] These trials have demonstrated significantly lower rates of access site-related bleeding, surgical access site repair, and blood transfusion with radial compared with femoral access. In the MATRIX trial, no significant interaction was observed between the type of ACS and the benefit associated with the radial approach, suggesting that the results of this trial can be extended to patients across the entire spectrum of ACS.[453] In a cost-effectiveness analysis of the MATRIX trial, radial access was also associated with significant savings in terms of quality-adjusted life years and PCI-related costs.[454] Therefore, radial access is recommended as the preferred approach in ACS patients undergoing invasive assessment with or without PCI. However, femoral access may still be selectively chosen instead of radial access in certain patients (i.e. depending on the haemodynamic situation and other technical aspects during the index PCI procedure).

9.1.2. Intravascular imaging/physiology of the infarct-related artery

9.1.2.1. Intravascular imaging

As a diagnostic tool, intravascular imaging is useful in ACS patients without significant obstructive CAD on coronary angiography. Excluding an atherothrombotic cause in the main coronary arteries for the ACS may have important clinical implications, not only for immediate invasive management but also for potentially lifelong antithrombotic therapies. Intravascular imaging is also useful in cases where there is ambiguity regarding the culprit lesion. Culprit lesion ambiguity can be present in more than 30% of patients with suspected NSTE-ACS and over 10% of patients may have multiple culprit lesions.[455,456] The recommendations for intravascular imaging in ACS are presented in *Figure 13*.

The role of intravascular imaging is well established as a tool to guide and optimize PCI. Evidence in support of intravascular ultrasound (IVUS) guidance in ACS generally derives from subgroup analyses of all-comers trials. Meta-analysis of available randomized trials confirms the superiority of IVUS guidance in the reduction of MACE, although a definitive, large-scale, multinational trial is missing.[457–459] Smaller RCTs have evaluated the role of optical coherence tomography (OCT) (see Supplementary data online).[460]

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

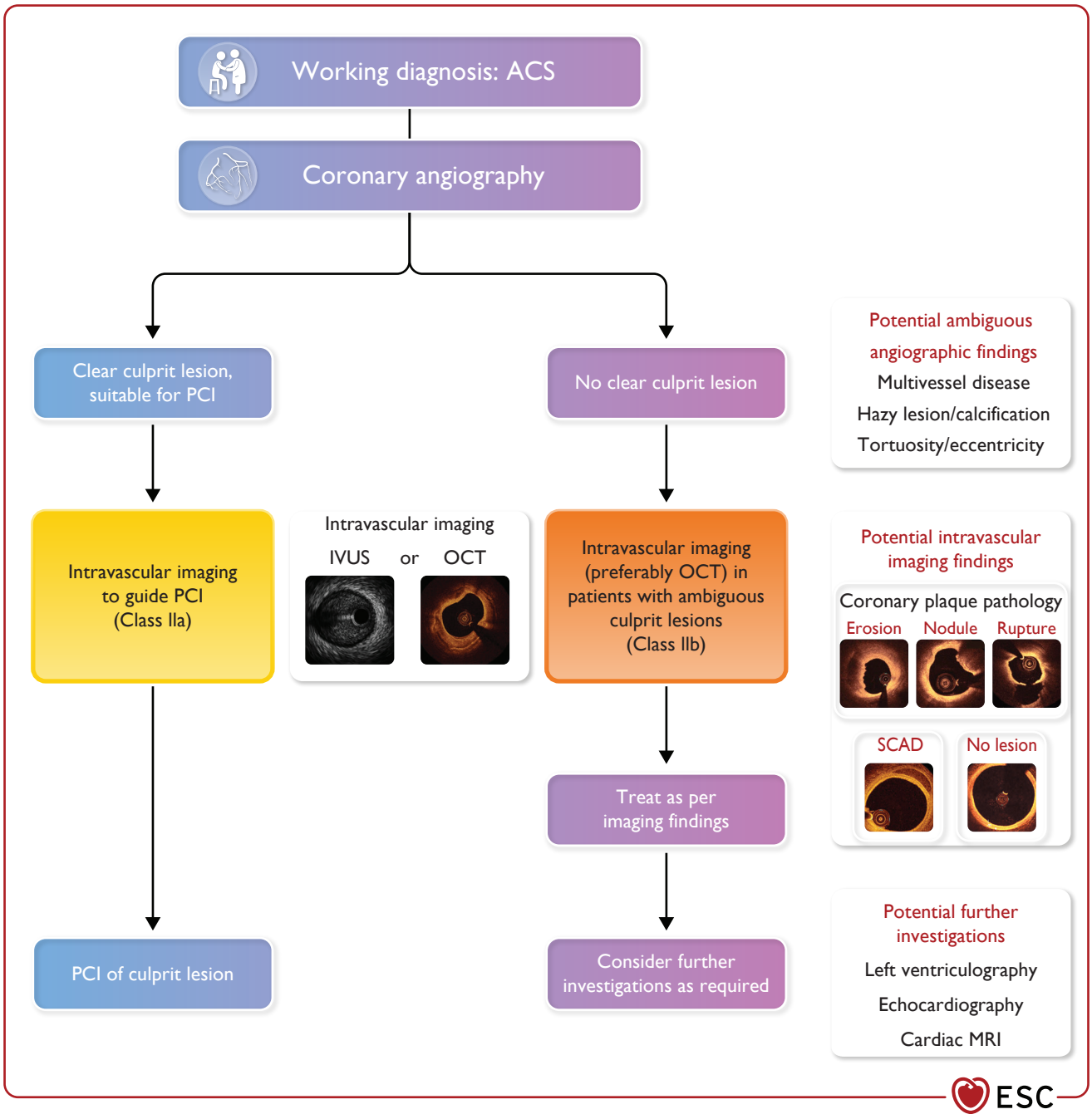


Figure 13 A practical algorithm to guide intravascular imaging in acute coronary syndrome patients. ACS, acute coronary syndrome; IVUS, intravascular ultrasound; MRI, magnetic resonance imaging; OCT, optical coherence tomography; PCI, percutaneous coronary intervention; SCAD, spontaneous coronary artery dissection.

9.1.2.2. Intravascular physiology

Intracoronary physiology is increasingly being used in patients with ACS to assess the haemodynamic significance of intermediate severity non-IRA stenoses (see *Section 10*). However, PCI of the IRA should not be deferred based on invasive epicardial functional assessment in patients with ACS. The coronary microcirculation begins to recover within 24 h of PPCI and acute functional assessment of the IRA may underestimate the true haemodynamic severity of the coronary stenosis.[461] Beyond 1 week from the acute event, fractional flow reserve (FFR) measurement has been reported to reliably predict abnormal nuclear imaging results.[462] Additional information about the role of intracoronary physiology in the IRA is presented in the Supplementary data online.

9.1.3. Timing of revascularization with percutaneous coronary intervention

In some patients with ACS undergoing ICA, an initial conservative management strategy with optimized guideline-directed medical therapy may be considered on a case-by-case basis. The specific circumstances include ACS patients with small calibre vessels, an occluded small side branch, or concerns regarding non-compliance with antithrombotic

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

therapy. In the context of complex CAD and anticipated complex PCI, an initial conservative strategy in medically stabilized patients without ongoing symptoms allows time for Heart Team discussion regarding the optimal revascularization strategy.

9.1.4. Balloons and stents

New-generation DES are associated with superior safety and improved efficacy compared with bare metal stents (BMS) and first-generation DES. The Norwegian Coronary Stent Trial (NORSTENT)—the largest clinical trial comparing outcomes of patients treated with new-generation DES or BMS—reported that the primary endpoint of death or MI was comparable in both treatment groups. Both target lesion revascularization (TLR) and stent thrombosis were reduced in the DES group and there was no treatment effect by ACS presentation interaction for the primary endpoint.[463] The COMFORTABLE-AMI (Comparison of Biolimus Eluted From an Erodible Stent Coating With Bare Metal Stents in Acute ST-Elevation Myocardial Infarction) and EXAMINATION (Everolimus-Eluting Stents Versus Bare-Metal Stents in ST Segment Elevation Myocardial Infarction) trials have also reported the clinical superiority of DES over BMS in terms of lower rates of re-infarction, target lesion revascularization, and stent thrombosis.[464,465] This clinical benefit was preserved at longer-term follow-up.[466–468]

A strategy of drug-coated balloon (DCB) angioplasty without stenting has also been proposed for patients with NSTE-ACS. In the small, prospective, randomized, single-centre REVELATION (REVascularization With PaclitaxEL-Coated Balloon Angioplasty Versus Drug-Eluting Stenting in Acute Myocardial InfarcTION) trial, DCB PCI vs. DES PCI was investigated in 120 patients undergoing PPCI. The primary endpoint of target vessel FFR at 9 months was not significantly different between the two groups.[469] In the small PEPCAD NSTEMI (Bare Metal Stent Versus Drug Coated Balloon With Provisional Stenting in Non-ST-Elevation Myocardial Infarction) trial, 210 patients were randomized to compare a DCB with primary stent treatment (BMS or DES).[470] During a mean follow-up period of 9.2 months, DCB treatment was non-inferior to treatment with a stent, with a target lesion failure (primary study endpoint) rate of 3.8% vs. 6.6% ($P = 0.53$). Given the limitations of these studies (in particular, the relatively small sample sizes), the use of DCB in NSTE-ACS requires further investigation in order to better inform future guideline recommendations.[471]

9.1.5. Embolic protection and microvascular salvage strategies

9.1.5.1. Thrombus aspiration

Large RCTs have failed to demonstrate a clinical benefit with routine manual thrombus aspiration in comparison to conventional PPCI.[472–474] In an individual patient data meta-analysis, thrombus aspiration was associated with fewer CV deaths and with more strokes or transient ischaemic attacks in the subgroup of patients with high thrombus burden (TIMI thrombus Grade 3).[475] However, in a sub-analysis from TOTAL (a Trial of routine aspiration ThrOmbecTomy with PCI vs. PCI ALone in patients with STEMI), routine thrombus aspiration did not improve outcomes at 1 year and was also associated with an increased rate of stroke in patients with high thrombus burden.[476] In patients with NSTE-ACS and thrombus-containing lesions, PCI with adjunctive thrombus aspiration was not associated with a reduction in MVO 4 days after the index procedure or with fewer MACE after up to 1 year of follow-up.[477] Based on these data, routine thrombus aspiration is not recommended, but in cases of large residual thrombus burden after opening the vessel with a guide wire or a balloon, thrombus aspiration may be considered.

9.1.5.2. Interventions to protect the microcirculation

The damage inflicted on the myocardium during AMI is the result of ischaemia and subsequent reperfusion (ischaemia/reperfusion injury). In patient-level pooled analyses, infarct size and MVO are independent predictors of long-term mortality and HF in survivors of STEMI.[436,478] Strategies to reduce ischaemia/reperfusion injury in general (and MVO in particular) remain an unmet clinical need. Further information regarding interventions to protect the microcirculation that are under clinical or experimental investigation is presented in the Supplementary data online.

9.2. Coronary artery bypass grafting

9.2.1. Indication and timing of coronary artery bypass grafting in acute coronary syndrome patients

There are no dedicated RCTs comparing percutaneous vs. surgical revascularization in patients with ACS. In the setting of STEMI, CABG should be considered only when PPCI is not feasible, particularly in the presence of ongoing ischaemia or large areas of jeopardized myocardium.[479]

In patients requiring immediate revascularization in the setting of very high-risk NSTE-ACS, PCI is usually preferred for reasons of timeliness, unless concomitant mechanical complications dictate a preference for surgical intervention.

In other patients with ACS, the choice of revascularization modality should be made according to the number of diseased vessels and the general principles of myocardial revascularization.[250] In patients with MVD, the choice of revascularization modality will be influenced by the overall anatomical disease complexity and the presence of comorbidities (including diabetes) in patients with low predicted surgical risk and mortality who are considered suitable for either modality. This is based on data from two large-scale individual patient meta-analyses.[480,481]

9.2.2. Technical considerations specific to acute coronary syndrome patients

The patient profile, including the need for emergency or extremely expeditious revascularization, may influence both the technique of CABG (including on-pump beating heart CABG) and the choice and use of CABG conduits. The need for prompt surgical revascularization in emergency circumstances does not facilitate the use of full arterial revascularization due to the prolonged period required for graft harvesting. Accordingly, the use of total venous graft-based CABG or the use of single left internal mammary artery plus additional venous grafts may be useful in this setting.[397]

9.3. Spontaneous coronary artery dissection

Spontaneous coronary artery dissection (SCAD) is an infrequent cause of ACS in general but accounts for a significant proportion of ACS cases in young/middle-aged women.[482] The pathophysiology underlying SCAD is different to that of Type 1 MI and there are some differences in its management and outcomes. For these reasons, it is of paramount importance that an accurate diagnosis is established. Until evidence from ongoing prospective trials becomes available, patients with SCAD should receive the same pharmacological therapy as other ACS patients.[483]

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

9.3.1. Intravascular imaging

There are no RCTs to guide management strategies in patients with SCAD. The use of intravascular imaging is based on observations reported from clinical cohort studies and expert opinion.[482,484,485] In cases of diagnostic uncertainty after angiography, the use of intracoronary imaging with OCT or IVUS has to be carefully considered. There should be sufficient diagnostic uncertainty to justify coronary instrumentation, and even if this is the case, other factors like vessel tortuosity, vessel diameter, and a distal lesion location may prohibitively increase the risk.[482] If the decision is made to perform intravascular imaging, it is imperative to ensure the guide wire is located within the true lumen of the coronary artery before advancing the imaging catheter.[482] In patients with a diagnosis of SCAD on angiography and a plan for medical therapy, additional coronary instrumentation and intravascular imaging is not recommended on safety grounds.[482,484,485]

9.3.2. Revascularization

Conservative medical management, as opposed to PCI, is generally recommended for patients with SCAD.[482] In an international case series, coronary complications following PCI occurred in >30% of patients.[486–488] In a pooled analysis of three SCAD-PCI cohorts including 215 patients (94% female) drawn from Dutch, Spanish, and UK registries, and a matched cohort of conservatively managed SCAD patients ($n = 221$), PCI was associated with complications in ≈40% of cases (including 13% with serious complications). PCI is recommended only for SCAD with associated symptoms and signs of ongoing myocardial ischaemia, a large area of myocardium in jeopardy, and reduced antegrade flow. Useful strategies for these patients may include minimal plain balloon angioplasty to restore flow, followed by a conservative strategy, targeted stenting to seal the proximal and distal ends of the dissection, and/or extended stent lengths to prevent propagation of the haematoma. In patients with SCAD, CABG is recommended when dissection affects the left main or two proximal vessels, if PCI is not feasible or unsuccessful, and if there are symptoms and signs of ongoing myocardial ischaemia. In a small observational study, patients with SCAD treated with CABG had favourable early clinical outcomes, with an event rate up to 5 years similar to that of patients treated conservatively, despite a significant (68%) rate of graft occlusion at 5 years.[486] The rate of graft occlusion over time can be explained by the fact that CABG in these patients may be technically challenging as the dissected coronary artery is more prone to anastomosis failure, and because spontaneous healing over time may restore the flow in the anastomosed vessel.[486,489] For this reason, vein grafts should be considered in these patients in order to preserve arterial conduits for future use.[485]

Recommendation Table 11 — Recommendations for technical aspects of invasive strategies

Recommendations	Class[a]	Level[b]
Radial access is recommended as the standard approach, unless there are overriding procedural considerations.[451,452]	I	A
PCI with stent deployment in the IRA during the index procedure is recommended in patients undergoing PPCI.[490–494]	I	A
Drug-eluting stents are recommended in preference to bare metal stents in all cases.[463,466,468]	I	A
In patients with spontaneous coronary artery dissection, PCI is recommended only for patients with symptoms and signs of ongoing myocardial ischaemia, a large area of myocardium in jeopardy, and reduced antegrade flow.	I	C
Intravascular imaging should be considered to guide PCI.[495–499]	IIa	A
Coronary artery bypass grafting should be considered in patients with an occluded IRA when PPCI is not feasible/unsuccessful and there is a large area of myocardium in jeopardy.	IIa	C
Intravascular imaging (preferably optical coherence tomography) may be considered in patients with ambiguous culprit lesions.	IIb	C
The routine use of thrombus aspiration is not recommended.[472–474]	III	A

IRA, infarct-related artery; PCI, percutaneous coronary intervention; PPCI, primary percutaneous coronary intervention.
[a]Class of recommendation.
[b]Level of evidence.

10. Management of patients with multivessel disease

Approximately half of ACS patients have coronary MVD.[500] Management of non-IRA disease varies depending on the clinical setting.

10.1. Management of multivessel disease in acute coronary syndrome complicated by cardiogenic shock

Cardiogenic shock may occur in up to 4–11% of ACS patients, and occurs more frequently in the presence of complete coronary occlusion.[501,502] Ischaemia-related HF, acute severe mitral regurgitation, and mechanical complications are the major precipitating causes of CS in ACS. Irrespective of the mode of presentation (i.e. with or without ST-segment elevation or equivalent ECG patterns), these patients should be transferred as soon as possible to a tertiary care centre (e.g. a shock centre) where ICA can be performed, supported by specialists with relevant experience (the Shock Team).[503,504]

In the SHOCK trial, which compared emergency revascularization with initial medical stabilization in 302 patients with acute MI complicated by CS, ~60% had anterior MI and 85% had MVD.[394] Among the patients assigned to emergency revascularization, 64% underwent PCI and 36% underwent CABG. There were no differences in mortality at 30 days (primary endpoint), but at 6 months mortality was lower in the group assigned to revascularization than in the group assigned to medical therapy. Based on this evidence, immediate coronary angiography, and PCI if feasible, is recommended in patients with acute MI complicated by CS. In patients with coronary anatomy unsuitable for PCI, emergency CABG is recommended.[394]

Nearly 80% of ACS patients with CS have MVD. Based on the Culprit Lesion Only PCI versus Multivessel PCI in Cardiogenic Shock (CULPRIT-SHOCK) trial including ACS patients (both with and without ST-segment elevation or equivalent), PCI during the index

© ESC 2023
Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

procedure should be restricted to the IRA only.[404] In the CULPRIT-SHOCK trial, IRA-only PCI was associated with a significant reduction in all-cause death or renal replacement therapy at 30-day follow-up (RR 0.83, 95% CI, 0.71–0.96).[404] At 1-year follow-up, mortality did not differ significantly between the two groups.[505]

For patients undergoing emergency CABG, appropriate perioperative strategies (particularly in relation to prophylactic or on-demand mechanical circulatory support) may be considered based on pre-operative clinical status (e.g. age, comorbidities, electrical instability, the extent of jeopardized myocardium, the duration of ischaemia from the time of symptom onset, right ventricular involvement, and the feasibility of cardiac surgery from technical/logistical perspectives). *Figure 14* shows the algorithm for the management of patients with ACS and MVD.

10.2. Patients with multivessel coronary artery disease undergoing primary percutaneous coronary intervention

Multivessel disease is evident in approximately half of patients undergoing PPCI and is associated with an adverse prognosis.[506,507]

Over the past decade, a series of RCTs have provided clinical evidence that supports preventive revascularization of non-IRA after successful PPCI of the IRA. The pivotal clinical trials (in chronological order) include PRAMI (Preventive Angioplasty in Myocardial Infarction), CvLPRIT (Complete versus Lesion-only Primary PCI Trial), DANAMI-3–PRIMULTI (Third Danish Study of Optimal Acute Treatment of Patients with ST-Segment Elevation Myocardial Infarction—Primary PCI in Multivessel Disease), COMPARE-ACUTE (Comparison

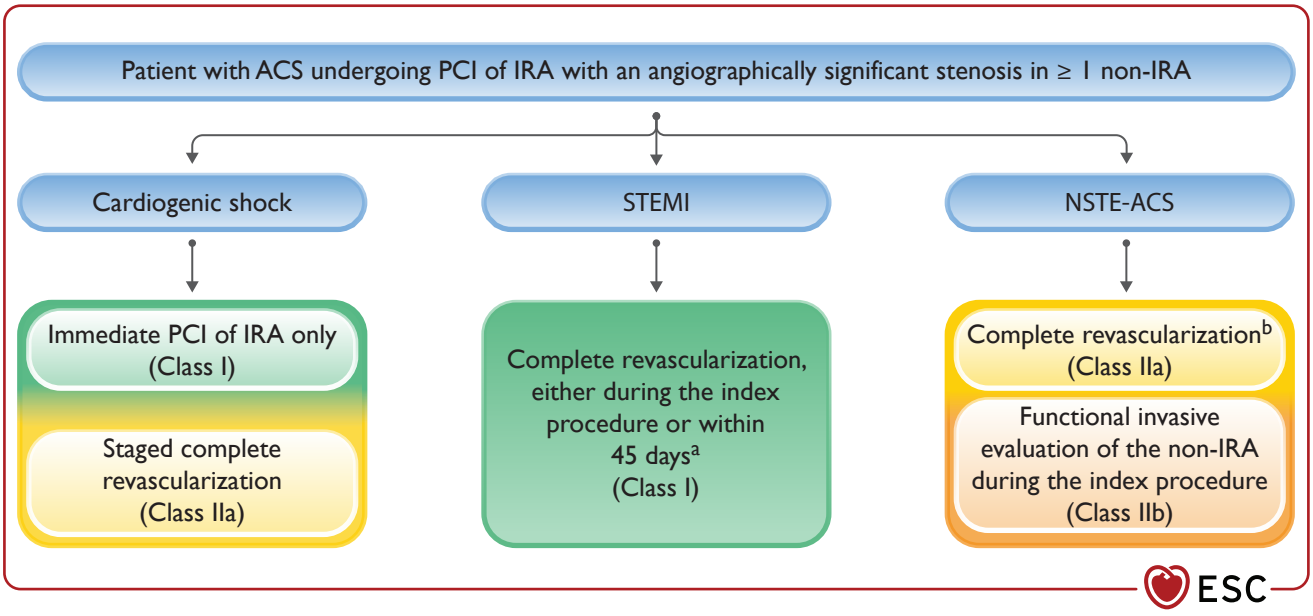


Figure 14 Algorithm for the management of acute coronary syndrome patients with multivessel coronary artery disease. CABG, coronary artery bypass grafting; IRA, infarct-related artery; MVD, multivessel disease; NSTE-ACS, non-ST-elevation acute coronary syndrome; PCI, percutaneous coronary intervention; STEMI, ST-elevation myocardial infarction; TIMI, Thrombolysis In Myocardial Infarction.[a]In patients presenting with STEMI and MVD without CS, complete revascularization either during the index PCI procedure or within 45 days, with PCI of non-IRA based on angiographic severity, is recommended. [b]In patients presenting with NSTE-ACS and MVD, complete revascularization, preferably during the index procedure should be considered. Functional invasive evaluation of non-IRA severity during the index procedure may be considered.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Between FFR Guided Revascularization Versus Conventional Strategy in Acute STEMI Patients With MVD), and COMPLETE (Complete vs. Culprit-only Revascularization to Treat Multivessel Disease After Early PCI for STEMI) (further details on these trials is provided in the Supplementary data online evidence tables).[508–511]

In a systematic review of 10 randomized trials that included 7030 patients with STEMI and MVD, complete revascularization was associated with reduced CV mortality compared with IRA-only PCI.[512] All-cause mortality was comparable in both groups. Complete revascularization was also associated with a reduced composite of CV death or new MI, supporting complete revascularization in patients with STEMI and MVD.[512]

10.3. Timing of non-infarct-related artery revascularization in acute coronary syndrome

10.3.1. Patients presenting with ST-elevation myocardial infarction and multivessel coronary artery disease

The previous ESC STEMI Guidelines recommended non-IRA PCI during the index procedure. The primary rationale for this recommendation was that all trials available until then had performed MVD PCI in that time frame. However, in the COMPLETE trial, non-IRA PCI in patients allocated to complete revascularization was performed either during hospitalization (67% of cases) or after discharge (33% of cases), at a mean time of 23 days after discharge but always within 45 days.[511] No treatment effect by timing of PCI interaction was observed. Given that the optimal timing of revascularization (immediate vs. staged) has still not been investigated in adequately sized randomized trials with a superiority design, no recommendation in favour of an immediate vs. a staged (i.e. either during index hospitalization or within 45 days of discharge) non-IRA PCI strategy can be formulated. No surgical studies have specifically investigated non-IRA revascularization.

10.3.2. Patients presenting with non-ST-elevation acute coronary syndrome and multivessel coronary artery disease

While there are a large number of studies providing evidence for patients presenting with STEMI and MVD, there are fewer data guiding the management of patients presenting with NSTE-ACS and MVD.[513] Currently, there is no dedicated trial comparing complete revascularization against IRA-only PCI for these patients. Observational studies and meta-analyses of non-randomized studies suggest that complete revascularization is associated with fewer deaths and MACE during follow-up in comparison to IRA-only PCI.[514,515] However, given that these are analyses of treatment effects based on non-randomized studies, the results should be considered as hypothesis-generating at best and this remains a gap in evidence.

10.4. Evaluation of non-infarct-related artery stenosis severity (angiography vs. physiology)

Overestimation of the severity of non-IRA lesions during the PPCI procedure when assessed by quantitative coronary angiography as compared with a repeated angiogram performed within 9 months has been reported.[516] Microvascular constriction may also occur in the non-IRAs, leading to some variation in functional measurements between baseline and follow-up, although the impact on decision-making may be modest.[517–520] A sub-analysis of the FAME (Fractional Flow Reserve versus Angiography for Multivessel Evaluation) trial reported that 65% of lesions in the angiographic severity range of 50–70% diameter stenosis, and 20% of lesions in the range 71–90%, have an FFR value above 0.80.[521]

The PRIME-FFR registry included 533 ACS patients and reported that systematic FFR measurement led to a change in the management strategy in 38% of cases (e.g. from CABG to PCI or to medical treatment), without an impact on MACE, death/MI, or angina symptoms at 1 year.[522] A subgroup analysis of the FAME trial in 328 patients with ACS (UA or NSTEMI) and MVD reported that the adoption of FFR to guide PCI resulted in similar risk reductions of MACE compared with patients with stable angina, with a lower number of stents implanted and less contrast media use.[523] The FAMOUS-NSTEMI (Fractional Flow Reserve Versus Angiographically Guided Management to Optimise Outcomes in Unstable Coronary Syndromes) trial randomized 350 patients with NSTE-ACS and at least one coronary stenosis (with diameter stenosis >30%) to either angiography-guided or FFR-guided management (medical therapy, PCI, or CABG), and demonstrated that a higher proportion of patients in the FFR-guided management group were initially treated with medical therapy. The FLOWER-MI (Flow Evaluation to Guide Revascularization in Multivessel ST-Elevation Myocardial Infarction) study randomized 1171 patients undergoing PPCI with MVD to complete revascularization guided by FFR or angiography. Compared with an angiography-guided approach, an FFR-guided strategy did not reduce the risk of death, MI, or urgent revascularization at 1 year.[524] PCI was performed in 66.2% of patients in the FFR-guided group and in 97.1% of the angiography-guided group. In FLOWER-MI, complete revascularization during the index procedure was only performed in 4% of patients in both groups, and functional evaluation was mainly undertaken at the time of the second procedure.[524] However, based on the study design, complete revascularization could also be performed during a separate staged procedure as early as possible before hospital discharge and within 5 days of the initial procedure.

A meta-analysis of 10 RCTs (including 3031 patients undergoing PPCI) assessed outcomes in patients with complete revascularization vs. IRA-only PCI according to whether the decision to carry out non-IRA preventive PCI was based on angiography alone or on angiography plus FFR.[525] Preventive PCI of the non-IRA was associated with a significant reduction in cardiac death and non-fatal MI only when the decision to proceed with non-IRA PCI was based solely on angiography. Similar findings were reported in another meta-analysis of seven RCTs including a total of 6597 patients undergoing PPCI.[526] In patients randomised to the complete revascularization arm, an angiography-guided strategy (≥70% diameter stenosis) for non-IRA lesions was associated with lower rates of recurrent MI, whereas an FFR-guided (≤0.80 for lesions with ≤90% diameter stenosis) guided approach was not. In another meta-analysis, which pre-dated the FLOWER-MI trial, there was no heterogeneity in the primary outcome when complete revascularization was performed using an FFR-guided strategy (OR 0.78, 95% CI, 0.43–1.44) or an angiography-guided strategy (OR 0.61, 95% CI, 0.38–0.97; $P = 0.52$ for interaction).[512] A pooled *post-hoc* patient-level analysis of three RCTs (FAME, DANAMI-3–PRIMULTI, and FAMOUS-NSTEMI) in ACS patients treated with a functionally complete revascularization strategy (i.e. PCI of the stenosis with FFR ≤0.80, deferral to medical therapy stenosis with FFR >0.80) reported that the residual SYNTAX score (a proxy of the residual coronary stenosis deferred to medical therapy) was not associated with MACE at 2 years, suggesting that it may be safe to defer the management of functionally non-significant stenoses in the non-IRA.[527] The FRAME AMI (FFR Versus Angiography-Guided Strategy for Management of AMI With Multivessel Disease) trial compared selective PCI guided by FFR (PCI if FFR ≤0.80) to routine PCI

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

guided by angiography (PCI if diameter stenosis >50%) of the non-IRA(s) in patients presenting with AMI who had undergone successful PCI of the IRA (47% STEMI, 53% NSTEMI).[528] This study reported that at a median follow-up of 3.5 years, the primary endpoint (death, MI, or repeat revascularization) occurred less frequently in patients randomized to the FFR-guided strategy, mainly driven by differences in patients presenting with NSTEMI. However, the trial was terminated early, with only 562 out of an intended 1292 patients enrolled, and there was a relatively small number of primary outcome events.

10.5. Hybrid revascularization

Hybrid coronary revascularization (HCR) is defined as combined or consecutive procedures consisting of an internal mammary artery graft to the left anterior descending artery (LAD) and PCI to the other non-LAD vessels for the treatment of MVD.[529] The preferred surgical technique for HCR is a minimally invasive left anterior mini-thoracotomy or robotic-assisted left internal mammary artery (LIMA)-LAD. The rationale for HCR is to combine the prognostic benefits of a LIMA for grafting of the LAD with the potential benefits of contemporary PCI with DES for disease in arteries that would otherwise be revascularized using vein grafts (which are prone to occlusion).[530] There is limited evidence from RCTs to support hybrid revascularization. Clinical decision-making in this regard should involve the Heart Team. Clinical criteria supporting an HCR strategy in ACS patients with an indication for CABG may include MVD with LAD suitable for CABG and non-LAD lesions suitable for PCI, atheroma in the ascending aorta, an unprotected left main coronary artery that is unsuitable for PCI, complex LAD disease, advanced age, low LVEF (≤30%), frailty, diabetes mellitus, renal failure, prior sternotomy, and the lack of available bypass conduits.

Recommendation Table 12 — Recommendations for management of patients with multivessel disease

Recommendations	Class[a]	Level[b]
It is recommended to base the revascularization strategy (IRA PCI, multivessel PCI/CABG) on the patient's clinical status and comorbidities, as well as their disease complexity, according to the principles of management of myocardial revascularization.[480,481]	I	B
Multivessel disease in ACS patients presenting in cardiogenic shock		
IRA-only PCI during the index procedure is recommended.[404,505]	I	B
Staged PCI of non-IRA should be considered.[c]	IIa	C
Multivessel disease in haemodynamically stable STEMI patients undergoing PPCI		
Complete revascularization is recommended either during the index PCI procedure or within 45 days.[508–511,531]	I	A
It is recommended that PCI of the non-IRA is based on angiographic severity.[511,524]	I	B
Invasive epicardial functional assessment of non-culprit segments of the IRA is not recommended during the index procedure.	III	C
Multivessel disease in haemodynamically stable NSTE-ACS patients undergoing PCI		
In patients presenting with NSTE-ACS and MVD, complete revascularization should be considered, preferably during the index procedure.[513,514]	IIa	C
Functional invasive evaluation of non-IRA severity during the index procedure may be considered.[518,527,528,532]	IIb	B

ACS, acute coronary syndrome; CABG, coronary artery bypass grafting; IRA, infarct-related artery; MVD, multivessel disease; NSTE-ACS, non-ST-elevation acute coronary syndrome; PCI, percutaneous coronary intervention; PPCI, primary percutaneous coronary intervention; STEMI, ST-elevation myocardial infarction.
[a]Class of recommendation.
[b]Level of evidence.
[c]Based on ischaemia, symptoms, patient comorbidities, and clinical condition.

11. Myocardial infarction with non-obstructive coronary arteries

Myocardial infarction with non-obstructive coronary arteries (MINOCA) refers to the clinical situation when a patient presents with symptoms suggestive of ACS, demonstrates troponin elevation, and has non-obstructive coronary arteries at the time of coronary angiography (defined as coronary artery stenosis <50% in any major epicardial vessel). The reported prevalence of MINOCA varies widely across studies (from around 1% to 14% of patients with ACS undergoing angiography).[533] MINOCA can be considered as an umbrella term that encompasses a heterogeneous group of underlying causes. This includes both coronary and non-coronary pathologies, with the latter including both cardiac and extra-cardiac disorders (*Figure 15*).[4,18,534–537]

© ESC 2023

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

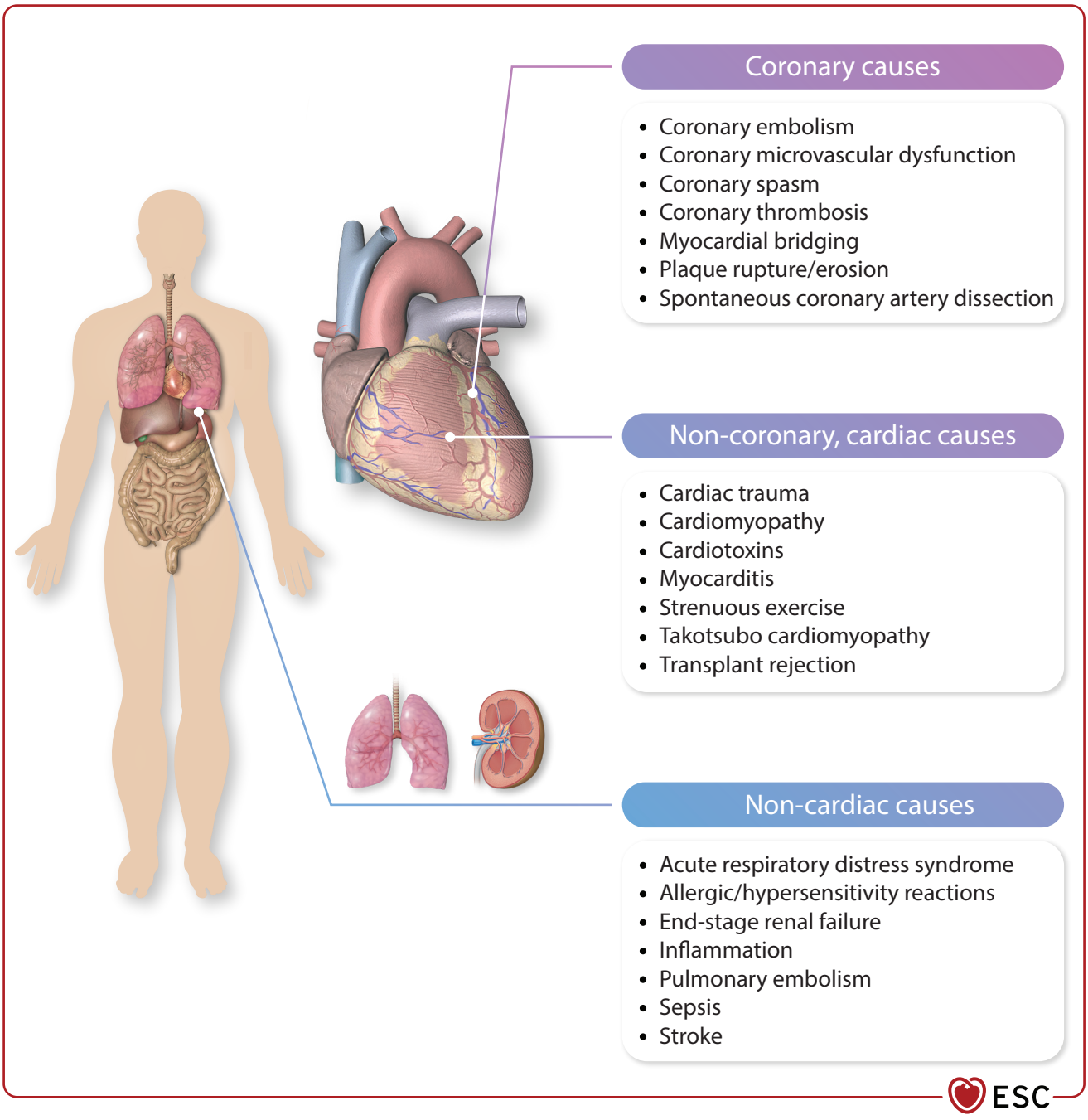


Figure 15 Underlying causes for patients with a working diagnosis of myocardial infarction with non-obstructive coronary arteries. This figure outlines some of the potential differential diagnoses in patients with a working diagnosis of MINOCA after coronary angiography, but this list is not exhaustive.

When a diagnosis is not established following coronary angiography, MINOCA represents a working diagnosis as opposed to a final diagnosis. It is vital for clinicians to perform further assessments and investigations to establish the underlying cause of the MINOCA, which will allow a final diagnosis to be established and patients to be managed appropriately. Failure to identify the underlying cause of MINOCA may result in inadequate or inappropriate therapy.

ICA is the recommended definitive diagnostic test for ACS patients. If the underlying cause of MINOCA is not established using ICA alone, further evaluation using left ventriculography (including measurement of LV end-diastolic pressure), functional assessment with measurement of microvascular function/coronary reactivity, and intravascular imaging can be useful to identify the underlying cause.[456,538,539] The term 'functional coronary angiography' refers to the combination of coronary angiography with adjunctive tests (e.g. testing for coronary microvascular dysfunction and vasoreactivity) (*Figure 16*).

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

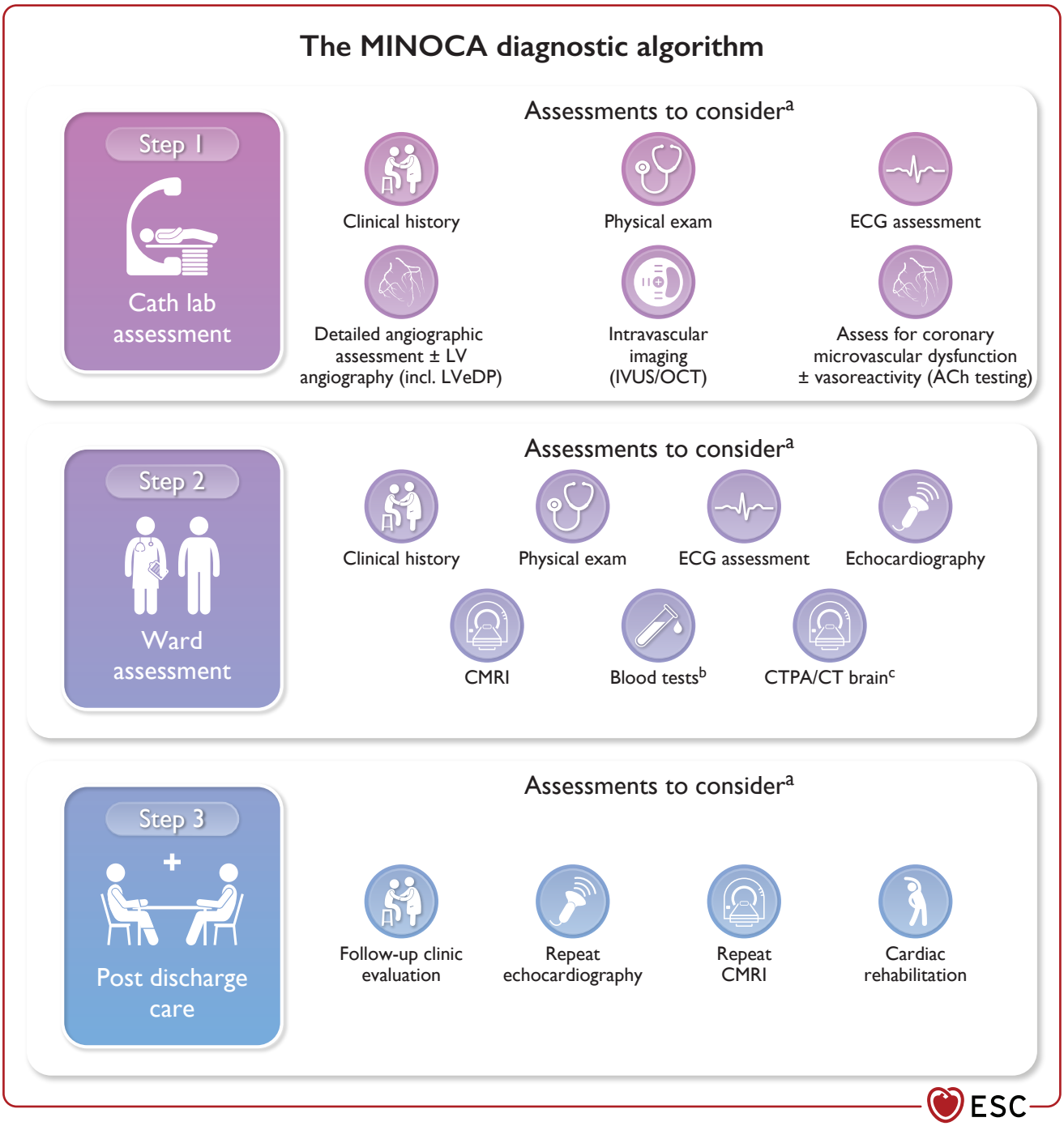


Figure 16 Evaluation of patients with a working diagnosis of MINOCA. ACh, acetylcholine; CMRI, cardiac magnetic resonance imaging; CT, computed tomography; CTPA, computed tomography pulmonary angiogram; ECG, electrocardiogram; IVUS, intravascular ultrasound; LV, left ventricular; LVeDP, left ventricular end-diastolic pressure; MINOCA, myocardial infarction with non-obstructive coronary arteries; NSTE-ACS, Non-ST elevation acute coronary syndrome; NTpro BNP, N terminal pro brain natriuretic peptide; OCT, optical coherence tomography; STEMI, ST-elevation myocardial infarction; UA, unstable angina. Patients presenting with STEMI present directly to catheter lab as per the current standard of care pathway **(1)**. In this context, when non-obstructive coronary arteries are identified then further assessment should be considered. When patients are subsequently admitted to the ward then investigations as shown in **(2)** should be considered. Patients presenting with NSTE-ACS or UA are often stabilized on the ward **(2)** prior to transfer to the cath lab **(1)**. In this context the order in which the investigations are carried out will vary depending on the location these patients are managed during first contact. MINOCA patients require follow-up review **(3)** and may require repeat assessment using echocardiography and magnetic resonance imaging, depending on the initial findings. [a]Options for adjunctive tests. Patients will not require all investigations but instead the appropriate tests should be selected based on their presentation and clinical course. [b]Examples of potential blood tests include: full blood count, renal profile, troponin, C-reactive protein, D-dimer, NT-pro BNP. [c]A CT scan of the brain should be considered if a cranial pathology (i.e. intracranial bleed) is suspected that might have resulted in ST elevation.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

If the underlying cause of MINOCA is not established using functional coronary angiography, then non-invasive imaging (i.e. echocardiography, CMR, CT) is recommended, as clinically appropriate. CMR is one of the key diagnostic tools to determine the underlying cause of MINOCA.[540–544] CMR can identify the underlying cause in up to 87% of patients with a working diagnosis of MINOCA and should be performed as soon as possible after presentation in these patients to maximize its diagnostic yield, ideally during the index admission.[545]

Diagnosis of the underlying cause of MINOCA will enable the appropriate treatment to be initiated based on the final diagnosis. Secondary prevention therapies should be considered for those with evidence of coronary atherosclerotic disease and to control risk factors. The management of takotsubo syndrome is not informed by any prospective RCTs, and treatment is largely supportive and empiric.[546,547] The treatment of patients with myocarditis has been covered by previous ESC documents.[548,549] Ischemia with non-obstructive coronary arteries (INOCA) has also been described in the context of CCS.[550,551] Additional information about MINOCA is provided in the Supplementary data online, including *Table S13*.

Recommendation Table 13 — Recommendations for myocardial infarction with non-obstructive coronary arteries

Recommendations	Class[a]	Level[b]
In patients with a working diagnosis of MINOCA, CMR imaging is recommended after invasive angiography if the final diagnosis is not clear.[544,545]	I	B
Management of MINOCA according to the final established underlying diagnosis is recommended, consistent with the appropriate disease-specific guidelines.[546,550,552]	I	B
In all patients with an initial working diagnosis of MINOCA, it is recommended to follow a diagnostic algorithm to determine the underlying final diagnosis.	I	C

© ESC 2023

CMR, cardiac magnetic resonance; MINOCA, myocardial infarction with non-obstructive coronary arteries.
[a]Class of recommendation.
[b]Level of evidence.

12. Special situations

12.1. Type 2 myocardial infarction and acute myocardial injury

Pathological processes other than atherothrombosis commonly underlie the presentation of patients with acute chest pain with troponin elevation. These include Type 2 MI and myocardial injury as defined in the fourth universal definition of MI.[1] Type 2 MI is an ischaemic myocardial injury in the context of a mismatch between oxygen supply and demand that is not related to acute coronary atherothrombosis. This may occur in the context of atherosclerosis and an oxygen supply/demand imbalance, with an oxygen supply/demand imbalance alone, secondary to vasospasm or coronary microvascular dysfunction, or secondary to non-atherosclerotic coronary dissection. These causes of Type 2 MI can be divided into those with underlying coronary (e.g. coronary embolus, dissection, spasm, microvascular dysfunction) or non-coronary mechanisms (supply demand mismatch due to hypoxia, hypotension, anaemia, tachycardia, bradycardia).[1] Type 2 MI is common and associated with a prognosis similar to Type 1 MI.[12]

Myocardial injury is characterized by myocyte necrosis and troponin elevation due to mechanisms other than myocardial ischaemia and can be acute (e.g. sepsis, myocarditis, takotsubo) or chronic (e.g. HF, cardiomyopathies, severe valve heart disease). Myocardial injury is increasingly appreciated in the era of hs-cTn assays, which are not specific for MI. In patients who have elevated hs-cTn values and do not have evidence of acute myocardial ischaemia, a diagnosis of myocardial injury can be made. It is important to recognize that this diagnosis can change if subsequent investigations indicate that the patient meets the criteria for MI.

Despite some common risk factors, the pathophysiology of Type 2 MI is different to that of Type 1 MI. Therefore, the natural history and appropriate management strategy of these two conditions also differs in some important respects. Type 2 and Type 1 MI require diagnostic distinction, which is best achieved by following an algorithmic approach.[1,553] Once patients with suspected Type 2 MI and myocardial injury have been stabilized and any precipitating illnesses have been treated, targeted echocardiography and/or coronary angiography (invasive or CCTA) can be used to identify contributory (and prognostically important) cardiac conditions and to guide appropriate long-term cardiovascular treatments.[12] Due to the lack of robust scientific evidence investigating management strategies and the wide range of precipitating causes, there are currently no specific recommended pharmacological interventions for patients with Type 2 MI. Therefore, management should instead focus on identifying and treating any precipitating conditions (e.g. anaemia, hypoxia) alongside strict control of CV risk factors.

12.2. Complications

12.2.1. Heart failure

Acute HF may occur as a complication of ACS. Acute HF as a result of ACS significantly increases the risk of other in-hospital complications, including worsening of renal function, respiratory failure, pneumonia, and death. *De novo* acute HF complicating ACS should be distinguished from pre-existing HF exacerbated by ACS.[554–556] This can be challenging and the presence of acute HF may impede the straightforward diagnosis of ACS. Patients with ACS and acute HF are more likely to present with resting dyspnoea and clinical signs/symptoms of fluid overload. In some clinical scenarios, increased troponin levels in patients with acute HF may reflect myocardial injury due to HF rather than myocardial necrosis due to ischaemia.

Patients with ACS complicated by acute HF require urgent and coordinated management of both conditions. The management of acute HF should follow current recommendations included in the ESC Guidelines on HF and ancillary documents.[557–559] The use of diuretics, vasodilators, inotropic agents, and vasopressors should be considered according to the established algorithms. Mechanical circulatory support may also be considered in selected cases. Invasive respiratory support and/or renal replacement therapy may be required in some circumstances.[557–559]

Patients presenting with acute HF (including patients with CS) complicating ACS require immediate ICA.[250,394,396] These patients should also undergo emergency echocardiography/chest ultrasonography to gather information about LV and RV function, regional wall motion abnormalities, valvular function, and possible mechanical complications.[250,557,560] In patients with ACS, CS may occur as a result of

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

extensive ischaemia due to MVD, acute severe mitral regurgitation, and mechanical complications. Patients with ACS and CS should be transferred as soon as possible to a PCI centre where immediate coronary angiography, and PCI of the IRA if needed, can be performed.[404,505] In patients with CS complicating ACS in whom the coronary anatomy is not suitable for PCI, emergency CABG is recommended. Management of MVD in this context is detailed in *Section 10*.

The clinical benefit of percutaneous MCS devices and/or VA-ECMO in the context of ACS remains unclear.[402,561] Micro-axial MCS devices have not been associated with lower 30-day mortality in comparison to IABP in observational studies.[400] In a large retrospective registry of 48 306 patients (>80% ACS) undergoing PCI with MCS, micro-axial MCS support was associated with higher mortality and bleeding rates in comparison to IABP.[562] Similar results were observed in another propensity-matched registry analysis confined to patients with CS, where micro-axial MCS support was also associated with more complications and higher mortality than IABP.[563] In the IABP-SHOCK II trial, the routine use of IABP in patients with ACS and CS did not reduce 30-day, 1-year, or 6-year mortality.[399,405,407] Based on these data, a benefit of LVAD in patients with ACS has not been demonstrated, and given that observational data have suggested that this may be associated with harm, caution is advised in this regard until further RCT evidence is available.

12.2.2. Mechanical complications

Mechanical complications may occur in the first days following MI, most commonly in patients presenting with STEMI. The incidence of mechanical complications has fallen significantly in the era of PPCI.[564] A recent large epidemiological investigation including almost 9 million ACS patients reported an overall prevalence of mechanical complications in 0.27% of STEMI cases and 0.06% of NSTEMI cases, with in-hospital mortality rates of 42.4% and 18%, respectively.[564] Mechanical complications are life-threatening and therefore require prompt identification and management (Supplementary data online, *Table S14*). Sudden hypotension, the recurrence of chest pain, new cardiac murmurs suggestive of acute mitral regurgitation or a ventricular septal defect, pulmonary congestion, or jugular vein distension should raise suspicion of a mechanical complication. Immediate echocardiographic assessment is indicated when mechanical complications are suspected.

The use of temporary MCS for mechanical complications, either to improve pre-operative clinical/haemodynamic status or prophylactically, represents a new trend in management. However, this approach requires more data and evidence in order to determine if it provides a clinical benefit.[565–568] Surgery is currently regarded as the treatment of choice for patients with ACS and mechanical complications, although percutaneous strategies are occasionally used in selected candidates with a prohibitive risk profile or contraindications to a surgical approach.[569–572] A multidisciplinary approach to the management of these patients is of paramount importance, and should apply to all stages of care, from the initial stabilization of the patient to discussion and application of the therapeutic strategy, including palliative care.[573,574] Patients with ACS-related mechanical complications should be considered for IABP while awaiting surgery.

12.2.3. Left ventricular thrombus

While the incidence of LV thrombus following AMI has declined due to advances in reperfusion and antithrombotic therapies, it remains relatively common, particularly following anterior STEMI, where it can be present in >9% of patients according to a large meta-analysis.[575,576]

Echocardiography remains the first-line imaging test for the detection of LV thrombus. In patients where the apex is not well visualized on regular echocardiography, contrast echocardiography may be considered for improved image quality. CMR is the gold standard imaging modality for the diagnosis and assessment of LV thrombi. Contemporary CMR data report LV thrombi in up to 6.3% of all STEMI patients and in 12.2% of those with anterior STEMI, suggesting that the incidence of LV thrombi may be underestimated with echocardiography.[577] Patients with LV thrombi that were not evident on echocardiography but were detected by CMR appear to have similar clinical outcomes to patients with LV thrombi that were evident on echocardiography.[578] Therefore, CMR should be considered in patients with equivocal echocardiographic images or in patients considered to be at a particularly high risk of LV thrombus.

The timing of imaging for LV thrombus may also be relevant, given that the identification of LV thrombus has been reported to increase in the first 2 weeks post-MI.[579] While more contemporary data are required, these data suggest that a high proportion of LV thrombi may develop following hospital discharge, indicating that delayed imaging at 2 weeks in high-risk patients may be of value.

Once an LV thrombus has been diagnosed, OAC therapy (warfarin or NOAC) should be considered for 3–6 months, guided by repeated echocardiography or CMR and with consideration of bleeding risk and the need for concomitant antiplatelet therapy.[580,581] However, there are a lack of prospective randomized data on the optimal anticoagulation regimen, anticoagulation duration, and the combination of oral anticoagulation with antiplatelet agents in patients with LV thrombus following MI.[581] The choice of therapy should be tailored to the patient's clinical status and the results of follow-up investigations.

12.2.4. Post-acute coronary syndrome pericarditis

Pericardial complications that may develop after an AMI include early infarct-associated pericarditis (occurring from a few hours to 4 days after AMI, mostly transient), late pericarditis or post-cardiac injury (Dressler) syndrome (typically occurring 1–2 weeks after AMI), and pericardial effusion.[548,582] This topic is discussed further in the Supplementary data online.

12.2.5. Arrhythmias

12.2.5.1. Atrial fibrillation

Atrial fibrillation is the most frequent supraventricular arrhythmia in patients with ACS.[583] AF may be pre-existing, first time detected, or of new onset during ACS management. Patients with AF have a greater number of comorbidities compared with patients without AF and are at higher risk of complications.[584] In most cases, AF is well tolerated and no specific treatment is required, apart from anticoagulation.[585] Prompt treatment is required for AF causing acute haemodynamic instability, with electrical cardioversion being the preferred approach. Adequate rate control can be achieved by administration of beta-blockers depending on the presence of HF and low ejection fraction. For patients with depressed LVEF, amiodarone or digoxin could be used (preferably amiodarone). In cases of hypotension, digoxin is preferred over amiodarone or beta-blockers. Patients with AF and risk factors for thrombo-embolism should be adequately treated with chronic oral anticoagulation.[585] ACS patients with documented AF of any

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

length have worse short- and long-term prognoses when compared with patients in sinus rhythm.[584,586] There is some evidence to suggest that transient, self-terminating AF during STEMI may be a predictor of an increased risk of stroke during long-term follow-up.[584,587]

12.2.5.2. Ventricular arrhythmias

With the widespread increased uptake of emergency reperfusion therapies for patients with STEMI, the incidence of malignant arrhythmias (ventricular tachycardia [VT] and ventricular fibrillation [VF]) has significantly declined. Nevertheless, 6–8% of patients with STEMI develop haemodynamically significant VT or VF.[588] The typical arrhythmia presentation is unstable, frequently polymorphic, and relatively fast VT, often degenerating into VF. Urgent reperfusion is most important as ischaemia is often the trigger for these arrhythmias. Early administration of i.v. or oral beta-blockers reduces the incidence of malignant arrhythmias.[163,164,169,589] Beta-blockers or amiodarone are recommended if malignant arrhythmias occur and lidocaine may be considered if these are contraindicated.[163,164,169,589,590] The prognostic role of early VT/VF within the first 48 h of STEMI is still controversial. Several studies have suggested that patients with early VT/VF have increased 30-day mortality but no increase in long-term arrhythmic risk.[591–593] Another study has suggested that while malignant ventricular arrhythmias occurring at the time of reperfusion do not confer poor prognosis, sustained VT or VF occurring during ongoing ischaemia or late after reperfusion (>48 h) is associated with an increase in long-term mortality.[594] Sustained VT/VF late after reperfusion (>48 h) requires an evaluation for ICD implantation for secondary prevention of sudden cardiac death. Ventricular premature beats are very frequent during the first 24 h after reperfusion for STEMI and no specific therapy is required.

Primary prevention of sudden cardiac death with ICD implantation within 40 days after MI is generally not indicated. Patients should be re-evaluated for ICD implantation post-revascularization after a period of 6–12 weeks on evidence-based treatments, although patients with a pre-existing impaired LVEF may be considered for ICD implantation for primary prevention even within the early post-infarction period. Some patients may develop electrical storm and/or incessant VT despite complete revascularization and treatment with anti-arrhythmic drugs. Overdrive stimulation may help to control this situation; however, recurrence of VT/VF upon cessation of stimulation is frequent and catheter ablation of such triggers appears to be the preferred treatment option in centres with that expertise. Successful radiofrequency ablation has been shown to abolish recurrent VT/VF.[595]

Non-sustained monomorphic VT is the most common form of ventricular arrhythmia in the early phase of ACS, and usually does not require anti-arrhythmic treatment. Accelerated idioventricular rhythm at reperfusion is frequent and does not require intervention given its benign nature.[596]

12.2.6. Bleeding

Bleeding is associated with a poor prognosis in ACS patients.[231,597,598] The mechanisms by which bleeding increases the risk of death are complex and multifactorial.[599] While intracranial or massive haemorrhage directly threatens life through fatal brain damage or sudden cardiocirculatory collapse, other less severe forms of haemorrhage may increase the risk of death through indirect mechanisms. Blood transfusion may increase systemic inflammation and represents one of the possible links between bleeding and subsequent mortality.[600] Bleeding is also a major driver of unplanned DAPT discontinuation and the interruption of other medication (e.g. statins, beta-blockers).[601,602]

12.2.6.1. Management of bleeding

See Supplementary data online, *Section 12.1.3.1*.

Recommendation Table 14 — Recommendations for acute coronary syndrome complications

Recommendations	Class[a]	Level[b]
Heart failure		
IABP should be considered in patients with haemodynamic instability/cardiogenic shock due to ACS-related mechanical complications.	IIa	C
LV thrombus		
CMR imaging should be considered in patients with equivocal echocardiographic images or in cases of high clinical suspicion of LV thrombus.[577,578]	IIa	C
Oral anticoagulant therapy (VKA or NOAC) should be considered for 3–6 months in patients with confirmed LV thrombus.[603]	IIa	C
Following an acute anterior MI, a contrast echocardiogram may be considered for the detection of LV thrombus if the apex is not well visualized on echocardiography.[604]	IIb	C
Atrial fibrillation		
Intravenous beta-blockers are recommended when rate control is needed in the absence of acute HF or hypotension.[605]	I	C
Intravenous amiodarone is recommended when rate control is needed in the presence of acute HF and no hypotension.[606]	I	C
Immediate electrical cardioversion is recommended in patients with ACS and haemodynamic instability and when adequate rate control cannot be achieved promptly with pharmacological agents.	I	C
Intravenous amiodarone is recommended to facilitate electrical cardioversion and/or decrease risk for early recurrence of AF after electrical cardioversion in unstable patients with recent-onset AF.[607,608]	I	C
In patients with documented *de novo* AF during the acute phase of ACS, long-term oral anticoagulation should be considered depending on the CHA_2DS_2-VASc score, after taking the HAS-BLED score and the need for concomitant antiplatelet therapy into consideration. NOACs are the preferred drugs.[583,584,587]	IIa	C
Ventricular arrythmias		
ICD therapy is recommended to reduce sudden cardiac death in patients with symptomatic HF (NYHA Class II–III) and LVEF ≤35% despite optimal medical therapy for >3 months and at least 6 weeks after MI who are expected to survive for at least 1 year with good functional status.[434,609,610]	I	A

Continued

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Intravenous beta-blocker and/or amiodarone treatment is recommended for patients with polymorphic VT and/or VF unless contraindicated.[611–614]	I	B
Prompt and complete revascularization is recommended to treat myocardial ischaemia that may be present in patients with recurrent VT and/or VF.[368,388]	I	C
Transvenous catheter pacing termination and/or overdrive pacing should be considered if VT cannot be controlled by repeated electrical cardioversion.	IIa	C
Radiofrequency catheter ablation at a specialized ablation centre followed by ICD implantation should be considered in patients with recurrent VT, VF, or electrical storm despite complete revascularization and optimal medical therapy.	IIa	C
Treatment of recurrent VT with haemodynamic relevance (despite repeated electrical cardioversion) with lidocaine may be considered if beta-blockers, amiodarone, and overdrive stimulation are not effective/applicable.[615]	IIb	C
In patients with recurrent life-threatening ventricular arrhythmias, sedation or general anaesthesia to reduce sympathetic drive may be considered.[616]	IIb	C
ICD implantation or the temporary use of a wearable cardioverter defibrillator may be considered <40 days after MI in selected patients (incomplete revascularization, pre-existing LVEF dysfunction, occurrence of arrhythmias >48 h after STEMI onset, polymorphic VT or VF).	IIb	C
Treatment of asymptomatic and haemodynamically irrelevant ventricular arrhythmias with anti-arrhythmic drugs is not recommended.	III	C
Bradyarrhythmias		
In cases of sinus bradycardia with haemodynamic intolerance or high-degree AV block without stable escape rhythm:		
• i.v. positive chronotropic medication (adrenaline, vasopressin, and/or atropine) is recommended.[617,618]	I	C
• temporary pacing is recommended in cases of failure to respond to atropine.	I	C
• urgent angiography with a view to revascularization is recommended if the patient has not received previous reperfusion therapy.	I	C
Implantation of a permanent pacemaker is recommended when high-degree AV block does not resolve within a waiting period of at least 5 days after MI.	I	C
In selected patients with high-degree AV block in the context of an anterior wall MI and acute HF, early device implantation (CRT-D/CRT-P) may be considered.[619,620]	IIb	C
Pacing is not recommended if high-degree AV block resolves after revascularization or spontaneously.[620–622]	III	B

Continued

© ESC 2023

ACS, acute coronary syndrome; AF, atrial fibrillation; AV, atrioventricular; CHA_2DS_2-VASc, Congestive heart failure, Hypertension, Age ≥75 (doubled), Diabetes, previous Stroke/transient ischaemic attack/thrombo-embolism (doubled), Vascular disease, Age: 65–74, Sex (female); CMR, cardiac magnetic resonance; CRT-D/CRT-P, cardiac resynchronization therapy—defibrillator/pacemaker; HAS-BLED, Hypertension, Abnormal liver/renal function, Stroke history, Bleeding history or predisposition, Labile INR, Elderly, Drug/alcohol usage; HF, heart failure; IABP, intra-aortic balloon pump; ICD, implantable cardioverter defibrillator; LV, left ventricular; LVEF, left ventricular ejection fraction; MI, myocardial infarction; NOAC, non-vitamin K antagonist oral anticoagulant; NYHA, New York Heart Association; STEMI, ST-elevation myocardial infarction; VF, ventricular fibrillation; VKA, vitamin K antagonist; VT, ventricular tachycardia.
[a]Class of recommendation.
[b]Level of evidence.

12.3. Comorbid conditions

12.3.1. Patients at high bleeding risk and with blood disorders (anaemia and thrombocytopaenia)

Anaemia is more prevalent in elderly/frail ACS patients and in patients with multimorbidity (i.e. HF, chronic kidney disease [CKD], diabetes mellitus, cancer, and autoimmune diseases). In some cases, severe anaemia may precipitate Type 2 MI. Persistent or worsening anaemia in patients with ACS is associated with an increased risk of recurrent ischaemic events, death, and major bleeding.[623–625] According to the ARC-HBR, haemoglobin <11 g/dL at the time of PCI constitutes a major criterion for HBR, whereas haemoglobin between 11 and 13 g/dL (12 g/dL for women) is a minor criterion.

There is no established strategy for treating anaemia in patients with ACS. The efficacy and safety of blood transfusion in this clinical scenario remains unknown. In the majority of studies investigating different transfusion protocols, a liberal blood transfusion strategy has been defined as any red blood cell transfusion at a haemoglobin level <9–10 g/dL, while a restrictive blood transfusion strategy has been defined as any transfusion at a haemoglobin level <7–8 g/dL. Observational data suggest that a liberal blood transfusion strategy may be associated with an increase in all-cause mortality.[626–630] The open-label Restrictive and Liberal Transfusion Strategies in Patients With Acute Myocardial Infarction (REALITY) trial enrolled 668 ACS patients who were randomized to management with a restrictive (triggered by haemoglobin ≤8) or a liberal (triggered by haemoglobin ≤10) transfusion strategy.[631] The composite outcome (all-cause death, stroke, recurrent MI, or emergency revascularization) at 30 days occurred in a comparable number of patients in both arms (11% vs. 14%, RR 0.79, with a one-sided 97.5% CI of 0.00–1.19), meeting the pre-specified non-inferiority criterion. All components of the composite endpoint were numerically higher in the liberal transfusion strategy arm. The trial was not powered to detect superiority of the restrictive strategy, and the CI included what may be a clinically important harm. The pre-specified 1-year follow-up of the REALITY trial yielded contradictory conclusions to the 30-day outcomes: at 1 year, the restrictive transfusion strategy (vs. a liberal

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

approach) did not achieve non-inferiority in terms of MACE. In addition, a *post-hoc* analysis of MACE between day 30 and 1 year demonstrated an increased risk in the restrictive transfusion strategy group.[632] Therefore, no formal recommendation as to the optimal transfusion strategy (liberal vs. restrictive) in patients with ACS can be made at present.

Although there are several classifications to grade the severity of thrombocytopaenia, clinically relevant thrombocytopaenia can be defined as a platelet count <100 000/μL or a relative drop in platelet count of 50% from baseline in the context of ACS. Thrombocytopaenia increases the risk of death, major bleeding events, and life-threatening thrombotic events.[633,634] The ARC-HBR criteria define a platelet count <100 000/μL as a major criterion for HBR. Management of GP IIb/IIIa inhibitor- and heparin-induced thrombocytopaenia is discussed in the Supplementary data online.

12.3.2. Chronic kidney disease

Moderate to severe CKD (stages III–V) is present in more than 30% of ACS patients.[635] Patients with ACS and concomitant CKD receive less interventional and pharmacological treatment and have a worse prognosis than patients with normal kidney function.[636–638] Likely contributing factors to this worse prognosis include a larger number of comorbidities and an increased risk of in-hospital complications, including serious bleeding complications.[639] Although evidence from RCTs is lacking, data from observational and registry-based studies indicate that ACS patients with moderate to severe CKD have a better prognosis with early revascularization than with medical therapy alone.[640,641]

The type and dose of antithrombotic agent (see Supplementary data online, *Table S15*) and the amount of contrast agent should be considered based on kidney function.[635,642] In relation to supplementary i.v. hydration during and after revascularization, the evidence around choice, timing, and duration of treatment is somewhat conflicting.[643] Taking the clinical circumstances and patient characteristics into consideration, i.v. hydration should be considered as part of the management of ACS patients with a low eGFR undergoing invasive management to minimize the risk of contrast-induced nephropathy.[250,635,642,644,645] For recommendations on long-term treatment in patients with ACS and concomitant CKD, please refer to the 2021 ESC Guidelines on cardiovascular disease prevention.[646]

12.3.3. Diabetes mellitus

ACS patients with diabetes mellitus (DM) may more commonly present with non-specific symptoms, which can lead to delays in both diagnosis and access to treatment.[647,648] Both treatment in the acute phase and risk factor management post-ACS is poorer in patients with DM and these patients tend to have more advanced CAD at diagnosis. These factors likely contribute to the worse long-term prognosis associated with ACS in patients with DM, particularly in patients requiring insulin treatment.[649–651]

All patients with ACS, regardless of a history of DM, should have their glycaemic status evaluated during hospitalization. Given that the ACS itself may give rise to hyperglycaemia due to catecholamine-induced stress, a diagnosis of DM made during hospitalization should be subsequently confirmed. While several studies have shown the benefits of managing hyperglycaemia (>11.0 mmol/L or 200 mg/dL) in hospitalized ACS patients, the risk of hypoglycaemia-related events when using intensive insulin therapy should not be neglected.[652–654]

Glucose lowering is important in order to prevent microvascular complications in patients with DM. However, recent trial evidence has shown that the reduction in the risk of new ACS events, HF, and renal impairment with glucose-lowering medications like sodium–glucose co-transporter 2 (SGLT2) inhibitors or glucagon-like peptide-1 receptor agonists (GLP-1-RA) is independent of baseline glycosylated haemoglobin (HbA1c) levels.[655–657] This should be taken into consideration when choosing glucose-lowering therapy for patients with DM and concomitant CAD. For further details, please refer to the 2023 ESC Guidelines on diabetes and cardiovascular diseases and the 2021 ESC Guidelines on cardiovascular disease prevention.[646,658]

12.3.4. Older adults with frailty and multimorbidity

12.3.4.1. The older person

Older adults represent an increasing proportion of ACS patients. One of the major predictors of adverse outcomes following ACS is age, but patients aged ≥75 years are often excluded from or under-represented in clinical trials.[659,660] Older age is associated with frailty, multimorbidity, and a greater risk of both ischaemic and bleeding events in patients with ACS.[661] Hs-cTn assays have an excellent diagnostic performance in the older person, but the specificity of the test is lower than in younger patients, and elevated cTn levels are more commonly associated with conditions other than ACS in older patients.[662]

There are limited data on the optimal management of older adults with ACS.[663] A small RCT enrolling older patients (≥80 years) with NSTE-ACS reported the superiority of an invasive vs. a conservative strategy in the reduction of the composite of MI, need for urgent revascularization, stroke, and death. No treatment effect was shown for all-cause death and the benefit associated with the invasive strategy was diluted with increasing age.[664] In the absence of robust clinical trial evidence, decisions regarding how to manage older patients should be individualized based on patient characteristics (i.e. ischaemic and bleeding risks, estimated life expectancy, comorbidities, the need for non-cardiac surgery, quality of life, frailty, cognitive and functional impairment, patient values and preferences, and the estimated risks and benefits of an invasive strategy).

In the context of STEMI, PPCI has drastically improved outcomes for all ages. However, data are limited in the 'very old' cohort, with lack of formal assessment of frailty or comorbidity.[665] In the context of CS and cardiac arrest, age is an independent predictor of mortality following PCI.[666,667] In the absence of robust RCT data, PPCI should be considered for all patients with STEMI. When PPCI cannot be performed in a timely manner, fibrinolysis may be a reasonable strategy in these patients. For details regarding pharmacotherapy in older patients, please see the Supplementary data online.

12.3.4.2. Frailty and multimorbidity

Geriatric syndromes (i.e. frailty and multimorbidity) are associated with adverse outcomes in older patients with ACS.[668,669] Frailty is a syndrome characterized by reduced biological reserve, leading to a failure of homeostatic mechanisms following stressor events, including ACS. There is a lack of consensus on which frailty assessment tool is optimal in older patients with CV disease.[670,671]

Frail patients with NSTE-ACS less frequently receive ACS pharmacotherapies and invasive assessment, have more complex coronary disease, have longer durations of hospital stay, and are at higher risk of death.[672] Specifically, frail patients are reported to have a higher rate of a composite of all-cause mortality, MI, stroke, unplanned

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

revascularization, and major bleeding.[673] Frail older adults with NSTE-ACS have poor health-related quality of life (HRQoL) at baseline. Invasive management appears to be associated with modest improvements in HRQoL through to 1 year follow-up in these patients. This improvement in HRQoL is most marked in frail and pre-frail patients, who receive a proportionally larger benefit than robust patients.[674] In older adults with NSTE-ACS referred for coronary angiography, the presence of multimorbidity is associated with an increased risk of long-term adverse CV events, driven by a higher risk of all-cause mortality.[675] Undiagnosed cognitive impairment is also common in older patients with NSTE-ACS undergoing ICA, and these patients are more likely to experience MACE at 1 year.[676]

In the absence of robust RCT data to inform healthcare professionals about the management of frail patients presenting with ACS, it is recommended to adapt a holistic approach to individualize interventional and pharmacological treatments after careful evaluation of risks vs. benefits. To aid in decision-making, the routine assessment of frailty (e.g. Rockwood Frailty Score) and comorbidity (e.g. Charlson index) in ACS patients is recommended. Following risk stratification using frailty assessment and evaluation of the comorbidity burden, it may be reasonable to offer optimal medical therapy plus an invasive strategy to frail patients at high risk of future CV events and low risk of complications, and to offer optimal medical therapy alone to those who are deemed to be at low risk of future events with a high risk of developing procedural complications. For those patients for whom any form of treatment might be futile, then a palliative end-of-life care approach should be considered.

12.3.5. Pregnancy

Acute coronary syndrome diagnostic criteria are the same for pregnant and non-pregnant patients.[677] Pregnant women with STEMI should not be managed differently to non-pregnant women. Given the high mortality associated with STEMI in pregnancy, PPCI is the preferred reperfusion therapy.[678] The management plan for pregnant women with ACS should be determined by a multidisciplinary team consisting of cardiologists, obstetricians, anaesthesiologists, and neonatologists, and these patients should be treated in an intensive care unit that can provide maternal monitoring and obstetric care.[678,679] ACS treatment should not be delayed for delivery. Delivery should be ideally postponed for at least 2 weeks post-ACS as there is increased risk of maternal mortality during this time.[678] It has been demonstrated that SCAD is the most common cause of AMI in pregnancy, and this tends to occur mainly in the late pregnancy or early post-partum periods.[680,681] Further details are provided in the Supplementary data online.

12.3.6. Drug abuse

Acute coronary syndrome in the setting of drug abuse is covered in the Supplementary data online.

12.3.7. Patients with cancer

The four most common types of cancer in patients with ACS are prostate, breast, colon, and lung.[682] Patients with a history of cancer should be treated like all other ACS patients, but the management of ACS patients with active cancer has some specific issues that need to be taken into consideration. Outcomes vary across types of cancer and the balance between the ischaemic and bleeding risks should be considered on an individual basis.

The percentage of ACS patients with a current diagnosis of cancer is rising, and currently constitutes ~3% of patients in large observational studies.[683] Patients with active cancer presenting with ACS pose important challenges as there are significant gaps in scientific knowledge. Therefore, recommendations based on solid evidence are scarce. Patients with active cancer presenting with ACS tend to be older, with a larger number of comorbidities and more extensive CAD. These patients often have concomitant haematologic and coagulation abnormalities that may present a challenge with respect to both the use of antithrombotic therapy and the performing of PCI.[684] Observational studies have reported that ACS in patients with cancer is associated with increased risk of major CV events, bleeding, and cardiac and non-cardiac mortality.[682,683,685,686] As per the ARC-HBR criteria, patients with active cancer diagnosed in the past 12 months are considered as HBR.

The diagnosis of ACS in patients with cancer should be based on the same principles as in patients without cancer. The management of ACS in patients with cancer can be challenging because of frailty, increased bleeding risk, thrombocytopaenia, and increased thrombotic risk.[687] Temporary interruption of cancer treatment and an urgent multidisciplinary approach is recommended.[688] Cancer patients with ACS have been reported to less frequently undergo invasive management; however, invasive management (and PCI with DES if needed) is recommended in ACS patients with cancer, as long as the prognosis is >6 months or, irrespective of the prognosis, if the patient is unstable.[689] Retrospective data have reported both a lower use of invasive management in cancer patients with STEMI, and better outcomes in patients who do undergo invasive management.[682,686,689] Invasive management in patients with advanced cancer or life expectancy <6 months has been reported to not demonstrate a mortality benefit compared with a conservative approach and therefore a conservative strategy should be considered in these patients.[690] When the coronary anatomy is not amenable for PCI, CABG surgery can be considered after a multidisciplinary team discussion and where the cancer prognosis is >12 months. Given that they are considered to be HBR, the preferred $P2Y_{12}$ inhibitor for ACS patients with active cancer is clopidogrel.[687] Potential drug–drug interactions with cancer therapies should be checked when using ticagrelor or clopidogrel, since some pharmacokinetic-based drug–drug interactions via CYP450 may occur.

When acute ischaemia is provoked by cancer therapy, alternative cancer therapies should be considered after a multidisciplinary team discussion. Some specific cancer treatments can have cardiotoxic vascular effects that can lead to ACS (Supplementary data online, *Table S16*). Following ACS, a review of the cancer medications is recommended, and any cancer drug associated with thrombosis and MI should be stopped. Cancer therapies that are not associated with MI can be restarted once revascularization (when indicated) has been completed and the patient is stabilized on ACS medical therapy without complications. Additional information can be found in the Supplementary data online, including Supplementary data online, *Table S16* and in the 2022 ESC Guidelines on cardio-oncology.[684]

12.3.8. Coronavirus disease (COVID-19)

A section on the impact of Coronavirus disease (COVID-19) on ACS management is presented in the Supplementary data online.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Recommendation Table 15 — Recommendations for acute coronary syndrome comorbid conditions

Recommendations	Class[a]	Level[b]
Chronic kidney disease		
The use of low- or iso-osmolar contrast media (at the lowest possible volume) is recommended for invasive strategies.[691–693]	I	A
It is recommended to assess kidney function using eGFR in all patients with ACS.	I	C
It is recommended to apply the same diagnostic and therapeutic strategies in patients with CKD (dose adjustment may be necessary) as in patients with normal kidney function.	I	C
Hydration during and after angiography should be considered in patients at risk of contrast-induced nephropathy, especially in patients with acute kidney injury and/or CKD with eGFR <30 mL/min/1.73 m^2.[694–697]	IIa	B
Diabetes		
It is recommended to base the choice of long-term glucose-lowering treatment on the presence of comorbidities, including heart failure, CKD, and obesity.[698–704]	I	A
It is recommended to assess glycaemic status at initial evaluation in all patients with ACS.[705–707]	I	B
It is recommended to frequently monitor blood glucose levels in patients with known diabetes mellitus or hyperglycaemia (defined as glucose levels ≥11.1 mmol/L or ≥200 mg/dL).	I	C
Glucose-lowering therapy should be considered in patients with ACS with persistent hyperglycaemia, while episodes of hypoglycaemia should be avoided.[708,709]	IIa	C
Older adults		
It is recommended to apply the same diagnostic and treatment strategies in older patients as in younger patients.[662,664,665,710,711]	I	B
It is recommended to adapt the choice and dosage of antithrombotic agent, as well as of secondary prevention medications, to renal function, co-medications, comorbidities, frailty, cognitive function, and specific contraindications.[363,712]	I	B
For frail older patients with comorbidities, a holistic approach is recommended to individualize interventional and pharmacological treatments after careful evaluation of the risks and benefits.[668,673,676]	I	B
Patients with cancer		
An invasive strategy is recommended in cancer patients presenting with high-risk ACS with expected survival ≥6 months.[682,689,690]	I	B
A temporary interruption of cancer therapy is recommended in patients in whom the cancer therapy is suspected to be a contributing cause of ACS[c].[713,714]	I	C
A conservative non-invasive strategy should be considered in ACS patients with poor cancer prognosis[d] (i.e. with expected survival <6 months) and/or very high bleeding risk.[690]	IIa	C
Aspirin is not recommended in cancer patients with a platelet count <10 000/μL.[715]	III	C
Clopidogrel is not recommended in cancer patients with a platelet count <30 000/μL.	III	C
In ACS patients with cancer and <50 000/μL platelet count, prasugrel or ticagrelor are not recommended.	III	C

Continued

ACS, acute coronary syndrome; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate.
[a]Class of recommendation.
[b]Level of evidence.
[c]Anticancer therapies associated with high risk of ACS (very common [>10%]) include: capecitabine, paclitaxel, cisplatin, carfilzomib, bevacizumab, ramucirumab, aflibercept, axitinib, sorafenib, pazopanib, cabozantinib, lenvatinib, ponatinib, and erlotinib.
[d]Related to advanced cancer stage and/or severe irreversible non-CV comorbidities.

13. Long-term treatment

Secondary prevention after ACS is central to increase quality of life and to decrease morbidity and mortality. This should start as early as possible after the index event.[716–718] The topic is covered in detail in the 2019 CCS Guidelines and the 2021 Prevention Guidelines.[195,646] Optimal medical therapy and treatment targets are well defined and are summarized in *Figure 17*. A figure aimed at educating patients on improving their 'heart health' after an ACS event is provided in the Supplementary data online, *Figure S5*.

© ESC 2023

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

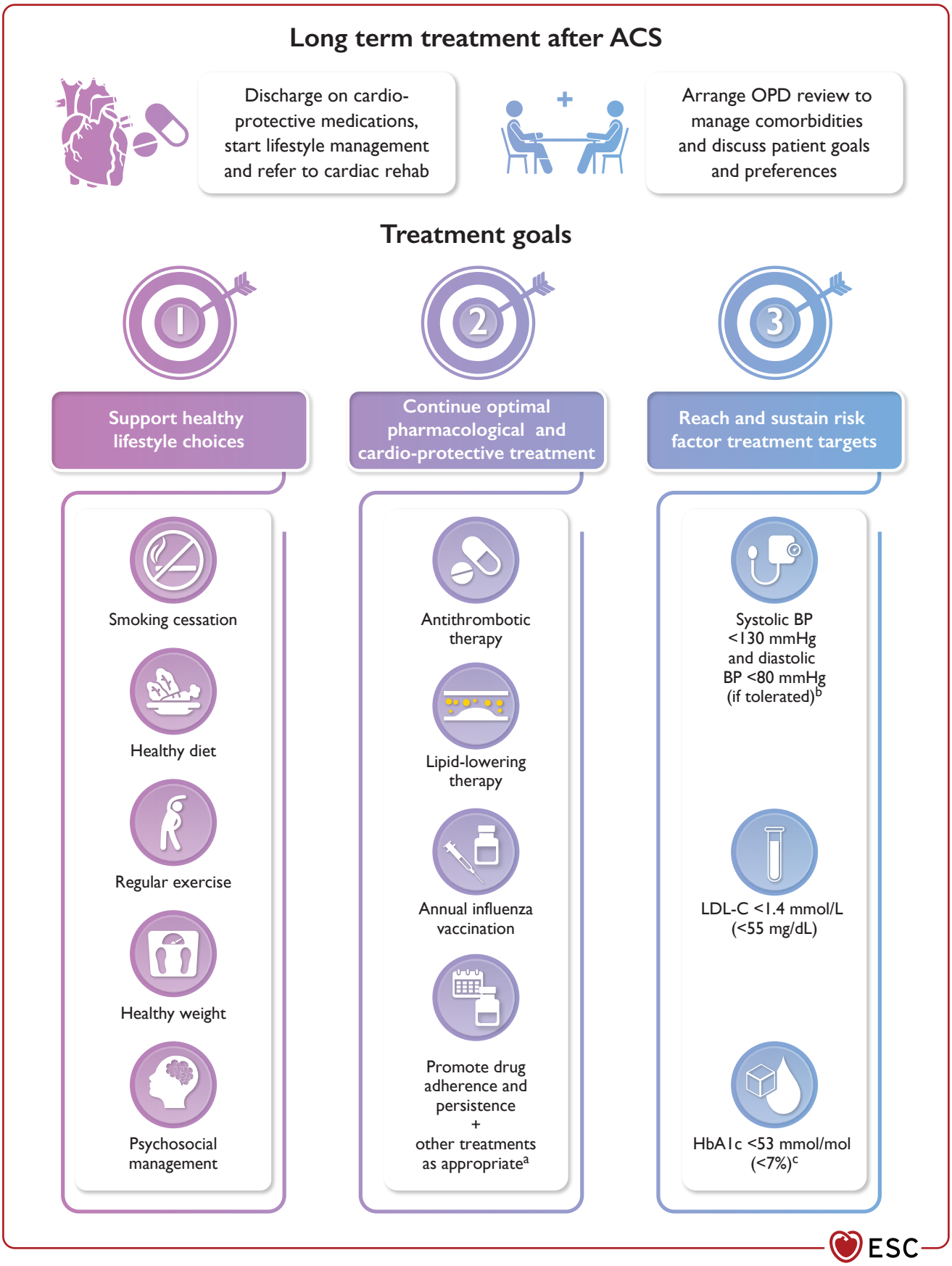


Figure 17 Long-term management after acute coronary syndrome. ACS, acute coronary syndrome; HbA1c, glycosylated haemoglobin; LDL-C, low-density lipoprotein cholesterol; OPD, outpatient department. [a]See Recommendation Table 16 for other pharmacological treatments after ACS. [b]For patients ≥70 years of age the systolic target should be <140 mmHg and down to 130 mmHg if tolerated.[646] [c]For patients with diabetes mellitus.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

13.1. Cardiac rehabilitation

13.1.1. Comprehensive cardiac rehabilitation

Secondary prevention is most effectively provided through cardiac rehabilitation (CR).[716,717] All ACS patients should participate in a comprehensive CR programme, which should start as early as possible after the ACS event.[716,717,719] CR may be performed in inpatient or outpatient settings, taking age, frailty, results of prognostic risk stratification, and comorbidities into account.[716] Comprehensive CR is a multidisciplinary intervention, supervised and performed by a team and usually co-ordinated by a cardiologist.[716] The core components of CR include patient assessment, management and control of CV risk factors, physical activity counselling, prescription of exercise training, dietary advice, tobacco counselling, patient education, psychosocial management, and vocational support.[716] Several studies have found that CR programmes after atherosclerotic cardiovascular disease (ASCVD) events or revascularization reduce CV hospitalizations, MI, CV mortality and, in some studies, all-cause mortality.[720–725] Despite proven benefits, the rates of referral to, participation in, and implementation of CR programmes are low.[726–730] Another identified issue is that many patients adopt healthier lifestyles during CR but relapse to pre-morbid habits when returning to everyday life.[731] Therefore, there is an unmet need for complementary pathways to the classical centre-based CR model. In addition to alternatives to CR, there is also a need for stronger endorsement of CR by physicians, cardiologists, and healthcare professionals.[732,733] It is also important to initiate and establish a strong partnership between patients and healthcare professionals as early as possible.[732–734]

13.1.2. Digital health

Telerehabilitation may be an effective strategy to maintain a healthy lifestyle over time and can support or even partially replace conventional, centre-based CR.[729] Telerehabilitation means rehabilitation from a distance, covering all CR core components, including telecoaching, social interaction, telemonitoring, and e-learning.[735,736] Studies in patients with CAD have shown that telerehabilitation can be equivalent to traditional CR in terms of achieving functional improvement, managing risk factors, and increasing patient well-being.[737–741] Few data are available about the effect of telerehabilitation on recurrent events.[742] Nevertheless, in a meta-analysis no significant difference was found between mortality following telehealth interventions and centre-based supervised CR.[743] Also, most trials have only focused on one of the CR core components—exercise training and/or physical activity.[742] Therefore, more research on the impact of telerehabilitation on outcomes is still needed, as are investigations into health and digital literacy in CR.

13.1.3. Adherence and persistence

Promotion of both adherence (the extent to which a patient adheres to a prescribed treatment or lifestyle advice) and persistence (the length of time between initiation and discontinuation of a prescribed treatment or lifestyle advice) are key in preventing recurrent CV events after ACS. Adherence to medication has been shown to be sub-optimal, ranging from 50% in primary prevention to 66% in secondary prevention. It is estimated that 9% of ASCVD events in Europe occur as a result of sub-optimal medication adherence.[646] Contributors to sub-optimal adherence and persistence are multidimensional and include: polypharmacy, drug regimen complexity, the doctor–patient relationship, a lack of patient-centred care and disease acceptance, concern regarding side effects, cognitive ability, mental and physical disorders, financial aspects, living alone, and depression.[646,744–749] Polypills, which include guideline-recommended treatments for secondary prevention, have been shown to increase adherence in post-ACS patients and may improve therapeutic targets.[750–752] The Secondary Prevention of Cardiovascular Disease in the Elderly (SECURE) study is the only RCT testing the impact of a strategy based on a polypill (containing aspirin, ramipril, and atorvastatin) vs. usual care on hard outcomes in ACS patients. The polypill strategy was associated with a significant reduction in major CV events, driven by a significant 33% reduction in CV mortality.[753] The use of technology to improve medication adherence is also generating interest: mobile phone applications and mobile health (mHealth) tools may improve medication adherence, but clinical trials of sufficient size and duration are needed.[754–756] Finally, it is important to recognize that adherence has complex underlying psychological drivers, and therefore a whole-systems approach is mandatory. This should include the education of health professionals, the use of patient-reported outcomes and experience measures, patient education, and patient-centred care.[734,757,758]

13.2. Lifestyle management

Lifestyle management is one of the cornerstones of comprehensive CR.[716] While most of the evidence regarding the benefits of a healthy lifestyle on prognosis comes from primary prevention, studies in secondary prevention settings indicate similar beneficial effects.[716,724,759–763]

13.2.1. Tobacco

Tobacco abstinence is associated with a reduced risk of re-infarction (30–40%) and death (35–45%) after ACS.[763–765] Measures to promote cessation of smoking are therefore a priority after ACS. Interventions for smoking cessation should begin during hospitalization using a combination of behavioural interventions, pharmacotherapy, and counselling.[18,766] Many patients continue or resume smoking after ACS, in particular patients with depression and environmental exposures.[646] During encounters with smokers, the 'very brief advice' evidence-based intervention should be used to facilitate dialogue between the patient and healthcare worker.[646] Drug interventions, including nicotine-replacement therapy (NRT), bupropion and varenicline, should be considered along with behavioural support. All forms of NRT are effective, and the anti-depressant bupropion aids in long-term smoking cessation with similar efficacy to NRT.[646,766] Varenicline is the most effective medical treatment to support smoking cessation and is safe to use in ACS patients.[767–770] An average weight gain of 5 kg can be expected when a person quits smoking, but it is important to recognize that the CV risk from continued smoking outweighs the CV risk from gaining weight.[646]

E-cigarettes have been used to help smokers quit, but evidence on their impact on successful smoking cessation is insufficient, particularly with regard to whether using e-cigarettes actually helps the person remain tobacco free. While e-cigarettes do contain nicotine, they do not contain as many tobacco chemicals as cigarettes. Caution should be given with respect to the use of e-cigarettes, as current evidence suggests they are harmful to CV health by increasing arterial stiffness, heart rate and blood pressure, and by causing endothelial dysfunction.[771]

13.2.2. Nutrition and alcohol

A healthy diet and eating habits influence CV risk. Adopting a Mediterranean-style diet can help reduce CV risk in all individuals, including persons at high CV risk and patients with ASCVD.[761,762,772]

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Supplementary data online, *Table S17* summarizes the characteristics of a healthy diet that should be adhered to. For further details on nutrition, please refer to the 2021 ESC Guidelines on cardiovascular disease prevention.[646]

With regard to alcohol consumption, recent data suggest that alcohol abstainers have the lowest risk of CVD outcomes, that any amount of alcohol uniformly increases blood pressure and body mass index, and that a weekly consumption of >100 g of alcohol is associated with decreased life expectancy.[773–775] Accordingly, it is recommended to restrict alcohol consumption to a maximum of 100 g per week (same limit for men and women).[646]

13.2.3. Physical activity and exercise

Based on extensive data from the general population, sedentary behaviour, defined as time spent sitting or lying with low energy expenditure, while awake, is an independent risk factor for all-cause mortality.[776,777] According to recommendations from the World Health Organization, adults with chronic conditions should limit their amount of sedentary time, replacing it with physical activity of any intensity (including light intensity).[646,778] General physical activity recommendations include a combination of regular aerobic physical activity and resistance exercise throughout the week, which also forms the basis of recommendations for patients post-ACS.[646,778] However, it is important to recognize that daily physical activity does not replace participation in exercise-based CR. With support from multiple randomized trials, exercise training is a pivotal part of comprehensive CR and participation in exercise-based CR should be offered to all patients after ACS.[779] Cardiorespiratory fitness is a strong predictor of future prognosis both in the general population and in post-ACS patients.[780]

13.2.4. Psychological considerations

There is a two-fold risk of anxiety and mood disorders in patients with heart disease. Depression, anxiety, and psychological stress are associated with worse outcomes. Psychological and pharmacological interventions can have a beneficial effect and should be considered for ACS patients with depression, anxiety, and stress.[781] It is recommended that all patients have their mental well-being assessed using validated tools before discharge, with consideration of onward psychological referral when appropriate.[782] For further details, please refer to the 2021 ESC Guidelines on cardiovascular disease prevention.[646]

13.2.5. Resumption of activities

Information on the resumption of activities, sexual activity, and environmental factors is presented in the Supplementary data online, *Section 13.1.2*.

13.3. Pharmacological treatment

13.3.1. Antithrombotic therapy

Recommendations for antithrombotic therapy are included in *Section 6*.

13.3.2. Lipid-lowering therapy

Dyslipidaemia should be managed according to the current dyslipidaemia guidelines, with a combination of lifestyle and pharmacological interventions.[783] Trials have consistently demonstrated that lower low-density lipoprotein-cholesterol (LDL-C) levels after ACS are associated with lower CV event rates.[784] The current treatment goal for secondary prevention is to lower LDL-C to <1.4 mmol/L (<55 mg/dL) and to achieve a ≥50% LDL-C reduction from baseline. For patients who experience a second CV event within 2 years (not necessarily of the same type as the first event), an LDL-C goal of <1.0 mmol/L (<40 mg/dL) appears to confer additional benefit.[783,785,786]

After an ACS event, lipid-lowering treatment should be initiated as early as possible, both for prognostic benefit and to increase patient adherence after discharge. It is recommended that a high-intensity statin (e.g. atorvastatin or rosuvastatin) is initiated as early as possible after hospital admission, preferably before planned PCI, and prescribed up to the highest tolerated dose in order to reach the LDL-C goals.[783,787] The intensity of statin therapy should be increased in patients who were receiving low- or moderate-intensity statin treatment before the ACS event. In IMPROVE-IT (Improved Reduction of Outcomes: Vytorin Efficacy International Trial), ezetimibe treatment early after ACS (within 10 days) was added on top of prior statin therapy or initiated concomitantly in statin-naïve patients (two-thirds of patients) and compared with statin monotherapy.[788] Treatment with ezetimibe was shown to be safe and provided long-term benefits for CV outcomes. As such, if patients are on a maximally tolerated statin dose, or have no prior statin treatment, and have LDL-C levels which indicate it is unlikely that targets will be reached with statin therapy alone, initiating ezetimibe in addition to a statin (or statin plus ezetimibe combination treatment) may be considered during the ACS hospitalization.[783,788] In the ODYSSEY OUTCOMES (Evaluation of Cardiovascular Outcomes After an Acute Coronary Syndrome During Treatment With Alirocumab) trial, treatment with the proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitor alirocumab was initiated as early as 1 month after ACS.[786] Treatment with PCSK9 inhibitors has been shown to be safe and effective in lowering LDL-C in patients hospitalized with ACS.[789–791] Recent data have also shown improvements in plaque phenotype and plaque regression in ACS patients treated with PCSK9 inhibitors.[792,793] Combined with the data from trials on the long-term benefits of PCSK9 inhibitors and observational data on the importance of lowering LDL-C early after ACS, PCSK9 inhibitor treatment should be initiated during ACS hospitalization in patients who were not at their LDL-C goal despite being on statin and ezetimibe treatment before admission.[785,786,794–796]

In all cases, lipid levels should be re-evaluated 4–6 weeks after each treatment or dose adjustment to determine whether treatment goals have been achieved and to check for any safety issues; the therapeutic regimen can then be adapted accordingly. If the LDL-C goals are not achieved with the maximum tolerated dose of a statin alone after 4–6 weeks following ACS, adding ezetimibe is recommended.[783,788] Initiation of PCSK9 inhibitor treatment is recommended in patients who do not reach their LDL-C goal despite maximum tolerated statin and ezetimibe therapy.[783,785,786] Finally, icosapent ethyl, at a dose of 2 g b.i.d., can be used in combination with a statin in patients with ACS and triglyceride levels of 1.5–5.6 mmol/L (135–499 mg/dL) despite statin treatment.[783,797] An algorithm for lipid-lowering management in ACS patients is outlined in *Figure 18*.

For a detailed description of the different lipid-lowering drug classes and respective trial data, please refer to the Supplementary data online.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

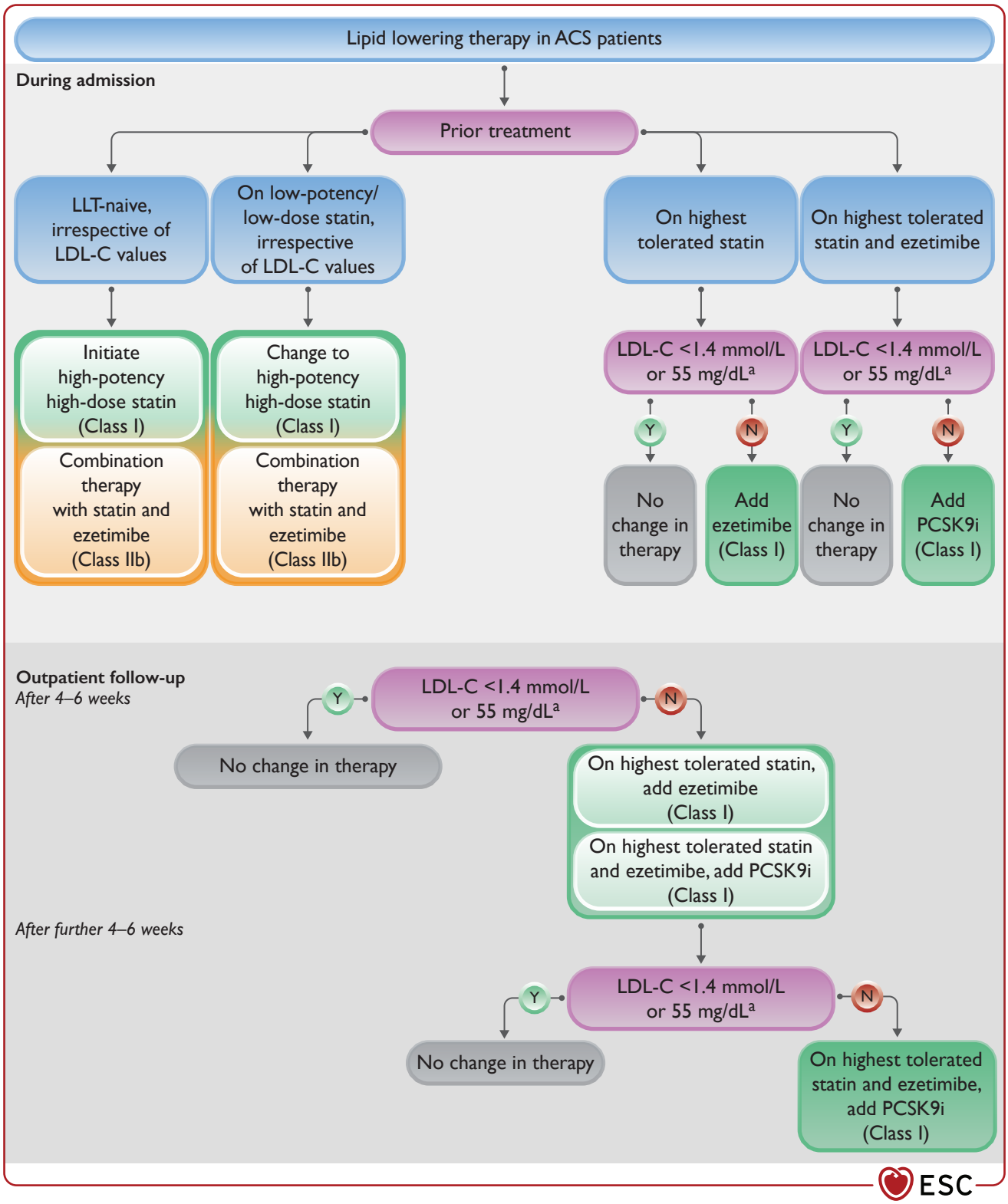


Figure 18 Lipid-lowering therapy in ACS patients. ACS, acute coronary syndrome; LDL-C, low-density lipoprotein cholesterol; LLT, lipid-lowering therapy; PCSK9i, proprotein convertase subtilisin/kexin type 9 inhibitor. [a]Consider LDL-C <1.0 mmol/L if recurrent event.

13.3.3. Beta-blockers

The clinical benefit of beta-blockers after ACS in patients with reduced LVEF is supported by evidence from contemporary trials.[557,798–800] However, the evidence for prescribing beta-blockers after uncomplicated ACS in patients with LVEF >40% is less well established. With the exception of the CAPRICORN (CArvedilol Post-infaRct survIval COntRolled evaluatioN) trial, which only recruited patients with LVEF ≤40%, all large RCTs testing the benefits of post-MI beta-blocker maintenance were performed in the pre-reperfusion era.[801] Pooled data demonstrated that post-MI beta-blocker therapy reduced the

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

risk of death by >20%. These trials mostly enrolled patients with STEMI, making the evidence for their benefit in NSTEMI less robust. In addition, since these trials were performed, the clinical scenario has changed dramatically, with improvements in invasive strategies and associated pharmacotherapy resulting in an improved prognosis for patients with ACS.[718] Modern observational studies and meta-analyses of these trials have yielded mixed results, with some studies suggesting a benefit of beta-blocker therapy irrespective of LVEF, and others reaching the opposite conclusion.[557,800,802–804]

There is only one small, open-label trial, CAPITAL-RCT (Carvedilol Post-Intervention Long-Term Administration in Large-scale Randomized Controlled Trial), that randomized 801 STEMI patients with successful PPCI and preserved LVEF to carvedilol or control.[805] During a 3-year follow-up, the incidence of a composite of all-cause death, MI, hospitalization for HF, and hospitalization for ACS was not significantly different between the two groups. However, the trial was underpowered and therefore this scientific question remains open. There are four ongoing pragmatic prospective large-scale RCTs in Europe randomizing ACS patients without reduced LVEF to beta-blocker or control: REBOOT-CNIC (TREatment With Beta-blockers After myOcardial Infarction withOut Reduced Ejection fracTion), 8468 ACS patients with LVEF >40%; REDUCE-SWEDEHEART (Evaluation of Decreased Usage of Betablockers After Myocardial Infarction in the SWEDEHEART Registry), 5000 ACS patients with LVEF ≥50% (NCT03278509); BETAMI (BEtablocker Treatment After Acute Myocardial Infarction in Patients Without Reduced Left Ventricular Systolic Function), 10 000 ACS patients with LVEF >40%; and DANBLOCK (Danish Trial of Beta Blocker Treatment After Myocardial Infarction Without Reduced Ejection Fraction), 3570 ACS patients with LVEF >40%.[806–808]

The duration of beta-blocker therapy after uncomplicated ACS is also another controversial topic. There are some observational studies suggesting that the clinical benefit of beta-blocker therapy is restricted to the first year after the index ACS event, but the non-randomized nature of the studies limits their conclusions.[809] There are two ongoing large-scale RCTs testing the impact of beta-blocker withdrawal after 6–12 months following uncomplicated ACS in patients with preserved LVEF: AβYSS (Beta Blocker Interruption After Uncomplicated Myocardial Infarction; NCT03498066) and SMART-DECISION (Long-term Beta-blocker Therapy After Acute Myocardial Infarction; NCT04769362).[810]

13.3.4. Nitrates and calcium channel blockers

Intravenous nitrates may be useful during the acute phase in STEMI patients with hypertension or HF, provided there is no hypotension or RV infarction. In the ISIS-4 (Fourth International Study of Infarct Survival) trial, oral nitrates had no survival benefit in MI patients.[811] Their use is therefore restricted to the control of residual angina, as recommended in the 2019 ESC Guidelines for the diagnosis and management of chronic coronary syndromes.[195] Calcium channel blocker use was not associated with prognostic benefit in a systematic review including 28 trials.[812] Calcium channel blocker use can be considered in the context of residual angina and for blood pressure control as recommended in the 2021 ESC Guidelines on CVD prevention and the 2019 ESC Guidelines for the diagnosis and management of CCS.[195,646]

13.3.5. Renin–angiotensin–aldosterone system inhibitors

Angiotensin-converting enzyme (ACE) inhibitors have been demonstrated to improve outcomes in post-MI patients with additional conditions, such as clinical HF and/or LVEF ≤40%, diabetes, CKD, and/or hypertension.[813–817] A systematic overview of (old) trials of ACE inhibition early in STEMI showed that their use is associated with a small but significant reduction in 30-day mortality, especially in anterior MIs.[818]

In the VALsartan In Acute myocardial iNfarcTion (VALIANT) trial, valsartan was found to be non-inferior to captopril in patients with a recent MI plus HF and/or LVEF ≤40%.[819]

There is established evidence that patients with heart failure with reduced ejection fraction (HFrEF), regardless of aetiology, benefit from ACE inhibitors.[820–823] Angiotensin receptor/neprilysin inhibitors (ARNI) have been shown to be superior to ACE inhibitors in patients with established HF (of different aetiologies) and LVEF ≤40%.[824] However, in the more recent PARADISE-MI (Prospective ARNI vs ACE Inhibitor Trial to Determine Superiority in Reducing Heart Failure Events After MI), a dedicated study in patients with recent ACS (1–7 days) complicated by HF and/or LVEF ≤40%, an ARNI combination (sacubitril plus valsartan) was not associated with a significantly lower incidence of death from CV causes or incident HF in comparison to the active comparator ramipril.[825]

In general, ACE inhibitors (or sacubitril plus valsartan as a replacement for them) are recommended for patients with established HFrEF regardless of the aetiology.[557] These agents may be considered for patients with HF with mildly reduced ejection fraction.[557] Patients who tolerate neither ACE inhibitors nor ARNI are recommended to be treated with an angiotensin receptor blocker.

In the Eplerenone Post-AMI Heart failure Efficacy and SUrvival Study (EPHESUS), the mineralocorticoid receptor antagonist (MRA) eplerenone was associated with reduced mortality and CV hospitalizations in patients with a recent MI and LV dysfunction with symptoms of either HF or diabetes.[826] The Double-Blind, Randomized, Placebo-Controlled Trial Evaluating The Safety And Efficacy Of Early Treatment With Eplerenone In Patients With Acute Myocardial Infarction (REMINDER) trial randomized 1012 patients with acute STEMI without HF to eplerenone or placebo within 24 h of symptom onset.[827] The primary endpoint was the composite of CV mortality, rehospitalization, or extended initial hospital stay due to diagnosis of HF, sustained VT or VF, ejection fraction ≤40%, or elevated BNP/NT-pro BNP at 1 month or more after randomization. Eplerenone was associated with a significant reduction in the primary composite endpoint, although this difference was primarily driven by BNP levels.[827]

13.3.6. Medications for diabetes

13.3.6.1. Sodium–glucose co-transporter 2 inhibitors

Pharmacological blockade of SGLT2 induces glycosuria with lowering of plasma glucose levels, improving glycaemic control without hypoglycaemia, and leading to reductions in weight and blood pressure.[828] In patients with type 2 diabetes and established ASCVD, three trials (with empagliflozin, canagliflozin, and dapagliflozin) have demonstrated significant CV benefits.[656,829,830] In a meta-analysis of these three trials, MACE were reduced by 11%, with no clear effect on stroke or MI. This benefit was only seen in patients with established ASCVD.[698] The benefits of SGLT2 inhibitors may relate more to cardio-renal haemodynamic effects than to atherosclerosis.[646] Further recommendations for patients with diabetes can be found in the current ESC Guidelines on diabetes, pre-diabetes,and cardiovascular diseases.[831]

In patients with HF regardless of their LVEF, dapagliflozin and empagliflozin have been shown to significantly reduce the risk of worsening HF or CV death, both in the presence or absence of type 2

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

diabetes.[702,703,832,833] In the EMMY (EMpagliflozin in patients with acute MYocardial infarction) trial, empagliflozin led to a significant improvement in NT-pro BNP reduction over 26 weeks post-MI, accompanied by a significant improvement in echocardiographic functional and structural parameters.[834] Ongoing outcome trials in ACS populations will be useful to better define the role of these agents in the absence of HF.[835]

13.3.6.2. Glucagon-like peptide-1 receptor agonists

In a systematic review and meta-analysis of seven trials (56 004 patients with type 2 diabetes) testing different GLP1-RAs, their use was associated with reductions in the incidence of MACE, CV death, all-cause mortality, MI, and stroke.[699]

13.3.7. Proton pump inhibitors

Proton pump inhibitors (PPIs) reduce the risk of upper gastroduodenal bleeding in patients treated with antiplatelet agents.[287,836,837] Therapy with a PPI is indicated for patients receiving any antithrombotic regimen who are at high risk of gastrointestinal bleeding (for details see *Section 8.2.2.3*, Bleeding risk assessment, in the Supplementary data online).

PPIs that inhibit CYP2C19, particularly omeprazole and esomeprazole, may reduce the pharmacodynamic response to clopidogrel, though there is no strong evidence that this results in an increased risk of ischaemic events or stent thrombosis in clinical trials and propensity score-matched studies.[287,288,838–842] Importantly, no interaction between the concomitant use of PPIs and aspirin, prasugrel or ticagrelor has been observed.

13.3.8. Vaccination

An annual influenza vaccination in patients with stable ASCVD appears to be associated with reduced incidence of MI, an improved prognosis in patients with HF, and decreased CV risk in adults aged 65 years and older.[843,844] In addition, influenza vaccination given early after an MI or in high-risk CAD has been shown to result in a lower risk of all-cause death and CV death at 12 months.[845–847] Therefore, influenza vaccination is recommended for all ACS patients and should be given preferentially during index hospitalization during influenza season for those not protected by a seasonal influenza vaccination.

13.3.9. Anti-inflammatory drugs

Inflammation plays a central role in the pathogenesis of atherosclerosis and acute coronary events. Several recent trials have tested the role of the anti-inflammatory agent colchicine in acute and chronic coronary syndromes.[848,849] In the Colchicine Cardiovascular Outcomes Trial (COLCOT), which enrolled 4745 patients with a recent ACS event, low-dose colchicine (0.5 mg daily) was associated with a significant reduction of the primary composite endpoint (CV death, resuscitated cardiac arrest, MI, stroke, or urgent revascularization) in comparison to placebo.[850] Of note, pneumonia was more frequent in the colchicine group. The Low-dose Colchicine trial-2 (LoDoCo2) enrolled 5522 patients with CCS (84% of whom had prior ACS) who were randomized to colchicine (0.5 mg daily) or placebo.[851] The primary endpoint (composite of CV death, MI, stroke, or ischaemia-driven coronary revascularization) rate was significantly lower in the colchicine group; however, the incidence of non-CV death was higher in the colchicine group. The benefits of colchicine in reducing CV events have been shown to be consistent irrespective of history and timing of prior ACS.[852]

13.3.10. Hormone replacement therapy

For further information on hormone replacement therapy in patients with ACS, please see the Supplementary data online.

Recommendation Table 16 — Recommendations for long-term management

Recommendations	Class[a]	Level[b]
Cardiac rehabilitation		
It is recommended that all ACS patients participate in a medically supervised, structured, comprehensive, multidisciplinary exercise-based cardiac rehabilitation and prevention programme.[721–724,853,854]	I	A
Lifestyle management		
It is recommended that ACS patients adopt a healthy lifestyle, including: • stopping all smoking of tobacco • healthy diet (Mediterranean style) • alcohol restriction • regular aerobic physical activity and resistance exercise • reduced sedentary time[724,761,763,772,773,776,777,855–858]	I	B
In smokers, offering follow-up support, nicotine replacement therapy, varenicline or bupropion, individually or in combination, should be considered.[859–864]	IIa	A
Pharmacological treatment		
Lipid-lowering therapy		
It is recommended that high-dose statin therapy is initiated or continued as early as possible, regardless of initial LDL-C values.[787,865–867]	I	A
It is recommended to aim to achieve an LDL-C level of <1.4 mmol/L (<55 mg/dL) and to reduce LDL-C by ≥50% from baseline.[868,869]	I	A
If the LDL-C goal is not achieved despite maximally tolerated statin therapy after 4–6 weeks, the addition of ezetimibe is recommended.[788]	I	B
If the LDL-C goal is not achieved despite maximally tolerated statin therapy and ezetimibe after 4–6 weeks, the addition of a PCSK9 inhibitor is recommended.[785,786,795,796]	I	A
It is recommended to intensify lipid-lowering therapy[c] during the index ACS hospitalization for patients who were on lipid-lowering therapy before admission.	I	C
For patients with a recurrent atherothrombotic event (recurrence within 2 years of first ACS episode) while taking maximally tolerated statin-based therapy, an LDL-C goal of <1.0 mmol/L (<40 mg/dL) may be considered.[785,786]	IIb	B
Combination therapy with high-dose statin plus ezetimibe may be considered during index hospitalization.[788]	IIb	B

Continued

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Beta-blockers		
Beta-blockers are recommended in ACS patients with LVEF ≤40% regardless of HF symptoms.[801,870–872]	I	A
Routine beta-blockers for all ACS patients regardless of LVEF should be considered.[798,873–878]	IIa	B
RAAS system inhibitors		
Angiotensin-converting enzyme (ACE) inhibitors[d] are recommended in ACS patients with HF symptoms, LVEF ≤40%, diabetes, hypertension, and/or CKD.[195,813–817,879]	I	A
Mineralocorticoid receptor antagonists are recommended in ACS patients with an LVEF ≤40% and HF or diabetes.[826,880]	I	A
Routine ACE inhibitors for all ACS patients regardless of LVEF should be considered.[816,817]	IIa	A
Adherence to medication		
A polypill should be considered as an option to improve adherence and outcomes in secondary prevention after ACS.[753]	IIa	B
Imaging		
In patients with pre-discharge LVEF ≤40%, repeat evaluation of the LVEF 6–12 weeks after an ACS (and after complete revascularization and the institution of optimal medical therapy) is recommended to assess the potential need for sudden cardiac death primary prevention ICD implantation.	I	C
Cardiac magnetic resonance imaging should be considered as an adjunctive imaging modality in order to assess the potential need for primary prevention ICD implantation.	IIa	C
Vaccination		
Influenza vaccination is recommended for all ACS patients.[843,845–847]	I	A
Anti-inflammatory drugs		
Low-dose colchicine (0.5 mg once daily) may be considered, particularly if other risk factors are insufficiently controlled or if recurrent cardiovascular disease events occur under optimal therapy.[850,851]	IIb	A

Continued

ACS, acute coronary syndrome; CKD, chronic kidney disease; HF, heart failure; ICD, implantable cardioverter defibrillator; LDL-C, low-density lipoprotein-cholesterol; LVEF, left ventricular ejection fraction; PCSK9, proprotein convertase subtilisin/kexin type 9; RAAS, renin–angiotensin–aldosterone system.
[a]Class of recommendation.
[b]Level of evidence.
[c]Increase statin potency/dose if the patient was on low-potency/low-dose statin, add ezetimibe if the patient was only on statins at highest tolerated dose, or add PCSK9 inhibitor if the patient was on statins plus ezetimibe.
[d]Angiotensin receptor blockers in cases of intolerance.

© ESC 2023

14. Patient perspectives

14.1. Patient-centred care

The management of patients with ACS should not only consider the best available evidence with regard to clinical management strategies, but also should be mindful of the provision of care that is respectful of and responsive to individual patient preferences, needs, and values, ensuring that these values are included in clinical decision-making.[881]

Patient-centred care should be guided by ethical values when considering a patient's physical, emotional, and psychological needs. Adopting a person-centred care approach after an ACS event improves patient outcomes and enhances quality of life.[882] Patients who are regarded as equal partners in their ACS medical management are more likely to actively engage and participate in their own healthcare.[883]

Educating and involving patients in their care should be seen as a continuous process. Engaging and educating the patient is a key component of ACS care and should take place throughout their patient journey, from admission to hospital discharge and cardiac rehabilitation (*Figure 19*).

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026



Figure 19 A person-centred approach to the ACS journey. ACS, acute coronary syndrome.

14.2. Shared decision-making

Shared decision-making is a process, during which the patient and a healthcare professional work together to make an informed decision about the patient's care.[884] During this process, information is provided, comprehension checked, and the patient is given an opportunity to ask questions in order to equip them with the tools needed to make an informed decision.

Using a shared decision-making approach during the consent process allows the patient's preferences to be established.[884] Discovery of the patient's concerns, goals, preferences, and values should be a central component of this process. The use of validated decision aids and audio-visual tools may also be helpful to facilitate informed consent and promote patient involvement.[884–887]

14.3. Informed consent

Informed consent should include the components listed in Supplementary data online, *Table S18*.[885,888] Informed consent is an opportunity to educate patients about the proposed procedure, the associated risks and benefits, and any available alternative interventions or treatments.[886,887] Assessment of the patient's understanding of the information given to them during the informed consent process using the 'teach back' technique should be considered (Supplementary data online, *Figure S6*).[885,889–891] The teach back method assesses understanding by asking patients to state in their own words what they need to know or do about their health.

Informed consent is an ethical and legal obligation for medical practitioners and is required before any invasive procedure. The information

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

should be provided in a simple and clear format. In patients undergoing emergency invasive angiography, a shortened informed consent process is appropriate. If a shortened informed consent process has been used, it is important that there is contact with the patient and/or family member after the intervention when the patient is physically and psychologically stable or following the death of the patient.[892] Further information can be found in the Supplementary data online.

14.4. Research participation and consent in the acute setting

With unstable ACS patients, it is often challenging to obtain their consent for emergency procedures—and even more challenging to enrol in clinical trials due to a number of factors, including the necessity for prompt clinical care, increased pain and stress levels, and impairment of consciousness. Where clinical trials are conducted, patient involvement in enrolment decisions is paramount, if possible.[893,894] A short witnessed verbal consent, followed by written consent after the acute phase, has been shown to be less stressful and more positively received than written consent in the acute setting.[894] The research and consent process must follow the ethical and legal requirements in the relevant country. Further information can be found in the Supplementary data online.

14.5. Patient satisfaction and expectations

Focusing healthcare around the needs and preferences of patients has the potential to improve clinical outcomes, quality of care and patient satisfaction, while decreasing healthcare costs and health disparities.[881] Patient perception of care is built on interpersonal interactions, the quality of clinical communication, delivery of care, and the administrative management of care. ACS patient expectations are summarized in *Figure 20* and further information can be found in the Supplementary data online, *Table S19*.



Figure 20 Acute coronary syndrome patient expectations. ACS, acute coronary syndrome.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

14.6. Patient-reported outcome measures and patient-reported experience measures

Understanding and measuring patient expectations and health outcomes using patient-reported outcome measures (PROMs) and patient-reported experience measures (PREMs) is central to improving patient satisfaction and delivering patient-centred care.[895] The quality of care for ACS patients should be measured during the patient's journey from initial presentation until discharge. Further information can be found in the Supplementary data online. Further information on PROMs and PREMs is also provided in the Supplementary data online.

14.7. Preparation for discharge

Many ACS patients may not be fully aware of what has happened to them and how to best manage their healthcare after discharge, leading to them both wanting and needing more information upon discharge.[896] Cognitive impairment can occur as a complication of ACS and some patients may have difficulty with instructions for care when transitioning towards discharge home.[897] Therefore, discharge information should be provided in both verbal and written formats and should include a discharge letter outlining the key components of the evidence-based discharge plan (Supplementary data online, *Table S20*).[898–901] Some important messages aimed at patients on how to improve their heart health after ACS are demonstrated in Supplementary data online, *Figure S5*. Moreover, following an ACS event, anxiety and depression are frequently encountered and confer an increased risk of non-adherence to medications and lifestyle changes, subsequent MACE, and death.[902–904] Non-adherence also generally increases over time, which has additional impact on clinical outcomes.[905] Assessing and identifying these patients and intervening with onward psychological referral is recommended.[858] Further information can be found in the Supplementary data online. A summary of patient concerns and educational needs throughout their ACS journey is also provided in Supplementary data online, *Figure S7*.

Recommendation Table 17 — Recommendations for patient perspectives in acute coronary syndrome care

Recommendations	Class[a]	Level[b]
Patient-centred care is recommended by assessing and adhering to individual patient preferences, needs and beliefs, ensuring that patient values are used to inform all clinical decisions.[744,881,906,907]	I	B
It is recommended to include ACS patients in decision-making (as much as their condition allows) and to inform them about the risk of adverse events, radiation exposure, and alternative options. Decision aids can be used to facilitate the discussion.[908,909]	I	B
It is recommended to assess symptoms using methods that help patients to describe their experience.[910]	I	C
Use of the 'teach back' technique for decision support during the securing of informed consent should be considered.[885,889–891]	IIa	B
Patient discharge information should be provided in both written and verbal formats prior to discharge. Adequate preparation and education for patient discharge using the teach back technique and/or motivational interviewing, giving information in chunks, and checking for understanding should be considered.[885,896,911]	IIa	B
Assessment of mental well-being using a validated tool and onward psychological referral when appropriate should be considered.[903,904,912,913]	IIa	B

Continued

© ESC 2023

ACS, acute coronary syndrome.
[a]Class of recommendation.
[b]Level of evidence.

15. Key messages

Epidemiology of ACS

Acute coronary syndromes encompass a spectrum of conditions that include patients with a recent change in clinical symptoms or signs, with or without changes on 12-lead ECG and with or without acute elevations in cardiac troponin concentrations. ACS are commonly classified based on ECG at presentation and the presence or absence of troponin elevation into UA, NSTEMI, or STEMI. The incidence of STEMI is decreasing whereas the incidence of NSTEMI is increasing. While there are some sex differences in the epidemiology of ACS, women and men receive equal benefit from invasive and non-invasive management strategies and, in general, should be managed similarly.

Diagnostic tools (ECG, troponin, and non-invasive imaging)

Chest pain/discomfort is the most common symptom initiating the ACS diagnostic and therapeutic pathway. High-sensitivity troponin measurements and rapid 'rule-in' and 'rule-out' algorithms should be used in patients with suspected NSTE-ACS. MI is not the only condition resulting in cardiomyocyte injury and cardiac troponin elevation, and other conditions should also be considered in the differential diagnosis. Non-invasive imaging can be useful to increase diagnostic accuracy and optimize risk assessment.

STEMI management networks

Co-ordination between EMS and hospitals with common written protocols is central to the management of STEMI. EMS should transfer patients immediately to 24/7 high-volume PCI centres regardless of the initial treatment strategy (PPCI or pre-hospital fibrinolysis). EMS should always alert the PCI centre immediately after selection of the reperfusion strategy, and patient transfer to the PCI centre should bypass the ED.

Invasive strategy and reperfusion therapy

An invasive strategy is recommended for patients with ACS. Invasive strategies are time sensitive. For STEMI and very high-risk NSTE-ACS, an immediate invasive strategy is recommended. For patients with NSTE-ACS an inpatient invasive strategy is recommended; in NSTE-ACS patients with high-risk characteristics, an early invasive strategy (<24 h) should be considered. If timely (within 120 min from time of diagnosis) PPCI cannot be performed in patients with STEMI, fibrinolytic therapy is indicated within 12 h of symptom onset in patients without contraindications.

Antithrombotic therapy

Antithrombotic therapy is indicated in all ACS patients, regardless of the management strategy. This consists of both antiplatelet and anticoagulant therapy. Aspirin is recommended for all ACS patients at an initial loading dose and a longer-term maintenance dose. In addition to aspirin, a $P2Y_{12}$ receptor inhibitor is recommended, and should be maintained over 12

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

months unless there are concerns regarding HBR. Regarding $P2Y_{12}$ receptor inhibitor choice, prasugrel and ticagrelor are recommended in preference to clopidogrel, and prasugrel should be considered in preference to ticagrelor for ACS patients who undergo PCI. Pre-treatment (i.e. treatment with a $P2Y_{12}$ receptor inhibitor prior to coronary angiography) in patients with NSTE-ACS is not recommended routinely but may be considered for patients with STEMI undergoing PPCI. Parenteral anticoagulation is recommended for all patients at the time of diagnosis. Discontinuation of parenteral anticoagulation should be considered immediately after the invasive procedure. Some patients with ACS will also have an indication for long-term OAC, most commonly AF. In these patients, TAT for up to 1 week, followed by DAT using a NOAC at the recommended dose for stroke prevention and a single oral antiplatelet agent (preferably clopidogrel), is recommended as the default strategy.

ACS with unstable presentation

A PPCI strategy is recommended in patients with resuscitated cardiac arrest and an ECG with persistent ST elevation (or ST elevation equivalents), whereas routine immediate angiography is not recommended in patients with an ECG without persistent ST elevation (or equivalents). Temperature control (i.e. continuous monitoring of core temperature and active prevention of fever [i.e. >37.7°C]) is recommended in patients with OHCA who remain unresponsive after ROSC. In patients with CS complicating ACS, emergency coronary angiography is recommended, whereas the routine use of IABP in ACS patients with CS and no mechanical complications is not.

Early care

Following reperfusion, it is recommended to admit high-risk ACS patients, including all STEMI patients, to a CCU/ICCU. ECG monitoring for arrhythmias and ST-segment changes is recommended for at least 24 h after symptom onset in all high-risk patients with ACS. It is recommended that all hospitals participating in the care of high-risk ACS patients have an ICCU/CCU equipped to provide all required aspects of care including treatment of ischaemia, severe HF, arrhythmias, and common comorbidities. It is also recommended that the LVEF is determined before hospital discharge in all patients with ACS. Discharge of high-risk ACS patients within 48–72 h should be considered in selected patients if early rehabilitation and adequate follow-up are arranged.

Technical aspects during PPCI

Routine radial access and use of DES are the standard of care during PCI for ACS. Intravascular imaging should be considered to guide PCI and may be considered in patients with ambiguous culprit lesions. Routine thrombus aspiration is not recommended. CABG should be considered in patients with an occluded IRA when PCI is not feasible or unsuccessful and there is a large area of myocardium in jeopardy. In patients presenting with SCAD, PCI is recommended only for patients with symptoms and signs of ongoing myocardial ischaemia, a large area of myocardium in jeopardy, and reduced antegrade flow.

Management of patients with MVD

For patients with MVD, it is recommended to base the revascularization strategy (IRA PCI, multivessel PCI/CABG) on the patient's clinical status and comorbidities, as well as their disease complexity, according to the principles of management of myocardial revascularization. For patients with MVD presenting with CS, IRA-only PCI during the index procedure is recommended. For patients with STEMI undergoing PPCI, complete revascularization is recommended either during the index PCI or within 45 days. In patients presenting with NSTE-ACS and MVD, complete revascularization should be considered, preferably during the index procedure. For patients with STEMI, it is recommended that decisions regarding PCI of non-IRA are based on angiographic severity, whereas for patients with NSTE-ACS, functional invasive evaluation of non-IRA severity during the index procedure may be considered.

MINOCA

The term MINOCA refers to the situation where patients present with symptoms suggestive of ACS and demonstrate troponin elevation and non-obstructive coronary arteries at the time of coronary angiography, i.e. coronary artery stenosis <50% in any major epicardial vessel. MINOCA is best considered as a working diagnosis that encompasses a heterogenous group of underlying causes (both cardiac and extra-cardiac) and is found in 1–14% of patients with ACS. In all patients with an initial working diagnosis of MINOCA, it is recommended to follow a diagnostic algorithm to determine the underlying cause. CMR imaging is a key diagnostic tool in patients with a working diagnosis of MINOCA.

Special patient subsets

Chronic kidney disease: moderate to severe CKD is present in >30% of ACS patients. These patients receive less interventional and pharmacological treatment and have a worse prognosis in comparison to patients with normal kidney function. It is recommended to apply the same diagnostic and therapeutic strategies in patients with CKD (dose adjustment may be necessary) as for patients with normal kidney function.

Older adults: in general, older adults should undergo the same diagnostic and treatment strategies, including invasive angiography and revascularization, as younger patients.

Patients with cancer: management of ACS in patients with cancer can be challenging for several reasons, including frailty, increased bleeding risk, thrombocytopaenia, and increased thrombotic risk. An invasive strategy is recommended in cancer patients presenting with high-risk ACS with expected survival ≥6 months. A conservative non-invasive strategy should be considered in ACS patients with poor cancer prognosis (with expected survival <6 months) and/or very high bleeding risk.

Long-term treatment

Secondary prevention after ACS should be offered to every patient and should start as early as possible after the index event. This includes cardiac rehabilitation, lifestyle management, and pharmacological treatment, and has been shown to both increase quality of life and decrease morbidity and mortality.

Patient perspectives

Some of the key first steps in the timely diagnosis and treatment of ACS are reliant on a comprehensive assessment of symptoms. An incomplete history or poorly elicited symptoms can result in delay or misdiagnosis. Patient-centred care is recommended as a critical tenet of routine clinical management and involves consideration of a patient's physical, emotional, and psychological needs.

The provision of care that is respectful of, and responsive to, individual patient preferences, needs and values, is important in the management of patients with ACS. It is recommended, as much as possible, to include ACS patients in decision-making. Preparing for discharge begins on admission. Educating and informing the patient using the teach back method and educationally appropriate material should be integrated into the patient care pathway.

Quality indicators

Acute coronary syndrome QIs aim to audit practice and improve clinical outcomes in real-life patients by demonstrating the gap between optimal guideline-based treatment and actual care of ACS patients. Subsequent measures to improve QI attainment can be implemented based on the local, regional, and global assessment of QIs.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

16. Gaps in evidence

Table 8 Gaps in evidence

	Section	Gaps in evidence	Research recommendations to address these gaps
3	**Triage and diagnosis**	• Observe zone: how can we improve the guidance for and management of patients assigned to the observe zone of the 0 h/1 h and 0 h/2 h ESC algorithms to improve their poor outcome? • No testing rule: what is the added value of biomarkers other than hs-cTn for rapid rule-out of NSTE-ACS compared with usual care? • There is insufficient evidence to set sex-specific thresholds for troponin levels. • The role of non-invasive anatomy (e.g. CCTA) or functional imaging (e.g. stress testing strategies) for low-risk NSTE-ACS patients should be further evaluated.	• Observe zone: prospectively evaluate changes in the 0 h/1 h and 0 h/2 h ESC algorithms to improve the outcomes of patients assigned to the observe zone. • No testing rule: randomization of patients to strategies with and without new biomarkers to evaluate whether their use improves clinical outcomes. • Prospectively evaluate the impact of using sex-specific cut-offs on the diagnosis, treatment, and outcomes of patients presenting to the ED with suspected ACS. • Adequately powered RCTs testing whether non-invasive imaging improves clinical outcomes in patients presenting with NSTE-ACS.
4	**Initial measures for patients presenting with suspected STEMI | Initial treatment**	• The impact of early i.v. beta-blockers on clinical outcomes in patients with a working diagnosis of STEMI remains unclear. • Infarct size and microvascular obstruction are the main determinants of long-term prognosis. Interventions which serve to limit infarct size are needed.	• Patients randomized to i.v. beta-blockers (ideally metoprolol) or placebo before PPCI, with hard endpoints evaluated. • Translate cardio-protective therapies from experimental to clinical setting by executing adequately powered trials.
5	**Acute-phase management of patients with NSTE-ACS**	• The comparison of routine or selective invasive assessment in low-risk NSTE-ACS has not been adequately evaluated. • The optimal timing of invasive angiography in high-risk NSTE-ACS patients remains uncertain.	• Low-risk patients should be randomized to routine or selective invasive strategy. • RCTs testing different time intervals to perform angiography within the 72 h window after the initial presentation.
6	**Antithrombotic therapy**	• Whether pre-treatment with oral $P2Y_{12}$ receptor inhibitors prior to ICA improves clinical outcomes in NSTE-ACS patients is uncertain. • Whether platelet function testing or genetic testing to guide de-escalation of oral $P2Y_{12}$ receptor inhibitors after the first month of therapy following PCI improves clinical management and outcomes remains unclear. • The optimal long-term antithrombotic regimen in NSTE-ACS patients who have undergone PCI is unknown. • After stopping DAPT, a head-to-head comparison based on superiority between aspirin monotherapy and clopidogrel monotherapy is required.	• Randomize patients to pre-treatment with oral $P2Y_{12}$ receptor inhibitors or no pre-treatment, prior to ICA. • Randomize ACS patients to prasugrel or ticagrelor, both without pre-treatment. • A strategy based on platelet function testing or genetic testing should be prospectively tested in patients who may benefit from de-escalating antithrombotic therapy. • RCTs evaluating the benefit-risk balance for ischaemic bleeding events for different periods of antithrombotic duration. • A head-to-head randomized comparison testing for superiority is needed to compare aspirin vs. clopidogrel monotherapy after DAPT.
7	**Acute coronary syndrome with unstable presentation**	• The role of percutaneous MCS devices in patients presenting with ACS and CS remains unclear.	• Randomized comparisons between standard of care and percutaneous MCS devices in ACS with CS.
8	**Management of acute coronary syndrome during hospitalization**	• Clinical improvement through the use of risk stratification based on risk prediction models.	• Patients randomized to a particular intervention or to usual care based on validated risk prediction models.
9	**Technical aspects of invasive strategies**	• Does intravascular imaging-guided revascularization strategy improve clinical outcomes in patients with ACS? • Does intracoronary physiology assessment of myocardial reperfusion after PPCI improve risk stratification and/or stratified medicine for limiting microvascular dysfunction and reperfusion injury/MVO post ACS?	• RCTs evaluating the efficacy of an intravascular imaging-guided revascularization strategy to improve meaningful clinical outcomes in patients with ACS. • Prospectively evaluate whether intracoronary physiology assessment of myocardial reperfusion better stratifies patient risk.

Continued

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

		• In ACS patients with an IRA that is unsuitable for stent implantation, does drug-coated balloon treatment of the IRA improve clinical outcomes? • Microvascular obstruction associated with PPCI represents an unmet clinical need in patients with ACS. Development of therapies for the prevention and treatment of MVO is urgently needed. • Does early implementation of MCS in the management of high-risk ACS patients improve clinical outcomes? • Does intracoronary hypothermia reduce infarct size and improve clinical outcomes in STEMI patients undergoing PPCI? • What is the optimal antiplatelet strategy in patients presenting with SCAD? Specific gaps in knowledge surround antithrombotic treatment in the acute and post-ACS periods, including the optimal combination and duration of treatment.	• Patients with an IRA unsuitable for stent implantation randomized to drug-coated balloon treatment or usual care to evaluate clinical outcomes. • Pre-clinical and clinical research is needed to evaluate cardio-protective therapies aimed at reducing microvascular obstruction. • RCTs evaluating the benefit of using MCS in high-risk patients. • Randomized trials are needed to demonstrate both whether intracoronary hypothermia reduces myocardial infarct size, and if this translates into clinical improvement. • RCTs evaluating several antiplatelet strategies in patients with SCAD with the aim of determining which results in the greatest clinical benefit.
10	**Management of patients with multivessel disease**	• Does complete revascularization of NSTE-ACS with multivessel CAD improve clinical outcomes vs. culprit-only PCI? • Does management of non-infarct-related CAD with intravascular imaging guidance to identify rupture-prone atherosclerotic plaque improve clinical outcomes? • Does FFR-guided management improve clinical outcomes vs. standard angiography-guided management in NSTE-ACS? • What is the optimal timing of coronary revascularization (immediate vs. index hospitalization vs. staged) for non-IRA revascularization in STEMI and NSTE-ACS? • Does intensive medical therapy improve outcomes in patients with MVD compared with standard secondary prevention? • The clinical utility of hybrid coronary revascularization in ACS patients with multivessel CAD is uncertain.	• Patients with NSTE-ACS and MVD randomized to complete vs. culprit-only PCI. • RCTs testing whether the use of intravascular imaging to guide the management of non-infarct-related lesions improves clinical outcomes. • Patients randomized to FFR-guided management vs. standard angiography-guided management in NSTE-ACS. • Three-arm study comparing the clinical benefits of immediate, in-hospital and staged coronary revascularization strategies. • Patients with MVD randomized to intensive secondary prevention vs. usual care to evaluate whether the former strategy improves clinical outcomes. • RCTs assessing the clinical benefit of hybrid revascularization.
12	**Special situations**	• How to better differentiate Type 2 from Type 1 MI before invasive assessment. • The optimal management strategy in older adults with NSTE-ACS is not known. • The optimal management strategy in older frail, comorbid adults with NSTE-ACS is not known. • The optimal management strategy in older frail, comorbid adults with STEMI is not known. • Optimal antiplatelet therapy and its duration to manage ACS in pregnant patients are not known. • The optimal management strategy for pregnant women with NSTE-ACS is not known. • There is a need to further evaluate the contribution of social determinants of health.	• Prospective evaluation of diagnostic strategies aimed at better classifying patients according to their type of MI (Type 1 vs. Type 2). • Further studies recruiting older adults should be conducted to evaluate whether the current standard of care also benefits this subset of patients. • Older frail, comorbid patients should not be systematically excluded from RCTs. • Prospective data are needed to better understand which antiplatelet therapy regimen is best for pregnant women. • Observational data are needed in patients with ACS to evaluate the real impact of social determinants of health on clinical outcomes. Randomized interventions aimed at reducing social inequalities are needed to evaluate how to reduce this gap.
13	**Long-term treatment**	• To evaluate the uptake, safety, and outcomes for alternative forms of cardiac rehabilitation, with a focus on telemedicine and eHealth. • How to improve referral for and uptake of CR, especially for groups with low participation, including women, older persons, and ethnic minorities.	• Remote cardiac rehabilitation methods need randomized data to evaluate their true potential. • Further monitoring is needed to increase the participation of historically under-represented patients in CR.

Continued

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

		• The role of personalized medicine in the short- and long-term treatment of ACS needs to be further studied. • How to address additional risk from non-traditional risk factors, e.g. cardio-obstetrics, cardio-oncology, and inflammatory conditions, needs further attention. • Inflammation as a treatment target in patients with atherosclerosis still needs unravelling, as well as the use of biomarkers of inflammation (high-sensivity C-reactive protein, interleukins 1 and 6) to guide treatment of residual risk. • The role of lipoprotein (a) in guiding treatment and as an independent treatment target needs to be studied further. • The added cardio-protective role of beta-blockers in post-ACS patients without reduced LVEF on otherwise optimal medical therapy needs to be clarified. • The added cardio-protective role of ACE inhibitors/ARBs in post-ACS patients without reduced LVEF on otherwise optimal medical therapy needs to be clarified. • The future role of new treatment options, using mRNA- and siRNA-based therapies targeting lipid metabolism and inflammation, needs to be explored. • It has to be determined whether SGLT2 inhibitors—in the specific group of patients with ACS without heart failure or diabetes—improve clinical outcomes, regardless of diabetes status.	• Patients randomized to personalized strategies vs. usual care are needed to determine the role of precision medicine in ACS. • Prospective cohorts are needed to evaluate non-traditional risk factors and residual risk. • RCTs testing whether management based on the use of biomarkers of inflammation improves clinical outcomes. • RCTs testing whether lipoprotein (a) measurement to guide medical management further improves clinical outcomes. • Patients randomized to beta-blocker and no beta-blocker use to evaluate treatment efficacy in patients with ACS and LVEF >40%. • RCTs evaluating the benefit of using ACE inhibitors/ARBs vs. placebo on top of standard care in ACS patients with LVEF >40%. • Randomized data are needed to evaluate the role of mRNA- and siRNA-based therapies in the current context of lipid management and lipid targets. • ACS patients without HF or diabetes should be randomized to SGLT2 inhibitors vs. standard of care.
14	**Patient perspectives**	• The feasibility of performing short witnessed verbal consent followed by written consent after the acute phase needs further evaluation. • There is a need to assess the contribution of social determinants of health on ACS incidence and prognosis. • The use of validated patient-reported outcome and experience measures in evidence-based medicine should be increased. • Quality of life is a relevant outcome not captured in most trials. • Use of validated decision aids and audio-visual tools can be helpful to make informed choices that consider patients' values and preferences and promote patient involvement.	• Studies comparing verbal vs. written consent to evaluate safety endpoints and any ethical concerns. • The influence of social determinants of health on clinical outcomes should be evaluated, as well as those interventions aimed at reducing social inequalities. • PROMs/PREMs should have a more prominent role in RCTs evaluating patients with ACS. • Include quality of life as a prominent outcome in clinical trials. • Testing the use of validated decision aids and audio-visual tools to improve decisions around informed choices.
19	**Quality indicators**	• There is a lack of implementation studies evaluating whether prospectively monitoring and reporting ESC QIs for ACS improve clinical outcomes.	• Implementation studies evaluating a quality of care programme based on the evaluation of ESC QIs for ACS.
	General	• Patients included in clinical trials represent a relatively small proportion of real-life patients.	• Conduct clinical trials that enrol more representative patient populations (e.g. pragmatic clinical trials).

Ongoing trials addressing some of these gaps in evidence are presented in the Supplementary data online.

ACE, angiotensin-converting enzyme; ACS, acute coronary syndrome; ARB, angiotensin receptor blocker; CAD, coronary artery disease; CCTA, coronary computed tomography angiography; CR, cardiac rehabilitation; CS, cardiogenic shock; DAPT, dual antiplatelet therapy; ED, emergency department; ESC, European Society of Cardiology; FFR, fractional flow reserve; HF, heart failure; ICA, invasive coronary angiography; IRA, infarct-related artery; i.v., intravenous; LVEF, left ventricular ejection fraction; MCS, mechanical circulatory support; MI, myocardial infarction; MINOCA, myocardial infarction with non-obstructive coronary arteries; mRNA, messenger ribonucleic acid; MVD, multivessel disease; MVO, microvascular obstruction; NSTE-ACS, non-ST-elevation acute coronary syndrome; PCI, percutaneous coronary intervention; PPCI, primary percutaneous coronary intervention; PREM, patient-reported experience measure; PROM, patient-reported outcome measure; QI, quality indicator; RCT, randomized controlled trial; SCAD, spontaneous coronary artery dissection; SGLT2, sodium-glucose co-transporter 2; siRNA, small interfering ribodeoxynucleic acid; STEMI, ST-elevation myocardial infarction.

17. Sex differences

There are currently no data supporting the differential management of ACS based on sex. However, several studies have reported that women presenting with ACS are treated differently than men.[914–918] This includes being less likely than men to receive ICA, timely revascularization, CR, and secondary prevention medications.[914–918]

Healthcare providers and policymakers should be conscious of this potential gender bias in the management of ACS and make a concerted effort to ensure that women with ACS receive evidence-based care.

In order to ensure the generalizability of the findings yielded by RCTs, patient recruitment should be reflective of real-world populations from different socioeconomic backgrounds.[919] Several studies have reported that a disproportionately low proportion of women are recruited to CV trials.[920–922] Alongside historic under-representation of other subsets of patients, including older patients and ethnic minorities, this suggests an underlying recruitment bias.[923] Increased representation of female patients in future clinical trials is required to better inform the optimal management of women with ACS.[924]

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

© ESC 2023

18. 'What to do' and 'What not to do' messages from the Guidelines

Table 9 'What to do' and 'What not to do'

Recommendations for clinical and diagnostic tools for patients with suspected acute coronary syndrome	Class[a]	Level[b]
It is recommended that patients with suspected STEMI are immediately triaged for an emergency reperfusion strategy.	I	A
It is recommended to base the diagnosis and initial short-term risk stratification of ACS on a combination of clinical history, symptoms, vital signs, other physical findings, ECG, and hs-cTn.	I	B
Twelve-lead ECG recording and interpretation is recommended as soon as possible at the point of first medical contact, with a target of <10 min.	I	B
Continuous ECG monitoring and the availability of defibrillator capacity is recommended as soon as possible in all patients with suspected STEMI, in suspected ACS with other ECG changes or ongoing chest pain, and once the diagnosis of MI is made.	I	B
The use of additional ECG leads (V3R, V4R, and V7–V9) is recommended in cases of inferior STEMI or if total vessel occlusion is suspected and standard leads are inconclusive.	I	B
An additional 12-lead ECG is recommended in cases with recurrent symptoms or diagnostic uncertainty.	I	C
It is recommended to measure cardiac troponins with high-sensitivity assays immediately after presentation and to obtain the results within 60 min of blood sampling.	I	B
It is recommended to use an ESC algorithmic approach with serial hs-cTn measurements (0 h/1 h or 0 h/2 h) to rule in and rule out NSTEMI.	I	B
Additional testing after 3 h is recommended if the first two hs-cTn measurements of the 0 h/1 h algorithm are inconclusive and no alternative diagnoses explaining the condition have been made.	I	B
Recommendations for non-invasive imaging in the initial assessment of patients with suspected acute coronary syndrome		
Emergency TTE is recommended in patients with suspected ACS presenting with cardiogenic shock or suspected mechanical complications.	I	C
Routine, early coronary computed tomography angiography in patients with suspected ACS is not recommended.	III	B
Recommendations for the initial management of patients with acute coronary syndrome		
It is recommended that the pre-hospital management of patients with a working diagnosis of STEMI is based on regional networks designed to deliver reperfusion therapy expeditiously and effectively, with efforts made to make PPCI available to as many patients as possible.	I	B
It is recommended that PPCI-capable centres deliver a 24/7 service and are able to perform PPCI without delay.	I	B
It is recommended that patients transferred for PPCI bypass the emergency department and CCU/ICU and are transferred directly to the catheterization laboratory.	I	B
Oxygen is recommended in patients with hypoxaemia (SaO_2 <90%).	I	C
It is recommended that EMS transfer patients with suspected STEMI to a PCI-capable centre, bypassing non-PCI centres.	I	C
It is recommended that ambulance teams are trained and equipped to identify ECG patterns suggestive of acute coronary occlusion and to administer initial therapy, including defibrillation, and fibrinolysis when applicable.	I	C
It is recommended that all hospitals and EMS participating in the care of patients with suspected STEMI record and audit delay times and work together to achieve and maintain quality targets.	I	C
Routine oxygen is not recommended in patients with oxygen saturation >90%.	III	A
Recommendations for reperfusion therapy and timing of invasive strategy		
Recommendations for reperfusion therapy for patients with STEMI		
Reperfusion therapy is recommended in all patients with a working diagnosis of STEMI (persistent ST-segment elevation or equivalents) and symptoms of ischaemia of ≤12 h duration.	I	A
A PPCI strategy is recommended over fibrinolysis if the anticipated time from diagnosis to PCI is <120 min.	I	A
If timely PPCI (<120 min) cannot be performed in patients with a working diagnosis of STEMI, fibrinolytic therapy is recommended within 12 h of symptom onset in patients without contraindications.	I	A
Rescue PCI is recommended for failed fibrinolysis (i.e. ST-segment resolution <50% within 60–90 min of fibrinolytic administration) or in the presence of haemodynamic or electrical instability, worsening ischaemia, or persistent chest pain.	I	A

Continued

Continued

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

In patients with a working diagnosis of STEMI and a time from symptom onset >12 h, a PPCI strategy is recommended in the presence of ongoing symptoms suggestive of ischaemia, haemodynamic instability, or life-threatening arrhythmias.	I	C
Routine PCI of an occluded IRA is not recommended in STEMI patients presenting >48 h after symptom onset and without persistent symptoms.	III	A
Transfer/interventions after fibrinolysis		
Transfer to a PCI-capable centre is recommended in all patients immediately after fibrinolysis.	I	A
Emergency angiography and PCI of the IRA, if indicated is recommended in patients with new-onset or persistent heart failure/shock after fibrinolysis.	I	A
Angiography and PCI of the IRA, if indicated, is recommended between 2 and 24 h after successful fibrinolysis.	I	A
Invasive strategy in NSTE-ACS		
An invasive strategy during hospital admission is recommended in NSTE-ACS patients with high-risk criteria or with a high index of suspicion for unstable angina.	I	A
A selective invasive approach is recommended in patients without very high- or high-risk features and a low index of suspicion for NSTE-ACS.	I	A
An immediate invasive strategy is recommended in patients with a working diagnosis of NSTE-ACS and with at least one of the following very high-risk criteria: • Haemodynamic instability or cardiogenic shock • Recurrent or refractory chest pain despite medical treatment • In-hospital life-threatening arrhythmias • Mechanical complications of MI • Acute heart failure presumed secondary to ongoing myocardial ischaemia *Recurrent* dynamic ST-segment or T wave changes, particularly intermittent ST-segment elevation.	I	C
Recommendations for antiplatelet and anticoagulant therapy in acute coronary syndrome		
Antiplatelet therapy		
Aspirin is recommended for all patients without contraindications at an initial oral LD of 150–300 mg (or 75–250 mg i.v.) and an MD of 75–100 mg o.d. for long-term treatment.	I	A
In all ACS patients, a $P2Y_{12}$ receptor inhibitor is recommended in addition to aspirin, given as an initial oral LD followed by an MD for 12 months unless there is high bleeding risk.	I	A
A proton pump inhibitor in combination with dual antiplatelet therapy is recommended in patients at high risk of gastrointestinal bleeding.	I	A

Continued

Prasugrel is recommended in $P2Y_{12}$ receptor inhibitor-naïve patients proceeding to PCI (60 mg LD, 10 mg o.d. MD, 5 mg o.d. MD for patients aged ≥75 years or with a body weight <60 kg).	I	B
Ticagrelor is recommended irrespective of the treatment strategy (invasive or conservative) (180 mg LD, 90 mg twice a day MD).	I	B
Clopidogrel (300–600 mg LD, 75 mg o.d. MD) is recommended when prasugrel or ticagrelor are not available, cannot be tolerated, or are contraindicated.	I	C
If patients presenting with ACS stop DAPT to undergo CABG, it is recommended they resume DAPT after surgery for at least 12 months.	I	C
Pre-treatment with a glycoprotein IIb/IIIa antagonist is not recommended.	III	A
Routine pre-treatment with a $P2Y_{12}$ receptor inhibitor in NSTE-ACS patients in whom coronary anatomy is not known and early invasive management (<24 h) is planned is not recommended.	III	A
Anticoagulant therapy		
Parenteral anticoagulation is recommended for all patients with ACS at the time of diagnosis.	I	A
Routine use of a UFH bolus (weight-adjusted i.v. bolus during PCI of 70–100 IU/kg) is recommended in patients undergoing PCI.	I	C
Patients with STEMI		
Fondaparinux is not recommended in patients with STEMI undergoing PPCI.	III	B
Patients with NSTE-ACS		
For patients with NSTE-ACS in whom early invasive angiography (i.e. within 24 h) is not anticipated, fondaparinux is recommended.	I	B
Combining antiplatelets and OAC		
As the default strategy for patients with atrial fibrillation and CHA_2DS_2-VASc score ≥1 in men and ≥2 in women, after up to 1 week of triple antithrombotic therapy following the ACS event, dual antithrombotic therapy using a non-vitamin K antagonist oral anticoagulant at the recommended dose for stroke prevention and a single oral antiplatelet agent (preferably clopidogrel) for up to 12 months is recommended.	I	A
During PCI, a UFH bolus is recommended in any of the following circumstances: • if the patient is on a NOAC • if the INR is <2.5 in VKA-treated patients.	I	C
The use of ticagrelor or prasugrel as part of triple antithrombotic therapy is not recommended.	III	C

Continued

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Recommendations for alternative antithrombotic therapy regimens		
Discontinuation of antiplatelet treatment in patients treated with an oral anticoagulant is recommended after 12 months.	I	B
De-escalation of antiplatelet therapy in the first 30 days after ACS is not recommended.	III	B
Recommendations for fibrinolytic therapy		
When fibrinolysis is the reperfusion strategy, it is recommended to initiate this treatment as soon as possible after diagnosis in the pre-hospital setting (aim for target of <10 min to lytic bolus).	I	A
A fibrin-specific agent (i.e. tenecteplase, alteplase, or reteplase) is recommended.	I	B
Antiplatelet co-therapy with fibrinolysis		
Aspirin and clopidogrel are recommended.	I	A
Anticoagulation co-therapy with fibrinolysis		
Anticoagulation is recommended in patients treated with fibrinolysis until revascularization (if performed) or for the duration of hospital stay (up to 8 days).	I	A
Enoxaparin i.v. followed by subcutaneous is recommended as the preferred anticoagulant.	I	A
When enoxaparin is not available, unfractionated heparin is recommended as a weight-adjusted i.v. bolus, followed by infusion.	I	B
Recommendations for cardiac arrest and out-of-hospital cardiac arrest		
A PPCI strategy is recommended in patients with resuscitated cardiac arrest and an ECG with persistent ST-segment elevation (or equivalents).	I	B
Temperature control (i.e. continuous monitoring of core temperature and active prevention of fever [i.e. >37.7°C]) is recommended after either out-of-hospital or in-hospital cardiac arrest for adults who remain unresponsive after return of spontaneous circulation.	I	B
Routine immediate angiography after resuscitated cardiac arrest is not recommended in haemodynamically stable patients without persistent ST-segment elevation (or equivalents).	III	A
Systems of care		
It is recommended that healthcare systems implement strategies to facilitate transfer of all patients in whom ACS is suspected after resuscitated cardiac arrest directly to a hospital offering 24/7 PPCI via one specialized EMS.	I	C
Evaluation of neurological prognosis		
Evaluation of neurological prognosis (no earlier than 72 h after admission) is recommended in all comatose survivors after cardiac arrest.	I	C

Continued

Recommendations for cardiogenic shock		
Immediate coronary angiography and PCI of the IRA (if indicated) is recommended in patients with CS complicating ACS.	I	B
Emergency CABG is recommended for ACS-related CS if PCI of the IRA is not feasible/ unsuccessful.	I	B
In cases of haemodynamic instability, emergency surgical/catheter-based repair of mechanical complications of ACS is recommended, based on Heart Team discussion.	I	C
The routine use of an intra-aortic balloon pump in ACS patients with CS and without mechanical complications is not recommended.	III	B
Recommendations for in-hospital management		
It is recommended that all hospitals participating in the care of high-risk patients have an ICCU/CCU equipped to provide all required aspects of care, including treatment of ischaemia, severe heart failure, arrhythmias, and common comorbidities.	I	C
It is recommended that high-risk patients (including all STEMI patients and very high-risk NSTE-ACS patients) have ECG monitoring for a minimum of 24 h.	I	C
It is recommended that high-risk patients with successful reperfusion therapy and an uncomplicated clinical course (including all STEMI patients and very high-risk NSTE-ACS patients) are kept in the CCU/ ICCU for a minimum of 24 h whenever possible, after which they may be moved to a step-down monitored bed for an additional 24–48 h.	I	C
Recommendations for technical aspects of invasive strategies		
Radial access is recommended as the standard approach, unless there are over-riding procedural considerations.	I	A
PCI with stent deployment in the IRA during the index procedure is recommended for patients undergoing PPCI.	I	A
Drug-eluting stents are recommended in preference to bare metal stents in all cases.	I	A
In patients with spontaneous coronary artery dissection, PCI is recommended only for patients with symptoms and signs of ongoing myocardial ischaemia, a large area of myocardium in jeopardy, and reduced antegrade flow.	I	C
The routine use of thrombus aspiration is not recommended.	III	A
Recommendations for management of patients with multivessel disease		
It is recommended to base the revascularization strategy (IRA PCI, multivessel PCI/CABG) on the patient's clinical status and comorbidities, as well as their disease complexity, according to the principles of management of myocardial revascularization.	I	B

Continued

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Multivessel disease in ACS patients presenting in cardiogenic shock		
IRA-only PCI during the index procedure is recommended.	I	B
Multivessel disease in haemodynamically stable STEMI patients undergoing PPCI		
Complete revascularization is recommended either during the index PCI procedure or within 45 days.	I	A
It is recommended that PCI of the non-IRA is based on angiographic severity.	I	B
Invasive epicardial functional assessment of non-culprit segments of the IRA is not recommended during the index procedure.	III	C
Recommendations for myocardial infarction with non-obstructive coronary arteries		
In patients with a working diagnosis of MINOCA CMR imaging is recommended after invasive angiography if the final diagnosis is not clear	I	B
Management of MINOCA according to the final established underlying diagnosis is recommended, consistent with the appropriate disease-specific guidelines.	I	B
In all patients with an initial working diagnosis of MINOCA, it is recommended to follow a diagnostic algorithm to determine the underlying final diagnosis.	I	C
Recommendations for acute coronary syndrome complications		
Atrial fibrillation		
Intravenous beta-blockers are recommended when rate control is needed in the absence of acute HF or hypotension.	I	C
Intravenous amiodarone is recommended when rate control is needed in the presence of acute HF and no hypotension.	I	C
Immediate electrical cardioversion is recommended in patients with ACS and haemodynamic instability and when adequate rate control cannot be achieved promptly with pharmacological agents.	I	C
Intravenous amiodarone is recommended to facilitate electrical cardioversion and/or decrease risk of early recurrence of AF after electrical cardioversion in unstable patients with recent-onset AF.	I	C
Ventricular arrythmias		
Implantable cardioverter defibrillator use is recommended to reduce sudden cardiac death in patients with symptomatic HF (NYHA Class II–III) and LVEF ≤35% despite optimal medical therapy for >3 months and at least 6 weeks after MI who are expected to survive for at least 1 year with good functional status.	I	A
Intravenous beta-blocker and/or amiodarone treatment is recommended for patients with polymorphic VT and/or VF unless contraindicated.	I	B
Prompt and complete revascularization is recommended to treat myocardial ischaemia that may be present in patients with recurrent VT and/or VF.	I	C
Bradyarrhythmias		
In cases of sinus bradycardia with haemodynamic intolerance or high-degree AV block without stable escape rhythm: • i.v. positive chronotropic medication (adrenaline, vasopressin, and/or atropine) is recommended. • temporary pacing is recommended in cases of failure to respond to atropine. • urgent angiography with a view to revascularization is recommended if the patient has not received previous reperfusion therapy.	I	C
Implantation of a permanent pacemaker is recommended when high-degree AV block does not resolve within a waiting period of at least 5 days after MI.	I	C
Pacing is not recommended if high-degree AV block resolves after revascularization or spontaneously.	III	B
Treatment of asymptomatic and haemodynamically irrelevant ventricular arrhythmias with anti-arrhythmic drugs is not recommended.	III	C
Recommendations for acute coronary syndrome comorbid conditions		
Chronic kidney disease		
The use of low- or iso-osmolar contrast media (at the lowest possible volume) is recommended for invasive strategies.	I	A
It is recommended to assess kidney function using eGFR in all patients with ACS.	I	C
It is recommended to apply the same diagnostic and therapeutic strategies in patients with CKD (dose adjustment may be necessary) as in patients with normal kidney function.	I	C
Diabetes		
It is recommended to base the choice of long-term glucose-lowering treatment on the presence of comorbidities, including heart failure, CKD, and obesity.	I	A
It is recommended to assess glycaemic status at initial evaluation in all patients with ACS.	I	B
It is recommended to frequently monitor blood glucose levels in patients with known diabetes mellitus or hyperglycaemia (defined as glucose levels ≥11.1 mmol/L or ≥200 mg/dL).	I	C
Older adults		
It is recommended to apply the same diagnostic and treatment strategies in older patients as in younger patients.	I	B

Continued

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

It is recommended to adapt the choice and dosage of antithrombotic agent, as well as of secondary prevention medications, to renal function, co-medications, comorbidities, frailty, cognitive function, and specific contraindications.	I	B
For frail older patients with comorbidities, a holistic approach is recommended to individualize interventional and pharmacological treatments after careful evaluation of the risks and benefits.	I	B
An invasive strategy is recommended in cancer patients presenting with high-risk ACS with expected life survival ≥6 months.	I	B
A temporary interruption of cancer therapy is recommended in patients in whom the cancer therapy is suspected to be a contributing cause of ACS.	I	C
Aspirin is not recommended in cancer patients with a platelet count <10 000/μL.	III	C
Clopidogrel is not recommended in cancer patients with a platelet count <30 000/μL.	III	C
In ACS patients with cancer and <50 000/μL platelet count, prasugrel or ticagrelor are not recommended.	III	C
Recommendations for long-term management		
It is recommended that all ACS patients participate in a medically supervised, structured, comprehensive, multidisciplinary exercise-based cardiac rehabilitation and prevention programme.	I	A
It is recommended that ACS patients adopt a healthy lifestyle, including: • stopping all smoking of tobacco • healthy diet (Mediterranean style) • alcohol restriction • regular aerobic physical activity and resistance exercise • reduced sedentary time	I	B
Pharmacological treatment		
Lipid-lowering therapy		
It is recommended that high-dose statin therapy is initiated or continued as early as possible, regardless of initial LDL-C values.	I	A
It is recommended to aim to achieve an LDL-C level of <1.4 mmol/L (<55 mg/dL) and to reduce LDL-C by ≥50% from baseline.	I	A
If the LDL-C goal is not achieved despite maximally tolerated statin therapy and ezetimibe after 4–6 weeks, the addition of a PCSK9 inhibitor is recommended	I	A
If the LDL-C goal is not achieved despite maximally tolerated statin therapy after 4–6 weeks, the addition of ezetimibe is recommended.	I	B
It is recommended to intensify lipid-lowering therapy during the index ACS hospitalization for patients who were on lipid-lowering therapy before admission.	I	C
Beta-blockers		
Beta-blockers are recommended in ACS patients with LVEF ≤40% regardless of HF symptoms.	I	A
RAAS system inhibitors		
Angiotensin-converting enzyme inhibitors are recommended in ACS patients with HF symptoms, LVEF ≤40%, diabetes, hypertension, and/or CKD.	I	A
Mineralocorticoid receptor antagonists are recommended in ACS patients with an LVEF ≤40% and HF or diabetes.	I	A
Imaging		
In patients with pre-discharge LVEF ≤40%, repeat evaluation of the LVEF 6–12 weeks after an ACS (and after complete revascularization and the institution of optimal medical therapy) is recommended to assess the potential need for sudden cardiac death primary prevention ICD implantation.	I	C
Vaccination		
Influenza vaccination is recommended for all ACS patients.	I	A
Recommendations for patient perspectives in acute coronary syndrome care		
Patient-centred care is recommended by assessing and adhering to individual patient preferences, needs and beliefs, ensuring that patient values are used to inform all clinical decisions.	I	B
It is recommended to include ACS patients in decision-making (as much as their condition allows) and to inform them about the risk of adverse events, radiation exposure, and alternative options. Decision aids should be used to facilitate the discussion.	I	B
It is recommended to assess symptoms using methods that help patients to describe their experience.	I	C

ACS, acute coronary syndrome; AV, atrioventricular; CABG, coronary artery bypass grafting; CCU, cardiac care unit; CHA_2DS_2-VASc, Congestive heart failure, Hypertension, Age ≥75 years, Diabetes mellitus, Stroke or transient ischaemic attack, Vascular disease; CKD, chronic kidney disease; CMR, cardiac magnetic resonance; CS, cardiogenic shock; DAPT, dual antiplatelet therapy; ECG, electrocardiogram; ESC, European Society of Cardiology; HF, heart failure; hs-cTn, high-sensitivity cardiac troponin; ICU, intensive care unit; IRA, infarct-related artery; i.v., intravenous; LD, loading dose; LDL-C, low-density lipoprotein cholesterol; MD, maintenance dose; MINOCA, myocardial infarction with non-obstructive coronary arteries; LVEF, left ventricular ejection fraction; MI, myocardial infarction; NSTE-ACS, non-ST-elevation acute coronary syndrome; NSTEMI, non-ST-elevation myocardial infarction; NYHA, New York Heart Association; o.d., once daily; OAC, oral anticoagulant; PCI, percutaneous coronary intervention; PCSK9, proprotein convertase subtilisin/kexin type 9; PPCI, primary percutaneous coronary intervention; RAAS, renin–angiotensin–aldosterone system; STEMI, ST-elevation myocardial infarction; UFH, unfractionated heparin; VKA, vitamin K antagonist.

[a]Class of recommendation.

[b]Level of evidence.

© ESC 2023

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

19. Quality indicators

Quality indicators are tools that can be used to evaluate care quality, including structures, processes, and outcomes of care.[925] They may also serve as a mechanism for enhancing adherence to guideline recommendations, through associated quality improvement initiatives and the benchmarking of care providers.[926,927] As such, the role of QIs in improving care and outcomes for CVD is increasingly recognized by healthcare authorities, professional organizations, payers, and the public.[925]

The ESC understands the need for measuring and reporting quality and outcomes of CV care and has established methods for the development of the ESC QIs for the quantification of care and outcomes for CVDs.[925] To date, the ESC has developed QI suites for a number of CVDs in parallel with the writing of the ESC Clinical Practice Guidelines. Previous QIs for the management of AMI have been tested in numerous large registries.[928–933] A systematic review of these studies has shown that there is room for improvement in terms of levels of attainment of QIs.[934]

The ESC aims to harmonize its QIs for various CV conditions and integrate them with ESC registries.[935,936] This integrative approach provides 'real-world' data about the patterns and outcomes of care for CVD across Europe.

20. Supplementary data

Supplementary data are available at *European Heart Journal* online.

21. Data availability statement

No new data were generated or analysed in support of this research.

22. Author information

Author/Task Force Member Affiliations: Xavier Rossello, Cardiology Department, Hospital Universitari Son Espases, Palma de Mallorca, Spain, Clinical Research Department Centro Nacional de Investigaciones Cardiovasculares (CNIC), Madrid, Spain, Health Research Institute of the Balearic Islands (IdiSBa), Universitat de les Illes Balears (UIB), Palma de Mallorca, Spain; **J.J. Coughlan**, Cardiovascular Research Institute, Mater Private Network, Dublin, Ireland; **Emanuele Barbato**, Clinical and Molecular Medicine, Sapienza University, Rome, Italy; **Colin Berry**, British Heart Foundation, Glasgow Cardiovascular Research Centre, University of Glasgow, Glasgow, United Kingdom, Cardiology, NHS Golden Jubilee, Clydebank, United Kingdom, Cardiology, NHS Greater Glasgow and Clyde Health Board, Glasgow, United Kingdom; **Alaide Chieffo**, IRCCS San Raffaele Scientific Institute, IRCCS San Raffaele Scientific Institute, Milan, Italy, University Vita-Salute San Raffaele, Milan, Italy; **Marc J. Claeys**, Cardiology, Antwerp University Hospital, Edegem, Belgium; **Gheorghe-Andrei Dan**, Colentina University Hospital, Cardiology Dpt., "Carol Davila" University of Medicine Bucharest, Romania; **Marc R. Dweck**, British Heart Foundation Centre for Cardiovascular Sciences, Chancellors Building, Little France Crescent, Little France, Edinburgh, United Kingdom; **Mary Galbraith** (United Kingdom), ESC Patient Forum, Sophia Antipolis, France; **Martine Gilard**, Cardiology, INSERM UMR 1304 GETBO- Brest University, Brest, France; **Lynne Hinterbuchner**, Department of Cardiology, Clinic of Internal Medicine II, Paracelsus Medical University of Salzburg, Salzburg, Austria; **Ewa A. Jankowska**, Institute of Heart Diseases, Wroclaw Medical University, Wroclaw, Poland, Institute of Heart Diseases, University Hospital in Wroclaw, Wroclaw, Poland; **Peter Jüni**, Nuffield Department of Population Health, University of Oxford, Oxford, United Kingdom; **Takeshi Kimura**, Department of Cardiology, Hirakata Kohsai Hospital, Osaka, Japan; **Vijay Kunadian**, Translational and Clinical Research Institute, Faculty of Medical Sciences, Newcastle University, Newcastle upon Tyne, United Kingdom, Cardiothoracic Centre, Freeman Hospital, Newcastle upon Tyne Hospitals NHS Foundation Trust, Newcastle upon Tyne, United Kingdom; **Margret Leosdottir**, Department of Cardiology, Skane University Hospital, Malmö, Sweden, Department of Clinical Sciences, Malmö, Lund University, Malmö, Sweden; **Roberto Lorusso**, Cardio-Thoracic Surgery, Maastricht University Medical Centre, Maastricht, Netherlands, Cardiovascular Research Institute Maastricht, Maastricht, Netherlands; **Roberto F.E. Pedretti**, Cardiovascular Department, IRCCS MultiMedica, Milan, Italy; **Angelos G. Rigopoulos**, Adult Cardiology, Mitera General Hospital-Hygeia Group, Athens, Greece; **Maria Rubini Gimenez**, Department of Internal Medicine and Cardiology, Heart Center Leipzig at the University of Leipzig, Leipzig, Germany, Cardiovascular Research Institute Basel, University Hospital Basel, Basel, Switzerland; **Holger Thiele**, Internal Medicine/Cardiology; Heart Center Leipzig at University of Leipzig, Leipzig, Germany, Leipzig Heart Science, Leipzig, Germany; **Pascal Vranckx**, Cardiology and Critical Care Medicine, Jessa Ziekenhuis, Hasselt, Belgium, Faculty of Medicine and Life Sciences, University of Hasselt, Hasselt, Belgium; **Sven Wassmann**, Cardiology, Cardiology Pasing, Munich, Germany; Faculty of Medicine, University of the Saarland, Homburg/Saar, Germany; **Nanette Kass Wenger**, Department of Medicine (Cardiology), Emory University School of Medicine, Atlanta, GA, United States of America, Consultant, Emory Heart and Vascular Center Emory University, School of Medicine, Atlanta, GA, United States of America, Founding Consultant, Emory Women's Heart Center, Emory University School of Medicine, Atlanta, GA, United States of America.

23. Appendix

ESC Scientific Document Group

Includes Document Reviewers and ESC National Cardiac Societies.

Document Reviewers: Sigrun Halvorsen (CPG Review Co-ordinator) (Norway), Stefan James (CPG Review Co-ordinator) (Sweden), Magdy Abdelhamid (Egypt), Victor Aboyans (France), Nina Ajmone Marsan (Netherlands), Sotiris Antoniou (United Kingdom), Riccardo Asteggiano (Italy), Maria Bäck (Sweden), Davide Capodanno (Italy), Ruben Casado-Arroyo (Belgium), Salvatore Cassese (Germany), Jelena Čelutkienė (Lithuania), Maja Cikes (Croatia), Jean-Philippe Collet (France), Gregory Ducrocq (France), Volkmar Falk (Germany), Laurent Fauchier (France), Tobias Geisler (Germany), Diana A. Gorog (United Kingdom), Lene Holmvang (Denmark), Tiny Jaarsma (Sweden), Hywel Wynne Jones (United Kingdom), Lars Køber (Denmark), Konstantinos C. Koskinas (Switzerland), Dipak Kotecha (United Kingdom), Konstantin A. Krychtiuk (Austria), Ulf Landmesser (Germany), George Lazaros (Greece), Basil S. Lewis (Israel), Bertil Lindahl (Sweden), Ales Linhart (Czech Republic), Maja-Lisa Løchen (Norway), Mamas A. Mamas

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

(United Kingdom), John William McEvoy (Ireland), Borislava Mihaylova (United Kingdom), Richard Mindham (United Kingdom), Christian Mueller (Switzerland), Lis Neubeck (United Kingdom), Josef Niebauer (Austria), Jens Cosedis Nielsen (Denmark), Alexander Niessner (Austria), Valeria Paradies (Netherlands), Agnes A. Pasquet (Belgium), Steffen E. Petersen (United Kingdom), Eva Prescott (Denmark), Amina Rakisheva (Kazakhstan), Bianca Rocca (Italy), Giuseppe M. C. Rosano (Italy), Leyla Elif Sade (United States of America/ Türkiye), François Schiele (France), Jolanta M. Siller-Matula (Austria), Christian Sticherling (Switzerland), Robert F. Storey (United Kingdom), Matthias Thielmann (Germany), Christiaan Vrints (Belgium), Stephan Windecker (Switzerland), Rune Wiseth (Norway), and Adam Witkowski (Poland).

ESC National Cardiac Societies actively involved in the review process of the 2023 ESC Guidelines for the management of acute coronary syndromes:

Algeria: Algerian Society of Cardiology, Mohammed El Amine Bouzid; **Armenia:** Armenian Cardiologists Association, Hamlet Hayrapetyan; **Austria:** Austrian Society of Cardiology, Bernhard Metzler; **Belgium:** Belgian Society of Cardiology, Patrizio Lancellotti; **Bosnia and Herzegovina:** Association of Cardiologists of Bosnia and Herzegovina, Mugdim Bajrić; **Bulgaria:** Bulgarian Society of Cardiology, Kiril Karamfiloff; **Cyprus:** Cyprus Society of Cardiology, Andreas Mitsis; **Czechia:** Czech Society of Cardiology, Petr Ostadal; **Denmark:** Danish Society of Cardiology, Rikke Sørensen; **Egypt:** Egyptian Society of Cardiology, Tamer Elwasify; **Estonia:** Estonian Society of Cardiology, Toomas Marandi; **Finland:** Finnish Cardiac Society, Essi Ryödi; **France:** French Society of Cardiology, Jean-Philippe Collet; **Georgia:** Georgian Society of Cardiology, Archil Chukhrukidze; **Germany:** German Cardiac Society, Julinda Mehilli; **Greece:** Hellenic Society of Cardiology, Periklis Davlouros; **Hungary:** Hungarian Society of Cardiology, Dávid Becker; **Iceland:** Icelandic Society of Cardiology, Ingibjörg Jóna Guðmundsdóttir; **Ireland:** Irish Cardiac Society, James Crowley; **Israel:** Israel Heart Society, Yigal Abramowitz; **Italy:** Italian Federation of Cardiology, Ciro Indolfi; **Kazakhstan:** Association of Cardiologists of Kazakhstan, Orazbek Sakhov; **Kosovo (Republic of):** Kosovo Society of Cardiology, Shpend Elezi; **Kyrgyzstan:** Kyrgyz Society of Cardiology, Medet Beishenkulov; **Latvia:** Latvian Society of Cardiology, Andrejs Erglis; **Lebanon:** Lebanese Society of Cardiology, Nicolas Moussallem; **Libya:** Libyan Cardiac Society, Hisham Benlamin; **Lithuania:** Lithuanian Society of Cardiology, Olivija Dobilienė; **Luxembourg:** Luxembourg Society of Cardiology, Philippe Degrell; **Malta:** Maltese Cardiac Society, Matthew Mercieca Balbi; **Moldova (Republic of):** Moldavian Society of Cardiology, Aurel Grosu; **Morocco:** Moroccan Society of Cardiology, Zouhair Lakhal; **Netherlands**: Netherlands Society of Cardiology, Jurriën ten Berg; **North Macedonia**: The National Society of Cardiology of North Macedonia, Hristo Pejkov; **Norway:** Norwegian Society of Cardiology, Kristin Angel; **Poland:** Polish Cardiac Society, Adam Witkowski; **Portugal:** Portuguese Society of Cardiology, Manuel De Sousa Almeida; **Romania:** Romanian Society of Cardiology, Ovidiu Chioncel; **San Marino:** San Marino Society of Cardiology, Luca Bertelli; **Serbia:** Cardiology Society of Serbia, Sinisa Stojkovic; **Slovakia:** Slovak Society of Cardiology, Martin Studenčan; **Slovenia:** Slovenian Society of Cardiology, Peter Radšel; **Spain:** Spanish Society of Cardiology, Jose Luis Ferreiro; **Sweden:** Swedish Society of Cardiology, Annica Ravn-Fischer; **Switzerland:** Swiss Society of Cardiology, Lorenz Räber; **Syrian Arab Republic:** Syrian Cardiovascular Association, Mohammed Yassin Bani Marjeh; **Tunisia:** Tunisian Society of Cardiology and Cardiovascular Surgery, Majed Hassine; **Türkiye:** Turkish Society of Cardiology, Aylin Yildirir; **Ukraine:** Ukrainian Association of Cardiology, Alexander Parkhomenko; **United Kingdom of Great Britain and Northern Ireland:** British Cardiovascular Society, Adrian Paul Banning.

ESC Clinical Practice Guidelines (CPG) Committee: Eva Prescott (Chairperson) (Denmark), Stefan James (Co-Chairperson) (Sweden), Elena Arbelo (Spain), Colin Baigent (United Kingdom), Michael A. Borger (Germany), Sergio Buccheri (Sweden), Borja Ibanez (Spain), Lars Køber (Denmark), Konstantinos C. Koskinas (Switzerland), John William McEvoy (Ireland), Borislava Mihaylova (United Kingdom), Richard Mindham (United Kingdom), Lis Neubeck (United Kingdom), Jens Cosedis Nielsen (Denmark), Agnes A. Pasquet (Belgium), Amina Rakisheva (Kazakhstan), Bianca Rocca (Italy), Xavier Rossello (Spain), Ilonca Vaartjes (Netherlands), Christiaan Vrints (Belgium), Adam Witkowski (Poland), and Katja Zeppenfeld (Netherlands).

24. References

1. Thygesen K, Alpert JS, Jaffe AS, Chaitman BR, Bax JJ, Morrow DA, *et al.* Fourth universal definition of myocardial infarction (2018). *Eur Heart J* 2019;**40**:237–269. https://doi.org/10.1093/eurheartj/ehy462
2. GBD 2017 Causes of Death Collaborators. Global, regional, and national age-sex-specific mortality for 282 causes of death in 195 countries and territories, 1980–2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet* 2018;**392**:1736–1788. https://doi.org/10.1016/s0140-6736(18)32203-7
3. Timmis A, Vardas P, Townsend N, Torbica A, Katus H, De Smedt D, *et al.* European Society of Cardiology: cardiovascular disease statistics 2021. *Eur Heart J* 2022;**43**:716–799. https://doi.org/10.1093/eurheartj/ehab892
4. Ibanez B, James S, Agewall S, Antunes MJ, Bucciarelli-Ducci C, Bueno H, *et al.* 2017 ESC Guidelines for the management of acute myocardial infarction in patients presenting with ST-segment elevation: the Task Force for the management of acute myocardial infarction in patients presenting with ST-segment elevation of the European Society of Cardiology (ESC). *Eur Heart J* 2018;**39**:119–177. https://doi.org/10.1093/eurheartj/ehx393
5. Diercks DB, Peacock WF, Hiestand BC, Chen AY, Pollack CV, Kirk JD, *et al.* Frequency and consequences of recording an electrocardiogram >10 minutes after arrival in an emergency room in non-ST-segment elevation acute coronary syndromes (from the CRUSADE Initiative). *Am J Cardiol* 2006;**97**:437–442. https://doi.org/10.1016/j.amjcard.2005.09.073
6. Lopez-Sendon J, Coma-Canella I, Alcasena S, Seoane J, Gamallo C. Electrocardiographic findings in acute right ventricular infarction: sensitivity and specificity of electrocardiographic alterations in right precordial leads V4R, V3R, V1, V2, and V3. *J Am Coll Cardiol* 1985;**6**:1273–1279. https://doi.org/10.1016/s0735-1097(85)80213-8
7. Schmitt C, Lehmann G, Schmieder S, Karch M, Neumann F-J, Scho¨mig A. Diagnosis of acute myocardial infarction in angiographically documented occluded infarct vessel: limitations of ST-segment elevation in standard and extended ECG leads. *Chest* 2001;**120**:1540–1546. https://doi.org/10.1378/chest.120.5.1540
8. Kosuge M, Kimura K, Ishikawa T, Hongo Y, Shigemasa T, Sugiyama M, *et al.* Implications of the absence of ST-segment elevation in lead V4R in patients who have inferior wall acute myocardial infarction with right ventricular involvement. *Clin Cardiol* 2001;**24**:225–230. https://doi.org/10.1002/clc.4960240310
9. Yan AT, Yan RT, Kennelly BM, Anderson FA, Budaj A, López-Sendón J, *et al.* Relationship of ST elevation in lead aVR with angiographic findings and outcome in non-ST elevation acute coronary syndromes. *Am Heart J* 2007;**154**:71–78. https://doi.org/10.1016/j.ahj.2007.03.037
10. Hirano T, Tsuchiya K, Nishigaki K, Sou K, Kubota T, Ojio S, *et al.* Clinical features of emergency electrocardiography in patients with acute myocardial infarction caused by left main trunk obstruction. *Circ J* 2006;**70**:525–529. https://doi.org/10.1253/circj.70.525
11. Yamaji H, Iwasaki K, Kusachi S, Murakami T, Hirami R, Hamamoto H, *et al.* Prediction of acute left main coronary artery obstruction by 12-lead electrocardiography. ST segment elevation in lead aVR with less ST segment elevation in lead V(1). *J Am Coll Cardiol* 2001;**38**:1348–1354. doi: doi:10.1016/S0735-1097(01)01563-7
12. Chapman AR, Shah ASV, Lee KK, Anand A, Francis O, Adamson P, *et al.* Long-term outcomes in patients with type 2 myocardial infarction and myocardial injury.

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Circulation 2018;**137**:1236–1245. https://doi.org/10.1161/CIRCULATIONAHA.117.031806
13. Nestelberger T, Boeddinghaus J, Badertscher P, Twerenbold R, Wildi K, Breitenbücher D, *et al.* Effect of definition on incidence and prognosis of type 2 myocardial infarction. *J Am Coll Cardiol* 2017;**70**:1558–1568. https://doi.org/10.1016/j.jacc.2017.07.774
14. Neumann JT, Sorensen NA, Rubsamen N, Ojeda F, Renné T, Qaderi V, *et al.* Discrimination of patients with type 2 myocardial infarction. *Eur Heart J* 2017;**38**:3514–3520. https://doi.org/10.1093/eurheartj/ehx457
15. Shah ASV, Anand A, Strachan FE, Ferry AV, Lee KK, Chapman AR, *et al.* High-sensitivity troponin in the evaluation of patients with suspected acute coronary syndrome: a stepped-wedge, cluster-randomised controlled trial. *Lancet* 2018;**392**:919–928. https://doi.org/10.1016/S0140-6736(18)31923-8
16. Mandelzweig L, Battler A, Boyko V, Bueno H, Danchin N, Filippatos G, *et al.* The second Euro Heart Survey on acute coronary syndromes: characteristics, treatment, and outcome of patients with ACS in Europe and the Mediterranean Basin in 2004. *Eur Heart J* 2006;**27**:2285–2293. https://doi.org/10.1093/eurheartj/ehl196
17. Twerenbold R, Boeddinghaus J, Nestelberger T, Wildi K, Rubini Gimenez M, Badertscher P, *et al.* Clinical use of high-sensitivity cardiac troponin in patients with suspected myocardial infarction. *J Am Coll Cardiol* 2017;**70**:996–1012. https://doi.org/10.1016/j.jacc.2017.07.718
18. Collet JP, Thiele H, Barbato E, Barthélémy O, Bauersachs J, Bhatt DL, *et al.* 2020 ESC Guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation. *Eur Heart J* 2021;**42**:1289–1367. https://doi.org/10.1093/eurheartj/ehaa575
19. Rokos IC, French WJ, Koenig WJ, Stratton SJ, Nighswonger B, Strunk B, *et al.* Integration of pre-hospital electrocardiograms and ST-elevation myocardial infarction receiving center (SRC) networks: impact on door-to-balloon times across 10 independent regions. *JACC Cardiovasc Interv* 2009;**2**:339–346. https://doi.org/10.1016/j.jcin.2008.11.013
20. O'Doherty M, Tayler DI, Quinn E, Vincent R, Chamberlain DA. Five hundred patients with myocardial infarction monitored within one hour of symptoms. *Br Med J (Clin Res Ed)* 1983;**286**:1405–1408. https://doi.org/10.1136/bmj.286.6375.1405
21. Mehta RH, Starr AZ, Lopes RD, Hochman JS, Widimsky P, Pieper KS, *et al.* Incidence of and outcomes associated with ventricular tachycardia or fibrillation in patients undergoing primary percutaneous coronary intervention. *JAMA* 2009;**301**:1779–1789. https://doi.org/10.1001/jama.2009.600
22. Kulkarni AU, Brown R, Ayoubi M, Banka VS. Clinical use of posterior electrocardiographic leads: a prospective electrocardiographic analysis during coronary occlusion. *Am Heart J* 1996;**131**:736–741. https://doi.org/10.1016/s0002-8703(96)90280-x
23. Casas RE, Marriott HJL, Glancy DL. Value of leads V7-V9 in diagnosing posterior wall acute myocardial infarction and other causes of tall R waves in V1-V2. *Am J Cardiol* 1997;**80**:508–509. https://doi.org/10.1016/s0002-9149(97)00404-9
24. Zalenski RJ, Rydman RJ, Sloan EP, Hahn KH, Cooke D, Fagan J, *et al.* Value of posterior and right ventricular leads in comparison to the standard 12-lead electrocardiogram in evaluation of ST-segment elevation in suspected acute myocardial infarction. *Am J Cardiol* 1997;**79**:1579–1585. https://doi.org/10.1016/s0002-9149(97)00202-6
25. Reichlin T, Hochholzer W, Bassetti S, Steuer S, Stelzig C, Hartwiger S, *et al.* Early diagnosis of myocardial infarction with sensitive cardiac troponin assays. *N Engl J Med* 2009;**361**:858–867. https://doi.org/10.1056/NEJMoa0900428
26. Reichlin T, Twerenbold R, Reiter M, Steuer S, Bassetti S, Balmelli C, *et al.* Introduction of high-sensitivity troponin assays: impact on myocardial infarction incidence and prognosis. *Am J Med* 2012;**125**:1205–1213.e1. https://doi.org/10.1016/j.amjmed.2012.07.015
27. Neumann JT, Twerenbold R, Ojeda F, Sörensen NA, Chapman AR, Shah ASV, *et al.* Application of high-sensitivity troponin in suspected myocardial infarction. *N Engl J Med* 2019;**380**:2529–2540. https://doi.org/10.1056/NEJMoa1803377
28. Boeddinghaus J, Twerenbold R, Nestelberger T, Badertscher P, Wildi K, Puelacher C, *et al.* Clinical validation of a novel high-sensitivity cardiac troponin I assay for early diagnosis of acute myocardial infarction. *Clin Chem* 2018;**64**:1347–1360. https://doi.org/10.1373/clinchem.2018.286906
29. Neumann JT, Sorensen NA, Rubsamen N, Ojeda F, Schock A, Seddighizadeh P, *et al.* Evaluation of a new ultra-sensitivity troponin I assay in patients with suspected myocardial infarction. *Int J Cardiol* 2019;**283**:35–40. https://doi.org/10.1016/j.ijcard.2018.12.001
30. Neumann JT, Sorensen NA, Schwemer T, Ojeda F, Bourry R, Sciacca V, *et al.* Diagnosis of myocardial infarction using a high-sensitivity troponin I 1-hour algorithm. *JAMA Cardiol* 2016;**1**:397–404. https://doi.org/10.1001/jamacardio.2016.0695
31. Reichlin T, Schindler C, Drexler B, Twerenbold R, Reiter M, Zellweger C, *et al.* One-hour rule-out and rule-in of acute myocardial infarction using high-sensitivity cardiac troponin T. *Arch Intern Med* 2012;**172**:1211–1218. https://doi.org/10.1001/archinternmed.2012.3698
32. Boeddinghaus J, Twerenbold R, Nestelberger T, Koechlin L, Wussler D, Meier M, *et al.* Clinical use of a new high-sensitivity cardiac troponin I assay in patients with suspected myocardial infarction. *Clin Chem* 2019;**65**:1426–1436. https://doi.org/10.1373/clinchem.2019.304725
33. Boeddinghaus J, Lopez-Ayala P, Nestelberger T, Koechlin L, Münch T, Miro O, *et al.* Prospective validation of the ESC 0/1h-algorithm using high-sensitivity cardiac troponin I. *Am J Cardiol* 2021;**158**:152–153. https://doi.org/10.1016/j.amjcard.2021.08.007
34. Boeddinghaus J, Nestelberger T, Twerenbold R, Neumann JT, Lindahl B, Giannitsis E, *et al.* Impact of age on the performance of the ESC 0/1h-algorithms for early diagnosis of myocardial infarction. *Eur Heart J* 2018;**39**:3780–3794. https://doi.org/10.1093/eurheartj/ehy514
35. Twerenbold R, Badertscher P, Boeddinghaus J, Nestelberger T, Wildi K, Puelacher C, *et al.* 0/1-Hour triage algorithm for myocardial infarction in patients with renal dysfunction. *Circulation* 2018;**137**:436–451. https://doi.org/10.1161/CIRCULATIONAHA.117.028901
36. Boeddinghaus J, Reichlin T, Cullen L, Greenslade JH, Parsonage WA, Hammett C, *et al.* Two-hour algorithm for triage toward rule-out and rule-in of acute myocardial infarction by use of high-sensitivity cardiac troponin I. *Clin Chem* 2016;**62**:494–504. https://doi.org/10.1373/clinchem.2015.249508
37. Wildi K, Cullen L, Twerenbold R, Greenslade JH, Parsonage W, Boeddinghaus J, *et al.* Direct comparison of 2 rule-out strategies for acute myocardial infarction: 2-h accelerated diagnostic protocol vs 2-h algorithm. *Clin Chem* 2017;**63**:1227–1236. https://doi.org/10.1373/clinchem.2016.268359
38. Nestelberger T, Boeddinghaus J, Greenslade J, Parsonage WA, Than M, Wussler D, *et al.* Two-hour algorithm for rapid triage of suspected acute myocardial infarction using a high-sensitivity cardiac troponin I assay. *Clin Chem* 2019;**65**:1437–1447. https://doi.org/10.1373/clinchem.2019.305193
39. Koechlin L, Boeddinghaus J, Nestelberger T, Lopez-Ayala P, Wussler D, Shrestha S, *et al.* Performance of the ESC 0/2h-algorithm using high-sensitivity cardiac troponin I in the early diagnosis of myocardial infarction. *Am Heart J* 2021;**242**:132–137. https://doi.org/10.1016/j.ahj.2021.08.008
40. Wildi K, Nelles B, Twerenbold R, Rubini Giménez M, Reichlin T, Singeisen H, *et al.* Safety and efficacy of the 0 h/3 h protocol for rapid rule out of myocardial infarction. *Am Heart J* 2016;**181**:16–25. https://doi.org/10.1016/j.ahj.2016.07.013
41. Badertscher P, Boeddinghaus J, Twerenbold R, Nestelberger T, Wildi K, Wussler D, *et al.* Direct comparison of the 0/1h and 0/3h algorithms for early rule-out of acute myocardial infarction. *Circulation* 2018;**137**:2536–2538. https://doi.org/10.1161/CIRCULATIONAHA.118.034260
42. Chapman AR, Anand A, Boeddinghaus J, Ferry AV, Sandeman D, Adamson PD, *et al.* Comparison of the efficacy and safety of early rule-out pathways for acute myocardial infarction. *Circulation* 2017;**135**:1586–1596. https://doi.org/10.1161/CIRCULATIONAHA.116.025021
43. Chapman AR, Fujisawa T, Lee KK, Andrews JP, Anand A, Sandeman D, *et al.* Novel high-sensitivity cardiac troponin I assay in patients with suspected acute coronary syndrome. *Heart* 2019;**105**:616–622. https://doi.org/10.1136/heartjnl-2018-314093
44. Chew DP, Lambrakis K, Blyth A, Seshadri A, Edmonds MJR, Briffa T, *et al.* A randomized trial of a 1-hour troponin T protocol in suspected acute coronary syndromes: the rapid assessment of possible acute coronary syndrome in the emergency department with high-sensitivity troponin T study (RAPID-TnT). *Circulation* 2019;**140**:1543–1556. https://doi.org/10.1161/CIRCULATIONAHA.119.042891
45. Nestelberger T, Wildi K, Boeddinghaus J, Twerenbold R, Reichlin T, Giménez MR, *et al.* Characterization of the observe zone of the ESC 2015 high-sensitivity cardiac troponin 0h/1h-algorithm for the early diagnosis of acute myocardial infarction. *Int J Cardiol* 2016;**207**:238–245. https://doi.org/10.1016/j.ijcard.2016.01.112
46. Lopez-Ayala P, Nestelberger T, Boeddinghaus J, Koechlin L, Ratmann PD, Strebel I, *et al.* Novel criteria for the observe-zone of the ESC 0/1h-hs-cTnT algorithm. *Circulation* 2021;**144**:773–787. https://doi.org/10.1161/CIRCULATIONAHA.120.052982
47. Granger CB, Goldberg RJ, Dabbous O, Pieper KS, Eagle KA, Cannon CP, *et al.* Predictors of hospital mortality in the global registry of acute coronary events. *Arch Intern Med* 2003;**163**:2345–2353. https://doi.org/10.1001/archinte.163.19.2345
48. Fox KA, Fitzgerald G, Puymirat E, Huang W, Carruthers K, Simon T, *et al.* Should patients with acute coronary disease be stratified for management according to their risk? Derivation, external validation and outcomes using the updated GRACE risk score. *BMJ Open* 2014;**4**:e004425. https://doi.org/10.1136/bmjopen-2013-004425
49. Wenzl FA, Kraler S, Ambler G, Weston C, Herzog SA, Räber L, *et al.* Sex-specific evaluation and redevelopment of the GRACE score in non-ST-segment elevation acute coronary syndromes in populations from the UK and Switzerland: a multinational analysis with external cohort validation. *Lancet* 2022;**400**:744–756. https://doi.org/10.1016/s0140-6736(22)01483-0
50. Effectiveness of intravenous thrombolytic treatment in acute myocardial infarction. *Lancet* 1986;**1**:397–402.
51. Boersma E, Maas AC, Deckers JW, Simoons ML. Early thrombolytic treatment in acute myocardial infarction: reappraisal of the golden hour. *Lancet* 1996;**348**:771–775. https://doi.org/10.1016/s0140-6736(96)02514-7
52. Keeley EC, Boura JA, Grines CL. Primary angioplasty versus intravenous thrombolytic therapy for acute myocardial infarction: a quantitative review of 23 randomised trials. *Lancet* 2003;**361**:13–20. https://doi.org/10.1016/s0140-6736(03)12113-7
53. Reichlin T, Twerenbold R, Maushart C, Reiter M, Moehring B, Schaub N, *et al.* Risk stratification in patients with unstable angina using absolute serial changes of 3 high-

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

sensitive troponin assays. *Am Heart J* 2013;**165**:371–378.e3. https://doi.org/10.1016/j.ahj.2012.11.010
54. Anand A, Lee KK, Chapman AR, Ferry AV, Adamson PD, Strachan FE, *et al.* High-sensitivity cardiac troponin on presentation to rule out myocardial infarction: a stepped-wedge cluster randomized controlled trial. *Circulation* 2021;**143**:2214–2224. https://doi.org/10.1161/CIRCULATIONAHA.120.052380
55. Mueller C. Biomarkers and acute coronary syndromes: an update. *Eur Heart J* 2014;**35**:552–556. https://doi.org/10.1093/eurheartj/eht530
56. Roffi M, Patrono C, Collet JP, Mueller C, Valgimigli M, Andreotti F, *et al.* 2015 ESC Guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation: Task Force for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation of the European Society of Cardiology (ESC). *Eur Heart J* 2016;**37**:267–315. https://doi.org/10.1093/eurheartj/ehv320
57. Mueller C, Giannitsis E, Mockel M, Huber K, Mair J, Plebani M, *et al.* Rapid rule out of acute myocardial infarction: novel biomarker-based strategies. *Eur Heart J Acute Cardiovasc Care* 2017;**6**:218–222. https://doi.org/10.1177/2048872616653229
58. Mockel M, Giannitsis E, Mueller C, Huber K, Jaffe AS, Mair J, *et al.* Rule-in of acute myocardial infarction: focus on troponin. *Eur Heart J Acute Cardiovasc Care* 2017;**6**:212–217. https://doi.org/10.1177/2048872616653228
59. Roe MT, Harrington RA, Prosper DM, Pieper KS, Bhatt DL, Lincoff AM, *et al.* Clinical and therapeutic profile of patients presenting with acute coronary syndromes who do not have significant coronary artery disease. The Platelet Glycoprotein IIb/IIIa in Unstable Angina: Receptor Suppression Using Integrilin Therapy (PURSUIT) trial investigators. *Circulation* 2000;**102**:1101–1106. https://doi.org/10.1161/01.cir.102.10.1101
60. Westermann D, Neumann JT, Sorensen NA, Blankenberg S. High-sensitivity assays for troponin in patients with cardiac disease. *Nat Rev Cardiol* 2017;**14**:472–483. https://doi.org/10.1038/nrcardio.2017.48
61. Rubini Gimenez M, Twerenbold R, Reichlin T, Wildi K, Haaf P, Schaefer M, *et al.* Direct comparison of high-sensitivity-cardiac troponin I vs. T for the early diagnosis of acute myocardial infarction. *Eur Heart J* 2014;**35**:2303–2311. https://doi.org/10.1093/eurheartj/ehu188
62. Boeddinghaus J, Nestelberger T, Twerenbold R, Koechlin L, Meier M, Troester V, *et al.* High-sensitivity cardiac troponin I assay for early diagnosis of acute myocardial infarction. *Clin Chem* 2019;**65**:893–904. https://doi.org/10.1373/clinchem.2018.300061
63. Keller T, Zeller T, Peetz D, Tzikas S, Roth A, Czyz E, *et al.* Sensitive troponin I assay in early diagnosis of acute myocardial infarction. *N Engl J Med* 2009;**361**:868–877. https://doi.org/10.1056/NEJMoa0903515
64. Collinson PO, Saenger AK, Apple FS; IFCC C-CB. High sensitivity, contemporary and point-of-care cardiac troponin assays: educational aids developed by the IFCC Committee on Clinical Application of Cardiac Bio-Markers. *Clin Chem Lab Med* 2019;**57**:623–632. https://doi.org/10.1515/cclm-2018-1211
65. Camaro C, Aarts GWA, Adang EMM, van Hout R, Brok G, Hoare A, *et al.* Rule-out of non-ST-segment elevation acute coronary syndrome by a single, pre-hospital troponin measurement: a randomized trial. *Eur Heart J* 2023;**44**:1705–1714. https://doi.org/10.1093/eurheartj/ehad056
66. Pickering JW, Young JM, George PM, Watson AS, Aldous SJ, Troughton RW, *et al.* Validity of a novel point-of-care troponin assay for single-test rule-out of acute myocardial infarction. *JAMA Cardiol* 2018;**3**:1108–1112. https://doi.org/10.1001/jamacardio.2018.3368
67. Apple FS, Schulz K, Schmidt CW, van Domburg TSY, Fonville JM, de Theije FK. Determination of sex-specific 99th percentile upper reference limits for a point of care high sensitivity cardiac troponin I assay. *Clin Chem Lab Med* 2021;**59**:1574–1578. https://doi.org/10.1515/cclm-2021-0262
68. Sorensen NA, Neumann JT, Ojeda F, Giannitsis E, Spanuth E, Blankenberg S, *et al.* Diagnostic evaluation of a high-sensitivity troponin I point-of-care assay. *Clin Chem* 2019;**65**:1592–1601. https://doi.org/10.1373/clinchem.2019.307405
69. Azmy C, Guerard S, Bonnet X, Gabrielli F, Skalli W. EOS orthopaedic imaging system to study patellofemoral kinematics: assessment of uncertainty. *Orthop Traumatol Surg Res* 2010;**96**:28–36. https://doi.org/10.1016/j.rcot.2009.12.003
70. Chapman AR, Lee KK, McAllister DA, Cullen L, Greenslade JH, Parsonage W, *et al.* Association of high-sensitivity cardiac troponin I concentration with cardiac outcomes in patients with suspected acute coronary syndrome. *JAMA* 2017;**318**:1913–1924. https://doi.org/10.1001/jama.2017.17488
71. Hillinger P, Twerenbold R, Wildi K, Rubini Gimenez M, Jaeger C, Boeddinghaus J, *et al.* Gender-specific uncertainties in the diagnosis of acute coronary syndrome. *Clin Res Cardiol* 2017;**106**:28–37. https://doi.org/10.1007/s00392-016-1020-y
72. Miller-Hodges E, Anand A, Shah ASV, Chapman AR, Gallacher P, Lee KK, *et al.* High-sensitivity cardiac troponin and the risk stratification of patients with renal impairment presenting with suspected acute coronary syndrome. *Circulation* 2018;**137**:425–435. https://doi.org/10.1161/CIRCULATIONAHA.117.030320
73. Twerenbold R, Neumann JT, Sorensen NA, Ojeda F, Karakas M, Boeddinghaus J, *et al.* Prospective validation of the 0/1-h algorithm for early diagnosis of myocardial infarction. *J Am Coll Cardiol* 2018;**72**:620–632. https://doi.org/10.1016/j.jacc.2018.05.040
74. Rubini Gimenez M, Twerenbold R, Boeddinghaus J, Nestelberger T, Puelacher C, Hillinger P, *et al.* Clinical effect of sex-specific cutoff values of high-sensitivity cardiac troponin T in suspected myocardial infarction. *JAMA Cardiol* 2016;**1**:912–920. https://doi.org/10.1001/jamacardio.2016.2882
75. Mueller-Hennessen M, Lindahl B, Giannitsis E, Biener M, Vafaie M, deFilippi CR, *et al.* Diagnostic and prognostic implications using age- and gender-specific cut-offs for high-sensitivity cardiac troponin T-sub-analysis from the TRAPID-AMI study. *Int J Cardiol* 2016;**209**:26–33. https://doi.org/10.1016/j.ijcard.2016.01.213
76. Sorensen NA, Neumann JT, Ojeda F, Schäfer S, Magnussen C, Keller T, *et al.* Relations of sex to diagnosis and outcomes in acute coronary syndrome. *J Am Heart Assoc* 2018;**7**:e007297. https://doi.org/10.1161/JAHA.117.007297
77. Rubini Gimenez M, Badertscher P, Twerenbold R, Boeddinghaus J, Nestelberger T, Wussler D, *et al.* Impact of the US Food and Drug Administration-approved sex-specific cutoff values for high-sensitivity cardiac troponin T to diagnose myocardial infarction. *Circulation* 2018;**137**:1867–1869. https://doi.org/10.1161/circulationaha.117.031940
78. Lee KK, Ferry AV, Anand A, Strachan FE, Chapman AR, Kimenai DM, *et al.* Sex-specific thresholds of high-sensitivity troponin in patients with suspected acute coronary syndrome. *J Am Coll Cardiol* 2019;**74**:2032–2043. https://doi.org/10.1016/j.jacc.2019.07.082
79. Kimenai DM, Lindahl B, Jernberg T, Bekers O, Meex SJR, Eggers KM. Sex-specific effects of implementing a high-sensitivity troponin I assay in patients with suspected acute coronary syndrome: results from SWEDEHEART registry. *Sci Rep* 2020;**10**:15227. https://doi.org/10.1038/s41598-020-72204-2
80. Peacock WF, Baumann BM, Rivers EJ, Davis TE, Handy B, Jones CW, *et al.* Using sex-specific cutoffs for high-sensitivity cardiac troponin T to diagnose acute myocardial infarction. *Acad Emerg Med* 2021;**28**:463–466. https://doi.org/10.1111/acem.14098
81. Zhao Y, Izadnegahdar M, Lee MK, Kavsak PA, Singer J, Scheuermeyer F, *et al.* High-sensitivity cardiac troponin-optimizing the diagnosis of acute myocardial infarction/injury in women (CODE-MI): rationale and design for a multicenter, stepped-wedge, cluster-randomized trial. *Am Heart J* 2020;**229**:18–28. https://doi.org/10.1016/j.ahj.2020.06.013
82. Reichlin T, Irfan A, Twerenbold R, Reiter M, Hochholzer W, Burkhalter H, *et al.* Utility of absolute and relative changes in cardiac troponin concentrations in the early diagnosis of acute myocardial infarction. *Circulation* 2011;**124**:136–145. https://doi.org/10.1161/CIRCULATIONAHA.111.023937
83. Wildi K, Boeddinghaus J, Nestelberger T, Twerenbold R, Badertscher P, Wussler D, *et al.* Comparison of fourteen rule-out strategies for acute myocardial infarction. *Int J Cardiol* 2019;**283**:41–47. https://doi.org/10.1016/j.ijcard.2018.11.140
84. Ambavane A, Lindahl B, Giannitsis E, Roiz J, Mendivil J, Frankenstein L, *et al.* Economic evaluation of the one-hour rule-out and rule-in algorithm for acute myocardial infarction using the high-sensitivity cardiac troponin T assay in the emergency department. *PLoS One* 2017;**12**:e0187662. https://doi.org/10.1371/journal.pone.0187662
85. Boeddinghaus J, Nestelberger T, Twerenbold R, Wildi K, Badertscher P, Cupa J, *et al.* Direct comparison of 4 very early rule-out strategies for acute myocardial infarction using high-sensitivity cardiac troponin I. *Circulation* 2017;**135**:1597–1611. https://doi.org/10.1161/CIRCULATIONAHA.116.025661
86. Ljung L, Lindahl B, Eggers KM, Frick M, Linder R, Löfmark HB, *et al.* A rule-out strategy based on high-sensitivity troponin and HEART score reduces hospital admissions. *Ann Emerg Med* 2019;**73**:491–499. https://doi.org/10.1016/j.annemergmed.2018.11.039
87. Odqvist M, Andersson PO, Tygesen H, Eggers KM, Holzmann MJ. High-sensitivity troponins and outcomes after myocardial infarction. *J Am Coll Cardiol* 2018;**71**:2616–2624. https://doi.org/10.1016/j.jacc.2018.03.515
88. Twerenbold R, Jaeger C, Rubini Gimenez M, Wildi K, Reichlin T, Nestelberger T, *et al.* Impact of high-sensitivity cardiac troponin on use of coronary angiography, cardiac stress testing, and time to discharge in suspected acute myocardial infarction. *Eur Heart J* 2016;**37**:3324–3332. https://doi.org/10.1093/eurheartj/ehw232
89. Greenslade J, Cho E, Van Hise C, Hawkins T, Parsonage W, Ungerer J, *et al.* Evaluating rapid rule-out of acute myocardial infarction using a high-sensitivity cardiac troponin I assay at presentation. *Clin Chem* 2018;**64**:820–829. https://doi.org/10.1373/clinchem.2017.283887
90. Pickering JW, Than MP, Cullen L, Aldous S, ter Avest E, Body R, *et al.* Rapid rule-out of acute myocardial infarction with a single high-sensitivity cardiac troponin T measurement below the limit of detection: a collaborative meta-analysis. *Ann Intern Med* 2017;**166**:715–724. https://doi.org/10.7326/M16-2562
91. Shah AS, Anand A, Sandoval Y, Lee KK, Smith SW, Adamson PD, *et al.* High-sensitivity cardiac troponin I at presentation in patients with suspected acute coronary syndrome: a cohort study. *Lancet* 2015;**386**:2481–2488. https://doi.org/10.1016/S0140-6736(15)00391-8
92. Boeddinghaus J, Nestelberger T, Lopez-Ayala P, Koechlin L, Buechi M, Miro O, *et al.* Diagnostic performance of the European Society of Cardiology 0/1-h algorithms in late presenters. *J Am Coll Cardiol* 2021;**77**:1264–1267. https://doi.org/10.1016/j.jacc.2021.01.004
93. Mueller C, Giannitsis E, Christ M, Ordóñez-Llanos J, deFilippi C, McCord J, *et al.* Multicenter evaluation of a 0-hour/1-hour algorithm in the diagnosis of myocardial

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

infarction with high-sensitivity cardiac troponin T. *Ann Emerg Med* 2016;**68**:76–87.e4. https://doi.org/10.1016/j.annemergmed.2015.11.013
94. Stoyanov KM, Hund H, Biener M, Gandowitz J, Riedle C, Löhr J, *et al.* RAPID-CPU: a prospective study on implementation of the ESC 0/1-hour algorithm and safety of discharge after rule-out of myocardial infarction. *Eur Heart J Acute Cardiovasc Care* 2019;**9**:39–51. https://doi.org/10.1177/2048872619861911
95. Twerenbold R, Costabel JP, Nestelberger T, Campos R, Wussler D, Arbucci R, *et al.* Outcome of applying the ESC 0/1-hour algorithm in patients with suspected myocardial infarction. *J Am Coll Cardiol* 2019;**74**:483–494. https://doi.org/10.1016/j.jacc.2019.05.046
96. Nestelberger T, Boeddinghaus J, Wussler D, Twerenbold R, Badertscher P, Wildi K, *et al.* Predicting major adverse events in patients with acute myocardial infarction. *J Am Coll Cardiol* 2019;**74**:842–854. https://doi.org/10.1016/j.jacc.2019.06.025
97. Rubini Gimenez M, Wildi K, Wussler D, Koechlin L, Boeddinghaus J, Nestelberger T, *et al.* Early kinetics of cardiac troponin in suspected acute myocardial infarction. *Rev Esp Cardiol (Engl Ed)* 2021;**74**:502–509. https://doi.org/10.1016/j.rec.2020.04.008
98. Chiang CH, Chiang CH, Lee GH, Gi W-T, Wu Y-K, Huang S-S, *et al.* Safety and efficacy of the European Society of Cardiology 0/1-hour algorithm for diagnosis of myocardial infarction: systematic review and meta-analysis. *Heart* 2020;**106**:985–991. https://doi.org/10.1136/heartjnl-2019-316343
99. Vigen R, Kutscher P, Fernandez F, Yu A, Bertulfo B, Hashim IA, *et al.* Evaluation of a novel rule-out myocardial infarction protocol incorporating high-sensitivity troponin T in a US hospital. *Circulation* 2018;**138**:2061–2063. https://doi.org/10.1161/circulationaha.118.033861
100. Katus H, Ziegler A, Ekinci O, Giannitsis E, Stough WG, Achenbach S, *et al.* Early diagnosis of acute coronary syndrome. *Eur Heart J* 2017;**38**:3049–3055. https://doi.org/10.1093/eurheartj/ehx492
101. Koechlin L, Boeddinghaus J, Lopez-Ayala P, Nestelberger T, Wussler D, Mais F, *et al.* Diagnostic discrimination of a novel high-sensitivity cardiac troponin I assay and derivation/validation of an assay-specific 0/1h-algorithm. *Am Heart J* 2023;**255**:58–70. https://doi.org/10.1016/j.ahj.2022.10.007
102. Kaier TE, Twerenbold R, Puelacher C, Marjot J, Imambaccus N, Boeddinghaus J, *et al.* Direct comparison of cardiac myosin-binding protein C with cardiac troponins for the early diagnosis of acute myocardial infarction. *Circulation* 2017;**136**:1495–1508. https://doi.org/10.1161/CIRCULATIONAHA.117.028084
103. Boeddinghaus J, Reichlin T, Nestelberger T, Twerenbold R, Meili Y, Wildi K, *et al.* Early diagnosis of acute myocardial infarction in patients with mild elevations of cardiac troponin. *Clin Res Cardiol* 2017;**106**:457–467. https://doi.org/10.1007/s00392-016-1075-9
104. Hillinger P, Twerenbold R, Jaeger C, Wildi K, Reichlin T, Gimenez MR, *et al.* Optimizing early rule-out strategies for acute myocardial infarction: utility of 1-hour copeptin. *Clin Chem* 2015;**61**:1466–1474. https://doi.org/10.1373/clinchem.2015.242743
105. Keller T, Tzikas S, Zeller T, Czyz E, Lillpopp L, Ojeda FM, *et al.* Copeptin improves early diagnosis of acute myocardial infarction. *J Am Coll Cardiol* 2010;**55**:2096–2106. https://doi.org/10.1016/j.jacc.2010.01.029
106. Möckel M, Searle J, Hamm C, Slagman A, Blankenberg S, Huber K, *et al.* Early discharge using single cardiac troponin and copeptin testing in patients with suspected acute coronary syndrome (ACS): a randomized, controlled clinical process study. *Eur Heart J* 2015;**36**:369–376. https://doi.org/10.1093/eurheartj/ehu178
107. Mueller C, Mockel M, Giannitsis E, Huber K, Mair J, Plebani M, *et al.* Use of copeptin for rapid rule-out of acute myocardial infarction. *Eur Heart J Acute Cardiovasc Care* 2018;**7**:570–576. https://doi.org/10.1177/2048872617710791
108. Mueller-Hennessen M, Lindahl B, Giannitsis E, Vafaie M, Biener M, Haushofer AC, *et al.* Combined testing of copeptin and high-sensitivity cardiac troponin T at presentation in comparison to other algorithms for rapid rule-out of acute myocardial infarction. *Int J Cardiol* 2019;**276**:261–267. https://doi.org/10.1016/j.ijcard.2018.10.084
109. Reichlin T, Hochholzer W, Stelzig C, Laule K, Freidank H, Morgenthaler NG, *et al.* Incremental value of copeptin for rapid rule out of acute myocardial infarction. *J Am Coll Cardiol* 2009;**54**:60–68. https://doi.org/10.1016/j.jacc.2009.01.076
110. Stallone F, Schoenenberger AW, Puelacher C, Rubini Gimenez M, Walz B, Naduvilekoot Devasia A, *et al.* Incremental value of copeptin in suspected acute myocardial infarction very early after symptom onset. *Eur Heart J Acute Cardiovasc Care* 2016;**5**:407–415. https://doi.org/10.1177/2048872616641289
111. Vargas KG, Kassem M, Mueller C, Wojta J, Huber K. Copeptin for the early rule-out of non-ST-elevation myocardial infarction. *Int J Cardiol* 2016;**223**:797–804. https://doi.org/10.1016/j.ijcard.2016.08.304
112. Wildi K, Zellweger C, Twerenbold R, Jaeger C, Reichlin T, Haaf P, *et al.* Incremental value of copeptin to highly sensitive cardiac troponin I for rapid rule-out of myocardial infarction. *Int J Cardiol* 2015;**190**:170–176. https://doi.org/10.1016/j.ijcard.2015.04.133
113. Zellweger C, Wildi K, Twerenbold R, Reichlin T, Naduvilekoot A, Neuhaus JD, *et al.* Use of copeptin and high-sensitive cardiac troponin T for diagnosis and prognosis in patients with diabetes mellitus and suspected acute myocardial infarction. *Int J Cardiol* 2015;**190**:190–197. https://doi.org/10.1016/j.ijcard.2015.04.134
114. Restan IZ, Sanchez AY, Steiro OT, Lopez-Ayala P, Tjora HL, Langørgen J, *et al.* Adding stress biomarkers to high-sensitivity cardiac troponin for rapid non-ST-elevation myocardial infarction rule-out protocols. *Eur Heart J Acute Cardiovasc Care* 2022;**11**:201–212. https://doi.org/10.1093/ehjacc/zuab124
115. Dedic A, Lubbers MM, Schaap J, Lammers J, Lamfers EJ, Rensing BJ, *et al.* Coronary CT angiography for suspected ACS in the era of high-sensitivity troponins: randomized multicenter study. *J Am Coll Cardiol* 2016;**67**:16–26. https://doi.org/10.1016/j.jacc.2015.10.045
116. Hoffmann U, Truong QA, Schoenfeld DA, Chou ET, Woodard PK, Nagurney JT, *et al.* Coronary CT angiography versus standard evaluation in acute chest pain. *N Engl J Med* 2012;**367**:299–308. https://doi.org/10.1056/NEJMoa1201161
117. Gray AJ, Roobottom C, Smith JE, Goodacre S, Oatey K, O'Brien R, *et al.* Early computed tomography coronary angiography in patients with suspected acute coronary syndrome: randomised controlled trial. *BMJ* 2021;**374**:n2106. https://doi.org/10.1136/bmj.n2106
118. Lee KK, Bularga A, O'Brien R, Ferry AV, Doudesis D, Fujisawa T, *et al.* Troponin-guided coronary computed tomographic angiography after exclusion of myocardial infarction. *J Am Coll Cardiol* 2021;**78**:1407–1417. https://doi.org/10.1016/j.jacc.2021.07.055
119. Kofoed KF, Engstrom T, Sigvardsen PE, Linde JJ, Torp-Pedersen C, de Knegt M, *et al.* Prognostic value of coronary CT angiography in patients with non-ST-segment elevation acute coronary syndromes. *J Am Coll Cardiol* 2021;**77**:1044–1052. https://doi.org/10.1016/j.jacc.2020.12.037
120. Linde JJ, Hove JD, Sorgaard M, Kelbæk H, Jensen GB, Kühl JT, *et al.* Long-term clinical impact of coronary CT angiography in patients with recent acute-onset chest pain: the randomized controlled CATCH trial. *JACC Cardiovasc Imaging* 2015;**8**:1404–1413. https://doi.org/10.1016/j.jcmg.2015.07.015
121. Linde JJ, Kelbæk H, Hansen TF, Sigvardsen PE, Torp-Pedersen C, Bech J, *et al.* Coronary CT angiography in patients with non-ST-segment elevation acute coronary syndrome. *J Am Coll Cardiol* 2020;**75**:453–463. https://doi.org/10.1016/j.jacc.2019.12.012
122. Samad Z, Hakeem A, Mahmood SS, Pieper K, Patel MR, Simel DL, *et al.* A meta-analysis and systematic review of computed tomography angiography as a diagnostic triage tool for patients with chest pain presenting to the emergency department. *J Nucl Cardiol* 2012;**19**:364–376. https://doi.org/10.1007/s12350-012-9520-2
123. Litt HI, Gatsonis C, Snyder B, Singh H, Miller CD, Entrikin DW, *et al.* CT angiography for safe discharge of patients with possible acute coronary syndromes. *N Engl J Med* 2012;**366**:1393–1403. https://doi.org/10.1056/NEJMoa1201163
124. Hulten E, Pickett C, Bittencourt MS, Villines TC, Petrillo S, Di Carli MF, *et al.* Outcomes after coronary computed tomography angiography in the emergency department: a systematic review and meta-analysis of randomized, controlled trials. *J Am Coll Cardiol* 2013;**61**:880–892. https://doi.org/10.1016/j.jacc.2012.11.061
125. Gaibazzi N, Reverberi C, Badano L. Usefulness of contrast stress-echocardiography or exercise-electrocardiography to predict long-term acute coronary syndromes in patients presenting with chest pain without electrocardiographic abnormalities or 12-hour troponin elevation. *Am J Cardiol* 2011;**107**:161–167. https://doi.org/10.1016/j.amjcard.2010.08.066
126. Lim SH, Anantharaman V, Sundram F, Chan ES-Y, Ang ES, Yo SL, *et al.* Stress myocardial perfusion imaging for the evaluation and triage of chest pain in the emergency department: a randomized controlled trial. *J Nucl Cardiol* 2013;**20**:1002–1012. https://doi.org/10.1007/s12350-013-9736-9
127. Nabi F, Kassi M, Muhyieddeen K, Chang SM, Xu J, Peterson LE, *et al.* Optimizing evaluation of patients with low-to-intermediate-risk acute chest pain: a randomized study comparing stress myocardial perfusion tomography incorporating stress-only imaging versus cardiac CT. *J Nucl Med* 2016;**57**:378–384. https://doi.org/10.2967/jnumed.115.166595
128. Jackson AM, Zhang R, Findlay I, Robertson K, Lindsay M, Morris T, *et al.* Healthcare disparities for women hospitalized with myocardial infarction and angina. *Eur Heart J Qual Care Clin Outcomes* 2020;**6**:156–165. https://doi.org/10.1093/ehjqcco/qcz040
129. Terkelsen CJ, Sørensen JT, Maeng M, Jensen LO, Tilsted HH, Trautner S, *et al.* System delay and mortality among patients with STEMI treated with primary percutaneous coronary intervention. *JAMA* 2010;**304**:763–771. https://doi.org/10.1001/jama.2010.1139
130. Jortveit J, Pripp AH, Halvorsen S. Outcomes after delayed primary percutaneous coronary intervention vs. pharmaco-invasive strategy in ST-segment elevation myocardial infarction in Norway. *Eur Heart J Cardiovasc Pharmacother* 2022;**8**:442–451. https://doi.org/10.1093/ehjcvp/pvab041
131. Larsen AI, Løland KH, Hovland S, Bleie Ø, Eek C, Fossum E, *et al.* Guideline-recommended time less than 90 minutes from ECG to primary percutaneous coronary intervention for ST-segment-elevation myocardial infarction is associated with major survival benefits, especially in octogenarians: a contemporary report in 11 226 patients from NORIC. *J Am Heart Assoc* 2022;**11**:e024849. https://doi.org/10.1161/jaha.122.024849
132. Fordyce CB, Al-Khalidi HR, Jollis JG, Roettig ML, Gu J, Bagai A, *et al.* Association of rapid care process implementation on reperfusion times across multiple ST-segment-elevation myocardial infarction networks. *Circ Cardiovasc Interv* 2017;**10:**e004061. https://doi.org/10.1161/circinterventions.116.004061
133. Stowens JC, Sonnad SS, Rosenbaum RA. Using EMS dispatch to trigger STEMI alerts decreases door-to-balloon times. *West J Emerg Med* 2015;**16**:472–480. https://doi.org/10.5811/westjem.2015.4.24248

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

134. Squire BT, Tamayo-Sarver JH, Rashi P, Koenig W, Niemann JT. Effect of prehospital cardiac catheterization lab activation on door-to-balloon time, mortality, and false-positive activation. *Prehosp Emerg Care* 2014;**18**:1–8. https://doi.org/10.3109/10903127.2013.836263
135. Shavadia JS, Roe MT, Chen AY, Lucas J, Fanaroff AC, Kochar A, *et al.* Association between cardiac catheterization laboratory pre-activation and reperfusion timing metrics and outcomes in patients with ST-segment elevation myocardial infarction undergoing primary percutaneous coronary intervention: a report from the ACTION registry. *JACC Cardiovasc Interv* 2018;**11**:1837–1847. https://doi.org/10.1016/j.jcin.2018.07.020
136. Kontos MC, Gunderson MR, Zegre-Hemsey JK, Lange DC, French WJ, Henry TD, *et al.* Prehospital activation of hospital resources (PreAct) ST-segment-elevation myocardial infarction (STEMI): a standardized approach to prehospital activation and direct to the catheterization laboratory for STEMI recommendations from the American Heart Association's mission: lifeline program. *J Am Heart Assoc* 2020;**9**:e011963. https://doi.org/10.1161/jaha.119.011963
137. Bagai A, Jollis JG, Dauerman HL, Peng SA, Rokos IC, Bates ER, *et al.* Emergency department bypass for ST-segment-elevation myocardial infarction patients identified with a prehospital electrocardiogram: a report from the American Heart Association Mission: Lifeline program. *Circulation* 2013;**128**:352–359. https://doi.org/10.1161/circulationaha.113.002339
138. Scholz KH, Friede T, Meyer T, Jacobshagen C, Lengenfelder B, Jung J, *et al.* Prognostic significance of emergency department bypass in stable and unstable patients with ST-segment elevation myocardial infarction. *Eur Heart J Acute Cardiovasc Care* 2020;**9**:34–44. https://doi.org/10.1177/2048872618813907
139. Meisel SR, Kleiner-Shochat M, Abu-Fanne R, Frimerman A, Danon A, Minha S, *et al.* Direct admission of patients with ST-segment-elevation myocardial infarction to the catheterization laboratory shortens pain-to-balloon and door-to-balloon time intervals but only the pain-to-balloon interval impacts short- and long-term mortality. *J Am Heart Assoc* 2021;**10**:e018343. https://doi.org/10.1161/jaha.120.018343
140. Wang TY, Nallamothu BK, Krumholz HM, Li S, Roe MT, Jollis JG, *et al.* Association of door-in to door-out time with reperfusion delays and outcomes among patients transferred for primary percutaneous coronary intervention. *JAMA* 2011;**305**:2540–2547. https://doi.org/10.1001/jama.2011.862
141. Huber K, De Caterina R, Kristensen SD, Verheugt FWA, Montalescot G, Maestro LB, *et al.* Pre-hospital reperfusion therapy: a strategy to improve therapeutic outcome in patients with ST-elevation myocardial infarction. *Eur Heart J* 2005;**26**:2063–2074. https://doi.org/10.1093/eurheartj/ehi413
142. Welsh RC, Chang W, Goldstein P, Adgey J, Granger CB, Verheugt FW, *et al.* Time to treatment and the impact of a physician on prehospital management of acute ST elevation myocardial infarction: insights from the ASSENT-3 PLUS trial. *Heart* 2005;**91**:1400–1406. https://doi.org/10.1136/hrt.2004.054510
143. Jollis JG, Al-Khalidi HR, Roettig ML, Berger PB, Corbett CC, Doerfler SM, *et al.* Impact of regionalization of ST-segment-elevation myocardial infarction care on treatment times and outcomes for emergency medical services-transported patients presenting to hospitals with percutaneous coronary intervention: mission: lifeline accelerator-2. *Circulation* 2018;**137**:376–387. https://doi.org/10.1161/circulationaha.117.032446
144. Fosbol EL, Granger CB, Jollis JG, Monk L, Lin L, Lytle BL, *et al.* The impact of a statewide pre-hospital STEMI strategy to bypass hospitals without percutaneous coronary intervention capability on treatment times. *Circulation* 2013;**127**:604–612. https://doi.org/10.1161/circulationaha.112.118463
145. Kalla K, Christ G, Karnik R, Malzer R, Norman G, Prachar H, *et al.* Implementation of guidelines improves the standard of care: the Viennese registry on reperfusion strategies in ST-elevation myocardial infarction (Vienna STEMI registry). *Circulation* 2006;**113**:2398–2405. https://doi.org/10.1161/circulationaha.105.586198
146. Henry TD, Sharkey SW, Burke MN, Chavez IJ, Graham KJ, Henry CR, *et al.* A regional system to provide timely access to percutaneous coronary intervention for ST-elevation myocardial infarction. *Circulation* 2007;**116**:721–728. https://doi.org/10.1161/circulationaha.107.694141
147. Le May MR, So DY, Dionne R, Glover CA, Froeschl MPV, Wells GA, *et al.* A citywide protocol for primary PCI in ST-segment elevation myocardial infarction. *N Engl J Med* 2008;**358**:231–240. https://doi.org/10.1056/NEJMoa073102
148. Hofmann R, James SK, Jernberg T, Lindahl B, Erlinge D, Witt N, *et al.* Oxygen therapy in suspected acute myocardial infarction. *N Engl J Med* 2017;**377**:1240–1249. https://doi.org/10.1056/NEJMoa1706222
149. Stewart RAH, Jones P, Dicker B, Jiang Y, Smith T, Swain A, *et al.* High flow oxygen and risk of mortality in patients with a suspected acute coronary syndrome: pragmatic, cluster randomised, crossover trial. *BMJ* 2021;**372**:n355. https://doi.org/10.1136/bmj.n355
150. Henrikson CA, Howell EE, Bush DE, Miles JS, Meininger GR, Friedlander T, *et al.* Chest pain relief by nitroglycerin does not predict active coronary artery disease. *Ann Intern Med* 2003;**139**:979–986. https://doi.org/10.7326/0003-4819-139-12-200312160-00007
151. Charpentier S, Galinski M, Bounes V, Ricard-Hibon A, El-Khoury C, Elbaz M, *et al.* Nitrous oxide/oxygen plus acetaminophen versus morphine in ST elevation myocardial infarction: open-label, cluster-randomized, non-inferiority study. *Scand J Trauma Resusc Emerg Med* 2020;**28**:36. https://doi.org/10.1186/s13049-020-00731-y
152. Silvain J, Storey RF, Cayla G, Esteve J-B, Dillinger J-G, Rousseau H, *et al.* P2Y12 receptor inhibition and effect of morphine in patients undergoing primary PCI for ST-segment elevation myocardial infarction. The PRIVATE-ATLANTIC study. *Thromb Haemost* 2016;**116**:369–378. https://doi.org/10.1160/th15-12-0944
153. Parodi G, Bellandi B, Xanthopoulou I, Capranzano P, Capodanno D, Valenti R, *et al.* Morphine is associated with a delayed activity of oral antiplatelet agents in patients with ST-elevation acute myocardial infarction undergoing primary percutaneous coronary intervention. *Circ Cardiovasc Interv* 2015;**8**:e001593. https://doi.org/10.1161/circinterventions.114.001593
154. Saad M, Meyer-Saraei R, de Waha-Thiele S, Stiermaier T, Graf T, Fuernau G, *et al.* Impact of morphine treatment with and without metoclopramide coadministration on ticagrelor-induced platelet inhibition in acute myocardial infarction: the randomized MonAMI trial. *Circulation* 2020;**141**:1354–1356. https://doi.org/10.1161/circulationaha.119.042816
155. Stiermaier T, Schaefer P, Meyer-Saraei R, Saad M, de Waha-Thiele S, Pöss J, *et al.* Impact of morphine treatment with and without metoclopramide coadministration on myocardial and microvascular injury in acute myocardial infarction: insights from the randomized MonAMI trial. *J Am Heart Assoc* 2021;**10**:e018881. https://doi.org/10.1161/jaha.120.018881
156. Kubica J, Adamski P, Ostrowska M, Sikora J, Kubica JM, Sroka WD, *et al.* Morphine delays and attenuates ticagrelor exposure and action in patients with myocardial infarction: the randomized, double-blind, placebo-controlled IMPRESSION trial. *Eur Heart J* 2016;**37**:245–252. https://doi.org/10.1093/eurheartj/ehv547
157. Zhang Y, Wang N, Gu Q. Effects of morphine on P2Y(12) platelet inhibitors in patients with acute myocardial infarction: a meta-analysis. *Am J Emerg Med* 2021;**41**:219–228. https://doi.org/10.1016/j.ajem.2020.11.003
158. Furtado RHM, Nicolau JC, Guo J, Im K, White JA, Sabatine MS, *et al.* Morphine and cardiovascular outcomes among patients with non-ST-segment elevation acute coronary syndromes undergoing coronary angiography. *J Am Coll Cardiol* 2020;**75**:289–300. https://doi.org/10.1016/j.jacc.2019.11.035
159. Kubica A, Kosobucka A, Niezgoda P, Adamski P, Buszko K, Lesiak M, *et al.* ANalgesic Efficacy and safety of MOrphiNe versus methoxyflurane in patients with acute myocardial infarction: the rationale and design of the ANEMON-SIRIO 3 study: a multicentre, open-label, phase II, randomised clinical trial. *BMJ Open* 2021;**11**:e043330. https://doi.org/10.1136/bmjopen-2020-043330
160. Batchelor R, Liu DH, Bloom J, Noaman S, Chan W. Association of periprocedural intravenous morphine use on clinical outcomes in ST-elevation myocardial infarction (STEMI) treated by primary percutaneous coronary intervention: systematic review and meta-analysis. *Catheter Cardiovasc Interv* 2020;**96**:76–88. https://doi.org/10.1002/ccd.28561
161. Bonin M, Mewton N, Roubille F, Morel O, Cayla G, Angoulvant D, *et al.* Effect and safety of morphine use in acute anterior ST-segment elevation myocardial infarction. *J Am Heart Assoc* 2018;**7**:e006833. https://doi.org/10.1161/jaha.117.006833
162. Clemente-Moragón A, Gómez M, Villena-Gutiérrez R, Lalama DV, García-Prieto J, Martínez F, *et al.* Metoprolol exerts a non-class effect against ischaemia-reperfusion injury by abrogating exacerbated inflammation. *Eur Heart J* 2020;**41**:4425–4440. https://doi.org/10.1093/eurheartj/ehaa733
163. Ibanez B, Macaya C, Sánchez-Brunete V, Pizarro G, Fernández-Friera L, Mateos A, *et al.* Effect of early metoprolol on infarct size in ST-segment-elevation myocardial infarction patients undergoing primary percutaneous coronary intervention: the effect of metoprolol in cardioprotection during an acute myocardial infarction (METOCARD-CNIC) trial. *Circulation* 2013;**128**:1495–1503. https://doi.org/10.1161/circulationaha.113.003653
164. Roolvink V, Ibáñez B, Ottervanger JP, Pizarro G, van Royen N, Mateos A, *et al.* Early intravenous beta-blockers in patients with ST-segment elevation myocardial infarction before primary percutaneous coronary intervention. *J Am Coll Cardiol* 2016;**67**:2705–2715. https://doi.org/10.1016/j.jacc.2016.03.522
165. Pizarro G, Fernández-Friera L, Fuster V, Fernández-Jiménez R, García-Ruiz JM, García-Álvarez A, *et al.* Long-term benefit of early pre-reperfusion metoprolol administration in patients with acute myocardial infarction: results from the METOCARD-CNIC trial (effect of metoprolol in cardioprotection during an acute myocardial infarction). *J Am Coll Cardiol* 2014;**63**:2356–2362. https://doi.org/10.1016/j.jacc.2014.03.014
166. García-Ruiz JM, Fernández-Jiménez R, García-Alvarez A, Pizarro G, Galán-Arriola C, Fernández-Friera L, *et al.* Impact of the timing of metoprolol administration during STEMI on infarct size and ventricular function. *J Am Coll Cardiol* 2016;**67**:2093–2104. https://doi.org/10.1016/j.jacc.2016.02.050
167. Hoedemaker NP, Roolvink V, de Winter RJ, van Royen N, Fuster V, García-Ruiz JM, *et al.* Early intravenous beta-blockers in patients undergoing primary percutaneous coronary intervention for ST-segment elevation myocardial infarction: a patient-pooled meta-analysis of randomized clinical trials. *Eur Heart J Acute Cardiovasc Care* 2020;**9**:469–477. https://doi.org/10.1177/2048872619830609
168. Sterling LH, Filion KB, Atallah R, Reynier P, Eisenberg MJ. Intravenous beta-blockers in ST-segment elevation myocardial infarction: a systematic review and meta-analysis. *Int J Cardiol* 2017;**228**:295–302. https://doi.org/10.1016/j.ijcard.2016.11.133

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

169. Chatterjee S, Chaudhuri D, Vedanthan R, Fuster V, Ibanez B, Bangalore S, *et al.* Early intravenous beta-blockers in patients with acute coronary syndrome–a meta-analysis of randomized trials. *Int J Cardiol* 2013;**168**:915–921. https://doi.org/10.1016/j.ijcard.2012.10.050
170. Elgendy IY, Elgendy AY, Mahmoud AN, Mansoor H, Mojadidi MK, Bavry AA, *et al.* Intravenous β-blockers for patients undergoing primary percutaneous coronary intervention: a meta-analysis of randomized trials. *Int J Cardiol* 2016;**223**:891–897. https://doi.org/10.1016/j.ijcard.2016.08.293
171. García-Prieto J, Villena-Gutiérrez R, Gómez M, Bernardo E, Pun-García A, García-Lunar I, *et al.* Neutrophil stunning by metoprolol reduces infarct size. *Nat Commun* 2017;**8**:14780. https://doi.org/10.1038/ncomms14780
172. Stub D, Smith K, Bernard S, Nehme Z, Stephenson M, Bray JE, *et al.* Air versus oxygen in ST-segment-elevation myocardial infarction. *Circulation* 2015;**131**:2143–2150. https://doi.org/10.1161/circulationaha.114.014494
173. Widimsky P, Wijns W, Fajadet J, de Belder M, Knot J, Aaberge L, *et al.* Reperfusion therapy for ST elevation acute myocardial infarction in Europe: description of the current situation in 30 countries. *Eur Heart J* 2010;**31**:943–957. https://doi.org/10.1093/eurheartj/ehp492
174. Hall M, Laut K, Dondo TB, Alabas OA, Brogan RA, Gutacker N, *et al.* Patient and hospital determinants of primary percutaneous coronary intervention in England, 2003–2013. *Heart* 2016;**102**:313–319. https://doi.org/10.1136/heartjnl-2015-308616
175. Widimsky P, Fajadet J, Danchin N, Wijns W. "Stent 4 Life" targeting PCI at all who will benefit the most. A joint project between EAPCI, Euro-PCR, EUCOMED and the ESC Working Group on Acute Cardiac Care. *EuroIntervention* 2009;**4**:555–557. doi: doi:10.4244/EIJV4I5A94
176. Pinto DS, Kirtane AJ, Nallamothu BK, Murphy SA, Cohen DJ, Laham RJ, *et al.* Hospital delays in reperfusion for ST-elevation myocardial infarction: implications when selecting a reperfusion strategy. *Circulation* 2006;**114**:2019–2025. https://doi.org/10.1161/circulationaha.106.638353
177. Steg PG, Cambou JP, Goldstein P, Durand E, Sauval P, Kadri Z, *et al.* Bypassing the emergency room reduces delays and mortality in ST elevation myocardial infarction: the USIC 2000 registry. *Heart* 2006;**92**:1378–1383. https://doi.org/10.1136/hrt.2006.101972
178. Baran KW, Kamrowski KA, Westwater JJ, Tschida VH, Alexander CF, Beahrs MM, *et al.* Very rapid treatment of ST-segment-elevation myocardial infarction: utilizing prehospital electrocardiograms to bypass the emergency department. *Circ Cardiovasc Qual Outcomes* 2010;**3**:431–437. https://doi.org/10.1161/circoutcomes.110.942631
179. Huynh T, Perron S, O'Loughlin J, Joseph L, Labrecque M, Tu JV, *et al.* Comparison of primary percutaneous coronary intervention and fibrinolytic therapy in ST-segment-elevation myocardial infarction: Bayesian hierarchical meta-analyses of randomized controlled trials and observational studies. *Circulation* 2009;**119**:3101–3109. https://doi.org/10.1161/circulationaha.108.793745
180. Nallamothu BK, Bates ER. Percutaneous coronary intervention versus fibrinolytic therapy in acute myocardial infarction: is timing (almost) everything? *Am J Cardiol* 2003;**92**:824–826. https://doi.org/10.1016/s0002-9149(03)00891-9
181. Betriu A, Masotti M. Comparison of mortality rates in acute myocardial infarction treated by percutaneous coronary intervention versus fibrinolysis. *Am J Cardiol* 2005;**95**:100–101. https://doi.org/10.1016/j.amjcard.2004.08.069
182. Boersma E. Does time matter? A pooled analysis of randomized clinical trials comparing primary percutaneous coronary intervention and in-hospital fibrinolysis in acute myocardial infarction patients. *Eur Heart J* 2006;**27**:779–788. https://doi.org/10.1093/eurheartj/ehi810
183. Pinto DS, Frederick PD, Chakrabarti AK, Kirtane AJ, Ullman E, Dejam A, *et al.* Benefit of transferring ST-segment-elevation myocardial infarction patients for percutaneous coronary intervention compared with administration of onsite fibrinolytic declines as delays increase. *Circulation* 2011;**124**:2512–2521. https://doi.org/10.1161/circulationaha.111.018549
184. Armstrong PW, Gershlick AH, Goldstein P, Wilcox R, Danays T, Lambert Y, *et al.* Fibrinolysis or primary PCI in ST-segment elevation myocardial infarction. *N Engl J Med* 2013;**368**:1379–1387. https://doi.org/10.1056/NEJMoa1301092
185. Gershlick AH, Stephens-Lloyd A, Hughes S, Abrams KR, Stevens SE, Uren NG, *et al.* Rescue angioplasty after failed thrombolytic therapy for acute myocardial infarction. *N Engl J Med* 2005;**353**:2758–2768. https://doi.org/10.1056/NEJMoa050849
186. Cantor WJ, Fitchett D, Borgundvaag B, Ducas J, Heffernan M, Cohen EA, *et al.* Routine early angioplasty after fibrinolysis for acute myocardial infarction. *N Engl J Med* 2009;**360**:2705–2718. https://doi.org/10.1056/NEJMoa0808276
187. Wijeysundera HC, Vijayaraghavan R, Nallamothu BK, Foody JoAnne M, Krumholz HM, Phillips CO, *et al.* Rescue angioplasty or repeat fibrinolysis after failed fibrinolytic therapy for ST-segment myocardial infarction: a meta-analysis of randomized trials. *J Am Coll Cardiol* 2007;**49**:422–430. https://doi.org/10.1016/j.jacc.2006.09.033
188. Sutton AG, Campbell PG, Graham R, Price DJA, Gray JC, Grech ED, *et al.* A randomized trial of rescue angioplasty versus a conservative approach for failed fibrinolysis in ST-segment elevation myocardial infarction: the Middlesbrough Early Revascularization to Limit INfarction (MERLIN) trial. *J Am Coll Cardiol* 2004;**44**:287–296. https://doi.org/10.1016/j.jacc.2003.12.059
189. Schömig A, Mehilli J, Antoniucci D, Ndrepepa G, Markwardt C, Di Pede F, *et al.* Mechanical reperfusion in patients with acute myocardial infarction presenting more than 12 hours from symptom onset: a randomized controlled trial. *JAMA* 2005;**293**:2865–2872. https://doi.org/10.1001/jama.293.23.2865
190. Ndrepepa G, Kastrati A, Mehilli J, Antoniucci D, Schömig A. Mechanical reperfusion and long-term mortality in patients with acute myocardial infarction presenting 12 to 48 hours from onset of symptoms. *JAMA* 2009;**301**:487–488. https://doi.org/10.1001/jama.2009.32
191. Bouisset F, Gerbaud E, Bataille V, Coste P, Puymirat E, Belle L, *et al.* Percutaneous myocardial revascularization in late-presenting patients with STEMI. *J Am Coll Cardiol* 2021;**78**:1291–1305. https://doi.org/10.1016/j.jacc.2021.07.039
192. Hochman JS, Lamas GA, Buller CE, Dzavik V, Reynolds HR, Abramsky SJ, *et al.* Coronary intervention for persistent occlusion after myocardial infarction. *N Engl J Med* 2006;**355**:2395–2407. https://doi.org/10.1056/NEJMoa066139
193. Menon V, Pearte CA, Buller CE, Steg PhG, Forman SA, White HD, *et al.* Lack of benefit from percutaneous intervention of persistently occluded infarct arteries after the acute phase of myocardial infarction is time independent: insights from Occluded Artery Trial. *Eur Heart J* 2009;**30**:183–191. https://doi.org/10.1093/eurheartj/ehn486
194. Ioannidis JPA, Katritsis DG. Percutaneous coronary intervention for late reperfusion after myocardial infarction in stable patients. *Am Heart J* 2007;**154**:1065–1071. https://doi.org/10.1016/j.ahj.2007.07.049
195. Knuuti J, Wijns W, Saraste A, Capodanno D, Barbato E, Funck-Brentano C, *et al.* 2019 ESC Guidelines for the diagnosis and management of chronic coronary syndromes. *Eur Heart J* 2020;**41**:407–477. https://doi.org/10.1093/eurheartj/ehz425
196. O'Donoghue M, Boden WE, Braunwald E, Cannon CP, Clayton TC, de Winter RJ, *et al.* Early invasive vs conservative treatment strategies in women and men with unstable angina and non-ST-segment elevation myocardial infarction: a meta-analysis. *JAMA* 2008;**300**:71–80. https://doi.org/10.1001/jama.300.1.71
197. Mehta SR, Cannon CP, Fox KAA, Wallentin L, Boden WE, Spacek R, *et al.* Routine vs selective invasive strategies in patients with acute coronary syndromes: a collaborative meta-analysis of randomized trials. *JAMA* 2005;**293**:2908–2917. https://doi.org/10.1001/jama.293.23.2908
198. Fox KA, Clayton TC, Damman P, Pocock SJ, de Winter RJ, Tijssen JGP, *et al.* Long-term outcome of a routine versus selective invasive strategy in patients with non-ST-segment elevation acute coronary syndrome a meta-analysis of individual patient data. *J Am Coll Cardiol* 2010;**55**:2435–2445. https://doi.org/10.1016/j.jacc.2010.03.007
199. Fanning JP, Nyong J, Scott IA, Aroney CN, Walters DL. Routine invasive strategies versus selective invasive strategies for unstable angina and non-ST elevation myocardial infarction in the stent era. *Cochrane Database Syst Rev* 2016;**2016**:CD004815. https://doi.org/10.1002/14651858.CD004815.pub4
200. Elgendy IY, Mahmoud AN, Wen X, Bavry AA. Meta-analysis of randomized trials of long-term all-cause mortality in patients with non-ST-elevation acute coronary syndrome managed with routine invasive versus selective invasive strategies. *Am J Cardiol* 2017;**119**:560–564. https://doi.org/10.1016/j.amjcard.2016.11.005
201. Navarese EP, Gurbel PA, Andreotti F, Tantry U, Jeong Y-H, Kozinski M, *et al.* Optimal timing of coronary invasive strategy in non-ST-segment elevation acute coronary syndromes: a systematic review and meta-analysis. *Ann Intern Med* 2013;**158**:261–270. https://doi.org/10.7326/0003-4819-158-4-201302190-00006
202. Jobs A, Mehta SR, Montalescot G, Vicaut E, van't Hof AWJ, Badings EA, *et al.* Optimal timing of an invasive strategy in patients with non-ST-elevation acute coronary syndrome: a meta-analysis of randomised trials. *Lancet* 2017;**390**:737–746. https://doi.org/10.1016/s0140-6736(17)31490-3
203. Kite TA, Kurmani SA, Bountziouka V, Cooper NJ, Lock ST, Gale CP, *et al.* Timing of invasive strategy in non-ST-elevation acute coronary syndrome: a meta-analysis of randomized controlled trials. *Eur Heart J* 2022;**43**:3148–3161. https://doi.org/10.1093/eurheartj/ehac213
204. Eggers KM, James SK, Jernberg T, Lindahl B. Timing of coronary angiography in patients with non-ST-elevation acute coronary syndrome: long-term clinical outcomes from the nationwide SWEDEHEART registry. *EuroIntervention* 2022;**18**:582–589. https://doi.org/10.4244/eij-d-21-00982
205. Indications for fibrinolytic therapy in suspected acute myocardial infarction: collaborative overview of early mortality and major morbidity results from all randomised trials of more than 1000 patients. *Lancet* 1994;**343**:311–322. https://doi.org/10.1016/S0140-6736(94)91161-4
206. Morrison LJ, Verbeek PR, McDonald AC, Sawadsky BV, Cook DJ. Mortality and prehospital thrombolysis for acute myocardial infarction: a meta-analysis. *JAMA* 2000;**283**:2686–2692. https://doi.org/10.1001/jama.283.20.2686
207. Steg PG, Bonnefoy E, Chabaud S, Lapostolle F, Dubien P-Y, Cristofini P, *et al.* Impact of time to treatment on mortality after prehospital fibrinolysis or primary angioplasty: data from the CAPTIM randomized clinical trial. *Circulation* 2003;**108**:2851–2856. https://doi.org/10.1161/01.Cir.0000103122.10021.F2
208. ASSENT-4 PCI Investigators. Primary versus tenecteplase-facilitated percutaneous coronary intervention in patients with ST-segment elevation acute myocardial infarction (ASSENT-4 PCI): randomised trial. *Lancet* 2006;**367**:569–578. https://doi.org/10.1016/s0140-6736(06)68147-6

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

209. Fazel R, Joseph TI, Sankardas MA, Pinto DS, Yeh RW, Kumbhani DJ, *et al.* Comparison of reperfusion strategies for ST-segment-elevation myocardial infarction: a multivariate network meta-analysis. *J Am Heart Assoc* 2020;**9**:e015186. https://doi.org/10.1161/jaha.119.015186
210. Sinnaeve PR, Armstrong PW, Gershlick AH, Goldstein P, Wilcox R, Lambert Y, *et al.* ST-segment-elevation myocardial infarction patients randomized to a pharmaco-invasive strategy or primary percutaneous coronary intervention: strategic Reperfusion Early After Myocardial Infarction (STREAM) 1-year mortality follow-up. *Circulation* 2014;**130**:1139–1145. https://doi.org/10.1161/circulationaha.114.009570
211. Arbel Y, Ko DT, Yan AT, Cantor WJ, Bagai A, Koh M, *et al.* Long-term follow-up of the trial of routine angioplasty and stenting after fibrinolysis to enhance reperfusion in acute myocardial infarction (TRANSFER-AMI). *Can J Cardiol* 2018;**34**:736–743. https://doi.org/10.1016/j.cjca.2018.02.005
212. Di Mario C, Dudek D, Piscione F, Mielecki W, Savonitto S, Murena E, *et al.* Immediate angioplasty versus standard therapy with rescue angioplasty after thrombolysis in the Combined Abciximab REteplase Stent Study in Acute Myocardial Infarction (CARESS-in-AMI): an open, prospective, randomised, multicentre trial. *Lancet* 2008;**371**:559–568. https://doi.org/10.1016/s0140-6736(08)60268-8
213. Bøhmer E, Hoffmann P, Abdelnoor M, Arnesen H, Halvorsen S. Efficacy and safety of immediate angioplasty versus ischemia-guided management after thrombolysis in acute myocardial infarction in areas with very long transfer distances results of the NORDISTEMI (NORwegian study on DIstrict treatment of ST-elevation myocardial infarction). *J Am Coll Cardiol* 2010;**55**:102–110. https://doi.org/10.1016/j.jacc.2009.08.007
214. Scheller B, Hennen B, Hammer B, Walle J, Hofer C, Hilpert V, *et al.* Beneficial effects of immediate stenting after thrombolysis in acute myocardial infarction. *J Am Coll Cardiol* 2003;**42**:634–641. https://doi.org/10.1016/s0735-1097(03)00763-0
215. Le May MR, Wells GA, Labinaz M, Davies RF, Turek M, Leddy D, *et al.* Combined angioplasty and pharmacological intervention versus thrombolysis alone in acute myocardial infarction (CAPITAL AMI study). *J Am Coll Cardiol* 2005;**46**:417–424. https://doi.org/10.1016/j.jacc.2005.04.042
216. Abdel-Qadir H, Yan AT, Tan M, Borgia F, Piscione F, Di Mario C, *et al.* Consistency of benefit from an early invasive strategy after fibrinolysis: a patient-level meta-analysis. *Heart* 2015;**101**:1554–1561. https://doi.org/10.1136/heartjnl-2015-307815
217. Madan M, Halvorsen S, Di Mario C, Tan M, Westerhout CM, Cantor WJ, *et al.* Relationship between time to invasive assessment and clinical outcomes of patients undergoing an early invasive strategy after fibrinolysis for ST-segment elevation myocardial infarction: a patient-level analysis of the randomized early routine invasive clinical trials. *JACC Cardiovasc Interv* 2015;**8**:166–174. https://doi.org/10.1016/j.jcin.2014.09.005
218. Andersen HR, Nielsen TT, Rasmussen K, Thuesen L, Kelbaek H, Thayssen P, *et al.* A comparison of coronary angioplasty with fibrinolytic therapy in acute myocardial infarction. *N Engl J Med* 2003;**349**:733–742. https://doi.org/10.1056/NEJMoa025142
219. Dalby M, Bouzamondo A, Lechat P, Montalescot G. Transfer for primary angioplasty versus immediate thrombolysis in acute myocardial infarction: a meta-analysis. *Circulation* 2003;**108**:1809–1814. https://doi.org/10.1161/01.Cir.0000091088.63921.8c
220. Gierlotka M, Gasior M, Wilczek K, Hawranek M, Szkodzinski J, Paczek P, *et al.* Reperfusion by primary percutaneous coronary intervention in patients with ST-segment elevation myocardial infarction within 12 to 24 hours of the onset of symptoms (from a prospective national observational study [PL-ACS]). *Am J Cardiol* 2011;**107**:501–508. https://doi.org/10.1016/j.amjcard.2010.10.008
221. Busk M, Kaltoft A, Nielsen SS, Bottcher M, Rehling M, Thuesen L, *et al.* Infarct size and myocardial salvage after primary angioplasty in patients presenting with symptoms for<12 h vs. 12–72 h. *Eur Heart J* 2009;**30**:1322–1330. https://doi.org/10.1093/eurheartj/ehp113
222. Borgia F, Goodman SG, Halvorsen S, Cantor WJ, Piscione F, Le May MR, *et al.* Early routine percutaneous coronary intervention after fibrinolysis vs. standard therapy in ST-segment elevation myocardial infarction: a meta-analysis. *Eur Heart J* 2010;**31**:2156–2169. https://doi.org/10.1093/eurheartj/ehq204
223. D'Souza SP, Mamas MA, Fraser DG, Fath-Ordoubadi F. Routine early coronary angioplasty versus ischaemia-guided angioplasty after thrombolysis in acute ST-elevation myocardial infarction: a meta-analysis. *Eur Heart J* 2011;**32**:972–982. https://doi.org/10.1093/eurheartj/ehq398
224. Fernandez-Avilés F, Alonso JJ, Castro-Beiras A, Vázquez N, Blanco J, Alonso-Briales J, *et al.* Routine invasive strategy within 24 hours of thrombolysis versus ischaemia-guided conservative approach for acute myocardial infarction with ST-segment elevation (GRACIA-1): a randomised controlled trial. *Lancet* 2004;**364**:1045–1053. https://doi.org/10.1016/s0140-6736(04)17059-1
225. Hochman JS, Sleeper LA, White HD, Dzavik V, Wong SC, Menon V, *et al.* One-year survival following early revascularization for cardiogenic shock. *JAMA* 2001;**285**:190–192. https://doi.org/10.1001/jama.285.2.190
226. Lemkes JS, Janssens GN, van der Hoeven NW, van de Ven PM, Marques KMJ, Nap A, *et al.* Timing of revascularization in patients with transient ST-segment elevation myocardial infarction: a randomized clinical trial. *Eur Heart J* 2019;**40**:283–291. https://doi.org/10.1093/eurheartj/ehy651
227. Mehta SR, Granger CB, Boden WE, Steg PG, Bassand J-P, Faxon DP, *et al.* Early versus delayed invasive intervention in acute coronary syndromes. *N Engl J Med* 2009;**360**:2165–2175. https://doi.org/10.1056/NEJMoa0807986
228. Kofoed KF, Kelbæk H, Hansen PR, Torp-Pedersen C, Høfsten D, Kløvgaard L, *et al.* Early versus standard care invasive examination and treatment of patients with non-ST-segment elevation acute coronary syndrome. *Circulation* 2018;**138**:2741–2750. https://doi.org/10.1161/circulationaha.118.037152
229. Butt JH, Kofoed KF, Kelbæk H, Hansen PR, Torp-Pedersen C, Høfsten D, *et al.* Importance of risk assessment in timing of invasive coronary evaluation and treatment of patients with non-ST-segment-elevation acute coronary syndrome: insights from the VERDICT trial. *J Am Heart Assoc* 2021;**10**:e022333. https://doi.org/10.1161/jaha.121.022333
230. Barbarawi M, Kheiri B, Zayed Y, Barbarawi O, Chahine A, Haykal T, *et al.* Meta-analysis of optimal timing of coronary intervention in non-ST-elevation acute coronary syndrome. *Catheter Cardiovasc Interv* 2020;**95**:185–193. https://doi.org/10.1002/ccd.28280
231. Vranckx P, White HD, Huang Z, Mahaffey KW, Armstrong PW, Van de Werf F, *et al.* Validation of BARC bleeding criteria in patients with acute coronary syndromes: the TRACER trial. *J Am Coll Cardiol* 2016;**67**:2135–2144. https://doi.org/10.1016/j.jacc.2016.02.056
232. Ndrepepa G, Berger PB, Mehilli J, Seyfarth M, Neumann F-J, Schömig A, *et al.* Periprocedural bleeding and 1-year outcome after percutaneous coronary interventions: appropriateness of including bleeding as a component of a quadruple end point. *J Am Coll Cardiol* 2008;**51**:690–697. https://doi.org/10.1016/j.jacc.2007.10.040
233. Urban P, Mehran R, Colleran R, Angiolillo DJ, Byrne RA, Capodanno D, *et al.* Defining high bleeding risk in patients undergoing percutaneous coronary intervention: a consensus document from the Academic Research Consortium for High Bleeding Risk. *Eur Heart J* 2019;**40**:2632–2653. https://doi.org/10.1093/eurheartj/ehz372
234. Doomun D, Doomun I, Schukraft S, Arroyo D, Cook S, Huwyler T, *et al.* Ischemic and bleeding outcomes according to the academic research consortium high bleeding risk criteria in all comers treated by percutaneous coronary interventions. *Front Cardiovasc Med* 2021;**8**:620354. https://doi.org/10.3389/fcvm.2021.620354
235. Antiplatelet Trialists' Collaboration. Collaborative overview of randomised trials of antiplatelet therapy-I: prevention of death, myocardial infarction, and stroke by prolonged antiplatelet therapy in various categories of patients. *BMJ* 1994;**308**:81–106. https://doi.org/10.1136/bmj.308.6921.81
236. Valgimigli M, Bueno H, Byrne RA, Collet J-P, Costa F, Jeppsson A, *et al.* 2017 ESC focused update on dual antiplatelet therapy in coronary artery disease developed in collaboration with EACTS: the Task Force for dual antiplatelet therapy in coronary artery disease of the European Society of Cardiology (ESC) and of the European Association for Cardio-Thoracic Surgery (EACTS). *Eur Heart J* 2018;**39**:213–260. https://doi.org/10.1093/eurheartj/ehx419
237. Jones WS, Mulder H, Wruck LM, Pencina MJ, Kripalani S, Muñoz D, *et al.* Comparative effectiveness of aspirin dosing in cardiovascular disease. *N Engl J Med* 2021;**384**:1981–1990. https://doi.org/10.1056/NEJMoa2102137
238. Wallentin L, Becker RC, Budaj A, Cannon CP, Emanuelsson H, Held C, *et al.* Ticagrelor versus clopidogrel in patients with acute coronary syndromes. *N Engl J Med* 2009;**361**:1045–1057. https://doi.org/10.1056/NEJMoa0904327
239. Wiviott SD, Braunwald E, McCabe CH, Montalescot G, Ruzyllo W, Gottlieb S, *et al.* Prasugrel versus clopidogrel in patients with acute coronary syndromes. *N Engl J Med* 2007;**357**:2001–2015. https://doi.org/10.1056/NEJMoa0706482
240. Aradi D, Kirtane A, Bonello L, Gurbel PA, Tantry US, Huber K, *et al.* Bleeding and stent thrombosis on P2Y12-inhibitors: collaborative analysis on the role of platelet reactivity for risk stratification after percutaneous coronary intervention. *Eur Heart J* 2015;**36**:1762–1771. https://doi.org/10.1093/eurheartj/ehv104
241. Aradi D, Storey RF, Komócsi A, Trenk D, Gulba D, Kiss RG, *et al.* Expert position paper on the role of platelet function testing in patients undergoing percutaneous coronary intervention. *Eur Heart J* 2014;**35**:209–215. https://doi.org/10.1093/eurheartj/eht375
242. Gimbel M, Qaderdan K, Willemsen L, Hermanides R, Bergmeijer T, de Vrey E, *et al.* Clopidogrel versus ticagrelor or prasugrel in patients aged 70 years or older with non-ST-elevation acute coronary syndrome (POPular AGE): the randomised, open-label, non-inferiority trial. *Lancet* 2020;**395**:1374–1381. https://doi.org/10.1016/s0140-6736(20)30325-1
243. Husted S, James S, Becker RC, Horrow J, Katus H, Storey RF, *et al.* Ticagrelor versus clopidogrel in elderly patients with acute coronary syndromes: a substudy from the prospective randomized PLATelet inhibition and patient Outcomes (PLATO) trial. *Circ Cardiovasc Qual Outcomes* 2012;**5**:680–688. https://doi.org/10.1161/circoutcomes.111.964395
244. Schüpke S, Neumann FJ, Menichelli M, Mayer K, Bernlochner I, Wöhrle J, *et al.* Ticagrelor or prasugrel in patients with acute coronary syndromes. *N Engl J Med* 2019;**381**:1524–1534. https://doi.org/10.1056/NEJMoa1908973
245. Montalescot G, van't Hof AW, Lapostolle F, Silvain J, Lassen JF, Bolognese L, *et al.* Prehospital ticagrelor in ST-segment elevation myocardial infarction. *N Engl J Med* 2014;**371**:1016–1027. https://doi.org/10.1056/NEJMoa1407024
246. Koul S, Smith JG, Götberg M, Omerovic E, Alfredsson J, Venetsanos D, *et al.* No benefit of ticagrelor pretreatment compared with treatment during percutaneous coronary

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

intervention in patients with ST-segment-elevation myocardial infarction undergoing primary percutaneous coronary intervention. *Circ Cardiovasc Interv* 2018;**11**: e005528. https://doi.org/10.1161/circinterventions.117.005528
247. Montalescot G, Bolognese L, Dudek D, Goldstein P, Hamm C, Tanguay J-F, *et al.* Pretreatment with prasugrel in non-ST-segment elevation acute coronary syndromes. *N Engl J Med* 2013;**369**:999–1010. https://doi.org/10.1056/NEJMoa1308075
248. Tarantini G, Mojoli M, Varbella F, Caporale R, Rigattieri S, Andò G, *et al.* Timing of oral P2Y(12) inhibitor administration in patients with non-ST-segment elevation acute coronary syndrome. *J Am Coll Cardiol* 2020;**76**:2450–2459. https://doi.org/10.1016/j.jacc.2020.08.053
249. Boersma E, Harrington RA, Moliterno DJ, White H, Théroux P, Van de Werf F, *et al.* Platelet glycoprotein IIb/IIIa inhibitors in acute coronary syndromes: a meta-analysis of all major randomised clinical trials. *Lancet* 2002;**359**:189–198. https://doi.org/10.1016/s0140-6736(02)07442-1
250. Neumann FJ, Sousa-Uva M, Ahlsson A, Alfonso F, Banning AP, Benedetto U, *et al.* 2018 ESC/EACTS Guidelines on myocardial revascularization. *Eur Heart J* 2019;**40**:87–165. https://doi.org/10.1093/eurheartj/ehy394
251. Harrington RA, Stone GW, McNulty S, White HD, Lincoff AM, Gibson CM, *et al.* Platelet inhibition with cangrelor in patients undergoing PCI. *N Engl J Med* 2009; **361**:2318–2329. https://doi.org/10.1056/NEJMoa0908628
252. Bhatt DL, Stone GW, Mahaffey KW, Gibson CM, Steg PG, Hamm CW, *et al.* Effect of platelet inhibition with cangrelor during PCI on ischemic events. *N Engl J Med* 2013; **368**:1303–1313. https://doi.org/10.1056/NEJMoa1300815
253. Bhatt DL, Lincoff AM, Gibson CM, Stone GW, McNulty S, Montalescot G, *et al.* Intravenous platelet blockade with cangrelor during PCI. *N Engl J Med* 2009;**361**: 2330–2341. https://doi.org/10.1056/NEJMoa0908629
254. Steg PG, Bhatt DL, Hamm CW, Stone GW, Gibson CM, Mahaffey KW, *et al.* Effect of cangrelor on periprocedural outcomes in percutaneous coronary interventions: a pooled analysis of patient-level data. *Lancet* 2013;**382**:1981–1992. https://doi.org/10.1016/s0140-6736(13)61615-3
255. Eikelboom JW, Anand SS, Malmberg K, Weitz JI, Ginsberg JS, Yusuf S. Unfractionated heparin and low-molecular-weight heparin in acute coronary syndrome without ST elevation: a meta-analysis. *Lancet* 2000;**355**:1936–1942. https://doi.org/10.1016/s0140-6736(00)02324-2
256. Ferguson JJ, Califf RM, Antman EM, Cohen M, Grines CL, Goodman S, *et al.* Enoxaparin vs unfractionated heparin in high-risk patients with non-ST-segment elevation acute coronary syndromes managed with an intended early invasive strategy: primary results of the SYNERGY randomized trial. *JAMA* 2004;**292**:45–54. https://doi.org/10.1001/jama.292.1.45
257. Cohen M, Mahaffey KW, Pieper K, Pollack CV, Antman EM, Hoekstra J, *et al.* A subgroup analysis of the impact of prerandomization antithrombin therapy on outcomes in the SYNERGY trial: enoxaparin versus unfractionated heparin in non-ST-segment elevation acute coronary syndromes. *J Am Coll Cardiol* 2006;**48**:1346–1354. https://doi.org/10.1016/j.jacc.2006.05.058
258. Montalescot G, Zeymer U, Silvain J, Boulanger B, Cohen M, Goldstein P, *et al.* Intravenous enoxaparin or unfractionated heparin in primary percutaneous coronary intervention for ST-elevation myocardial infarction: the international randomised open-label ATOLL trial. *Lancet* 2011;**378**:693–703. https://doi.org/10.1016/s0140-6736(11)60876-3
259. Li Y, Liang Z, Qin L, Wang M, Wang X, Zhang H, *et al.* Bivalirudin plus a high-dose infusion versus heparin monotherapy in patients with ST-segment elevation myocardial infarction undergoing primary percutaneous coronary intervention: a randomised trial. *Lancet* 2022;**400**:1847–1857. https://doi.org/10.1016/s0140-6736(22)01999-7
260. Yusuf S, Mehta SR, Chrolavicius S, Cohen M, Grines CL, Goodman S, *et al.* Effects of fondaparinux on mortality and reinfarction in patients with acute ST-segment elevation myocardial infarction: the OASIS-6 randomized trial. *JAMA* 2006;**295**: 1519–1530. https://doi.org/10.1001/jama.295.13.joc60038
261. Silvain J, Beygui F, Barthélémy O, Pollack C, Cohen M, Zeymer U, *et al.* Efficacy and safety of enoxaparin versus unfractionated heparin during percutaneous coronary intervention: systematic review and meta-analysis. *BMJ* 2012;**344**:e553. https://doi.org/10.1136/bmj.e553
262. The Fifth Organization to Assess Strategies in Acute Ischemic Syndromes Investigators. Comparison of fondaparinux and enoxaparin in acute coronary syndromes. *N Engl J Med* 2006;**354**:1464–1476. https://doi.org/10.1056/NEJMoa055443
263. Mehta SR, Yusuf S, Peters RJ, Bertrand ME, Lewis BS, Natarajan MK, *et al.* Effects of pretreatment with clopidogrel and aspirin followed by long-term therapy in patients undergoing percutaneous coronary intervention: the PCI-CURE study. *Lancet* 2001; **358**:527–533. https://doi.org/10.1016/s0140-6736(01)05701-4
264. Palmerini T, Della Riva D, Benedetto U, Bacchi Reggiani L, Feres F, Abizaid A, *et al.* Three, six, or twelve months of dual antiplatelet therapy after DES implantation in patients with or without acute coronary syndromes: an individual patient data pairwise and network meta-analysis of six randomized trials and 11 473 patients. *Eur Heart J* 2017;**38**:1034–1043. https://doi.org/10.1093/eurheartj/ehw627
265. Hahn J-Y, Song YB, Oh J-H, Cho D-K, Lee JB, Doh J-H, *et al.* 6-month versus 12-month or longer dual antiplatelet therapy after percutaneous coronary intervention in patients with acute coronary syndrome (SMART-DATE): a randomised, open-label, non-inferiority trial. *Lancet* 2018;**391**:1274–1284. https://doi.org/10.1016/s0140-6736(18)30493-8
266. Kedhi E, Fabris E, van der Ent M, Buszman P, von Birgelen C, Roolvink V, *et al.* Six months versus 12 months dual antiplatelet therapy after drug-eluting stent implantation in ST-elevation myocardial infarction (DAPT-STEMI): randomised, multicentre, non-inferiority trial. *BMJ* 2018;**363**:k3793. https://doi.org/10.1136/bmj.k3793
267. De Luca G, Damen SA, Camaro C, Benit E, Verdoia M, Rasoul S, *et al.* Final results of the randomised evaluation of short-term dual antiplatelet therapy in patients with acute coronary syndrome treated with a new-generation stent (REDUCE trial). *EuroIntervention* 2019;**15**:e990–e998. https://doi.org/10.4244/eij-d-19-00539
268. Hahn J-Y, Song YB, Oh J-H, Chun WJ, Park YH, Jang WJ, *et al.* Effect of P2Y12 inhibitor monotherapy vs dual antiplatelet therapy on cardiovascular events in patients undergoing percutaneous coronary intervention: the SMART-CHOICE randomized clinical trial. *JAMA* 2019;**321**:2428–2437. https://doi.org/10.1001/jama.2019.8146
269. Vranckx P, Valgimigli M, Jüni P, Hamm C, Steg PG, Heg D, *et al.* Ticagrelor plus aspirin for 1 month, followed by ticagrelor monotherapy for 23 months vs aspirin plus clopidogrel or ticagrelor for 12 months, followed by aspirin monotherapy for 12 months after implantation of a drug-eluting stent: a multicentre, open-label, randomised superiority trial. *Lancet* 2018;**392**:940–949. https://doi.org/10.1016/s0140-6736(18)31858-0
270. Watanabe H, Domei T, Morimoto T, Natsuaki M, Shiomi H, Toyota T, *et al.* Effect of 1-month dual antiplatelet therapy followed by clopidogrel vs 12-month dual antiplatelet therapy on cardiovascular and bleeding events in patients receiving PCI: the STOPDAPT-2 randomized clinical trial. *JAMA* 2019;**321**:2414–2427. https://doi.org/10.1001/jama.2019.8145
271. Mehran R, Baber U, Sharma SK, Cohen DJ, Angiolillo DJ, Briguori C, *et al.* Ticagrelor with or without aspirin in high-risk patients after PCI. *N Engl J Med* 2019;**381**: 2032–2042. https://doi.org/10.1056/NEJMoa1908419
272. Baber U, Dangas G, Angiolillo DJ, Cohen DJ, Sharma SK, Nicolas J, *et al.* Ticagrelor alone vs. ticagrelor plus aspirin following percutaneous coronary intervention in patients with non-ST-segment elevation acute coronary syndromes: TWILIGHT-ACS. *Eur Heart J* 2020;**41**:3533–3545. https://doi.org/10.1093/eurheartj/ehaa670
273. Kim BK, Hong SJ, Cho YH, Yun KHo, Kim YH, Suh Y, *et al.* Effect of ticagrelor monotherapy vs ticagrelor with aspirin on major bleeding and cardiovascular events in patients with acute coronary syndrome: the TICO randomized clinical trial. *JAMA* 2020;**323**:2407–2416. https://doi.org/10.1001/jama.2020.7580
274. Giacoppo D, Matsuda Y, Fovino LN, D'Amico G, Gargiulo G, Byrne RA, *et al.* Short dual antiplatelet therapy followed by P2Y12 inhibitor monotherapy vs. prolonged dual antiplatelet therapy after percutaneous coronary intervention with second-generation drug-eluting stents: a systematic review and meta-analysis of randomized clinical trials. *Eur Heart J* 2021;**42**:308–319. https://doi.org/10.1093/eurheartj/ehaa739
275. Watanabe H, Morimoto T, Natsuaki M, Yamamoto K, Obayashi Y, Ogita M, *et al.* Comparison of clopidogrel monotherapy after 1 to 2 months of dual antiplatelet therapy with 12 months of dual antiplatelet therapy in patients with acute coronary syndrome: the STOPDAPT-2 ACS randomized clinical trial. *JAMA Cardiol* 2022;**7**: 407–417. https://doi.org/10.1001/jamacardio.2021.5244
276. Valgimigli M, Frigoli E, Heg D, Tijssen J, Jüni P, Vranckx P, *et al.* Dual antiplatelet therapy after PCI in patients at high bleeding risk. *N Engl J Med* 2021;**385**:1643–1655. https://doi.org/10.1056/NEJMoa2108749
277. Zettler ME, Peterson ED, McCoy LA, Effron MB, Anstrom KJ, Henry TD, *et al.* Switching of adenosine diphosphate receptor inhibitor after hospital discharge among myocardial infarction patients: insights from the treatment with adenosine diphosphate receptor inhibitors: longitudinal assessment of treatment patterns and events after acute coronary syndrome (TRANSLATE-ACS) observational study. *Am Heart J* 2017;**183**:62–68. https://doi.org/10.1016/j.ahj.2016.10.006
278. Angiolillo DJ, Rollini F, Storey RF, Bhatt DL, James S, Schneider DJ, *et al.* International expert consensus on switching platelet P2Y(12) receptor-inhibiting therapies. *Circulation* 2017;**136**:1955–1975. https://doi.org/10.1161/circulationaha.117.031164
279. Sibbing D, Aradi D, Jacobshagen C, Gross L, Trenk D, Geisler T, *et al.* Guided de-escalation of antiplatelet treatment in patients with acute coronary syndrome undergoing percutaneous coronary intervention (TROPICAL-ACS): a randomised, open-label, multicentre trial. *Lancet* 2017;**390**:1747–1757. https://doi.org/10.1016/s0140-6736(17)32155-4
280. Claassens DMF, Vos GJA, Bergmeijer TO, Hermanides RS, van't Hof AWJ, van der Harst P, *et al.* A genotype-guided strategy for oral P2Y(12) inhibitors in primary PCI. *N Engl J Med* 2019;**381**:1621–1631. https://doi.org/10.1056/NEJMoa1907096
281. Cuisset T, Deharo P, Quilici J, Johnson TW, Deffarges S, Bassez C, *et al.* Benefit of switching dual antiplatelet therapy after acute coronary syndrome: the TOPIC (timing of platelet inhibition after acute coronary syndrome) randomized study. *Eur Heart J* 2017;**38**:3070–3078. https://doi.org/10.1093/eurheartj/ehx175
282. Kim CJ, Park MW, Kim MC, Choo EH, Hwang B-H, Lee KY, *et al.* Unguided de-escalation from ticagrelor to clopidogrel in stabilised patients with acute myocardial infarction undergoing percutaneous coronary intervention (TALOS-AMI): an investigator-initiated, open-label, multicentre, non-inferiority, randomised trial. *Lancet* 2021;**398**:1305–1316. https://doi.org/10.1016/s0140-6736(21)01445-8

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

283. Kim HS, Kang J, Hwang D, Han J-K, Yang H-M, Kang H-J, *et al.* Prasugrel-based de-escalation of dual antiplatelet therapy after percutaneous coronary intervention in patients with acute coronary syndrome (HOST-REDUCE-POLYTECH-ACS): an open-label, multicentre, non-inferiority randomised trial. *Lancet* 2020;**396**:1079–1089. https://doi.org/10.1016/s0140-6736(20)31791-8
284. Antithrombotic Trialists' Collaboration. Collaborative meta-analysis of randomised trials of antiplatelet therapy for prevention of death, myocardial infarction, and stroke in high risk patients. *BMJ* 2002;**324**:71–86. https://doi.org/10.1136/bmj.324.7329.71
285. Antithrombotic Trialists' (ATT) Collaboration. Aspirin in the primary and secondary prevention of vascular disease: collaborative meta-analysis of individual participant data from randomised trials. *Lancet* 2009;**373**:1849–1860. https://doi.org/10.1016/s0140-6736(09)60503-1
286. Yusuf S, Zhao F, Mehta SR, Chrolavicius S, Tognoni G, Fox KK, *et al.* Effects of clopidogrel in addition to aspirin in patients with acute coronary syndromes without ST-segment elevation. *N Engl J Med* 2001;**345**:494–502. https://doi.org/10.1056/NEJMoa010746
287. Bhatt DL, Cryer BL, Contant CF, Cohen M, Lanas A, Schnitzer TJ, *et al.* Clopidogrel with or without omeprazole in coronary artery disease. *N Engl J Med* 2010;**363**:1909–1917. https://doi.org/10.1056/NEJMoa1007964
288. Gargiulo G, Costa F, Ariotti S, Biscaglia S, Campo G, Esposito G, *et al.* Impact of proton pump inhibitors on clinical outcomes in patients treated with a 6- or 24-month dual-antiplatelet therapy duration: insights from the PROlonging Dual-antiplatelet treatment after Grading stent-induced Intimal hyperplasia studY trial. *Am Heart J* 2016;**174**:95–102. https://doi.org/10.1016/j.ahj.2016.01.015
289. Mehta SR, Tanguay JF, Eikelboom JW, Jolly SS, Joyner CD, Granger CB, *et al.* Double-dose versus standard-dose clopidogrel and high-dose versus low-dose aspirin in individuals undergoing percutaneous coronary intervention for acute coronary syndromes (CURRENT-OASIS 7): a randomised factorial trial. *Lancet* 2010;**376**:1233–1243. https://doi.org/10.1016/s0140-6736(10)61088-4
290. Coughlan JJ, Aytekin A, Lahu S, Ndrepepa G, Menichelli M, Mayer K, *et al.* Ticagrelor or prasugrel for patients with acute coronary syndrome treated with percutaneous coronary intervention: a prespecified subgroup analysis of a randomized clinical trial. *JAMA Cardiol* 2021;**6**:1121–1129. https://doi.org/10.1001/jamacardio.2021.2228
291. Szummer K, Montez-Rath ME, Alfredsson J, Erlinge D, Lindahl B, Hofmann R, *et al.* Comparison between ticagrelor and clopidogrel in elderly patients with an acute coronary syndrome: insights from the SWEDEHEART registry. *Circulation* 2020;**142**:1700–1708. https://doi.org/10.1161/circulationaha.120.050645
292. Giugliano RP, White JA, Bode C, Armstrong PW, Montalescot G, Lewis BS, *et al.* Early versus delayed, provisional eptifibatide in acute coronary syndromes. *N Engl J Med* 2009;**360**:2176–2190. https://doi.org/10.1056/NEJMoa0901316
293. Dworeck C, Redfors B, Angerås O, Haraldsson I, Odenstedt J, Ioanes D, *et al.* Association of pretreatment with P2Y12 receptor antagonists preceding percutaneous coronary intervention in non-ST-segment elevation acute coronary syndromes with outcomes. *JAMA Netw Open* 2020;**3**:e2018735. https://doi.org/10.1001/jamanetworkopen.2020.18735
294. Steinhubl SR, Berger PB, Mann JT III, Fry ETA, DeLago A, Wilmer C, *et al.* Early and sustained dual oral antiplatelet therapy following percutaneous coronary intervention: a randomized controlled trial. *JAMA* 2002;**288**:2411–2420. https://doi.org/10.1001/jama.288.19.2411
295. Dawson LP, Chen D, Dagan M, Bloom J, Taylor A, Duffy SJ, *et al.* Assessment of pretreatment with oral P2Y12 inhibitors and cardiovascular and bleeding outcomes in patients with non-ST elevation acute coronary syndromes: a systematic review and meta-analysis. *JAMA Netw Open* 2021;**4**:e2134322. https://doi.org/10.1001/jamanetworkopen.2021.34322
296. Oler A, Whooley MA, Oler J, Grady D. Adding heparin to aspirin reduces the incidence of myocardial infarction and death in patients with unstable angina. A meta-analysis. *JAMA* 1996;**276**:811–815. doi: doi:10.1001/jama.1996.03540100055028
297. Antman EM, Cohen M, Radley D, McCabe C, Rush J, Premmereur J, *et al.* Assessment of the treatment effect of enoxaparin for unstable angina/non-Q-wave myocardial infarction: TIMI 11B-ESSENCE meta-analysis. *Circulation* 1999;**100**:1602–1608. https://doi.org/10.1161/01.cir.100.15.1602
298. Collet JP, Huber K, Cohen M, Zeymer U, Goldstein P, Pollack C, *et al.* A direct comparison of intravenous enoxaparin with unfractionated heparin in primary percutaneous coronary intervention (from the ATOLL trial). *Am J Cardiol* 2013;**112**:1367–1372. https://doi.org/10.1016/j.amjcard.2013.07.003
299. Stone GW, McLaurin BT, Cox DA, Bertrand ME, Lincoff AM, Moses JW, *et al.* Bivalirudin for patients with acute coronary syndromes. *N Engl J Med* 2006;**355**:2203–2216. https://doi.org/10.1056/NEJMoa062437
300. Valgimigli M, Frigoli E, Leonardi S, Rothenbühler M, Gagnor A, Calabrò P, *et al.* Bivalirudin or unfractionated heparin in acute coronary syndromes. *N Engl J Med* 2015;**373**:997–1009. https://doi.org/10.1056/NEJMoa1507854
301. Kastrati A, Neumann FJ, Schulz S, Massberg S, Byrne RA, Ferenc M, *et al.* Abciximab and heparin versus bivalirudin for non-ST-elevation myocardial infarction. *N Engl J Med* 2011;**365**:1980–1989. https://doi.org/10.1056/NEJMoa1109596
302. Erlinge D, Omerovic E, Fröbert O, Linder R, Danielewicz M, Hamid M, *et al.* Bivalirudin versus heparin monotherapy in myocardial infarction. *N Engl J Med* 2017;**377**:1132–1142. https://doi.org/10.1056/NEJMoa1706443
303. Han Y, Guo J, Zheng Y, Zang H, Su X, Wang Y, *et al.* Bivalirudin vs heparin with or without tirofiban during primary percutaneous coronary intervention in acute myocardial infarction: the BRIGHT randomized clinical trial. *JAMA* 2015;**313**:1336–1346. https://doi.org/10.1001/jama.2015.2323
304. Steg PG, Jolly SS, Mehta SR, Afzal R, Xavier D, Rupprecht H-J, *et al.* Low-dose vs standard-dose unfractionated heparin for percutaneous coronary intervention in acute coronary syndromes treated with fondaparinux: the FUTURA/OASIS-8 randomized trial. *JAMA* 2010;**304**:1339–1349. https://doi.org/10.1001/jama.2010.1320
305. Cannon CP, Bhatt DL, Oldgren J, Lip GYH, Ellis SG, Kimura T, *et al.* Dual antithrombotic therapy with dabigatran after PCI in atrial fibrillation. *N Engl J Med* 2017;**377**:1513–1524. https://doi.org/10.1056/NEJMoa1708454
306. Dewilde WJ, Oirbans T, Verheugt FW, Kelder JC, De Smet BJGL, Herrman J-P, *et al.* Use of clopidogrel with or without aspirin in patients taking oral anticoagulant therapy and undergoing percutaneous coronary intervention: an open-label, randomised, controlled trial. *Lancet* 2013;**381**:1107–1115. https://doi.org/10.1016/s0140-6736(12)62177-1
307. Gibson CM, Mehran R, Bode C, Halperin J, Verheugt FW, Wildgoose P, *et al.* Prevention of bleeding in patients with atrial fibrillation undergoing PCI. *N Engl J Med* 2016;**375**:2423–2434. https://doi.org/10.1056/NEJMoa1611594
308. Lopes RD, Heizer G, Aronson R, Vora AN, Massaro T, Mehran R, *et al.* Antithrombotic therapy after acute coronary syndrome or PCI in atrial fibrillation. *N Engl J Med* 2019;**380**:1509–1524. https://doi.org/10.1056/NEJMoa1817083
309. Vranckx P, Valgimigli M, Eckardt L, Tijssen J, Lewalter T, Gargiulo G, *et al.* Edoxaban-based versus vitamin K antagonist-based antithrombotic regimen after successful coronary stenting in patients with atrial fibrillation (ENTRUST-AF PCI): a randomised, open-label, phase 3b trial. *Lancet* 2019;**394**:1335–1343. https://doi.org/10.1016/s0140-6736(19)31872-0
310. Gargiulo G, Goette A, Tijssen J, Eckardt L, Lewalter T, Vranckx P, *et al.* Safety and efficacy outcomes of double vs. triple antithrombotic therapy in patients with atrial fibrillation following percutaneous coronary intervention: a systematic review and meta-analysis of non-vitamin K antagonist oral anticoagulant-based randomized clinical trials. *Eur Heart J* 2019;**40**:3757–3767. https://doi.org/10.1093/eurheartj/ehz732
311. Dewilde WJ, Janssen PW, Kelder JC, Verheugt FWA, De Smet BJGL, Adriaenssens T, *et al.* Uninterrupted oral anticoagulation versus bridging in patients with long-term oral anticoagulation during percutaneous coronary intervention: subgroup analysis from the WOEST trial. *EuroIntervention* 2015;**11**:381–390. https://doi.org/10.4244/eijy14m06_07
312. Windecker S, Lopes RD, Massaro T, Jones-Burton C, Granger CB, Aronson R, *et al.* Antithrombotic therapy in patients with atrial fibrillation and acute coronary syndrome treated medically or with percutaneous coronary intervention or undergoing elective percutaneous coronary intervention: insights from the AUGUSTUS trial. *Circulation* 2019;**140**:1921–1932. https://doi.org/10.1161/circulationaha.119.043308
313. Smits PC, Frigoli E, Tijssen J, Jüni P, Vranckx P, Ozaki Y, *et al.* Abbreviated antiplatelet therapy in patients at high bleeding risk with or without oral anticoagulant therapy after coronary stenting: an open-label, randomized, controlled trial. *Circulation* 2021;**144**:1196–1211. https://doi.org/10.1161/circulationaha.121.056680
314. Eikelboom JW, Connolly SJ, Bosch J, Dagenais GR, Hart RG, Shestakovska O, *et al.* Rivaroxaban with or without aspirin in stable cardiovascular disease. *N Engl J Med* 2017;**377**:1319–1330. https://doi.org/10.1056/NEJMoa1709118
315. Connolly SJ, Eikelboom JW, Bosch J, Dagenais G, Dyal L, Lanas F, *et al.* Rivaroxaban with or without aspirin in patients with stable coronary artery disease: an international, randomised, double-blind, placebo-controlled trial. *Lancet* 2018;**391**:205–218. https://doi.org/10.1016/s0140-6736(17)32458-3
316. Mauri L, Kereiakes DJ, Yeh RW, Driscoll-Shempp P, Cutlip DE, Steg PG, *et al.* Twelve or 30 months of dual antiplatelet therapy after drug-eluting stents. *N Engl J Med* 2014;**371**:2155–2166. https://doi.org/10.1056/NEJMoa1409312
317. Bonaca MP, Bhatt DL, Steg PG, Storey RF, Cohen M, Im K, *et al.* Ischaemic risk and efficacy of ticagrelor in relation to time from P2Y12 inhibitor withdrawal in patients with prior myocardial infarction: insights from PEGASUS-TIMI 54. *Eur Heart J* 2016;**37**:1133–1142. https://doi.org/10.1093/eurheartj/ehv531
318. Bonaca MP, Bhatt DL, Cohen M, Steg PG, Storey RF, Jensen EC, *et al.* Long-term use of ticagrelor in patients with prior myocardial infarction. *N Engl J Med* 2015;**372**:1791–1800. https://doi.org/10.1056/NEJMoa1500857
319. Steg PG, Bhatt DL, Simon T, Fox K, Mehta SR, Harrington RA, *et al.* Ticagrelor in patients with stable coronary disease and diabetes. *N Engl J Med* 2019;**381**:1309–1320. https://doi.org/10.1056/NEJMoa1908077
320. Valgimigli M, Gragnano F, Branca M, Franzone A, Baber U, Jang Y, *et al.* P2Y12 inhibitor monotherapy or dual antiplatelet therapy after coronary revascularisation: individual patient level meta-analysis of randomised controlled trials. *BMJ* 2021;**373**:n1332. https://doi.org/10.1136/bmj.n1332
321. Shoji S, Kuno T, Fujisaki T, Takagi H, Briasoulis A, Deharo P, *et al.* De-escalation of dual antiplatelet therapy in patients with acute coronary syndromes. *J Am Coll Cardiol* 2021;**78**:763–777. https://doi.org/10.1016/j.jacc.2021.06.012

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

322. Laudani C, Greco A, Occhipinti G, Ingala S, Calderone D, Scalia L, *et al.* Short duration of DAPT versus de-escalation after percutaneous coronary intervention for acute coronary syndromes. *JACC Cardiovasc Interv* 2022;**15**:268–277. https://doi.org/10.1016/j.jcin.2021.11.028
323. Giustino G, Mehran R, Dangas GD, Kirtane AJ, Redfors B, Généreux P, *et al.* Characterization of the average daily ischemic and bleeding risk after primary PCI for STEMI. *J Am Coll Cardiol* 2017;**70**:1846–1857. https://doi.org/10.1016/j.jacc.2017.08.018
324. Yasuda S, Kaikita K, Akao M, Ako J, Matoba T, Nakamura M, *et al.* Antithrombotic therapy for atrial fibrillation with stable coronary disease. *N Engl J Med* 2019;**381**:1103–1113. https://doi.org/10.1056/NEJMoa1904143
325. Matsumura-Nakano Y, Shizuta S, Komasa A, Morimoto T, Masuda H, Shiomi H, *et al.* Open-label randomized trial comparing oral anticoagulation with and without single antiplatelet therapy in patients with atrial fibrillation and stable coronary artery disease beyond 1 year after coronary stent implantation. *Circulation* 2019;**139**:604–616. https://doi.org/10.1161/circulationaha.118.036768
326. Chiarito M, Sanz-Sánchez J, Cannata F, Cao D, Sturla M, Panico C, *et al.* Monotherapy with a P2Y(12) inhibitor or aspirin for secondary prevention in patients with established atherosclerosis: a systematic review and meta-analysis. *Lancet* 2020;**395**:1487–1495. https://doi.org/10.1016/s0140-6736(20)30315-9
327. Koo BK, Kang J, Park KW, Rhee T-M, Yang H-M, Won KB, *et al.* Aspirin versus clopidogrel for chronic maintenance monotherapy after percutaneous coronary intervention (HOST-EXAM): an investigator-initiated, prospective, randomised, open-label, multicentre trial. *Lancet* 2021;**397**:2487–2496. https://doi.org/10.1016/s0140-6736(21)01063-1
328. Gilard M, Blanchard D, Helft G, Carrier D, Eltchaninoff H, Belle L, *et al.* Antiplatelet therapy in patients with anticoagulants undergoing percutaneous coronary stenting (from STENTIng and oral antiCOagulants [STENTICO]). *Am J Cardiol* 2009;**104**:338–342. https://doi.org/10.1016/j.amjcard.2009.03.053
329. Lip GY, Windecker S, Huber K, Kirchhof P, Marin F, Ten Berg JM, *et al.* Management of antithrombotic therapy in atrial fibrillation patients presenting with acute coronary syndrome and/or undergoing percutaneous coronary or valve interventions: a joint consensus document of the European Society of Cardiology Working Group on Thrombosis, European Heart Rhythm Association (EHRA), European Association of Percutaneous Cardiovascular Interventions (EAPCI) and European Association of Acute Cardiac Care (ACCA) endorsed by the Heart Rhythm Society (HRS) and Asia-Pacific Heart Rhythm Society (APHRS). *Eur Heart J* 2014;**35**:3155–3179. https://doi.org/10.1093/eurheartj/ehu298
330. Ruiz-Nodar JM, Marín F, Hurtado JA, Valencia J, Pinar E, Pineda J, *et al.* Anticoagulant and antiplatelet therapy use in 426 patients with atrial fibrillation undergoing percutaneous coronary intervention and stent implantation implications for bleeding risk and prognosis. *J Am Coll Cardiol* 2008;**51**:818–825. https://doi.org/10.1016/j.jacc.2007.11.035
331. Beyer-Westendorf J, Gelbricht V, Förster K, Ebertz F, Kohler C, Werth S, *et al.* Peri-interventional management of novel oral anticoagulants in daily care: results from the prospective Dresden NOAC registry. *Eur Heart J* 2014;**35**:1888–1896. https://doi.org/10.1093/eurheartj/eht557
332. Kiviniemi T, Karjalainen P, Pietilä M, Ylitalo A, Niemelä M, Vikman S, *et al.* Comparison of additional versus no additional heparin during therapeutic oral anticoagulation in patients undergoing percutaneous coronary intervention. *Am J Cardiol* 2012;**110**:30–35. https://doi.org/10.1016/j.amjcard.2012.02.045
333. Fiedler KA, Maeng M, Mehilli J, Schulz-Schüpke S, Byrne RA, Sibbing D, *et al.* Duration of triple therapy in patients requiring oral anticoagulation after drug-eluting stent implantation: the ISAR-TRIPLE trial. *J Am Coll Cardiol* 2015;**65**:1619–1629. https://doi.org/10.1016/j.jacc.2015.02.050
334. Lopes RD, Leonardi S, Wojdyla DM, Vora AN, Thomas L, Storey RF, *et al.* Stent thrombosis in patients with atrial fibrillation undergoing coronary stenting in the AUGUSTUS trial. *Circulation* 2020;**141**:781–783. https://doi.org/10.1161/circulationaha.119.044584
335. Alexander JH, Wojdyla D, Vora AN, Thomas L, Granger CB, Goodman SG, *et al.* Risk/benefit tradeoff of antithrombotic therapy in patients with atrial fibrillation early and late after an acute coronary syndrome or percutaneous coronary intervention: insights from AUGUSTUS. *Circulation* 2020;**141**:1618–1627. https://doi.org/10.1161/circulationaha.120.046534
336. Lopes RD, Hong H, Harskamp RE, Bhatt DL, Mehran R, Cannon CP, *et al.* Safety and efficacy of antithrombotic strategies in patients with atrial fibrillation undergoing percutaneous coronary intervention: a network meta-analysis of randomized controlled trials. *JAMA Cardiol* 2019;**4**:747–755. https://doi.org/10.1001/jamacardio.2019.1880
337. Lopes RD, Hong H, Harskamp RE, Bhatt DL, Mehran R, Cannon CP, *et al.* Optimal antithrombotic regimens for patients with atrial fibrillation undergoing percutaneous coronary intervention: an updated network meta-analysis. *JAMA Cardiol* 2020;**5**:582–589. https://doi.org/10.1001/jamacardio.2019.6175
338. Capodanno D, Di Maio M, Greco A, Bhatt DL, Gibson CM, Goette A, *et al.* Safety and efficacy of double antithrombotic therapy with non-vitamin K antagonist oral anticoagulants in patients with atrial fibrillation undergoing percutaneous coronary intervention: a systematic review and meta-analysis. *J Am Heart Assoc* 2020;**9**:e017212. https://doi.org/10.1161/jaha.120.017212
339. Lip GYH, Collet JP, Haude M, Byrne R, Chung EH, Fauchier L, *et al.* 2018 Joint European consensus document on the management of antithrombotic therapy in atrial fibrillation patients presenting with acute coronary syndrome and/or undergoing percutaneous cardiovascular interventions: a joint consensus document of the European Heart Rhythm Association (EHRA), European Society of Cardiology Working Group on Thrombosis, European Association of Percutaneous Cardiovascular Interventions (EAPCI), and European Association of Acute Cardiac Care (ACCA) endorsed by the Heart Rhythm Society (HRS), Asia-Pacific Heart Rhythm Society (APHRS), Latin America Heart Rhythm Society (LAHRS), and Cardiac Arrhythmia Society of Southern Africa (CASSA). *Europace* 2019;**21**:192–193. https://doi.org/10.1093/europace/euy174
340. ISIS-2 Collaborative Group. Randomised trial of intravenous streptokinase, oral aspirin, both, or neither among 17,187 cases of suspected acute myocardial infarction: ISIS-2. *Lancet* 1988;**2**:349–360.
341. Chen ZM, Jiang LX, Chen YP, Xie JX, Pan HC, Peto R, *et al.* Addition of clopidogrel to aspirin in 45,852 patients with acute myocardial infarction: randomised placebo-controlled trial. *Lancet* 2005;**366**:1607–1621. https://doi.org/10.1016/s0140-6736(05)67660-x
342. Sabatine MS, Cannon CP, Gibson CM, López-Sendón JL, Montalescot G, Theroux P, *et al.* Addition of clopidogrel to aspirin and fibrinolytic therapy for myocardial infarction with ST-segment elevation. *N Eng J Med* 2005;**352**:1179–1189. https://doi.org/10.1056/NEJMoa050522
343. Osman M, Kheiri B, Shigle AJ, Saleem M, Osman K, Sengupta PP, *et al.* Ticagrelor after pharmacological thrombolysis in patients with ST-segment elevation myocardial infarctions: insight from a trial sequential analysis. *J Thromb Thrombolysis* 2019;**48**:661–667. https://doi.org/10.1007/s11239-019-01953-3
344. Berwanger O, Lopes RD, Moia DDF, Fonseca FA, Jiang L, Goodman SG, *et al.* Ticagrelor versus clopidogrel in patients with STEMI treated with fibrinolysis: TREAT trial. *J Am Coll Cardiol* 2019;**73**:2819–2828. https://doi.org/10.1016/j.jacc.2019.03.011
345. Kheiri B, Osman M, Abdalla A, Haykal T, Barbarawi M, Zayed Y, *et al.* Ticagrelor versus clopidogrel after fibrinolytic therapy in patients with ST-elevation myocardial infarction: a systematic review and meta-analysis of randomized clinical trials. *J Thromb Thrombolysis* 2018;**46**:299–303. https://doi.org/10.1007/s11239-018-1706-2
346. Sánchez PL, Gimeno F, Ancillo P, Sanz JJ, Alonso-Briales JH, Bosa F, *et al.* Role of the paclitaxel-eluting stent and tirofiban in patients with ST-elevation myocardial infarction undergoing postfibrinolysis angioplasty: the GRACIA-3 randomized clinical trial. *Circ Cardiovasc Interv* 2010;**3**:297–307. https://doi.org/10.1161/circinterventions.109.920868
347. ASSENT-3 Investigators. Efficacy and safety of tenecteplase in combination with enoxaparin, abciximab, or unfractionated heparin: the ASSENT-3 randomised trial in acute myocardial infarction. *Lancet* 2001;**358**:605–613. https://doi.org/10.1016/s0140-6736(01)05775-0
348. Giraldez RR, Nicolau JC, Corbalan R, Gurfinkel EP, Juarez U, Lopez-Sendon J, *et al.* Enoxaparin is superior to unfractionated heparin in patients with ST elevation myocardial infarction undergoing fibrinolysis regardless of the choice of lytic: an ExTRACT-TIMI 25 analysis. *Eur Heart J* 2007;**28**:1566–1573. https://doi.org/10.1093/eurheartj/ehm179
349. White HD, Braunwald E, Murphy SA, Jacob AJ, Gotcheva N, Polonetsky L, *et al.* Enoxaparin vs. unfractionated heparin with fibrinolysis for ST-elevation myocardial infarction in elderly and younger patients: results from ExTRACT-TIMI 25. *Eur Heart J* 2007;**28**:1066–1071. https://doi.org/10.1093/eurheartj/ehm081
350. Peters RJ, Joyner C, Bassand JP, Afzal R, Chrolavicius S, Mehta SR, *et al.* The role of fondaparinux as an adjunct to thrombolytic therapy in acute myocardial infarction: a subgroup analysis of the OASIS-6 trial. *Eur Heart J* 2008;**29**:324–331. https://doi.org/10.1093/eurheartj/ehm616
351. White H. Thrombin-specific anticoagulation with bivalirudin versus heparin in patients receiving fibrinolytic therapy for acute myocardial infarction: the HERO-2 randomised trial. *Lancet* 2001;**358**:1855–1863. https://doi.org/10.1016/s0140-6736(01)06887-8
352. Fernández-Avilés F, Alonso JJ, Peña G, Blanco J, Alonso-Briales J, Lopez-Mesa J, *et al.* Primary angioplasty vs. early routine post-fibrinolysis angioplasty for acute myocardial infarction with ST-segment elevation: the GRACIA-2 non-inferiority, randomized, controlled trial. *Eur Heart J* 2007;**28**:949–960. https://doi.org/10.1093/eurheartj/ehl461
353. Björklund E, Stenestrand U, Lindbäck J, Svensson L, Wallentin L, Lindahl B. Pre-hospital thrombolysis delivered by paramedics is associated with reduced time delay and mortality in ambulance-transported real-life patients with ST-elevation myocardial infarction. *Eur Heart J* 2006;**27**:1146–1152. https://doi.org/10.1093/eurheartj/ehi886
354. Bonnefoy E, Steg PG, Boutitie F, Dubien P-Y, Lapostolle F, Roncalli J, *et al.* Comparison of primary angioplasty and pre-hospital fibrinolysis in acute myocardial infarction (CAPTIM) trial: a 5-year follow-up. *Eur Heart J* 2009;**30**:1598–1606. https://doi.org/10.1093/eurheartj/ehp156
355. Bonnefoy E, Lapostolle F, Leizorovicz A, Steg G, McFadden EP, Dubien PY, *et al.* Primary angioplasty versus prehospital fibrinolysis in acute myocardial infarction: a

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

randomised study. *Lancet* 2002;**360**:825–829. https://doi.org/10.1016/s0140-6736(02)09963-4
356. Assessment of the Safety and Efficacy of a New Thrombolytic (ASSENT-2) Investigators. Single-bolus tenecteplase compared with front-loaded alteplase in acute myocardial infarction: the ASSENT-2 double-blind randomised trial. *Lancet* 1999;**354**:716–722. https://doi.org/10.1016/s0140-6736(99)07403-6
357. GUSTO investigators. An international randomized trial comparing four thrombolytic strategies for acute myocardial infarction. *N Engl J Med* 1993;**329**:673–682. https://doi.org/10.1056/nejm199309023291001
358. Wallentin L, Goldstein P, Armstrong PW, Granger CB, Adgey AAJ, Arntz HR, *et al.* Efficacy and safety of tenecteplase in combination with the low-molecular-weight heparin enoxaparin or unfractionated heparin in the prehospital setting: the Assessment of the Safety and Efficacy of a New Thrombolytic Regimen (ASSENT)-3 PLUS randomized trial in acute myocardial infarction. *Circulation* 2003;**108**:135–142. https://doi.org/10.1161/01.Cir.0000081659.72985.A8
359. Ross AM, Molhoek P, Lundergan C, Knudtson M, Draoui Y, Regalado L, *et al.* Randomized comparison of enoxaparin, a low-molecular-weight heparin, with unfractionated heparin adjunctive to recombinant tissue plasminogen activator thrombolysis and aspirin: second trial of Heparin and Aspirin Reperfusion Therapy (HART II). *Circulation* 2001;**104**:648–652. https://doi.org/10.1161/hc3101.093866
360. Antman EM, Louwerenburg HW, Baars HF, Wesdorp JCL, Hamer B, Bassand J-P, *et al.* Enoxaparin as adjunctive antithrombin therapy for ST-elevation myocardial infarction: results of the ENTIRE-Thrombolysis in Myocardial Infarction (TIMI) 23 Trial. *Circulation* 2002;**105**:1642–1649. https://doi.org/10.1161/01.cir.0000013402.34759.46
361. James SK, Roe MT, Cannon CP, Cornel JH, Horrow J, Husted S, *et al.* Ticagrelor versus clopidogrel in patients with acute coronary syndromes intended for non-invasive management: substudy from prospective randomised PLATelet inhibition and patient Outcomes (PLATO) trial. *BMJ* 2011;**342**:d3527. https://doi.org/10.1136/bmj.d3527
362. Wiviott SD, White HD, Ohman EM, Fox KAA, Armstrong PW, Prabhakaran D, *et al.* Prasugrel versus clopidogrel for patients with unstable angina or non-ST-segment elevation myocardial infarction with or without angiography: a secondary, prespecified analysis of the TRILOGY ACS trial. *Lancet* 2013;**382**:605–613. https://doi.org/10.1016/s0140-6736(13)61451-8
363. Savonitto S, Ferri LA, Piatti L, Grosseto D, Piovaccari G, Morici N, *et al.* Comparison of reduced-dose prasugrel and standard-dose clopidogrel in elderly patients with acute coronary syndromes undergoing early percutaneous revascularization. *Circulation* 2018;**137**:2435–2445. https://doi.org/10.1161/circulationaha.117.032180
364. Patterson T, Perkins GD, Hassan Y, Moschonas K, Gray H, Curzen N, *et al.* Temporal trends in identification, management, and clinical outcomes after out-of-hospital cardiac arrest: insights from the myocardial ischaemia national audit project database. *Circ Cardiovasc Interv* 2018;**11**:e005346. https://doi.org/10.1161/circinterventions.117.005346
365. Byrne R, Constant O, Smyth Y, Callagy G, Nash P, Daly K, *et al.* Multiple source surveillance incidence and aetiology of out-of-hospital sudden cardiac death in a rural population in the West of Ireland. *Eur Heart J* 2008;**29**:1418–1423. https://doi.org/10.1093/eurheartj/ehn155
366. Kroupa J, Knot J, Ulman J, Bednar F, Dohnalova A, Motovska Z. Characteristics and survival determinants in patients after out-of-hospital cardiac arrest in the era of 24/7 coronary intervention facilities. *Heart Lung Circ* 2017;**26**:799–807. https://doi.org/10.1016/j.hlc.2016.11.012
367. Perkins GD, Graesner JT, Semeraro F, Olasveengen T, Soar J, Lott C, *et al.* European Resuscitation Council Guidelines 2021: executive summary. *Resuscitation* 2021;**161**:1–60. https://doi.org/10.1016/j.resuscitation.2021.02.003
368. Dumas F, Cariou A, Manzo-Silberman S, Grimaldi D, Vivien B, Rosencher J, *et al.* Immediate percutaneous coronary intervention is associated with better survival after out-of-hospital cardiac arrest: insights from the PROCAT (Parisian Region Out of hospital Cardiac ArresT) registry. *Circ Cardiovasc Interv* 2010;**3**:200–207. https://doi.org/10.1161/circinterventions.109.913665
369. Nolan JP, Sandroni C, Böttiger BW, Cariou A, Cronberg T, Friberg H, *et al.* European Resuscitation Council and European Society of Intensive Care Medicine guidelines 2021: post-resuscitation care. *Intensive Care Med* 2021;**47**:369–421. https://doi.org/10.1007/s00134-021-06368-4
370. Rab T, Kern KB, Tamis-Holland JE, Henry TD, McDaniel M, Dickert NW, *et al.* Cardiac arrest: a treatment algorithm for emergent invasive cardiac procedures in the resuscitated comatose patient. *J Am Coll Cardiol* 2015;**66**:62–73. https://doi.org/10.1016/j.jacc.2015.05.009
371. Gorjup V, Radsel P, Kocjancic ST, Erzen D, Noc M. Acute ST-elevation myocardial infarction after successful cardiopulmonary resuscitation. *Resuscitation* 2007;**72**:379–385. https://doi.org/10.1016/j.resuscitation.2006.07.013
372. Lemkes JS, Janssens GN, van der Hoeven NW, Jewbali LSD, Dubois EA, Meuwissen MM, *et al.* Coronary angiography after cardiac arrest without ST segment elevation: one-year outcomes of the COACT randomized clinical trial. *JAMA Cardiol* 2020;**5**:1358–1365. https://doi.org/10.1001/jamacardio.2020.3670
373. Desch S, Freund A, Akin I, Behnes M, Preusch MR, Zelniker TA, *et al.* Angiography after out-of-hospital cardiac arrest without ST-segment elevation. *N Engl J Med* 2021;**385**:2544–2553. https://doi.org/10.1056/NEJMoa2101909
374. Lemkes JS, Janssens GN, van der Hoeven NW, Jewbali LSD, Dubois EA, Meuwissen M, *et al.* Coronary angiography after cardiac arrest without ST-segment elevation. *N Engl J Med* 2019;**380**:1397–1407. https://doi.org/10.1056/NEJMoa1816897
375. Kern KB, Radsel P, Jentzer JC, Seder DB, Lee KS, Lotun K, *et al.* Randomized pilot clinical trial of early coronary angiography versus no early coronary angiography after cardiac arrest without ST-segment elevation: the PEARL study. *Circulation* 2020;**142**:2002–2012. https://doi.org/10.1161/circulationaha.120.049569
376. Hauw-Berlemont C, Lamhaut L, Diehl JL, Andreotti C, Varenne O, Leroux P, *et al.* Emergency vs delayed coronary angiogram in survivors of out-of-hospital cardiac arrest: results of the randomized, multicentric EMERGE trial. *JAMA Cardiol* 2022;**7**:700–707. https://doi.org/10.1001/jamacardio.2022.1416
377. Viana-Tejedor A, Andrea-Riba R, Scardino C, Ariza-Solé A, Bañeras J, García-García C, *et al.* Coronary angiography in patients without ST-segment elevation following out-of-hospital cardiac arrest. *Rev Esp Cardiol (Engl Ed)* 2022;**76**:94–102. https://doi.org/10.1016/j.rec.2022.05.013
378. The Hypothermia after Cardiac Arrest Study Group. Mild therapeutic hypothermia to improve the neurologic outcome after cardiac arrest. *N Engl J Med* 2002;**346**:549–556. https://doi.org/10.1056/NEJMoa012689
379. Bernard SA, Gray TW, Buist MD, Jones BM, Silvester W, Gutteridge G, *et al.* Treatment of comatose survivors of out-of-hospital cardiac arrest with induced hypothermia. *N Engl J Med* 2002;**346**:557–563. https://doi.org/10.1056/NEJMoa003289
380. Belliard G, Catez E, Charron C, Caille V, Aegerter P, Dubourg O, *et al.* Efficacy of therapeutic hypothermia after out-of-hospital cardiac arrest due to ventricular fibrillation. *Resuscitation* 2007;**75**:252–259. https://doi.org/10.1016/j.resuscitation.2007.04.014
381. Nielsen N, Wetterslev J, Cronberg T, Erlinge D, Gasche Y, Hassager C, *et al.* Targeted temperature management at 33°C versus 36°C after cardiac arrest. *N Engl J Med* 2013;**369**:2197–2206. https://doi.org/10.1056/NEJMoa1310519
382. Vaahersalo J, Hiltunen P, Tiainen M, Oksanen T, Kaukonen K-M, Kurola J, *et al.* Therapeutic hypothermia after out-of-hospital cardiac arrest in Finnish intensive care units: the FINNRESUSCI study. *Intensive Care Med* 2013;**39**:826–837. https://doi.org/10.1007/s00134-013-2868-1
383. Nolan JP, Sandroni C, Andersen LW, Böttiger BW, Cariou A, Cronberg T, *et al.* ERC-ESICM guidelines on temperature control after cardiac arrest in adults. *Resuscitation* 2022;**172**:229–236. https://doi.org/10.1016/j.resuscitation.2022.01.009
384. Hassager C, Schmidt H, Møller JE, Grand J, Mølstrøm S, Beske RP, *et al.* Duration of device-based fever prevention after cardiac arrest. *N Engl J Med* 2023;**388**:888–897. https://doi.org/10.1056/NEJMoa2212528
385. Wolfrum S, Roedl K, Hanebutte A, Pfeifer R, Kurowski V, Riessen R, *et al.* Temperature control after in-hospital cardiac arrest: a randomized clinical trial. *Circulation* 2022;**146**:1357–1366. https://doi.org/10.1161/circulationaha.122.060106
386. Sandroni C, Geocadin RG. Neurological prognostication after cardiac arrest. *Curr Opin Crit Care* 2015;**21**:209–214. https://doi.org/10.1097/mcc.0000000000000202
387. Garot P, Lefevre T, Eltchaninoff H, Morice M-C, Tamion F, Abry B, *et al.* Six-month outcome of emergency percutaneous coronary intervention in resuscitated patients after cardiac arrest complicating ST-elevation myocardial infarction. *Circulation* 2007;**115**:1354–1362. https://doi.org/10.1161/circulationaha.106.657619
388. Spaulding CM, Joly LM, Rosenberg A, Monchi M, Weber SN, Dhainaut J-FA, *et al.* Immediate coronary angiography in survivors of out-of-hospital cardiac arrest. *N Engl J Med* 1997;**336**:1629–1633. https://doi.org/10.1056/nejm199706053362302
389. Dankiewicz J, Cronberg T, Lilja G, Jakobsen JC, Levin H, Ullén S, *et al.* Hypothermia versus normothermia after out-of-hospital cardiac arrest. *N Engl J Med* 2021;**384**:2283–2294. https://doi.org/10.1056/NEJMoa2100591
390. Callaway CW, Schmicker R, Kampmeyer M, Powell J, Rea TD, Daya MR, *et al.* Receiving hospital characteristics associated with survival after out-of-hospital cardiac arrest. *Resuscitation* 2010;**81**:524–529. https://doi.org/10.1016/j.resuscitation.2009.12.006
391. Wnent J, Seewald S, Heringlake M, Lemke H, Brauer K, Lefering R, *et al.* Choice of hospital after out-of-hospital cardiac arrest – a decision with far-reaching consequences: a study in a large German city. *Crit Care* 2012;**16**:R164. https://doi.org/10.1186/cc11516
392. Kragholm K, Malta Hansen C, Dupre ME, Xian Y, Strauss B, Tyson C, *et al.* Direct transport to a percutaneous cardiac intervention center and outcomes in patients with out-of-hospital cardiac arrest. *Circ Cardiovasc Qual Outcomes* 2017;**10**:e003414. https://doi.org/10.1161/circoutcomes.116.003414
393. Yeo JW, Ng ZHC, Goh AXC, Gao JF, Liu N, Lam SWS, *et al.* Impact of cardiac arrest centers on the survival of patients with nontraumatic out-of-hospital cardiac arrest: a systematic review and meta-analysis. *J Am Heart Assoc* 2022;**11**:e023806. https://doi.org/10.1161/jaha.121.023806
394. Hochman JS, Sleeper LA, Webb JG, Sanborn TA, White HD, Talley JD, *et al.* Early revascularization in acute myocardial infarction complicated by cardiogenic shock. *N Engl J Med* 1999;**341**:625–634. https://doi.org/10.1056/nejm199908263410901
395. White HD, Assmann SF, Sanborn TA, Jacobs AK, Webb JG, Sleeper LA, *et al.* Comparison of percutaneous coronary intervention and coronary artery bypass

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

grafting after acute myocardial infarction complicated by cardiogenic shock: results from the Should We Emergently Revascularize Occluded Coronaries for Cardiogenic Shock (SHOCK) trial. *Circulation* 2005;**112**:1992–2001. https://doi.org/10.1161/circulationaha.105.540948
396. Hochman JS, Sleeper LA, Webb JG, Dzavik V, Buller CE, Aylward P, *et al.* Early revascularization and long-term survival in cardiogenic shock complicating acute myocardial infarction. *JAMA* 2006;**295**:2511–2515. https://doi.org/10.1001/jama.295.21.2511
397. Liakopoulos OJ, Schlachtenberger G, Wendt D, Choi Y-H, Slottosch I, Welp H, *et al.* Early clinical outcomes of surgical myocardial revascularization for acute coronary syndromes complicated by cardiogenic shock: a report from the north-Rhine-Westphalia surgical myocardial infarction registry. *J Am Heart Assoc* 2019;**8**:e012049. https://doi.org/10.1161/jaha.119.012049
398. Thielmann M, Wendt D, Slottosch I, Welp H, Schiller W, Tsagakis K, *et al.* Coronary artery bypass graft surgery in patients with acute coronary syndromes after primary percutaneous coronary intervention: a current report from the north-Rhine Westphalia surgical myocardial infarction registry. *J Am Heart Assoc* 2021;**10**:e021182. https://doi.org/10.1161/jaha.121.021182
399. Thiele H, Zeymer U, Neumann FJ, Ferenc M, Olbrich H-G, Hausleiter J, *et al.* Intraaortic balloon support for myocardial infarction with cardiogenic shock. *N Engl J Med* 2012;**367**:1287–1296. https://doi.org/10.1056/NEJMoa1208410
400. Schrage B, Ibrahim K, Loehn T, Werner N, Sinning J-M, Pappalardo F, *et al.* Impella support for acute myocardial infarction complicated by cardiogenic shock. *Circulation* 2019;**139**:1249–1258. https://doi.org/10.1161/circulationaha.118.036614
401. Miller PE, Bromfield SG, Ma Q, Crawford G, Whitney J, DeVries A, *et al.* Clinical outcomes and cost associated with an intravascular microaxial left ventricular assist device vs intra-aortic balloon pump in patients presenting with acute myocardial infarction complicated by cardiogenic shock. *JAMA Intern Med* 2022;**182**:926–933. https://doi.org/10.1001/jamainternmed.2022.2735
402. Ostadal P, Rokyta R, Karasek J, Kruger A, Vondrakova D, Janotka M, *et al.* Extracorporeal membrane oxygenation in the therapy of cardiogenic shock: results of the ECMO-CS randomized clinical trial. *Circulation* 2022;**147**:454–464. https://doi.org/10.1161/circulationaha.122.062949
403. Kim Y, Shapero K, Ahn SS, Goldsweig AM, Desai N, Altin SE. Outcomes of mechanical circulatory support for acute myocardial infarction complicated by cardiogenic shock. *Catheter Cardiovasc Interv* 2022;**99**:658–663. https://doi.org/10.1002/ccd.29834
404. Thiele H, Akin I, Sandri M, Fuernau G, de Waha S, Meyer-Saraei R, *et al.* PCI strategies in patients with acute myocardial infarction and cardiogenic shock. *N Engl J Med* 2017;**377**:2419–2432. https://doi.org/10.1056/NEJMoa1710261
405. Thiele H, Zeymer U, Neumann FJ, Ferenc M, Olbrich H-G, Hausleiter J, *et al.* Intra-aortic balloon counterpulsation in acute myocardial infarction complicated by cardiogenic shock (IABP-SHOCK II): final 12 month results of a randomised, open-label trial. *Lancet* 2013;**382**:1638–1645. https://doi.org/10.1016/s0140-6736(13)61783-3
406. Unverzagt S, Buerke M, de Waha A, Haerting J, Pietzner D, Seyfarth M, *et al.* Intra-aortic balloon pump counterpulsation (IABP) for myocardial infarction complicated by cardiogenic shock. *Cochrane Database Syst Rev* 2015;**2015**:Cd007398. https://doi.org/10.1002/14651858.CD007398.pub3
407. Thiele H, Zeymer U, Thelemann N, Neumann F-J, Hausleiter J, Abdel-Wahab M, *et al.* Intraaortic balloon pump in cardiogenic shock complicating acute myocardial infarction: long-term 6-year outcome of the randomized IABP-SHOCK II trial. *Circulation* 2018;**139**:395–403. https://doi.org/10.1161/circulationaha.118.038201
408. Bonnefoy-Cudraz E, Bueno H, Casella G, De Maria E, Fitzsimons D, Halvorsen S, *et al.* Editor's Choice – acute cardiovascular care association position paper on intensive cardiovascular care units: an update on their definition, structure, organisation and function. *Eur Heart J Acute Cardiovasc Care* 2018;**7**:80–95. https://doi.org/10.1177/2048872617724269
409. Winkler C, Funk M, Schindler DM, Hemsey JZ, Lampert R, Drew BJ. Arrhythmias in patients with acute coronary syndrome in the first 24 hours of hospitalization. *Heart Lung* 2013;**42**:422–427. https://doi.org/10.1016/j.hrtlng.2013.07.010
410. Wasfy JH, Kennedy KF, Masoudi FA, Ferris TG, Arnold SV, Kini V, *et al.* Predicting length of stay and the need for postacute care after acute myocardial infarction to improve healthcare efficiency. *Circ Cardiovasc Qual Outcomes* 2018;**11**:e004635. https://doi.org/10.1161/circoutcomes.118.004635
411. Melberg T, Jørgensen M, Ørn S, Solli T, Edland U, Dickstein K. Safety and health status following early discharge in patients with acute myocardial infarction treated with primary PCI: a randomized trial. *Eur J Prev Cardiol* 2015;**22**:1427–1434. https://doi.org/10.1177/2047487314559276
412. Berger AK, Duval S, Jacobs DR Jr, Barber C, Vazquez G, Lee S, *et al.* Relation of length of hospital stay in acute myocardial infarction to postdischarge mortality. *Am J Cardiol* 2008;**101**:428–434. https://doi.org/10.1016/j.amjcard.2007.09.090
413. Grines CL, Marsalese DL, Brodie B, Griffin J, Donohue B, Costantini CR, *et al.* Safety and cost-effectiveness of early discharge after primary angioplasty in low risk patients with acute myocardial infarction. *J Am Coll Cardiol* 1998;**31**:967–972. https://doi.org/10.1016/s0735-1097(98)00031-x
414. De Luca G, Suryapranata H, van't Hof AW, de Boer M-J, Hoorntje JCA, Dambrink J-HE, *et al.* Prognostic assessment of patients with acute myocardial infarction treated with primary angioplasty: implications for early discharge. *Circulation* 2004;**109**:2737–2743. https://doi.org/10.1161/01.Cir.0000131765.73959.87
415. Novobilsky K, Stipal R, Cerny P, Horak I, Kaucak V, Mrozek J, *et al.* Safety of early discharge in low risk patients after acute ST-segment elevation myocardial infarction, treated with primary percutaneous coronary intervention. Open label, randomized trial. *Biomed Pap Med Fac Univ Palacky Olomouc Czech Repub* 2019;**163**:61–66. https://doi.org/10.5507/bp.2018.041
416. Albanese M, Alpaslan K, Ouarrak T, Merguet P, Schneider S, Schöls W. In-hospital major arrhythmias, arrhythmic death and resuscitation after successful primary percutaneous intervention for acute transmural infarction: a retrospective single-centre cohort study. *BMC Cardiovasc Disord* 2018;**18**:116. https://doi.org/10.1186/s12872-018-0851-z
417. Yndigegn T, Gilje P, Dankiewicz J, Mokhtari A, Isma N, Holmqvist J, *et al.* Safety of early hospital discharge following admission with ST-elevation myocardial infarction treated with percutaneous coronary intervention: a nationwide cohort study. *EuroIntervention* 2022;**17**:1091–1099. https://doi.org/10.4244/eij-d-21-00501
418. Seto AH, Shroff A, Abu-Fadel M, Blankenship JC, Boudoulas KD, Cigarroa JE, *et al.* Length of stay following percutaneous coronary intervention: an expert consensus document update from the society for cardiovascular angiography and interventions. *Catheter Cardiovasc Interv* 2018;**92**:717–731. https://doi.org/10.1002/ccd.27637
419. Estévez-Loureiro R, Calviño-Santos R, Vázquez JM, Barge-Caballero E, Salgado-Fernández J, Piñeiro M, *et al.* Safety and feasibility of returning patients early to their originating centers after transfer for primary percutaneous coronary intervention. *Rev Esp Cardiol* 2009;**62**:1356–1364. https://doi.org/10.1016/s1885-5857(09)73529-7
420. Morrow DA, Antman EM, Charlesworth A, Cairns R, Murphy SA, de Lemos JA, *et al.* TIMI risk score for ST-elevation myocardial infarction: a convenient, bedside, clinical score for risk assessment at presentation: an intravenous nPA for treatment of infarcting myocardium early II trial substudy. *Circulation* 2000;**102**:2031–2037. https://doi.org/10.1161/01.cir.102.17.2031
421. Fox KA, Dabbous OH, Goldberg RJ, Pieper KS, Eagle KA, Van de Werf F, *et al.* Prediction of risk of death and myocardial infarction in the six months after presentation with acute coronary syndrome: prospective multinational observational study (GRACE). *BMJ* 2006;**333**:1091. https://doi.org/10.1136/bmj.38985.646481.55
422. Aragam KG, Tamhane UU, Kline-Rogers E, Li J, Fox KAA, Goodman SG, *et al.* Does simplicity compromise accuracy in ACS risk prediction? A retrospective analysis of the TIMI and GRACE risk scores. *PLoS One* 2009;**4**:e7947. https://doi.org/10.1371/journal.pone.0007947
423. D'Ascenzo F, Biondi-Zoccai G, Moretti C, Bollati M, Omedè P, Sciuto F, *et al.* TIMI, GRACE and alternative risk scores in Acute Coronary Syndromes: a meta-analysis of 40 derivation studies on 216,552 patients and of 42 validation studies on 31,625 patients. *Contemp Clin Trials* 2012;**33**:507–514. https://doi.org/10.1016/j.cct.2012.01.001
424. Gale CP, Manda SO, Weston CF, Birkhead JS, Batin PD, Hall AS. Evaluation of risk scores for risk stratification of acute coronary syndromes in the Myocardial Infarction National Audit Project (MINAP) database. *Heart* 2009;**95**:221–227. https://doi.org/10.1136/hrt.2008.144022
425. D'Ascenzo F, De Filippo O, Gallone G, Mittone G, Deriu MA, Iannaccone M, *et al.* Machine learning-based prediction of adverse events following an acute coronary syndrome (PRAISE): a modelling study of pooled datasets. *Lancet* 2021;**397**:199–207. https://doi.org/10.1016/s0140-6736(20)32519-8
426. Ng VG, Lansky AJ, Meller S, Witzenbichler B, Guagliumi G, Peruga JZ, *et al.* The prognostic importance of left ventricular function in patients with ST-segment elevation myocardial infarction: the HORIZONS-AMI trial. *Eur Heart J Acute Cardiovasc Care* 2014;**3**:67–77. https://doi.org/10.1177/2048872613507149
427. de Waha S, Eitel I, Desch S, Fuernau G, Lurz P, Stiermaier T, *et al.* Prognosis after ST-elevation myocardial infarction: a study on cardiac magnetic resonance imaging versus clinical routine. *Trials* 2014;**15**:249. https://doi.org/10.1186/1745-6215-15-249
428. Larose E, Côté J, Rodés-Cabau J, Noël B, Barbeau G, Bordeleau E, *et al.* Contrast-enhanced cardiovascular magnetic resonance in the hyperacute phase of ST-elevation myocardial infarction. *Int J Cardiovasc Imaging* 2009;**25**:519–527. https://doi.org/10.1007/s10554-009-9451-4
429. Stiermaier T, Jobs A, de Waha S, Fuernau G, Pöss J, Desch S, *et al.* Optimized prognosis assessment in ST-segment-elevation myocardial infarction using a cardiac magnetic resonance imaging risk score. *Circ Cardiovasc Imaging* 2017;**10**:e006774. https://doi.org/10.1161/circimaging.117.006774
430. de Waha S, Desch S, Eitel I, Fuernau G, Zachrau J, Leuschner A, *et al.* Impact of early vs. late microvascular obstruction assessed by magnetic resonance imaging on long-term outcome after ST-elevation myocardial infarction: a comparison with traditional prognostic markers. *Eur Heart J* 2010;**31**:2660–2668. https://doi.org/10.1093/eurheartj/ehq247
431. van Kranenburg M, Magro M, Thiele H, de Waha S, Eitel I, Cochet A, *et al.* Prognostic value of microvascular obstruction and infarct size, as measured by CMR in STEMI patients. *JACC Cardiovasc Imaging* 2014;**7**:930–939. https://doi.org/10.1016/j.jcmg.2014.05.010
432. van Loon RB, Veen G, Baur LHB, Kamp O, Bronzwaer JGF, Twisk JWR, *et al.* Improved clinical outcome after invasive management of patients with recent myocardial

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

infarction and proven myocardial viability: primary results of a randomized controlled trial (VIAMI-trial). *Trials* 2012;**13**:1. https://doi.org/10.1186/1745-6215-13-1
433. van Loon RB, Veen G, Baur LH, Twisk JW, van Rossum AC. Long-term follow-up of the viability guided angioplasty after acute myocardial infarction (VIAMI) trial. *Int J Cardiol* 2015;**186**:111–116. https://doi.org/10.1016/j.ijcard.2015.03.152
434. Priori SG, Blomström-Lundqvist C, Mazzanti A, Blom N, Borggrefe M, Camm J, *et al.* 2015 ESC Guidelines for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death: the Task Force for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death of the European Society of Cardiology (ESC). Endorsed by: Association for European Paediatric and Congenital Cardiology (AEPC). *Eur Heart J* 2015;**36**:2793–2867. https://doi.org/10.1093/eurheartj/ehv316
435. Ibanez B, Aletras AH, Arai AE, Arheden H, Bax J, Berry C, *et al.* Cardiac MRI endpoints in myocardial infarction experimental and clinical trials: JACC Scientific Expert Panel. *J Am Coll Cardiol* 2019;**74**:238–256. https://doi.org/10.1016/j.jacc.2019.05.024
436. Stone GW, Selker HP, Thiele H, Patel MR, Udelson JE, Ohman EM, *et al.* Relationship between infarct size and outcomes following primary PCI: patient-level analysis from 10 randomized trials. *J Am Coll Cardiol* 2016;**67**:1674–1683. https://doi.org/10.1016/j.jacc.2016.01.069
437. Ibáñez B, Heusch G, Ovize M, Van de Werf F. Evolving therapies for myocardial ischemia/reperfusion injury. *J Am Coll Cardiol* 2015;**65**:1454–1471. https://doi.org/10.1016/j.jacc.2015.02.032
438. Eitel I, de Waha S, Wöhrle J, Fuernau G, Lurz P, Pauschinger M, *et al.* Comprehensive prognosis assessment by CMR imaging after ST-segment elevation myocardial infarction. *J Am Coll Cardiol* 2014;**64**:1217–1226. https://doi.org/10.1016/j.jacc.2014.06.1194
439. Haaf P, Reichlin T, Twerenbold R, Hoeller R, Rubini Gimenez M, Zellweger C, *et al.* Risk stratification in patients with acute chest pain using three high-sensitivity cardiac troponin assays. *Eur Heart J* 2014;**35**:365–375. https://doi.org/10.1093/eurheartj/eht218
440. Tveit SH, Myhre PL, Hoff NJS, Le TM, Seljeflot I, Røysland R, *et al.* Superiority of high sensitivity cardiac troponin T vs. I for long-term prognostic value in patients with chest pain; data from the Akershus Cardiac Examination (ACE) 3 study. *Clin Biochem* 2020;**78**:10–17. https://doi.org/10.1016/j.clinbiochem.2019.12.016
441. Welsh P, Preiss D, Hayward C, Shah ASV, McAllister D, Briggs A, *et al.* Cardiac troponin T and troponin I in the general population. *Circulation* 2019;**139**:2754–2764. https://doi.org/10.1161/circulationaha.118.038529
442. Thygesen K, Mair J, Giannitsis E, Mueller C, Lindahl B, Blankenberg S, *et al.* How to use high-sensitivity cardiac troponins in acute cardiac care. *Eur Heart J* 2012;**33**:2252–2257. https://doi.org/10.1093/eurheartj/ehs154
443. Anavekar NS, McMurray JJ, Velazquez EJ, Solomon SD, Kober L, Rouleau J-L, *et al.* Relation between renal dysfunction and cardiovascular outcomes after myocardial infarction. *N Engl J Med* 2004;**351**:1285–1295. https://doi.org/10.1056/NEJMoa041365
444. Thygesen K, Mair J, Mueller C, Huber K, Weber M, Plebani M, *et al.* Recommendations for the use of natriuretic peptides in acute cardiac care: a position statement from the Study Group on Biomarkers in Cardiology of the ESC Working Group on Acute Cardiac Care. *Eur Heart J* 2012;**33**:2001–2006. https://doi.org/10.1093/eurheartj/ehq509
445. Ducrocq G, Schulte PJ, Budaj A, Cornel JH, Held C, Himmelmann A, *et al.* Balancing the risk of spontaneous ischemic and major bleeding events in acute coronary syndromes. *Am Heart J* 2017;**186**:91–99. https://doi.org/10.1016/j.ahj.2017.01.010
446. Valgimigli M, Costa F, Lokhnygina Y, Clare RM, Wallentin L, Moliterno DJ, *et al.* Trade-off of myocardial infarction vs. bleeding types on mortality after acute coronary syndrome: lessons from the Thrombin Receptor Antagonist for Clinical Event Reduction in Acute Coronary Syndrome (TRACER) randomized trial. *Eur Heart J* 2017;**38**:804–810. https://doi.org/10.1093/eurheartj/ehw525
447. Newby LK, Hasselblad V, Armstrong PW, Van de Werf F, Mark DB, White HD, *et al.* Time-based risk assessment after myocardial infarction. Implications for timing of discharge and applications to medical decision-making. *Eur Heart J* 2003;**24**:182–189. https://doi.org/10.1016/s0195-668x(02)00301-9
448. Kwok CS, Khan MA, Rao SV, Kinnaird T, Sperrin M, Buchan I, *et al.* Access and non-access site bleeding after percutaneous coronary intervention and risk of subsequent mortality and major adverse cardiovascular events: systematic review and meta-analysis. *Circ Cardiovasc Interv* 2015;**8**:e001645. https://doi.org/10.1161/circinterventions.114.001645
449. Ndrepepa G, Neumann FJ, Richardt G, Schulz S, Tölg R, Stoyanov KM, *et al.* Prognostic value of access and non-access sites bleeding after percutaneous coronary intervention. *Circ Cardiovasc Interv* 2013;**6**:354–361. https://doi.org/10.1161/circinterventions.113.000433
450. Rao SV, Cohen MG, Kandzari DE, Bertrand OF, Gilchrist IC. The transradial approach to percutaneous coronary intervention: historical perspective, current concepts, and future directions. *J Am Coll Cardiol* 2010;**55**:2187–2195. https://doi.org/10.1016/j.jacc.2010.01.039
451. Jolly SS, Yusuf S, Cairns J, Niemelä K, Xavier D, Widimsky P, *et al.* Radial versus femoral access for coronary angiography and intervention in patients with acute coronary syndromes (RIVAL): a randomised, parallel group, multicentre trial. *Lancet* 2011;**377**:1409–1420. https://doi.org/10.1016/s0140-6736(11)60404-2
452. Valgimigli M, Gagnor A, Calabró P, Frigoli E, Leonardi S, Zaro T, *et al.* Radial versus femoral access in patients with acute coronary syndromes undergoing invasive management: a randomised multicentre trial. *Lancet* 2015;**385**:2465–2476. https://doi.org/10.1016/s0140-6736(15)60292-6
453. Vranckx P, Frigoli E, Rothenbühler M, Tomassini F, Garducci S, Andò G, *et al.* Radial versus femoral access in patients with acute coronary syndromes with or without ST-segment elevation. *Eur Heart J* 2017;**38**:1069–1080. https://doi.org/10.1093/eurheartj/ehx048
454. Lee P, Liew D, Brennan A, Stub D, Lefkovits J, Reid CM, *et al.* Cost-effectiveness of radial access percutaneous coronary intervention in acute coronary syndrome. *Am J Cardiol* 2021;**156**:44–51. https://doi.org/10.1016/j.amjcard.2021.06.034
455. Kerensky RA, Wade M, Deedwania P, Boden WE, Pepine CJ. Revisiting the culprit lesion in non-Q-wave myocardial infarction. Results from the VANQWISH trial angiographic core laboratory. *J Am Coll Cardiol* 2002;**39**:1456–1463. https://doi.org/10.1016/s0735-1097(02)01770-9
456. Johnson TW, Räber L, di Mario C, Bourantas C, Jia H, Mattesini A, *et al.* Clinical use of intracoronary imaging. Part 2: acute coronary syndromes, ambiguous coronary angiography findings, and guiding interventional decision-making: an expert consensus document of the European Association of Percutaneous Cardiovascular Interventions. *Eur Heart J* 2019;**40**:2566–2584. https://doi.org/10.1093/eurheartj/ehz332
457. di Mario C, Koskinas KC, Räber L. Clinical benefit of IVUS guidance for coronary stenting: the ULTIMATE step toward definitive evidence? *J Am Coll Cardiol* 2018;**72**:3138–3141. https://doi.org/10.1016/j.jacc.2018.10.029
458. Gao XF, Wang ZM, Wang F, Gu Y, Ge Z, Kong X-Q, *et al.* Intravascular ultrasound guidance reduces cardiac death and coronary revascularization in patients undergoing drug-eluting stent implantation: results from a meta-analysis of 9 randomized trials and 4724 patients. *Int J Cardiovasc Imaging* 2019;**35**:239–247. https://doi.org/10.1007/s10554-019-01555-3
459. Darmoch F, Alraies MC, Al-Khadra Y, Moussa Pacha H, Pinto DS, Osborn EA. Intravascular ultrasound imaging-guided versus coronary angiography-guided percutaneous coronary intervention: a systematic review and meta-analysis. *J Am Heart Assoc* 2020;**9**:e013678. https://doi.org/10.1161/jaha.119.013678
460. Jia H, Dai J, He L, Xu Y, Shi Y, Zhao L, *et al.* EROSION III: a multicenter RCT of OCT-guided reperfusion in STEMI with early infarct artery patency. *JACC Cardiovasc Interv* 2022;**15**:846–856. https://doi.org/10.1016/j.jcin.2022.01.298
461. Cuculi F, De Maria GL, Meier P, Dall'Armellina E, de Caterina AR, Channon KM, *et al.* Impact of microvascular obstruction on the assessment of coronary flow reserve, index of microcirculatory resistance, and fractional flow reserve after ST-segment elevation myocardial infarction. *J Am Coll Cardiol* 2014;**64**:1894–1904. https://doi.org/10.1016/j.jacc.2014.07.987
462. De Bruyne B, Pijls NH, Bartunek J, Kulecki K, Bech J-W, De Winter H, *et al.* Fractional flow reserve in patients with prior myocardial infarction. *Circulation* 2001;**104**:157–162. https://doi.org/10.1161/01.cir.104.2.157
463. Bønaa KH, Mannsverk J, Wiseth R, Aaberge L, Myreng Y, Nygård O, *et al.* Drug-eluting or bare-metal stents for coronary artery disease. *N Engl J Med* 2016;**375**:1242–1252. https://doi.org/10.1056/NEJMoa1607991
464. Räber L, Kelbæk H, Ostojic M, Baumbach A, Heg D, Tüller D, *et al.* Effect of biolimus-eluting stents with biodegradable polymer vs bare-metal stents on cardiovascular events among patients with acute myocardial infarction: the COMFORTABLE AMI randomized trial. *JAMA* 2012;**308**:777–787. https://doi.org/10.1001/jama.2012.10065
465. Sabate M, Cequier A, Iñiguez A, Serra A, Hernandez-Antolin R, Mainar V, *et al.* Everolimus-eluting stent versus bare-metal stent in ST-segment elevation myocardial infarction (EXAMINATION): 1 year results of a randomised controlled trial. *Lancet* 2012;**380**:1482–1490. https://doi.org/10.1016/s0140-6736(12)61223-9
466. Sabaté M, Brugaletta S, Cequier A, Iñiguez A, Serra A, Jiménez-Quevedo P, *et al.* Clinical outcomes in patients with ST-segment elevation myocardial infarction treated with everolimus-eluting stents versus bare-metal stents (EXAMINATION): 5-year results of a randomised trial. *Lancet* 2016;**387**:357–366. https://doi.org/10.1016/s0140-6736(15)00548-6
467. Brugaletta S, Gomez-Lara J, Ortega-Paz L, Jimenez-Diaz V, Jimenez M, Jiménez-Quevedo P, *et al.* 10-Year follow-up of patients with everolimus-eluting versus bare-metal stents after ST-segment elevation myocardial infarction. *J Am Coll Cardiol* 2021;**77**:1165–1178. https://doi.org/10.1016/j.jacc.2020.12.059
468. Räber L, Yamaji K, Kelbæk H, Engstrøm T, Baumbach A, Roffi M, *et al.* Five-year clinical outcomes and intracoronary imaging findings of the COMFORTABLE AMI trial: randomized comparison of biodegradable polymer-based biolimus-eluting stents with bare-metal stents in patients with acute ST-segment elevation myocardial infarction. *Eur Heart J* 2019;**40**:1909–1919. https://doi.org/10.1093/eurheartj/ehz074
469. Vos NS, Fagel ND, Amoroso G, Herrman J-PR, Patterson MS, Piers LH, *et al.* Paclitaxel-coated balloon angioplasty versus drug-eluting stent in acute myocardial infarction: the REVELATION randomized trial. *JACC Cardiovasc Interv* 2019;**12**:1691–1699. https://doi.org/10.1016/j.jcin.2019.04.016
470. Scheller B, Ohlow MA, Ewen S, Kische S, Rudolph TK, Clever YP, *et al.* Bare metal or drug-eluting stent versus drug-coated balloon in non-ST-elevation myocardial

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

infarction: the randomised PEPCAD NSTEMI trial. *EuroIntervention* 2020;**15**:1527–1533. https://doi.org/10.4244/eij-d-19-00723
471. Belkacemi A, Agostoni P, Nathoe HM, Voskuil M, Shao CL, Van Belle E, *et al.* First results of the DEB-AMI (drug eluting balloon in acute ST-segment elevation myocardial infarction) trial: a multicenter randomized comparison of drug-eluting balloon plus bare-metal stent versus bare-metal stent versus drug-eluting stent in primary percutaneous coronary intervention with 6-month angiographic, intravascular, functional, and clinical outcomes. *J Am Coll Cardiol* 2012;**59**:2327–2337. https://doi.org/10.1016/j.jacc.2012.02.027
472. Fröbert O, Lagerqvist B, Olivecrona GK, Omerovic E, Gudnason T, Maeng M, *et al.* Thrombus aspiration during ST-segment elevation myocardial infarction. *N Engl J Med* 2013;**369**:1587–1597. https://doi.org/10.1056/NEJMoa1308789
473. Lagerqvist B, Fröbert O, Olivecrona GK, Gudnason T, Maeng M, Alström P, *et al.* Outcomes 1 year after thrombus aspiration for myocardial infarction. *N Engl J Med* 2014;**371**:1111–1120. https://doi.org/10.1056/NEJMoa1405707
474. Jolly SS, Cairns JA, Yusuf S, Meeks B, Pogue J, Rokoss MJ, *et al.* Randomized trial of primary PCI with or without routine manual thrombectomy. *N Engl J Med* 2015;**372**:1389–1398. https://doi.org/10.1056/NEJMoa1415098
475. Jolly SS, James S, Džavík V, Cairns JA, Mahmoud KD, Zijlstra F, *et al.* Thrombus aspiration in ST-segment-elevation myocardial infarction: an individual patient meta-analysis: thrombectomy trialists collaboration. *Circulation* 2017;**135**:143–152. https://doi.org/10.1161/circulationaha.116.025371
476. Jolly SS, Cairns JA, Lavi S, Cantor WJ, Bernat I, Cheema AN, *et al.* Thrombus aspiration in patients with high thrombus burden in the TOTAL trial. *J Am Coll Cardiol* 2018;**72**:1589–1596. https://doi.org/10.1016/j.jacc.2018.07.047
477. Thiele H, de Waha S, Zeymer U, Desch S, Scheller B, Lauer B, *et al.* Effect of aspiration thrombectomy on microvascular obstruction in NSTEMI patients: the TATORT-NSTEMI trial. *J Am Coll Cardiol* 2014;**64**:1117–1124. https://doi.org/10.1016/j.jacc.2014.05.064
478. de Waha S, Patel MR, Granger CB, Ohman EM, Maehara A, Eitel I, *et al.* Relationship between microvascular obstruction and adverse events following primary percutaneous coronary intervention for ST-segment elevation myocardial infarction: an individual patient data pooled analysis from seven randomized trials. *Eur Heart J* 2017;**38**:3502–3510. https://doi.org/10.1093/eurheartj/ehx414
479. Corrigendum to: 2018 ESC/EACTS Guidelines on myocardial revascularization. *Eur Heart J* 2019;**40**:3096. https://doi.org/10.1093/eurheartj/ehz507
480. Sabatine MS, Bergmark BA, Murphy SA, O'Gara PT, Smith PK, Serruys PW, *et al.* Percutaneous coronary intervention with drug-eluting stents versus coronary artery bypass grafting in left main coronary artery disease: an individual patient data meta-analysis. *Lancet* 2021;**398**:2247–2257. https://doi.org/10.1016/s0140-6736(21)02334-5
481. Head SJ, Milojevic M, Daemen J, Ahn J-M, Boersma E, Christiansen EH, *et al.* Mortality after coronary artery bypass grafting versus percutaneous coronary intervention with stenting for coronary artery disease: a pooled analysis of individual patient data. *Lancet* 2018;**391**:939–948. https://doi.org/10.1016/s0140-6736(18)30423-9
482. Adlam D, Alfonso F, Maas A, Vrints C. European Society of Cardiology, Acute Cardiovascular Care Association, SCAD study group: a position paper on spontaneous coronary artery dissection. *Eur Heart J* 2018;**39**:3353–3368. https://doi.org/10.1093/eurheartj/ehy080
483. Alfonso F, de la Torre Hernández JM, Ibáñez B, Sabaté M, Pan M, Gulati R, *et al.* Rationale and design of the BA-SCAD (Beta-blockers and Antiplatelet agents in patients with Spontaneous Coronary Artery Dissection) randomized clinical trial. *Rev Esp Cardiol (Engl Ed)* 2022;**75**:515–522. https://doi.org/10.1016/j.rec.2021.08.003
484. Jackson R, Al-Hussaini A, Joseph S, van Soest G, Wood A, Macaya F, *et al.* Spontaneous coronary artery dissection: pathophysiological insights from optical coherence tomography. *JACC Cardiovasc Imaging* 2019;**12**:2475–2488. https://doi.org/10.1016/j.jcmg.2019.01.015
485. Hayes SN, Kim ESH, Saw J, Adlam D, Arslanian-Engoren C, Economy KE, *et al.* Spontaneous coronary artery dissection: current state of the science: a scientific statement from the American Heart Association. *Circulation* 2018;**137**:e523–e557. https://doi.org/10.1161/cir.0000000000000564
486. Tweet MS, Eleid MF, Best PJM, Lennon RJ, Lerman A, Rihal CS, *et al.* Spontaneous coronary artery dissection: revascularization versus conservative therapy. *Circ Cardiovasc Interv* 2014;**7**:777–786. https://doi.org/10.1161/circinterventions.114.001659
487. Saw J, Aymong E, Sedlak T, Buller CE, Starovoytov A, Ricci D, *et al.* Spontaneous coronary artery dissection: association with predisposing arteriopathies and precipitating stressors and cardiovascular outcomes. *Circ Cardiovasc Interv* 2014;**7**:645–655. https://doi.org/10.1161/circinterventions.114.001760
488. Lettieri C, Zavalloni D, Rossini R, Morici N, Ettori F, Leonzi O, *et al.* Management and long-term prognosis of spontaneous coronary artery dissection. *Am J Cardiol* 2015;**116**:66–73. https://doi.org/10.1016/j.amjcard.2015.03.039
489. Hayes SN, Tweet MS, Adlam D, Kim ESH, Gulati R, Price JE, *et al.* Spontaneous coronary artery dissection: JACC State-of-the-Art Review. *J Am Coll Cardiol* 2020;**76**:961–984. https://doi.org/10.1016/j.jacc.2020.05.084
490. Nordmann AJ, Hengstler P, Harr T, Young J, Bucher HC. Clinical outcomes of primary stenting versus balloon angioplasty in patients with myocardial infarction: a meta-analysis of randomized controlled trials. *Am J Med* 2004;**116**:253–262. https://doi.org/10.1016/j.amjmed.2003.08.035
491. Stone GW, Grines CL, Cox DA, Garcia E, Tcheng JE, Griffin JJ, *et al.* Comparison of angioplasty with stenting, with or without abciximab, in acute myocardial infarction. *N Engl J Med* 2002;**346**:957–966. https://doi.org/10.1056/NEJMoa013404
492. Belle L, Motreff P, Mangin L, Rangé G, Marcaggi X, Marie A, *et al.* Comparison of immediate with delayed stenting using the minimalist immediate mechanical intervention approach in acute ST-segment-elevation myocardial infarction: the MIMI study. *Circ Cardiovasc Interv* 2016;**9**:e003388. https://doi.org/10.1161/circinterventions.115.003388
493. Kelbæk H, Høfsten DE, Køber L, Helqvist S, Kløvgaard L, Holmvang L, *et al.* Deferred versus conventional stent implantation in patients with ST-segment elevation myocardial infarction (DANAMI 3-DEFER): an open-label, randomised controlled trial. *Lancet* 2016;**387**:2199–2206. https://doi.org/10.1016/s0140-6736(16)30072-1
494. Carrick D, Oldroyd KG, McEntegart M, Haig C, Petrie MC, Eteiba H, *et al.* A randomized trial of deferred stenting versus immediate stenting to prevent no- or slow-reflow in acute ST-segment elevation myocardial infarction (DEFER-STEMI). *J Am Coll Cardiol* 2014;**63**:2088–2098. https://doi.org/10.1016/j.jacc.2014.02.530
495. Hong SJ, Kim BK, Shin DH, Nam CM, Kim JS, Ko Y-G, *et al.* Effect of intravascular ultrasound-guided vs angiography-guided everolimus-eluting stent implantation: the IVUS-XPL randomized clinical trial. *JAMA* 2015;**314**:2155–2163. https://doi.org/10.1001/jama.2015.15454
496. Zhang J, Gao X, Kan J, Ge Z, Han L, Lu S, *et al.* Intravascular ultrasound versus angiography-guided drug-eluting stent implantation: the ULTIMATE trial. *J Am Coll Cardiol* 2018;**72**:3126–3137. https://doi.org/10.1016/j.jacc.2018.09.013
497. Gao XF, Ge Z, Kong XQ, Kan J, Han L, Lu S, *et al.* 3-Year outcomes of the ULTIMATE trial comparing intravascular ultrasound versus angiography-guided drug-eluting stent implantation. *JACC Cardiovasc Interv* 2021;**14**:247–257. https://doi.org/10.1016/j.jcin.2020.10.001
498. Meneveau N, Souteyrand G, Motreff P, Caussin C, Amabile N, Ohlmann P, *et al.* Optical coherence tomography to optimize results of percutaneous coronary intervention in patients with non-ST-elevation acute coronary syndrome: results of the multicenter, randomized DOCTORS study (Does Optical Coherence Tomography Optimize Results of Stenting). *Circulation* 2016;**134**:906–917. https://doi.org/10.1161/circulationaha.116.024393
499. Kala P, Cervinka P, Jakl M, Kanovsky J, Kupec A, Spacek R, *et al.* OCT guidance during stent implantation in primary PCI: a randomized multicenter study with nine months of optical coherence tomography follow-up. *Int J Cardiol* 2018;**250**:98–103. https://doi.org/10.1016/j.ijcard.2017.10.059
500. Secemsky EA, Butala N, Raja A, Khera R, Wang Y, Curtis JP, *et al.* Temporal changes and institutional variation in use of percutaneous coronary intervention for ST-elevation myocardial infarction with multivessel coronary artery disease in the United States: an NCDR research to practice project. *JAMA Cardiol* 2021;**6**:574–580. https://doi.org/10.1001/jamacardio.2020.5354
501. Holmes DR Jr, Berger PB, Hochman JS, Granger CB, Thompson TD, Califf RM, *et al.* Cardiogenic shock in patients with acute ischemic syndromes with and without ST-segment elevation. *Circulation* 1999;**100**:2067–2073. https://doi.org/10.1161/01.cir.100.20.2067
502. Thiele H, Ohman EM, de Waha-Thiele S, Zeymer U, Desch S. Management of cardiogenic shock complicating myocardial infarction: an update 2019. *Eur Heart J* 2019;**40**:2671–2683. https://doi.org/10.1093/eurheartj/ehz363
503. Papolos AI, Kenigsberg BB, Berg DD, Alviar CL, Bohula E, Burke JA, *et al.* Management and outcomes of cardiogenic shock in cardiac ICUs with versus without shock teams. *J Am Coll Cardiol* 2021;**78**:1309–1317. https://doi.org/10.1016/j.jacc.2021.07.044
504. Rab T, Ratanapo S, Kern KB, Basir MB, McDaniel M, Meraj P, *et al.* Cardiac shock care centers: JACC review topic of the week. *J Am Coll Cardiol* 2018;**72**:1972–1980. https://doi.org/10.1016/j.jacc.2018.07.074
505. Thiele H, Akin I, Sandri M, de Waha-Thiele S, Meyer-Saraei R, Fuernau G, *et al.* One-year outcomes after PCI strategies in cardiogenic shock. *N Engl J Med* 2018;**379**:1699–1710. https://doi.org/10.1056/NEJMoa1808788
506. Sorajja P, Gersh BJ, Cox DA, McLaughlin MG, Zimetbaum P, Costantini C, *et al.* Impact of multivessel disease on reperfusion success and clinical outcomes in patients undergoing primary percutaneous coronary intervention for acute myocardial infarction. *Eur Heart J* 2007;**28**:1709–1716. https://doi.org/10.1093/eurheartj/ehm184
507. Dziewierz A, Siudak Z, Rakowski T, Zasada W, Dubiel JS, Dudek D. Impact of multivessel coronary artery disease and noninfarct-related artery revascularization on outcome of patients with ST-elevation myocardial infarction transferred for primary percutaneous coronary intervention (from the EUROTRANSFER Registry). *Am J Cardiol* 2010;**106**:342–347. https://doi.org/10.1016/j.amjcard.2010.03.029
508. Wald DS, Morris JK, Wald NJ, Chase AJ, Edwards RJ, Hughes LO, *et al.* Randomized trial of preventive angioplasty in myocardial infarction. *N Engl J Med* 2013;**369**:1115–1123. https://doi.org/10.1056/NEJMoa1305520
509. Engstrøm T, Kelbæk H, Helqvist S, Høfsten DE, Kløvgaard L, Holmvang L, *et al.* Complete revascularisation versus treatment of the culprit lesion only in patients with ST-segment elevation myocardial infarction and multivessel disease

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

(DANAMI-3—PRIMULTI): an open-label, randomised controlled trial. *Lancet* 2015; **386**:665–671. https://doi.org/10.1016/s0140-6736(15)60648-1
510. Smits PC, Abdel-Wahab M, Neumann FJ, Boxma-de Klerk BM, Lunde K, Schotborgh CE, *et al.* Fractional flow reserve-guided multivessel angioplasty in myocardial infarction. *N Engl J Med* 2017;**376**:1234–1244. https://doi.org/10.1056/NEJMoa1701067
511. Mehta SR, Wood DA, Storey RF, Mehran R, Bainey KR, Nguyen H, *et al.* Complete revascularization with multivessel PCI for myocardial infarction. *N Engl J Med* 2019; **381**:1411–1421. https://doi.org/10.1056/NEJMoa1907775
512. Bainey KR, Engstrøm T, Smits PC, Gershlick AH, James SK, Storey RF, *et al.* Complete vs culprit-lesion-only revascularization for ST-segment elevation myocardial infarction: a systematic review and meta-analysis. *JAMA Cardiol* 2020;**5**:881–888. https://doi.org/10.1001/jamacardio.2020.1251
513. Sardella G, Lucisano L, Garbo R, Pennacchi M, Cavallo E, Stio RE, *et al.* Single-staged compared with multi-staged PCI in multivessel NSTEMI patients: the SMILE trial. *J Am Coll Cardiol* 2016;**67**:264–272. https://doi.org/10.1016/j.jacc.2015.10.082
514. Siebert VR, Borgaonkar S, Jia X, Nguyen HL, Birnbaum Y, Lakkis NM, *et al.* Meta-analysis comparing multivessel versus culprit coronary arterial revascularization for patients with non-ST-segment elevation acute coronary syndromes. *Am J Cardiol* 2019;**124**:1501–1511. https://doi.org/10.1016/j.amjcard.2019.07.071
515. Rathod KS, Koganti S, Jain AK, Astroulakis Z, Lim P, Rakhit R, *et al.* Complete versus culprit-only lesion intervention in patients with acute coronary syndromes. *J Am Coll Cardiol* 2018;**72**:1989–1999. https://doi.org/10.1016/j.jacc.2018.07.089
516. Hanratty CG, Koyama Y, Rasmussen HH, Nelson GIC, Hansen PS, Ward MR. Exaggeration of nonculprit stenosis severity during acute myocardial infarction: implications for immediate multivessel revascularization. *J Am Coll Cardiol* 2002;**40**:911–916. https://doi.org/10.1016/s0735-1097(02)02049-1
517. Gibson CM, Ryan KA, Murphy SA, Mesley R, Marble SJ, Giugliano RP, *et al.* Impaired coronary blood flow in nonculprit arteries in the setting of acute myocardial infarction. *J Am Coll Cardiol* 1999;**34**:974–982. https://doi.org/10.1016/s0735-1097(99)00335-6
518. Ntalianis A, Sels JW, Davidavicius G, Tanaka N, Muller O, Trana C, *et al.* Fractional flow reserve for the assessment of nonculprit coronary artery stenoses in patients with acute myocardial infarction. *JACC Cardiovasc Interv* 2010;**3**:1274–1281. https://doi.org/10.1016/j.jcin.2010.08.025
519. Musto C, De Felice F, Rigattieri S, Chin D, Marra A, Nazzaro MS, *et al.* Instantaneous wave-free ratio and fractional flow reserve for the assessment of nonculprit lesions during the index procedure in patients with ST-segment elevation myocardial infarction: the WAVE study. *Am Heart J* 2017;**193**:63–69. https://doi.org/10.1016/j.ahj.2017.07.017
520. Erbay A, Penzel L, Abdelwahed YS, Klotsche J, Schatz A-S, Steiner J, *et al.* Feasibility and diagnostic reliability of quantitative flow ratio in the assessment of non-culprit lesions in acute coronary syndrome. *Int J Cardiovasc Imaging* 2021;**37**:1815–1823. https://doi.org/10.1007/s10554-021-02195-2
521. Tonino PA, Fearon WF, De Bruyne B, Oldroyd KG, Leesar MA, Ver Lee PN, *et al.* Angiographic versus functional severity of coronary artery stenoses in the FAME study fractional flow reserve versus angiography in multivessel evaluation. *J Am Coll Cardiol* 2010;**55**:2816–2821. https://doi.org/10.1016/j.jacc.2009.11.096
522. Van Belle E, Baptista SB, Raposo L, Henderson J, Rioufol G, Santos L, *et al.* Impact of routine fractional flow reserve on management decision and 1-year clinical outcome of patients with acute coronary syndromes: PRIME-FFR (Insights From the POST-IT [Portuguese Study on the Evaluation of FFR-Guided Treatment of Coronary Disease] and R3F [French FFR Registry] Integrated Multicenter Registries – Implementation of FFR [Fractional Flow Reserve] in Routine Practice). *Circ Cardiovasc Interv* 2017;**10**:e004296. https://doi.org/10.1161/circinterventions.116.004296
523. Sels JW, Tonino PA, Siebert U, Fearon WF, Van't Veer M, De Bruyne B, *et al.* Fractional flow reserve in unstable angina and non-ST-segment elevation myocardial infarction experience from the FAME (Fractional flow reserve versus Angiography for Multivessel Evaluation) study. *JACC Cardiovasc Interv* 2011;**4**:1183–1189. https://doi.org/10.1016/j.jcin.2011.08.008
524. Puymirat E, Cayla G, Simon T, Steg PG, Montalescot G, Durand-Zaleski I, *et al.* Multivessel PCI guided by FFR or angiography for myocardial infarction. *N Engl J Med* 2021;**385**:297–308. https://doi.org/10.1056/NEJMoa2104650
525. Wald DS, Hadyanto S, Bestwick JP. Should fractional flow reserve follow angiographic visual inspection to guide preventive percutaneous coronary intervention in ST-elevation myocardial infarction? *Eur Heart J Qual Care Clin Outcomes* 2020;**6**:186–192. https://doi.org/10.1093/ehjqcco/qcaa012
526. Gallone G, Angelini F, Fortuni F, Gnecchi M, De Filippo O, Baldetti L, *et al.* Angiography- vs. physiology-guided complete revascularization in patients with ST-elevation myocardial infarction and multivessel disease: who is the better gatekeeper in this setting? A meta-analysis of randomized controlled trials. *Eur Heart J Qual Care Clin Outcomes* 2020;**6**:199–200. https://doi.org/10.1093/ehjqcco/qcaa007
527. Kobayashi Y, Lønborg J, Jong A, Nishi T, De Bruyne B, Høfsten DE, *et al.* Prognostic value of the residual SYNTAX score after functionally complete revascularization in ACS. *J Am Coll Cardiol* 2018;**72**:1321–1329. https://doi.org/10.1016/j.jacc.2018.06.069
528. Lee JM, Kim HK, Park KH, Choo EH, Kim CJ, Lee SH, *et al.* Fractional flow reserve versus angiography-guided strategy in acute myocardial infarction with multivessel disease: a randomized trial. *Eur Heart J* 2023;**44**:473–484. https://doi.org/10.1093/eurheartj/ehac763
529. Harskamp RE, Bonatti JO, Zhao DX, Puskas JD, de Winter RJ, Alexander JH, *et al.* Standardizing definitions for hybrid coronary revascularization. *J Thorac Cardiovasc Surg* 2014;**147**:556–560. https://doi.org/10.1016/j.jtcvs.2013.10.019
530. Doenst T, Haverich A, Serruys P, Bonow RO, Kappetein P, Falk V, *et al.* PCI and CABG for treating stable coronary artery disease: JACC review topic of the week. *J Am Coll Cardiol* 2019;**73**:964–976. https://doi.org/10.1016/j.jacc.2018.11.053
531. Gershlick AH, Khan JN, Kelly DJ, Greenwood JP, Sasikaran T, Curzen N, *et al.* Randomized trial of complete versus lesion-only revascularization in patients undergoing primary percutaneous coronary intervention for STEMI and multivessel disease: the CvLPRIT trial. *J Am Coll Cardiol* 2015;**65**:963–972. https://doi.org/10.1016/j.jacc.2014.12.038
532. Layland J, Oldroyd KG, Curzen N, Sood A, Balachandran K, Das R, *et al.* Fractional flow reserve vs. angiography in guiding management to optimize outcomes in non-ST-segment elevation myocardial infarction: the British Heart Foundation FAMOUS-NSTEMI randomized trial. *Eur Heart J* 2015;**36**:100–111. https://doi.org/10.1093/eurheartj/ehu338
533. Pasupathy S, Air T, Dreyer RP, Tavella R, Beltrame JF. Systematic review of patients presenting with suspected myocardial infarction and nonobstructive coronary arteries. *Circulation* 2015;**131**:861–870. https://doi.org/10.1161/circulationaha.114.011201
534. Agewall S, Beltrame JF, Reynolds HR, Niessner A, Rosano G, Caforio ALP, *et al.* ESC working group position paper on myocardial infarction with non-obstructive coronary arteries. *Eur Heart J* 2017;**38**:143–153. https://doi.org/10.1093/eurheartj/ehw149
535. Tamis-Holland JE, Jneid H, Reynolds HR, Agewall S, Brilakis ES, Brown TM, *et al.* Contemporary diagnosis and management of patients with myocardial infarction in the absence of obstructive coronary artery disease: a scientific statement from the American Heart Association. *Circulation* 2019;**139**:e891–e908. https://doi.org/10.1161/cir.0000000000000670
536. Matta A, Nader V, Canitrot R, Delmas C, Bouisset F, Lhermusier T, *et al.* Myocardial bridging is significantly associated to myocardial infarction with non-obstructive coronary arteries. *Eur Heart J Acute Cardiovasc Care* 2022;**11**:501–507. https://doi.org/10.1093/ehjacc/zuac047
537. Pargaonkar VS, Kimura T, Kameda R, Tanaka S, Yamada R, Schwartz JG, *et al.* Invasive assessment of myocardial bridging in patients with angina and no obstructive coronary artery disease. *EuroIntervention* 2021;**16**:1070–1078. https://doi.org/10.4244/eij-d-20-00779
538. Ford TJ, Ong P, Sechtem U, Beltrame J, Camici PG, Crea F, *et al.* Assessment of vascular dysfunction in patients without obstructive coronary artery disease: why, how, and when. *JACC Cardiovasc Interv* 2020;**13**:1847–1864. https://doi.org/10.1016/j.jcin.2020.05.052
539. Occhipinti G, Bucciarelli-Ducci C, Capodanno D. Diagnostic pathways in myocardial infarction with non-obstructive coronary artery disease (MINOCA). *Eur Heart J Acute Cardiovasc Care* 2021;**10**:813–822. https://doi.org/10.1093/ehjacc/zuab049
540. Eitel I, Behrendt F, Schindler K, Kivelitz D, Gutberlet M, Schuler G, *et al.* Differential diagnosis of suspected apical ballooning syndrome using contrast-enhanced magnetic resonance imaging. *Eur Heart J* 2008;**29**:2651–2659. https://doi.org/10.1093/eurheartj/ehn433
541. Eitel I, von Knobelsdorff-Brenkenhoff F, Bernhardt P, Carbone I, Muellerleile K, Aldrovandi A, *et al.* Clinical characteristics and cardiovascular magnetic resonance findings in stress (takotsubo) cardiomyopathy. *JAMA* 2011;**306**:277–286. https://doi.org/10.1001/jama.2011.992
542. Ferreira VM, Schulz-Menger J, Holmvang G, Kramer CM, Carbone I, Sechtem U, *et al.* Cardiovascular magnetic resonance in nonischemic myocardial inflammation: expert recommendations. *J Am Coll Cardiol* 2018;**72**:3158–3176. https://doi.org/10.1016/j.jacc.2018.09.072
543. Lurz P, Luecke C, Eitel I, Föhrenbach F, Frank C, Grothoff M, *et al.* Comprehensive cardiac magnetic resonance imaging in patients with suspected myocarditis: the MyoRacer-trial. *J Am Coll Cardiol* 2016;**67**:1800–1811. https://doi.org/10.1016/j.jacc.2016.02.013
544. Reynolds HR, Maehara A, Kwong RY, Sedlak T, Saw J, Smilowitz NR, *et al.* Coronary optical coherence tomography and cardiac magnetic resonance imaging to determine underlying causes of myocardial infarction with nonobstructive coronary arteries in women. *Circulation* 2021;**143**:624–640. https://doi.org/10.1161/circulationaha.120.052008
545. Pathik B, Raman B, Mohd Amin NH, Mahadavan D, Rajendran S, McGavigan AD, *et al.* Troponin-positive chest pain with unobstructed coronary arteries: incremental diagnostic value of cardiovascular magnetic resonance imaging. *Eur Heart J Cardiovasc Imaging* 2016;**17**:1146–1152. https://doi.org/10.1093/ehjci/jev289
546. Lyon AR, Bossone E, Schneider B, Sechtem U, Citro R, Underwood SR, *et al.* Current state of knowledge on Takotsubo syndrome: a Position Statement from the Taskforce on Takotsubo Syndrome of the Heart Failure Association of the European Society of Cardiology. *Eur J Heart Fail* 2016;**18**:8–27. https://doi.org/10.1002/ejhf.424
547. Ghadri JR, Wittstein IS, Prasad A, Sharkey S, Dote K, Akashi YJ, *et al.* International expert consensus document on Takotsubo syndrome (Part II): diagnostic workup,

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

outcome, and management. *Eur Heart J* 2018;**39**:2047–2062. https://doi.org/10.1093/eurheartj/ehy077
548. Adler Y, Charron P, Imazio M, Badano L, Barón-Esquivias G, Bogaert J, *et al.* 2015 ESC Guidelines for the diagnosis and management of pericardial diseases: the Task Force for the diagnosis and management of pericardial diseases of the European Society of Cardiology (ESC) endorsed by: the European Association for Cardio-Thoracic Surgery (EACTS). *Eur Heart J* 2015;**36**:2921–2964. https://doi.org/10.1093/eurheartj/ehv318
549. Caforio AL, Pankuweit S, Arbustini E, Basso C, Gimeno-Blanes J, Felix SB, *et al.* Current state of knowledge on aetiology, diagnosis, management, and therapy of myocarditis: a position statement of the European Society of Cardiology Working Group on Myocardial and Pericardial Diseases. *Eur Heart J* 2013;**34**:2636–2648,2648a-2648d. https://doi.org/10.1093/eurheartj/eht210
550. Ford TJ, Stanley B, Good R, Rocchiccioli P, McEntegart M, Watkins S, *et al.* Stratified medical therapy using invasive coronary function testing in angina: the CorMicA trial. *J Am Coll Cardiol* 2018;**72**:2841–2855. https://doi.org/10.1016/j.jacc.2018.09.006
551. Kunadian V, Chieffo A, Camici PG, Berry C, Escaned J, Maas AHEM, *et al.* An EAPCI expert consensus document on ischaemia with non-obstructive coronary arteries in collaboration with European Society of Cardiology Working Group on Coronary Pathophysiology & Microcirculation Endorsed by Coronary Vasomotor Disorders International Study Group. *Eur Heart J* 2020;**41**:3504–3520. https://doi.org/10.1093/eurheartj/ehaa503
552. Cerrato E, Giacobbe F, Quadri G, Macaya F, Bianco M, Mori R, *et al.* Antiplatelet therapy in patients with conservatively managed spontaneous coronary artery dissection from the multicentre DISCO registry. *Eur Heart J* 2021;**42**:3161–3171. https://doi.org/10.1093/eurheartj/ehab372
553. McCarthy CP, Raber I, Chapman AR, Sandoval Y, Apple FS, Mills NL, *et al.* Myocardial injury in the era of high-sensitivity cardiac troponin assays: a practical approach for clinicians. *JAMA Cardiol* 2019;**4**:1034–1042. https://doi.org/10.1001/jamacardio.2019.2724
554. Bahit MC, Lopes RD, Clare RM, Newby LK, Pieper KS, Van de Werf F, *et al.* Heart failure complicating non-ST-segment elevation acute coronary syndrome: timing, predictors, and clinical outcomes. *JACC Heart Fail* 2013;**1**:223–229. https://doi.org/10.1016/j.jchf.2013.02.007
555. Chioncel O, Mebazaa A, Harjola VP, Coats AJ, Piepoli MF, Crespo-Leiro MG, *et al.* Clinical phenotypes and outcome of patients hospitalized for acute heart failure: the ESC Heart Failure Long-Term Registry. *Eur J Heart Fail* 2017;**19**:1242–1254. https://doi.org/10.1002/ejhf.890
556. Steg PG, Dabbous OH, Feldman LJ, Cohen-Solal A, Aumont MC, López-Sendón J, *et al.* Determinants and prognostic impact of heart failure complicating acute coronary syndromes: observations from the Global Registry of Acute Coronary Events (GRACE). *Circulation* 2004;**109**:494–499. https://doi.org/10.1161/01.Cir.0000109691.16944.Da
557. McDonagh TA, Metra M, Adamo M, Gardner RS, Baumbach A, Böhm M, *et al.* 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J* 2021;**42**:3599–3726. https://doi.org/10.1093/eurheartj/ehab368
558. Chioncel O, Parissis J, Mebazaa A, Thiele H, Desch S, Bauersachs J, *et al.* Epidemiology, pathophysiology and contemporary management of cardiogenic shock—a position statement from the Heart Failure Association of the European Society of Cardiology. *Eur J Heart Fail* 2020;**22**:1315–1341. https://doi.org/10.1002/ejhf.1922
559. Mullens W, Damman K, Harjola VP, Mebazaa A, Brunner-La Rocca HP, Martens P, *et al.* The use of diuretics in heart failure with congestion—a position statement from the Heart Failure Association of the European Society of Cardiology. *Eur J Heart Fail* 2019;**21**:137–155. https://doi.org/10.1002/ejhf.1369
560. Lancellotti P, Price S, Edvardsen T, Cosyns B, Neskovic AN, Dulgheru R, *et al.* The use of echocardiography in acute cardiovascular care: recommendations of the European Association of Cardiovascular Imaging and the Acute Cardiovascular Care Association. *Eur Heart J Cardiovasc Imaging* 2015;**16**:119–146. https://doi.org/10.1093/ehjci/jeu210
561. Thiele H, Jobs A, Ouweneel DM, Henriques JPS, Seyfarth M, Desch S, *et al.* Percutaneous short-term active mechanical support devices in cardiogenic shock: a systematic review and collaborative meta-analysis of randomized trials. *Eur Heart J* 2017;**38**:3523–3531. https://doi.org/10.1093/eurheartj/ehx363
562. Amin AP, Spertus JA, Curtis JP, Desai N, Masoudi FA, Bach RG, *et al.* The evolving landscape of impella use in the United States among patients undergoing percutaneous coronary intervention with mechanical circulatory support. *Circulation* 2020;**141**:273–284. https://doi.org/10.1161/circulationaha.119.044007
563. Dhruva SS, Ross JS, Mortazavi BJ, Hurley NC, Krumholz HM, Curtis JP, *et al.* Association of use of an intravascular microaxial left ventricular assist device vs intra-aortic balloon pump with in-hospital mortality and major bleeding among patients with acute myocardial infarction complicated by cardiogenic shock. *JAMA* 2020;**323**:734–745. https://doi.org/10.1001/jama.2020.0254
564. Elbadawi A, Elgendy IY, Mahmoud K, Barakat AF, Mentias A, Mohamed AH, *et al.* Temporal trends and outcomes of mechanical complications in patients with acute myocardial infarction. *JACC Cardiovasc Interv* 2019;**12**:1825–1836. https://doi.org/10.1016/j.jcin.2019.04.039
565. Schrage B, Becher PM, Goßling A, Savarese G, Dabboura S, Yan I, *et al.* Temporal trends in incidence, causes, use of mechanical circulatory support and mortality in cardiogenic shock. *ESC Heart Fail* 2021;**8**:1295–1303. https://doi.org/10.1002/ehf2.13202
566. Watkins AC, Maassel NL, Ghoreishi M, Dawood MY, Pham SM, Kon ZN, *et al.* Preoperative venoarterial extracorporeal membrane oxygenation slashes risk score in advanced structural heart disease. *Ann Thorac Surg* 2018;**106**:1709–1715. https://doi.org/10.1016/j.athoracsur.2018.07.038
567. Ronco D, Matteucci M, Ravaux JM, Marra S, Torchio F, Corazzari C, *et al.* Mechanical circulatory support as a bridge to definitive treatment in post-infarction ventricular septal rupture. *JACC Cardiovasc Interv* 2021;**14**:1053–1066. https://doi.org/10.1016/j.jcin.2021.02.046
568. Matteucci M, Fina D, Jiritano F, Meani P, Raffa GM, Kowalewski M, *et al.* The use of extracorporeal membrane oxygenation in the setting of postinfarction mechanical complications: outcome analysis of the Extracorporeal Life Support Organization Registry. *Interact Cardiovasc Thorac Surg* 2020;**31**:369–374. https://doi.org/10.1093/icvts/ivaa108
569. Assenza GE, McElhinney DB, Valente AM, Pearson DD, Volpe M, Martucci G, *et al.* Transcatheter closure of post-myocardial infarction ventricular septal rupture. *Circ Cardiovasc Interv* 2013;**6**:59–67. https://doi.org/10.1161/circinterventions.112.972711
570. Kilic A, Sultan I, Chu D, Wang Y, Gleason TG. Mitral valve surgery for papillary muscle rupture: outcomes in 1342 patients from the society of thoracic surgeons database. *Ann Thorac Surg* 2020;**110**:1975–1981. https://doi.org/10.1016/j.athoracsur.2020.03.097
571. Valle JA, Miyasaka RL, Carroll JD. Acute mitral regurgitation secondary to papillary muscle tear: is transcatheter edge-to-edge mitral valve repair a new paradigm? *Circ Cardiovasc Interv* 2017;**10**:e005050. https://doi.org/10.1161/circinterventions.117.005050
572. Terashima M, Fujiwara S, Yaginuma GY, Takizawa K, Kaneko U, Meguro T. Outcome of percutaneous intrapericardial fibrin-glue injection therapy for left ventricular free wall rupture secondary to acute myocardial infarction. *Am J Cardiol* 2008;**101**:419–421. https://doi.org/10.1016/j.amjcard.2007.09.086
573. Damluji AA, van Diepen S, Katz JN, Menon V, Tamis-Holland JE, Bakitas M, *et al.* Mechanical complications of acute myocardial infarction: a scientific statement from the American Heart Association. *Circulation* 2021;**144**:e16–e35. https://doi.org/10.1161/cir.0000000000000985
574. Gong FF, Vaitenas I, Malaisrie SC, Maganti K. Mechanical complications of acute myocardial infarction: a review. *JAMA Cardiol* 2021;**6**:341–349. https://doi.org/10.1001/jamacardio.2020.3690
575. Robinson AA, Jain A, Gentry M, McNamara RL. Left ventricular thrombi after STEMI in the primary PCI era: a systematic review and meta-analysis. *Int J Cardiol* 2016;**221**:554–559. https://doi.org/10.1016/j.ijcard.2016.07.069
576. Levine GN, McEvoy JW, Fang JC, Ibeh C, McCarthy CP, Misra A, *et al.* Management of patients at risk for and with left ventricular thrombus: a scientific statement from the American Heart Association. *Circulation* 2022;**146**:e205–e223. https://doi.org/10.1161/cir.0000000000001092
577. Bulluck H, Chan MHH, Paradies V, Yellon RL, Ho HH, Chan MY, *et al.* Incidence and predictors of left ventricular thrombus by cardiovascular magnetic resonance in acute ST-segment elevation myocardial infarction treated by primary percutaneous coronary intervention: a meta-analysis. *J Cardiovasc Magn Reson* 2018;**20**:72. https://doi.org/10.1186/s12968-018-0494-3
578. Velangi PS, Choo C, Chen KA, Kazmirczak F, Nijjar PS, Farzaneh-Far A, *et al.* Long-term embolic outcomes after detection of left ventricular thrombus by late gadolinium enhancement cardiovascular magnetic resonance imaging: a matched cohort study. *Circ Cardiovasc Imaging* 2019;**12**:e009723. https://doi.org/10.1161/circimaging.119.009723
579. Funke Küpper AJ, Verheugt FW, Peels CH, Galema TW, Roos JP. Left ventricular thrombus incidence and behavior studied by serial two-dimensional echocardiography in acute anterior myocardial infarction: left ventricular wall motion, systemic embolism and oral anticoagulation. *J Am Coll Cardiol* 1989;**13**:1514–1520. https://doi.org/10.1016/0735-1097(89)90341-0
580. Zhang Z, Si D, Zhang Q, Jin L, Zheng H, Qu M, *et al.* Prophylactic rivaroxaban therapy for left ventricular thrombus after anterior ST-segment elevation myocardial infarction. *JACC Cardiovasc Interv* 2022;**15**:861–872. https://doi.org/10.1016/j.jcin.2022.01.285
581. Dalia T, Lahan S, Ranka S, Goyal A, Zoubek S, Gupta K, *et al.* Warfarin versus direct oral anticoagulants for treating left ventricular thrombus: a systematic review and meta-analysis. *Thromb J* 2021;**19**:7. https://doi.org/10.1186/s12959-021-00259-w
582. Verma BR, Montane B, Chetrit M, Khayata M, Furqan MM, Ayoub C, *et al.* Pericarditis and post-cardiac injury syndrome as a sequelae of acute myocardial infarction. *Curr Cardiol Rep* 2020;**22**:127. https://doi.org/10.1007/s11886-020-01371-5
583. Schmitt J, Duray G, Gersh BJ, Hohnloser SH. Atrial fibrillation in acute myocardial infarction: a systematic review of the incidence, clinical features and prognostic implications. *Eur Heart J* 2009;**30**:1038–1045. https://doi.org/10.1093/eurheartj/ehn579
584. Batra G, Svennblad B, Held C, Jernberg T, Johanson P, Wallentin L, *et al.* All types of atrial fibrillation in the setting of myocardial infarction are associated with impaired outcome. *Heart* 2016;**102**:926–933. https://doi.org/10.1136/heartjnl-2015-308678
585. Hindricks G, Potpara T, Dagres N, Arbelo E, Bax JJ, Blomström-Lundqvist C, *et al.* 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation developed in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS). The Task Force for the diagnosis and management of atrial fibrillation of the European

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

Society of Cardiology (ESC). Developed with the special contribution of the European Heart Rhythm Association (EHRA) of the ESC. *Eur Heart J* 2021;**42**:373–498. https://doi.org/10.1093/eurheartj/ehaa612
586. Jabre P, Jouven X, Adnet F, Thabut G, Bielinski SJ, Weston SA, *et al.* Atrial fibrillation and death after myocardial infarction: a community study. *Circulation* 2011;**123**:2094–2100. https://doi.org/10.1161/circulationaha.110.990192
587. Siu CW, Jim MH, Ho HH, Miu R, Lee SWL, Lau CP, *et al.* Transient atrial fibrillation complicating acute inferior myocardial infarction: implications for future risk of ischemic stroke. *Chest* 2007;**132**:44–49. https://doi.org/10.1378/chest.06-2733
588. Piccini JP, Schulte PJ, Pieper KS, Mehta RH, White HD, Van de Werf F, *et al.* Antiarrhythmic drug therapy for sustained ventricular arrhythmias complicating acute myocardial infarction. *Crit Care Med* 2011;**39**:78–83. https://doi.org/10.1097/CCM.0b013e3181fd6ad7
589. Piccini JP, Hranitzky PM, Kilaru R, Rouleau JL, White HD, Aylward PE, *et al.* Relation of mortality to failure to prescribe beta blockers acutely in patients with sustained ventricular tachycardia and ventricular fibrillation following acute myocardial infarction (from the VALsartan In Acute myocardial iNfarcTion trial [VALIANT] Registry). *Am J Cardiol* 2008;**102**:1427–1432. https://doi.org/10.1016/j.amjcard.2008.07.033
590. Wolfe CL, Nibley C, Bhandari A, Chatterjee K, Scheinman M. Polymorphous ventricular tachycardia associated with acute myocardial infarction. *Circulation* 1991;**84**:1543–1551. https://doi.org/10.1161/01.cir.84.4.1543
591. Liang JJ, Fender EA, Cha YM, Lennon RJ, Prasad A, Barsness GW, *et al.* Long-term outcomes in survivors of early ventricular arrhythmias after acute ST-elevation and non-ST-elevation myocardial infarction treated with percutaneous coronary intervention. *Am J Cardiol* 2016;**117**:709–713. https://doi.org/10.1016/j.amjcard.2015.12.002
592. Mehta RH, Yu J, Piccini JP, Tcheng JE, Farkouh ME, Reiffel J, *et al.* Prognostic significance of postprocedural sustained ventricular tachycardia or fibrillation in patients undergoing primary percutaneous coronary intervention (from the HORIZONS-AMI Trial). *Am J Cardiol* 2012;**109**:805–812. https://doi.org/10.1016/j.amjcard.2011.10.043
593. Masuda M, Nakatani D, Hikoso S, Suna S, Usami M, Matsumoto S, *et al.* Clinical impact of ventricular tachycardia and/or fibrillation during the acute phase of acute myocardial infarction on in-hospital and 5-year mortality rates in the percutaneous coronary intervention era. *Circ J* 2016;**80**:1539–1547. https://doi.org/10.1253/circj.CJ-16-0183
594. Podolecki T, Lenarczyk R, Kowalczyk J, Jedrzejczyk-Patej E, Chodor P, Mazurek M, *et al.* Prognostic significance of complex ventricular arrhythmias complicating ST-segment elevation myocardial infarction. *Am J Cardiol* 2018;**121**:805–809. https://doi.org/10.1016/j.amjcard.2017.12.036
595. Komatsu Y, Hocini M, Nogami A, Maury P, Peichl P, Iwasaki YK, *et al.* Catheter ablation of refractory ventricular fibrillation storm after myocardial infarction. *Circulation* 2019;**139**:2315–2325. https://doi.org/10.1161/circulationaha.118.037997
596. Terkelsen CJ, Sørensen JT, Kaltoft AK, Nielsen SS, Thuesen L, Bøtker HE, *et al.* Prevalence and significance of accelerated idioventricular rhythm in patients with ST-elevation myocardial infarction treated with primary percutaneous coronary intervention. *Am J Cardiol* 2009;**104**:1641–1646. https://doi.org/10.1016/j.amjcard.2009.07.037
597. Mehran R, Pocock SJ, Nikolsky E, Clayton T, Dangas GD, Kirtane AJ, *et al.* A risk score to predict bleeding in patients with acute coronary syndromes. *J Am Coll Cardiol* 2010;**55**:2556–2566. https://doi.org/10.1016/j.jacc.2009.09.076
598. Mehran R, Pocock SJ, Stone GW, Clayton TC, Dangas GD, Feit F, *et al.* Associations of major bleeding and myocardial infarction with the incidence and timing of mortality in patients presenting with non-ST-elevation acute coronary syndromes: a risk model from the ACUITY trial. *Eur Heart J* 2009;**30**:1457–1466. https://doi.org/10.1093/eurheartj/ehp110
599. Rao SV. The conundrum of reducing ischemic and bleeding events after PCI. *J Am Coll Cardiol* 2015;**65**:1421–1423. https://doi.org/10.1016/j.jacc.2015.02.012
600. Kwok CS, Sherwood MW, Watson SM, Nasir SB, Sperrin M, Nolan J, *et al.* Blood transfusion after percutaneous coronary intervention and risk of subsequent adverse outcomes: a systematic review and meta-analysis. *JACC Cardiovasc Interv* 2015;**8**:436–446. https://doi.org/10.1016/j.jcin.2014.09.026
601. Mehran R, Baber U, Steg PG, Ariti C, Weisz G, Witzenbichler B, *et al.* Cessation of dual antiplatelet treatment and cardiac events after percutaneous coronary intervention (PARIS): 2 year results from a prospective observational study. *Lancet* 2013;**382**:1714–1722. https://doi.org/10.1016/s0140-6736(13)61720-1
602. Reidenberg MM. Drug discontinuation effects are part of the pharmacology of a drug. *J Pharmacol Exp Ther* 2011;**339**:324–328. https://doi.org/10.1124/jpet.111.183285
603. Abdelnabi M, Saleh Y, Fareed A, Nossikof A, Wang L, Morsi M, *et al.* Comparative study of oral anticoagulation in left ventricular thrombi (no-LVT trial). *J Am Coll Cardiol* 2021;**77**:1590–1592. https://doi.org/10.1016/j.jacc.2021.01.049
604. Weinsaft JW, Kim J, Medicherla CB, Ma CL, Codella NCF, Kukar N, *et al.* Echocardiographic algorithm for post-myocardial infarction LV thrombus: a gatekeeper for thrombus evaluation by delayed enhancement CMR. *JACC Cardiovasc Imaging* 2016;**9**:505–515. https://doi.org/10.1016/j.jcmg.2015.06.017
605. Segal JB, McNamara RL, Miller MR, Kim N, Goodman SN, Powe NR, *et al.* The evidence regarding the drugs used for ventricular rate control. *J Fam Pract* 2000;**49**:47–59
606. Hou ZY, Chang MS, Chen CY, Tu MS, Lin SL, Chiang HT, *et al.* Acute treatment of recent-onset atrial fibrillation and flutter with a tailored dosing regimen of intravenous amiodarone: a randomized, digoxin-controlled study. *Eur Heart J* 1995;**16**:521–528. https://doi.org/10.1093/oxfordjournals.eurheartj.a060945
607. Deedwania PC, Singh BN, Ellenbogen K, Fisher S, Fletcher R, Singh SN. Spontaneous conversion and maintenance of sinus rhythm by amiodarone in patients with heart failure and atrial fibrillation: observations from the veterans affairs Congestive Heart failure Survival Trial of Antiarrhythmic Therapy (CHF-STAT). *Circulation* 1998;**98**:2574–2579. https://doi.org/10.1161/01.cir.98.23.2574
608. Hofmann R, Steinwender C, Kammler J, Kypta A, Wimmer G, Leisch F, *et al.* Intravenous amiodarone bolus for treatment of atrial fibrillation in patients with advanced congestive heart failure or cardiogenic shock. *Wien Klin Wochenschr* 2004;**116**:744–749. https://doi.org/10.1007/s00508-004-0264-0
609. Moss AJ, Zareba W, Hall WJ, Klein H, Wilber DJ, Cannom DS, *et al.* Prophylactic implantation of a defibrillator in patients with myocardial infarction and reduced ejection fraction. *N Engl J Med* 2002;**346**:877–883. https://doi.org/10.1056/NEJMoa013474
610. Bardy GH, Lee KL, Mark DB, Poole JE, Packer DL, Boineau R, *et al.* Amiodarone or an implantable cardioverter-defibrillator for congestive heart failure. *N Engl J Med* 2005;**352**:225–237. https://doi.org/10.1056/NEJMoa043399
611. Nademanee K, Taylor R, Bailey WE, Rieders DE, Kosar EM. Treating electrical storm: sympathetic blockade versus advanced cardiac life support-guided therapy. *Circulation* 2000;**102**:742–747. https://doi.org/10.1161/01.cir.102.7.742
612. Miwa Y, Ikeda T, Mera H, Miyakoshi M, Hoshida K, Yanagisawa R, *et al.* Effects of landiolol, an ultra-short-acting beta1-selective blocker, on electrical storm refractory to class III antiarrhythmic drugs. *Circ J* 2010;**74**:856–863. https://doi.org/10.1253/circj.cj-09-0772
613. Kudenchuk PJ, Cobb LA, Copass MK, Cummins RO, Doherty AM, Fahrenbruch CE, *et al.* Amiodarone for resuscitation after out-of-hospital cardiac arrest due to ventricular fibrillation. *N Engl J Med* 1999;**341**:871–878. https://doi.org/10.1056/nejm199909163411203
614. Levine JH, Massumi A, Scheinman MM, Winkle RA, Platia EV, Chilson DA, *et al.* Intravenous amiodarone for recurrent sustained hypotensive ventricular tachyarrhythmias. *J Am Coll Cardiol* 1996;**27**:67–75. https://doi.org/10.1016/0735-1097(95)00427-0
615. Gorenek B, Blomström Lundqvist C, Brugada Terradellas J, Camm AJ, Hindricks G, Huber K, *et al.* Cardiac arrhythmias in acute coronary syndromes: position paper from the joint EHRA, ACCA, and EAPCI task force. *Europace* 2014;**16**:1655–1673. https://doi.org/10.1093/europace/euu208
616. Kalarus Z, Svendsen JH, Capodanno D, Dan GA, De Maria E, Gorenek B, *et al.* Cardiac arrhythmias in the emergency settings of acute coronary syndrome and revascularization: an European Heart Rhythm Association (EHRA) consensus document, endorsed by the European Association of Percutaneous Cardiovascular Interventions (EAPCI), and European Acute Cardiovascular Care Association (ACCA). *Europace* 2019;**21**:1603–1604. https://doi.org/10.1093/europace/euz163
617. Feigl D, Ashkenazy J, Kishon Y. Early and late atrioventricular block in acute inferior myocardial infarction. *J Am Coll Cardiol* 1984;**4**:35–38. https://doi.org/10.1016/s0735-1097(84)80315-0
618. Brady WJ, Swart G, DeBehnke DJ, Ma OJ, Aufderheide TP. The efficacy of atropine in the treatment of hemodynamically unstable bradycardia and atrioventricular block: prehospital and emergency department considerations. *Resuscitation* 1999;**41**:47–55. https://doi.org/10.1016/s0300-9572(99)00032-5
619. Kusumoto FM, Calkins H, Boehmer J, Buxton AE, Chung MK, Gold MR, *et al.* HRS/ACC/AHA expert consensus statement on the use of implantable cardioverter-defibrillator therapy in patients who are not included or not well represented in clinical trials. *Circulation* 2014;**130**:94–125. https://doi.org/10.1161/cir.0000000000000056
620. Glikson M, Nielsen JC, Kronborg MB, Michowitz Y, Auricchio A, Barbash IM, *et al.* 2021 ESC Guidelines on cardiac pacing and cardiac resynchronization therapy. *Eur Heart J* 2021;**42**:3427–3520. https://doi.org/10.1093/eurheartj/ehab364
621. Jim MH, Chan AO, Tse HF, Barold SS, Lau CP. Clinical and angiographic findings of complete atrioventricular block in acute inferior myocardial infarction. *Ann Acad Med Singap* 2010;**39**:185–190. https://doi.org/10.47102/annals-acadmedsg.V39N3p185
622. Gang UJ, Hvelplund A, Pedersen S, Iversen A, Jons C, Abildstrom SZ, *et al.* High-degree atrioventricular block complicating ST-segment elevation myocardial infarction in the era of primary percutaneous coronary intervention. *Europace* 2012;**14**:1639–1645. https://doi.org/10.1093/europace/eus161
623. Vicente-Ibarra N, Marín F, Pernías-Escrig V, Sandín-Rollán M, Núñez-Martínez L, Lozano T, *et al.* Impact of anemia as risk factor for major bleeding and mortality in patients with acute coronary syndrome. *Eur J Intern Med* 2019;**61**:48–53. https://doi.org/10.1016/j.ejim.2018.12.004
624. Younge JO, Nauta ST, Akkerhuis KM, Deckers JW, van Domburg RT. Effect of anemia on short- and long-term outcome in patients hospitalized for acute coronary syndromes. *Am J Cardiol* 2012;**109**:506–510. https://doi.org/10.1016/j.amjcard.2011.09.046
625. Bassand JP, Afzal R, Eikelboom J, Wallentin L, Peters R, Budaj A, *et al.* Relationship between baseline haemoglobin and major bleeding complications in acute coronary syndromes. *Eur Heart J* 2010;**31**:50–58. https://doi.org/10.1093/eurheartj/ehp401

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

626. Carson JL, Guyatt G, Heddle NM, Grossman BJ, Cohn CS, Fung MK, *et al.* Clinical practice guidelines from the AABB: red blood cell transfusion thresholds and storage. *JAMA* 2016;**316**:2025–2035. https://doi.org/10.1001/jama.2016.9185
627. Chatterjee S, Wetterslev J, Sharma A, Lichstein E, Mukherjee D. Association of blood transfusion with increased mortality in myocardial infarction: a meta-analysis and diversity-adjusted study sequential analysis. *JAMA Intern Med* 2013;**173**:132–139. https://doi.org/10.1001/2013.jamainternmed.1001
628. Carson JL, Stanworth SJ, Dennis JA, Trivella M, Roubinian N, Fergusson DA, *et al.* Transfusion thresholds for guiding red blood cell transfusion. *Cochrane Database Syst Rev* 2021;**12**:CD002042. https://doi.org/10.1002/14651858.CD002042.pub5
629. Cooper HA, Rao SV, Greenberg MD, Rumsey MP, McKenzie M, Alcorn KW, *et al.* Conservative versus liberal red cell transfusion in acute myocardial infarction (the CRIT Randomized Pilot Study). *Am J Cardiol* 2011;**108**:1108–1111. https://doi.org/10.1016/j.amjcard.2011.06.014
630. Alexander KP, Chen AY, Wang TY, Rao SV, Newby LK, LaPointe NMA, *et al.* Transfusion practice and outcomes in non-ST-segment elevation acute coronary syndromes. *Am Heart J* 2008;**155**:1047–1053. https://doi.org/10.1016/j.ahj.2008.01.009
631. Ducrocq G, Gonzalez-Juanatey JR, Puymirat E, Lemesle G, Cachanado M, Durand-Zaleski I, *et al.* Effect of a restrictive vs liberal blood transfusion strategy on major cardiovascular events among patients with acute myocardial infarction and anemia: the REALITY randomized clinical trial. *JAMA* 2021;**325**:552–560. https://doi.org/10.1001/jama.2021.0135
632. Gonzalez-Juanatey JR, Lemesle G, Puymirat E, Ducrocq G, Cachanado M, Arnaiz JA, *et al.* One-year major cardiovascular events after restrictive versus liberal blood transfusion strategy in patients with acute myocardial infarction and anemia: the REALITY randomized trial. *Circulation* 2022;**145**:486–488. https://doi.org/10.1161/circulationaha.121.057909
633. Gore JM, Spencer FA, Gurfinkel EP, López-Sendón J, Steg PG, Granger CB, *et al.* Thrombocytopenia in patients with an acute coronary syndrome (from the Global Registry of Acute Coronary Events [GRACE]). *Am J Cardiol* 2009;**103**:175–180. https://doi.org/10.1016/j.amjcard.2008.08.055
634. Vora AN, Chenier M, Schulte PJ, Goodman S, Peterson ED, Pieper K, *et al.* Long-term outcomes associated with hospital acquired thrombocytopenia among patients with non-ST-segment elevation acute coronary syndrome. *Am Heart J* 2014;**168**:189–196.e1. https://doi.org/10.1016/j.ahj.2014.04.010
635. Szummer K, Lundman P, Jacobson SH, Schön S, Lindbäck J, Stenestrand U, *et al.* Relation between renal function, presentation, use of therapies and in-hospital complications in acute coronary syndrome: data from the SWEDEHEART register. *J Intern Med* 2010;**268**:40–49. https://doi.org/10.1111/j.1365-2796.2009.02204.x
636. Berger AK, Duval S, Krumholz HM. Aspirin, beta-blocker, and angiotensin-converting enzyme inhibitor therapy in patients with end-stage renal disease and an acute myocardial infarction. *J Am Coll Cardiol* 2003;**42**:201–208. https://doi.org/10.1016/s0735-1097(03)00572-2
637. Panchal HB, Zheng S, Devani K, White CJ, Leinaar EF, Mukherjee D, *et al.* Impact of chronic kidney disease on revascularization and outcomes in patients with ST-elevation myocardial infarction. *Am J Cardiol* 2021;**150**:15–23. https://doi.org/10.1016/j.amjcard.2021.03.057
638. Matsushita K, van der Velde M, Astor BC, Woodward M, Levey AS, de Jong PE, *et al.* Association of estimated glomerular filtration rate and albuminuria with all-cause and cardiovascular mortality in general population cohorts: a collaborative meta-analysis. *Lancet* 2010;**375**:2073–2081. https://doi.org/10.1016/s0140-6736(10)60674-5
639. Santopinto JJ, Fox KA, Goldberg RJ, Budaj A, Piñero G, Avezum A, *et al.* Creatinine clearance and adverse hospital outcomes in patients with acute coronary syndromes: findings from the global registry of acute coronary events (GRACE). *Heart* 2003;**89**:1003–1008. https://doi.org/10.1136/heart.89.9.1003
640. Szummer K, Lundman P, Jacobson SH, Schön S, Lindbäck J, Stenestrand U, *et al.* Influence of renal function on the effects of early revascularization in non-ST-elevation myocardial infarction: data from the Swedish Web-System for Enhancement and Development of Evidence-Based Care in Heart Disease Evaluated According to Recommended Therapies (SWEDEHEART). *Circulation* 2009;**120**:851–858. https://doi.org/10.1161/circulationaha.108.838169
641. Huang HD, Alam M, Hamzeh I, Virani S, Deswal A, Aguilar D, *et al.* Patients with severe chronic kidney disease benefit from early revascularization after acute coronary syndrome. *Int J Cardiol* 2013;**168**:3741–3746. https://doi.org/10.1016/j.ijcard.2013.06.013
642. Kume K, Yasuoka Y, Adachi H, Noda Y, Hattori S, Araki R, *et al.* Impact of contrast-induced acute kidney injury on outcomes in patients with ST-segment elevation myocardial infarction undergoing primary percutaneous coronary intervention. *Cardiovasc Revasc Med* 2013;**14**:253–257. https://doi.org/10.1016/j.carrev.2013.07.009
643. Bangalore S, Briguori C. Preventive strategies for contrast-induced acute kidney injury: and the winner is…. *Circ Cardiovasc Interv* 2017;**10**:e005262. https://doi.org/10.1161/circinterventions.117.005262
644. Davenport MS, Perazella MA, Yee J, Dillman JR, Fine D, McDonald RJ, *et al.* Use of intravenous iodinated contrast media in patients with kidney disease: consensus statements from the American College of Radiology and the National Kidney Foundation. *Kidney Med* 2020;**2**:85–93. https://doi.org/10.1016/j.xkme.2020.01.001
645. Schweiger MJ, Chambers CE, Davidson CJ, Zhang S, Blankenship J, Bhalla NP, *et al.* Prevention of contrast induced nephropathy: recommendations for the high risk patient undergoing cardiovascular procedures. *Catheter Cardiovasc Interv* 2007;**69**:135–140. https://doi.org/10.1002/ccd.20964
646. Visseren FLJ, Mach F, Smulders YM, Carballo D, Koskinas KC, Bäck M, *et al.* 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice. *Eur Heart J* 2021;**42**:3227–3337. https://doi.org/10.1093/eurheartj/ehab484
647. Ängerud KH, Brulin C, Näslund U, Eliasson M. Longer pre-hospital delay in first myocardial infarction among patients with diabetes: an analysis of 4266 patients in the northern Sweden MONICA Study. *BMC Cardiovasc Disord* 2013;**13**:6. https://doi.org/10.1186/1471-2261-13-6
648. Fu R, Li S-D, Song C-X, Yang J-A, Xu H-Y, Gao X-J, *et al.* Clinical significance of diabetes on symptom and patient delay among patients with acute myocardial infarction–an analysis from China Acute Myocardial Infarction (CAMI) registry. *J Geriatr Cardiol* 2019;**16**:395–400. https://doi.org/10.11909/j.issn.1671-5411.2019.05.002
649. Rossello X, Ferreira JP, McMurray JJ, Aguilar D, Pfeffer MA, Pitt B, *et al.* Editor's choice–impact of insulin-treated diabetes on cardiovascular outcomes following high-risk myocardial infarction. *Eur Heart J Acute Cardiovasc Care* 2019;**8**:231–241. https://doi.org/10.1177/2048872618803701
650. Wallert J, Mitchell A, Held C, Hagström E, Leosdottir M, Olsson EMG. Cardiac rehabilitation goal attainment after myocardial infarction with versus without diabetes: a nationwide registry study. *Int J Cardiol* 2019;**292**:19–24. https://doi.org/10.1016/j.ijcard.2019.04.049
651. Ritsinger V, Jensen J, Ohm D, Omerovic E, Koul S, Fröbert O, *et al.* Elevated admission glucose is common and associated with high short-term complication burden after acute myocardial infarction: insights from the VALIDATE-SWEDEHEART study. *Diab Vasc Dis Res* 2019;**16**:582–584. https://doi.org/10.1177/1479164119871540
652. Weston C, Walker L, Birkhead J. Early impact of insulin treatment on mortality for hyperglycaemic patients without known diabetes who present with an acute coronary syndrome. *Heart* 2007;**93**:1542–1546. https://doi.org/10.1136/hrt.2006.108696
653. Ritsinger V, Malmberg K, Mårtensson A, Rydén L, Wedel H, Norhammar A. Intensified insulin-based glycaemic control after myocardial infarction: mortality during 20 year follow-up of the randomised Diabetes Mellitus Insulin Glucose Infusion in Acute Myocardial Infarction (DIGAMI 1) trial. *Lancet Diabetes Endocrinol* 2014;**2**:627–633. https://doi.org/10.1016/s2213-8587(14)70088-9
654. Finfer S, Chittock DR, Su SY, Blair D, Foster D, Dhingra V, *et al.* Intensive versus conventional glucose control in critically ill patients. *N Engl J Med* 2009;**360**:1283–1297. https://doi.org/10.1056/NEJMoa0810625
655. Rossello X, Yellon DM. A new era in the management of type 2 diabetes: is cardioprotection at long last a reality? *Int J Cardiol* 2017;**228**:198–200. https://doi.org/10.1016/j.ijcard.2016.11.246
656. Zinman B, Wanner C, Lachin JM, Fitchett D, Bluhmki E, Hantel S, *et al.* Empagliflozin, cardiovascular outcomes, and mortality in type 2 diabetes. *N Engl J Med* 2015;**373**:2117–2128. https://doi.org/10.1056/NEJMoa1504720
657. Marso SP, Daniels GH, Brown-Frandsen K, Kristensen P, Mann JFE, Nauck MA, *et al.* Liraglutide and cardiovascular outcomes in type 2 diabetes. *N Engl J Med* 2016;**375**:311–322. https://doi.org/10.1056/NEJMoa1603827
658. Cosentino F, Grant PJ, Aboyans V, Bailey CJ, Ceriello A, Delgado V, *et al.* 2019 ESC Guidelines on diabetes, pre-diabetes, and cardiovascular diseases developed in collaboration with the EASD. *Eur Heart J* 2020;**41**:255–323. https://doi.org/10.1093/eurheartj/ehz486
659. Sinclair H, Batty JA, Qiu W, Kunadian V. Engaging older patients in cardiovascular research: observational analysis of the ICON-1 study. *Open Heart* 2016;**3**:e000436. https://doi.org/10.1136/openhrt-2016-000436
660. Rosengren A, Wallentin L, Simoons M, Gitt AK, Behar S, Battler A, *et al.* Age, clinical presentation, and outcome of acute coronary syndromes in the Euroheart acute coronary syndrome survey. *Eur Heart J* 2006;**27**:789–795. https://doi.org/10.1093/eurheartj/ehi774
661. Lopes RD, White JA, Tricoci P, White HD, Armstrong PW, Braunwald E, *et al.* Age, treatment, and outcomes in high-risk non-ST-segment elevation acute coronary syndrome patients: insights from the EARLY ACS trial. *Int J Cardiol* 2013;**167**:2580–2587. https://doi.org/10.1016/j.ijcard.2012.06.053
662. Reiter M, Twerenbold R, Reichlin T, Haaf P, Peter F, Meissner J, *et al.* Early diagnosis of acute myocardial infarction in the elderly using more sensitive cardiac troponin assays. *Eur Heart J* 2011;**32**:1379–1389. https://doi.org/10.1093/eurheartj/ehr033
663. Mills GB, Ratcovich H, Adams-Hall J, Beska B, Kirkup E, Raharjo DE, *et al.* Is the contemporary care of the older persons with acute coronary syndrome evidence-based? *European Heart Journal Open* 2021;**2**:oeab044. https://doi.org/10.1093/ehjopen/oeab044
664. Tegn N, Abdelnoor M, Aaberge L, Endresen K, Smith P, Aakhus S, *et al.* Invasive versus conservative strategy in patients aged 80 years or older with non-ST-elevation myocardial infarction or unstable angina pectoris (After Eighty study): an open-label randomised controlled trial. *Lancet* 2016;**387**:1057–1065. https://doi.org/10.1016/s0140-6736(15)01166-6
665. Bueno H, Betriu A, Heras M, Alonso JJ, Cequier A, Garcia EJ, *et al.* Primary angioplasty vs. fibrinolysis in very old patients with acute myocardial infarction: TRIANA

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

(TRatamiento del Infarto Agudo de miocardio eN Ancianos) randomized trial and pooled analysis with previous studies. *Eur Heart J* 2011;**32**:51–60. https://doi.org/10.1093/eurheartj/ehq375
666. Kunadian V, Qiu W, Ludman P, Redwood S, Curzen N, Stables R, *et al.* Outcomes in patients with cardiogenic shock following percutaneous coronary intervention in the contemporary era: an analysis from the BCIS database (British Cardiovascular Intervention Society). *JACC Cardiovasc Interv* 2014;**7**:1374–1385. https://doi.org/10.1016/j.jcin.2014.06.017
667. Kunadian V, Bawamia B, Maznyczka A, Zaman A, Qiu W. Outcomes following primary percutaneous coronary intervention in the setting of cardiac arrest: a registry database study. *Eur Heart J Acute Cardiovasc Care* 2015;**4**:6–15. https://doi.org/10.1177/2048872614534079
668. Richter D, Guasti L, Walker D, Lambrinou E, Lionis C, Abreu A, *et al.* Frailty in cardiology: definition, assessment and clinical implications for general cardiology. A consensus document of the Council for Cardiology Practice (CCP), Association for Acute Cardio Vascular Care (ACVC), Association of Cardiovascular Nursing and Allied Professions (ACNAP), European Association of Preventive Cardiology (EAPC), European Heart Rhythm Association (EHRA), Council on Valvular Heart Diseases (VHD), Council on Hypertension (CHT), Council of Cardio-Oncology (CCO), Working Group (WG) Aorta and Peripheral Vascular Diseases, WG e-Cardiology, WG Thrombosis, of the European Society of Cardiology, European Primary Care Cardiology Society (EPCCS). *Eur J Prev Cardiol* 2022;**29**:216–227. https://doi.org/10.1093/eurjpc/zwaa167
669. Walker DM, Gale CP, Lip G, Martin-Sanchez FJ, McIntyre HF, Mueller C, *et al.* Editor's choice–frailty and the management of patients with acute cardiovascular disease: a position paper from the Acute Cardiovascular Care Association. *Eur Heart J Acute Cardiovasc Care* 2018;**7**:176–193. https://doi.org/10.1177/2048872618758931
670. Clegg A, Young J, Iliffe S, Rikkert MO, Rockwood K. Frailty in elderly people. *Lancet* 2013;**381**:752–762. https://doi.org/10.1016/s0140-6736(12)62167-9
671. Chung KJNC, Wilkinson C, Veerasamy M, Kunadian V. Frailty scores and their utility in older patients with cardiovascular disease. *Interv Cardiol* 2021;**16**:e05. https://doi.org/10.15420/icr.2020.18
672. Gu SZ, Qiu W, Batty JA, Sinclair H, Veerasamy M, Brugaletta S, *et al.* Coronary artery lesion phenotype in frail older patients with non-ST-elevation acute coronary syndrome undergoing invasive care. *EuroIntervention* 2019;**15**:e261–e268. https://doi.org/10.4244/eij-d-18-00848
673. Batty J, Qiu W, Gu S, Sinclair H, Veerasamy M, Beska B, *et al.* One-year clinical outcomes in older patients with non-ST elevation acute coronary syndrome undergoing coronary angiography: an analysis of the ICON1 study. *Int J Cardiol* 2019;**274**:45–51. https://doi.org/10.1016/j.ijcard.2018.09.086
674. Beska B, Coakley D, MacGowan G, Adams-Hall J, Wilkinson C, Kunadian V, *et al.* Frailty and quality of life after invasive management for non-ST elevation acute coronary syndrome. *Heart* 2022;**108**:203–211. https://doi.org/10.1136/heartjnl-2021-319064
675. Beska B, Mills GB, Ratcovich H, Wilkinson C, Damluji AA, Kunadian V, *et al.* Impact of multimorbidity on long-term outcomes in older adults with non-ST elevation acute coronary syndrome in the North East of England: a multi-centre cohort study of patients undergoing invasive care. *BMJ Open* 2022;**12**:e061830. https://doi.org/10.1136/bmjopen-2022-061830
676. Gu SZ, Beska B, Chan D, Neely D, Batty JA, Adams-Hall J, *et al.* Cognitive decline in older patients with non-ST elevation acute coronary syndrome. *J Am Heart Assoc* 2019;**8**:e011218. https://doi.org/10.1161/jaha.118.011218
677. Ismail S, Wong C, Rajan P, Vidovich MI. ST-elevation acute myocardial infarction in pregnancy: 2016 update. *Clin Cardiol* 2017;**40**:399–406. https://doi.org/10.1002/clc.22655
678. Regitz-Zagrosek V, Roos-Hesselink JW, Bauersachs J, Blomström-Lundqvist C, Cífková R, De Bonis M, *et al.* 2018 ESC Guidelines for the management of cardiovascular diseases during pregnancy. *Eur Heart J* 2018;**39**:3165–3241. https://doi.org/10.1093/eurheartj/ehy340
679. Roth A, Elkayam U. Acute myocardial infarction associated with pregnancy. *J Am Coll Cardiol* 2008;**52**:171–180. https://doi.org/10.1016/j.jacc.2008.03.049
680. Elkayam U, Jalnapurkar S, Barakkat MN, Khatri N, Kealey AJ, Mehra A, *et al.* Pregnancy-associated acute myocardial infarction: a review of contemporary experience in 150 cases between 2006 and 2011. *Circulation* 2014;**129**:1695–1702. https://doi.org/10.1161/circulationaha.113.002054
681. Elkayam U, Goland S, Pieper PG, Silverside CK. High-risk cardiac disease in pregnancy: Part I. *J Am Coll Cardiol* 2016;**68**:396–410. https://doi.org/10.1016/j.jacc.2016.05.048
682. Bharadwaj A, Potts J, Mohamed MO, Parwani P, Swamy P, Lopez-Mattei JC, *et al.* Acute myocardial infarction treatments and outcomes in 6.5 million patients with a current or historical diagnosis of cancer in the USA. *Eur Heart J* 2020;**41**:2183–2193. https://doi.org/10.1093/eurheartj/ehz851
683. Velders MA, Boden H, Hofma SH, Osanto S, van der Hoeven BL, Heestermans AACM, *et al.* Outcome after ST elevation myocardial infarction in patients with cancer treated with primary percutaneous coronary intervention. *Am J Cardiol* 2013;**112**:1867–1872. https://doi.org/10.1016/j.amjcard.2013.08.019
684. Lyon AR, López-Fernández T, Couch LS, Asteggiano R, Aznar MC, Bergler-Klein J, *et al.* 2022 ESC Guidelines on cardio-oncology developed in collaboration with the European Hematology Association (EHA), the European Society for Therapeutic Radiology and Oncology (ESTRO) and the International Cardio-Oncology Society (IC-OS): developed by the Task Force on cardio-oncology of the European Society of Cardiology (ESC). *Eur Heart J* 2022;**43**:4229–4361. https://doi.org/10.1093/eurheartj/ehac244
685. Potts JE, Iliescu CA, Lopez Mattei JC, Martinez SC, Holmvang L, Ludman P, *et al.* Percutaneous coronary intervention in cancer patients: a report of the prevalence and outcomes in the United States. *Eur Heart J* 2019;**40**:1790–1800. https://doi.org/10.1093/eurheartj/ehy769
686. Pothineni NV, Shah NN, Rochlani Y, Saad M, Kovelamudi S, Marmagkiolis K, *et al.* Temporal trends and outcomes of acute myocardial infarction in patients with cancer. *Ann Transl Med* 2017;**5**:482. https://doi.org/10.21037/atm.2017.11.29
687. Gevaert SA, Halvorsen S, Sinnaeve PR, Sambola A, Gulati G, Lancellotti P, *et al.* Evaluation and management of cancer patients presenting with acute cardiovascular disease: a Consensus Document of the Acute CardioVascular Care (ACVC) association and the ESC Council of Cardio-Oncology-Part 1: acute coronary syndromes and acute pericardial diseases. *Eur Heart J Acute Cardiovasc Care* 2021;**10**:947–959. https://doi.org/10.1093/ehjacc/zuab056
688. Lancellotti P, Suter TM, López-Fernández T, Galderisi M, Lyon AR, Van der Meer P, *et al.* Cardio-Oncology Services: rationale, organization, and implementation. *Eur Heart J* 2019;**40**:1756–1763. https://doi.org/10.1093/eurheartj/ehy453
689. Mohamed MO, Van Spall HGC, Kontopantelis E, Alkhouli M, Barac A, Elgendy IY, *et al.* Effect of primary percutaneous coronary intervention on in-hospital outcomes among active cancer patients presenting with ST-elevation myocardial infarction: a propensity score matching analysis. *Eur Heart J Acute Cardiovasc Care* 2021;**10**:829–839. https://doi.org/10.1093/ehjacc/zuaa032
690. Guddati AK, Joy PS, Kumar G. Analysis of outcomes of percutaneous coronary intervention in metastatic cancer patients with acute coronary syndrome over a 10-year period. *J Cancer Res Clin Oncol* 2016;**142**:471–479. https://doi.org/10.1007/s00432-015-2056-5
691. Aspelin P, Aubry P, Fransson SG, Strasser R, Willenbrock R, Berg KJ. Nephrotoxic effects in high-risk patients undergoing angiography. *N Engl J Med* 2003;**348**:491–499. https://doi.org/10.1056/NEJMoa021833
692. Jo SH, Youn TJ, Koo BK, Park JS, Kang HJ, Cho YS, *et al.* Renal toxicity evaluation and comparison between visipaque (Iodixanol) and hexabrix (Ioxaglate) in patients with renal insufficiency undergoing coronary angiography: the RECOVER study: a randomized controlled trial. *J Am Coll Cardiol* 2006;**48**:924–930. https://doi.org/10.1016/j.jacc.2006.06.047
693. Solomon RJ, Natarajan MK, Doucet S, Sharma SK, Staniloae CS, Katholi RE, *et al.* Cardiac Angiography in Renally Impaired Patients (CARE) study: a randomized double-blind trial of contrast-induced nephropathy in patients with chronic kidney disease. *Circulation* 2007;**115**:3189–3196. https://doi.org/10.1161/circulationaha.106.671644
694. Brar SS, Shen AY, Jorgensen MB, Kotlewski A, Aharonian VJ, Desai N, *et al.* Sodium bicarbonate vs sodium chloride for the prevention of contrast medium-induced nephropathy in patients undergoing coronary angiography: a randomized trial. *JAMA* 2008;**300**:1038–1046. https://doi.org/10.1001/jama.300.9.1038
695. Brar SS, Aharonian V, Mansukhani P, Moore N, Shen AYJ, Jorgensen M, *et al.* Haemodynamic-guided fluid administration for the prevention of contrast-induced acute kidney injury: the POSEIDON randomised controlled trial. *Lancet* 2014;**383**:1814–1823. https://doi.org/10.1016/s0140-6736(14)60689-9
696. Giacoppo D, Gargiulo G, Buccheri S, Aruta P, Byrne RA, Cassese S, *et al.* Preventive strategies for contrast-induced acute kidney injury in patients undergoing percutaneous coronary procedures: evidence from a hierarchical Bayesian network meta-analysis of 124 trials and 28240 patients. *Circ Cardiovasc Interv* 2017;**10**:e004383. https://doi.org/10.1161/circinterventions.116.004383
697. Nijssen EC, Rennenberg RJ, Nelemans PJ, Essers BA, Janssen MM, Vermeeren MA, *et al.* Prophylactic hydration to protect renal function from intravascular iodinated contrast material in patients at high risk of contrast-induced nephropathy (AMACING): a prospective, randomised, phase 3, controlled, open-label, non-inferiority trial. *Lancet* 2017;**389**:1312–1322. https://doi.org/10.1016/s0140-6736(17)30057-0
698. Zelniker TA, Wiviott SD, Raz I, Im K, Goodrich EL, Bonaca MP, *et al.* SGLT2 inhibitors for primary and secondary prevention of cardiovascular and renal outcomes in type 2 diabetes: a systematic review and meta-analysis of cardiovascular outcome trials. *Lancet* 2019;**393**:31–39. https://doi.org/10.1016/s0140-6736(18)32590-x
699. Kristensen SL, Rørth R, Jhund PS, Docherty KF, Sattar N, Preiss D, *et al.* Cardiovascular, mortality, and kidney outcomes with GLP-1 receptor agonists in patients with type 2 diabetes: a systematic review and meta-analysis of cardiovascular outcome trials. *Lancet Diabetes Endocrinol* 2019;**7**:776–785. https://doi.org/10.1016/s2213-8587(19)30249-9
700. Perkovic V, Jardine MJ, Neal B, Bompoint S, Heerspink HJL, Charytan DM, *et al.* Canagliflozin and renal outcomes in type 2 diabetes and nephropathy. *N Engl J Med* 2019;**380**:2295–2306. https://doi.org/10.1056/NEJMoa1811744
701. Heerspink HJL, Stefánsson BV, Correa-Rotter R, Chertow GM, Greene T, Hou FF, *et al.* Dapagliflozin in patients with chronic kidney disease. *N Engl J Med* 2020;**383**:1436–1446. https://doi.org/10.1056/NEJMoa2024816

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

702. McMurray JJV, Solomon SD, Inzucchi SE, Køber L, Kosiborod MN, Martinez FA, *et al.* Dapagliflozin in patients with heart failure and reduced ejection fraction. *N Engl J Med* 2019;**381**:1995–2008. https://doi.org/10.1056/NEJMoa1911303
703. Packer M, Anker SD, Butler J, Filippatos G, Pocock SJ, Carson P, *et al.* Cardiovascular and renal outcomes with empagliflozin in heart failure. *N Engl J Med* 2020;**383**: 1413–1424. https://doi.org/10.1056/NEJMoa2022190
704. Packer M, Butler J, Zannad F, Filippatos G, Ferreira JP, Pocock SJ, *et al.* Effect of empagliflozin on worsening heart failure events in patients with heart failure and a preserved ejection fraction: the EMPEROR-preserved trial. *Circulation* 2021;**144**:1284–1294. https://doi.org/10.1161/circulationaha.121.056824
705. Ferrannini G, De Bacquer D, De Backer G, Kotseva K, Mellbin L, Wood D, *et al.* Screening for glucose perturbations and risk factor management in dysglycemic patients with coronary artery disease–a persistent challenge in need of substantial improvement: a report from ESC EORP EUROASPIRE V. *Diabetes Care* 2020;**43**: 726–733. https://doi.org/10.2337/dc19-2165
706. Shahim B, De Bacquer D, De Backer G, Gyberg V, Kotseva K, Mellbin L, *et al.* The prognostic value of fasting plasma glucose, two-hour postload glucose, and HbA(1c) in patients with coronary artery disease: a report from EUROASPIRE IV: a survey from the European Society of Cardiology. *Diabetes Care* 2017;**40**:1233–1240. https://doi.org/10.2337/dc17-0245
707. Ritsinger V, Tanoglidi E, Malmberg K, Näsman P, Rydén L, Tenerz Å, *et al.* Sustained prognostic implications of newly detected glucose abnormalities in patients with acute myocardial infarction: long-term follow-up of the Glucose Tolerance in Patients with Acute Myocardial Infarction cohort. *Diab Vasc Dis Res* 2015;**12**:23–32. https://doi.org/10.1177/1479164114551746
708. Svensson AM, McGuire DK, Abrahamsson P, Dellborg M. Association between hyper- and hypoglycaemia and 2 year all-cause mortality risk in diabetic patients with acute coronary events. *Eur Heart J* 2005;**26**:1255–1261. https://doi.org/10.1093/eurheartj/ehi230
709. Pinto DS, Skolnick AH, Kirtane AJ, Murphy SA, Barron HV, Giugliano RP, *et al.* U-shaped relationship of blood glucose with adverse outcomes among patients with ST-segment elevation myocardial infarction. *J Am Coll Cardiol* 2005;**46**:178–180. https://doi.org/10.1016/j.jacc.2005.03.052
710. Cholesterol Treatment Trialists' Collaboration. Efficacy and safety of statin therapy in older people: a meta-analysis of individual participant data from 28 randomised controlled trials. *Lancet* 2019;**393**:407–415. https://doi.org/10.1016/s0140-6736(18)31942-1
711. Bach RG, Cannon CP, Giugliano RP, White JA, Lokhnygina Y, Bohula EA, *et al.* Effect of simvastatin-ezetimibe compared with simvastatin monotherapy after acute coronary syndrome among patients 75 years or older: a secondary analysis of a randomized clinical trial. *JAMA Cardiol* 2019;**4**:846–854. https://doi.org/10.1001/jamacardio.2019.2306
712. Roe MT, Armstrong PW, Fox KA, White HD, Prabhakaran D, Goodman SG, *et al.* Prasugrel versus clopidogrel for acute coronary syndromes without revascularization. *N Engl J Med* 2012;**367**:1297–1309. https://doi.org/10.1056/NEJMoa1205512
713. Chang HM, Okwuosa TM, Scarabelli T, Moudgil R, Yeh ETH. Cardiovascular complications of cancer therapy: best practices in diagnosis, prevention, and management: Part 2. *J Am Coll Cardiol* 2017;**70**:2552–2565. https://doi.org/10.1016/j.jacc.2017.09.1095
714. Herrmann J. Vascular toxic effects of cancer therapies. *Nat Rev Cardiol* 2020;**17**: 503–522. https://doi.org/10.1038/s41569-020-0347-2
715. Long M, Ye Z, Zheng J, Chen W, Li L. Dual anti-platelet therapy following percutaneous coronary intervention in a population of patients with thrombocytopenia at baseline: a meta-analysis. *BMC Pharmacol Toxicol* 2020;**21**:31. https://doi.org/10.1186/s40360-020-00409-2
716. Ambrosetti M, Abreu A, Corrà U, Davos CH, Hansen D, Frederix I, *et al.* Secondary prevention through comprehensive cardiovascular rehabilitation: from knowledge to implementation. 2020 update. A position paper from the Secondary Prevention and Rehabilitation Section of the European Association of Preventive Cardiology. *Eur J Prev Cardiol* 2020;**28**:460–495. https://doi.org/10.1177/2047487320913379
717. Abreu A, Frederix I, Dendale P, Janssen A, Doherty P, Piepoli MF, *et al.* Standardization and quality improvement of secondary prevention through cardiovascular rehabilitation programmes in Europe: the avenue towards EAPC accreditation programme: a position statement of the Secondary Prevention and Rehabilitation Section of the European Association of Preventive Cardiology (EAPC). *Eur J Prev Cardiol* 2020;**28**: 496–509. https://doi.org/10.1177/2047487320924912
718. Rossello X, Pocock SJ, Julian DG. Long-term use of cardiovascular drugs: challenges for research and for patient care. *J Am Coll Cardiol* 2015;**66**:1273–1285. https://doi.org/10.1016/j.jacc.2015.07.018
719. Frederix I, Dendale P, Schmid JP. Who needs secondary prevention? *Eur J Prev Cardiol* 2017;**24**:8–13. https://doi.org/10.1177/2047487317706112
720. Rea F, Ronco R, Pedretti RFE, Merlino L, Corrao G. Better adherence with out-of-hospital healthcare improved long-term prognosis of acute coronary syndromes: evidence from an Italian real-world investigation. *Int J Cardiol* 2020;**318**: 14–20. https://doi.org/10.1016/j.ijcard.2020.06.017
721. Salzwedel A, Jensen K, Rauch B, Doherty P, Metzendorf MI, Hackbusch M, *et al.* Effectiveness of comprehensive cardiac rehabilitation in coronary artery disease patients treated according to contemporary evidence based medicine: update of the Cardiac Rehabilitation Outcome Study (CROS-II). *Eur J Prev Cardiol* 2020;**27**: 1756–1774. https://doi.org/10.1177/2047487320905719
722. Santiago de Araújo Pio C, Marzolini S, Pakosh M, Grace SL. Effect of cardiac rehabilitation dose on mortality and morbidity: a systematic review and meta-regression analysis. *Mayo Clin Proc* 2017;**92**:1644–1659. https://doi.org/10.1016/j.mayocp.2017.07.019
723. van Halewijn G, Deckers J, Tay HY, van Domburg R, Kotseva K, Wood D. Lessons from contemporary trials of cardiovascular prevention and rehabilitation: a systematic review and meta-analysis. *Int J Cardiol* 2017;**232**:294–303. https://doi.org/10.1016/j.ijcard.2016.12.125
724. Dibben G, Faulkner J, Oldridge N, Rees K, Thompson DR, Zwisler AD, *et al.* Exercise-based cardiac rehabilitation for coronary heart disease. *Cochrane Database Syst Rev* 2021;**11**:Cd001800. https://doi.org/10.1002/14651858.CD001800.pub4
725. Anderson L, Thompson DR, Oldridge N, Zwisler AD, Rees K, Martin N, *et al.* Exercise-based cardiac rehabilitation for coronary heart disease. *Cochrane Database Syst Rev* 2016;**2016**:CD001800. https://doi.org/10.1002/14651858.CD001800.pub3
726. Benzer W, Rauch B, Schmid JP, Zwisler AD, Dendale P, Davos CH, *et al.* Exercise-based cardiac rehabilitation in twelve European countries: results of the European cardiac rehabilitation registry. *Int J Cardiol* 2017;**228**:58–67. https://doi.org/10.1016/j.ijcard.2016.11.059
727. Clark RA, Conway A, Poulsen V, Keech W, Tirimacco R, Tideman P. Alternative models of cardiac rehabilitation: a systematic review. *Eur J Prev Cardiol* 2015;**22**:35–74. https://doi.org/10.1177/2047487313501093
728. Kotseva K, De Backer G, De Bacquer D, Rydén L, Hoes A, Grobbee D, *et al.* Lifestyle and impact on cardiovascular risk factor control in coronary patients across 27 countries: results from the European Society of Cardiology ESC-EORP EUROASPIRE V registry. *Eur J Prev Cardiol* 2019;**26**:824–835. https://doi.org/10.1177/2047487318825350
729. Frederix I, Vanhees L, Dendale P, Goetschalckx K. A review of telerehabilitation for cardiac patients. *J Telemed Telecare* 2015;**21**:45–53. https://doi.org/10.1177/1357633x14562732
730. Conraads VM, Deaton C, Piotrowicz E, Santaularia N, Tierney S, Piepoli MF, *et al.* Adherence of heart failure patients to exercise: barriers and possible solutions: a position statement of the Study Group on Exercise Training in Heart Failure of the Heart Failure Association of the European Society of Cardiology. *Eur J Heart Fail* 2012;**14**: 451–458. https://doi.org/10.1093/eurjhf/hfs048
731. De Bacquer D, Astin F, Kotseva K, Pogosova N, De Smedt D, De Backer G, *et al.* Poor adherence to lifestyle recommendations in patients with coronary heart disease: results from the EUROASPIRE surveys. *Eur J Prev Cardiol* 2021;**29**:383–395. https://doi.org/10.1093/eurjpc/zwab115
732. Dalal HM, Taylor RS. Telehealth technologies could improve suboptimal rates of participation in cardiac rehabilitation. *Heart* 2016;**102**:1155–1156. https://doi.org/10.1136/heartjnl-2016-309429
733. Lavie CJ, Arena R, Franklin BA. Cardiac rehabilitation and healthy life-style interventions: rectifying program deficiencies to improve patient outcomes. *J Am Coll Cardiol* 2016;**67**:13–15. https://doi.org/10.1016/j.jacc.2015.09.103
734. Fors A, Taft C, Ulin K, Ekman I. Person-centred care improves self-efficacy to control symptoms after acute coronary syndrome: a randomized controlled trial. *Eur J Cardiovasc Nurs* 2016;**15**:186–194. https://doi.org/10.1177/1474515115623437
735. Frederix I, Caiani EG, Dendale P, Anker S, Bax J, Böhm A, *et al.* ESC e-Cardiology Working Group Position Paper: overcoming challenges in digital health implementation in cardiovascular medicine. *Eur J Prev Cardiol* 2019;**26**:1166–1177. https://doi.org/10.1177/2047487319832394
736. Rosselló X, Stanbury M, Beeri R, Kirchhof P, Casadei B, Kotecha D. Digital learning and the future cardiologist. *Eur Heart J* 2019;**40**:499–501. https://doi.org/10.1093/eurheartj/ehy884
737. Frederix I, Solmi F, Piepoli MF, Dendale P. Cardiac telerehabilitation: a novel cost-efficient care delivery strategy that can induce long-term health benefits. *Eur J Prev Cardiol* 2017;**24**:1708–1717. https://doi.org/10.1177/2047487317732274
738. Avila A, Claes J, Buys R, Azzawi M, Vanhees L, Cornelissen V. Home-based exercise with telemonitoring guidance in patients with coronary artery disease: does it improve long-term physical fitness? *Eur J Prev Cardiol* 2020;**27**:367–377. https://doi.org/10.1177/2047487319892201
739. Claes J, Cornelissen V, McDermott C, Moyna N, Pattyn N, Cornelis N, *et al.* Feasibility, acceptability, and clinical effectiveness of a technology-enabled cardiac rehabilitation platform (physical activity toward health-I): randomized controlled trial. *J Med Internet Res* 2020;**22**:e14221. https://doi.org/10.2196/14221
740. Kraal JJ, Peek N, Van den Akker-Van Marle ME, Kemps HM. Effects of home-based training with telemonitoring guidance in low to moderate risk patients entering cardiac rehabilitation: short-term results of the FIT@Home study. *Eur J Prev Cardiol* 2014;**21**: 26–31. https://doi.org/10.1177/2047487314552606
741. Maddison R, Rawstorn JC, Stewart RAH, Benatar J, Whittaker R, Rolleston A, *et al.* Effects and costs of real-time cardiac telerehabilitation: randomised controlled non-inferiority trial. *Heart* 2019;**105**:122–129. https://doi.org/10.1136/heartjnl-2018-313189

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

742. Scherrenberg M, Wilhelm M, Hansen D, Völler H, Cornelissen V, Frederix I, *et al.* The future is now: a call for action for cardiac telerehabilitation in the COVID-19 pandemic from the secondary prevention and rehabilitation section of the European Association of Preventive Cardiology. *Eur J Prev Cardiol* 2020;**28**:524–540. https://doi.org/10.1177/2047487320939671
743. Huang K, Liu W, He D, Huang B, Xiao D, Peng Y, *et al.* Telehealth interventions versus center-based cardiac rehabilitation of coronary artery disease: a systematic review and meta-analysis. *Eur J Prev Cardiol* 2015;**22**:959–971. https://doi.org/10.1177/2047487314561168
744. Wolf A, Vella R, Fors A. The impact of person-centred care on patients' care experiences in relation to educational level after acute coronary syndrome: secondary outcome analysis of a randomised controlled trial. *Eur J Cardiovasc Nurs* 2019;**18**:299–308. https://doi.org/10.1177/1474515118821242
745. Brown MT, Bussell J, Dutta S, Davis K, Strong S, Mathew S. Medication adherence: truth and consequences. *Am J Med Sci* 2016;**351**:387–399. https://doi.org/10.1016/j.amjms.2016.01.010
746. Arlt AD, Nestoriuc Y, Rief W. Why current drug adherence programs fail: addressing psychological risk factors of nonadherence. *Curr Opin Psychiatry* 2017;**30**:326–333. https://doi.org/10.1097/yco.0000000000000345
747. Easthall C, Taylor N, Bhattacharya D. Barriers to medication adherence in patients prescribed medicines for the prevention of cardiovascular disease: a conceptual framework. *Int J Pharm Pract* 2019;**27**:223–231. https://doi.org/10.1111/ijpp.12491
748. Seabury SA, Dougherty JS, Sullivan J. Medication adherence as a measure of the quality of care provided by physicians. *Am J Manag Care* 2019;**25**:78–83.
749. Pedretti RFE, Hansen D, Ambrosetti M, Back M, Berger T, Ferreira MC, *et al.* How to optimize the adherence to a guideline-directed medical therapy in the secondary prevention of cardiovascular diseases: a clinical consensus statement from the European Association of Preventive Cardiology (EAPC). *Eur J Prev Cardiol* 2022;**30**:149–166. https://doi.org/10.1093/eurjpc/zwac204
750. Castellano JM, Sanz G, Peñalvo JL, Bansilal S, Fernández-Ortiz A, Alvarez L, *et al.* A polypill strategy to improve adherence: results from the FOCUS project. *J Am Coll Cardiol* 2014;**64**:2071–2082. https://doi.org/10.1016/j.jacc.2014.08.021
751. Selak V, Webster R, Stepien S, Bullen C, Patel A, Thom S, *et al.* Reaching cardiovascular prevention guideline targets with a polypill-based approach: a meta-analysis of randomised clinical trials. *Heart* 2019;**105**:42–48. https://doi.org/10.1136/heartjnl-2018-313108
752. Castellano JM, Fuster V, Jennings C, Prescott E, Bueno H. Role of the polypill for secondary prevention in ischaemic heart disease. *Eur J Prev Cardiol* 2017;**24**:44–51. https://doi.org/10.1177/2047487317707324
753. Castellano JM, Pocock SJ, Bhatt DL, Quesada AJ, Owen R, Fernandez-Ortiz A, *et al.* Polypill strategy in secondary cardiovascular prevention. *N Engl J Med* 2022;**387**:967–977. https://doi.org/10.1056/NEJMoa2208275
754. Palmer MJ, Barnard S, Perel P, Free C. Mobile phone-based interventions for improving adherence to medication prescribed for the primary prevention of cardiovascular disease in adults. *Cochrane Database Syst Rev* 2018;**6**:CD012675. https://doi.org/10.1002/14651858.CD012675.pub2
755. Guerriero C, Cairns J, Roberts I, Rodgers A, Whittaker R, Free C. The cost-effectiveness of smoking cessation support delivered by mobile phone text messaging: Txt2stop. *Eur J Health Econ* 2013;**14**:789–797. https://doi.org/10.1007/s10198-012-0424-5
756. Gandapur Y, Kianoush S, Kelli HM, Misra S, Urrea B, Blaha MJ, *et al.* The role of mHealth for improving medication adherence in patients with cardiovascular disease: a systematic review. *Eur Heart J Qual Care Clin Outcomes* 2016;**2**:237–244. https://doi.org/10.1093/ehjqcco/qcw018
757. Fortuna RJ, Nagel AK, Rocco TA, Legette-Sobers S, Quigley DD. Patient experience with care and its association with adherence to hypertension medications. *Am J Hypertens* 2018;**31**:340–345. https://doi.org/10.1093/ajh/hpx200
758. Keenan J. Improving adherence to medication for secondary cardiovascular disease prevention. *Eur J Prev Cardiol* 2017;**24**:29–35. https://doi.org/10.1177/2047487317708145
759. Geidl W, Schlesinger S, Mino E, Miranda L, Pfeifer K. Dose-response relationship between physical activity and mortality in adults with noncommunicable diseases: a systematic review and meta-analysis of prospective observational studies. *Int J Behav Nutr Phys Act* 2020;**17**:109. https://doi.org/10.1186/s12966-020-01007-5
760. Ekblom O, Ek A, Cider Å, Hambraeus K, Börjesson M. Increased physical activity post-myocardial infarction is related to reduced mortality: results from the SWEDEHEART registry. *J Am Heart Assoc* 2018;**7**:e010108. https://doi.org/10.1161/jaha.118.010108
761. Delgado-Lista J, Alcala-Diaz JF, Torres-Peña JD, Quintana-Navarro GM, Fuentes F, Garcia-Rios A, *et al.* Long-term secondary prevention of cardiovascular disease with a Mediterranean diet and a low-fat diet (CORDIOPREV): a randomised controlled trial. *Lancet* 2022;**399**:1876–1885. https://doi.org/10.1016/s0140-6736(22)00122-2
762. de Lorgeril M, Salen P, Martin JL, Monjaud I, Delaye J, Mamelle N. Mediterranean diet, traditional risk factors, and the rate of cardiovascular complications after myocardial infarction: final report of the Lyon Diet Heart Study. *Circulation* 1999;**99**:779–785. https://doi.org/10.1161/01.cir.99.6.779
763. Critchley JA, Capewell S. Mortality risk reduction associated with smoking cessation in patients with coronary heart disease: a systematic review. *JAMA* 2003;**290**:86–97. https://doi.org/10.1001/jama.290.1.86
764. Chow CK, Jolly S, Rao-Melacini P, Fox KAA, Anand SS, Yusuf S. Association of diet, exercise, and smoking modification with risk of early cardiovascular events after acute coronary syndromes. *Circulation* 2010;**121**:750–758. https://doi.org/10.1161/circulationaha.109.891523
765. United States Public Health Service Office of the Surgeon General. Smoking Cessation: A Report of the Surgeon General. In: US Department of Health and Human Services; 2020.
766. Levine GN, Bates ER, Bittl JA, Brindis RG, Fihn SD, Fleisher LA, *et al.* 2016 ACC/AHA Guideline Focused Update on Duration of Dual Antiplatelet Therapy in Patients With Coronary Artery Disease: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines: an Update of the 2011 ACCF/AHA/SCAI Guideline for Percutaneous Coronary Intervention, 2011 ACCF/AHA Guideline for Coronary Artery Bypass Graft Surgery, 2012 ACC/AHA/ACP/AATS/PCNA/SCAI/STS Guideline for the Diagnosis and Management of Patients With Stable Ischemic Heart Disease, 2013 ACCF/AHA Guideline for the Management of ST-Elevation Myocardial Infarction, 2014 AHA/ACC Guideline for the Management of Patients With Non-ST-Elevation Acute Coronary Syndromes, and 2014 ACC/AHA Guideline on Perioperative Cardiovascular Evaluation and Management of Patients Undergoing Noncardiac Surgery. *Circulation* 2016;**134**:e123–e155. https://doi.org/10.1161/cir.0000000000000404
767. Cahill K, Stevens S, Perera R, Lancaster T. Pharmacological interventions for smoking cessation: an overview and network meta-analysis. *Cochrane Database Syst Rev* 2013;**2013**:CD009329. https://doi.org/10.1002/14651858.CD009329.pub2
768. Eisenberg MJ, Windle SB, Roy N, Old W, Grondin FR, Bata I, *et al.* Varenicline for smoking cessation in hospitalized patients with acute coronary syndrome. *Circulation* 2016;**133**:21–30. https://doi.org/10.1161/circulationaha.115.019634
769. Sterling LH, Windle SB, Filion KB, Touma L, Eisenberg MJ. Varenicline and adverse cardiovascular events: a systematic review and meta-analysis of randomized controlled trials. *J Am Heart Assoc* 2016;**5**:e002849. https://doi.org/10.1161/jaha.115.002849
770. Windle SB, Dehghani P, Roy N, Old W, Grondin FR, Bata I, *et al.* Smoking abstinence 1 year after acute coronary syndrome: follow-up from a randomized controlled trial of varenicline in patients admitted to hospital. *CMAJ* 2018;**190**:E347–E354. https://doi.org/10.1503/cmaj.170377
771. Kavousi M, Pisinger C, Barthelemy JC, De Smedt D, Koskinas K, Marques-Vidal P, *et al.* Electronic cigarettes and health with special focus on cardiovascular effects: position paper of the European Association of Preventive Cardiology (EAPC). *Eur J Prev Cardiol* 2020;**28**:1552–1566. https://doi.org/10.1177/2047487320941993
772. Estruch R, Ros E, Salas-Salvadó J, Covas MI, Corella D, Arós F, *et al.* Primary prevention of cardiovascular disease with a mediterranean diet supplemented with extra-virgin olive oil or nuts. *N Engl J Med* 2018;**378**:e34. https://doi.org/10.1056/NEJMoa1800389
773. Wood AM, Kaptoge S, Butterworth AS, Willeit P, Warnakula S, Bolton T, *et al.* Risk thresholds for alcohol consumption: combined analysis of individual-participant data for 599 912 current drinkers in 83 prospective studies. *Lancet* 2018;**391**:1513–1523. https://doi.org/10.1016/s0140-6736(18)30134-x
774. Holmes MV, Dale CE, Zuccolo L, Silverwood RJ, Guo Y, Ye Z, *et al.* Association between alcohol and cardiovascular disease: Mendelian randomisation analysis based on individual participant data. *BMJ* 2014;**349**:g4164. https://doi.org/10.1136/bmj.g4164
775. Millwood IY, Walters RG, Mei XW, Guo Y, Yang L, Bian Z, *et al.* Conventional and genetic evidence on alcohol and vascular disease aetiology: a prospective study of 500 000 men and women in China. *Lancet* 2019;**393**:1831–1842. https://doi.org/10.1016/s0140-6736(18)31772-0
776. Ekelund U, Tarp J, Steene-Johannessen J, Hansen BH, Jefferis B, Fagerland MW, *et al.* Dose-response associations between accelerometry measured physical activity and sedentary time and all cause mortality: systematic review and harmonised meta-analysis. *BMJ* 2019;**366**:l4570. https://doi.org/10.1136/bmj.l4570
777. Patterson R, McNamara E, Tainio M, de Sá TH, Smith AD, Sharp SJ, *et al.* Sedentary behaviour and risk of all-cause, cardiovascular and cancer mortality, and incident type 2 diabetes: a systematic review and dose response meta-analysis. *Eur J Epidemiol* 2018;**33**:811–829. https://doi.org/10.1007/s10654-018-0380-1
778. WHO Guidelines Review Committee. *WHO Guidelines on Physical Activity and Sedentary Behaviour*. World Health Organization, 2020.
779. Hansen D, Abreu A, Ambrosetti M, Cornelissen V, Gevaert A, Kemps H, *et al.* Exercise intensity assessment and prescription in cardiovascular rehabilitation and beyond: why and how: a position statement from the Secondary Prevention and Rehabilitation Section of the European Association of Preventive Cardiology. *Eur J Prev Cardiol* 2022;**29**:230–245. https://doi.org/10.1093/eurjpc/zwab007
780. De Schutter A, Kachur S, Lavie CJ, Menezes A, Shum KK, Bangalore S, *et al.* Cardiac rehabilitation fitness changes and subsequent survival. *Eur Heart J Qual Care Clin Outcomes* 2018;**4**:173–179. https://doi.org/10.1093/ehjqcco/qcy018
781. Baumeister H, Hutter N, Bengel J. Psychological and pharmacological interventions for depression in patients with coronary artery disease. *Cochrane Database Syst Rev* 2011;**2011**:Cd008012. https://doi.org/10.1002/14651858.CD008012.pub3

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

782. Richards SH, Anderson L, Jenkinson CE, Whalley B, Rees K, Davies P, *et al.* Psychological interventions for coronary heart disease: cochrane systematic review and meta-analysis. *Eur J Prev Cardiol* 2018;**25**:247–259. https://doi.org/10.1177/2047487317739978
783. Mach F, Baigent C, Catapano AL, Koskinas KC, Casula M, Badimon L, *et al.* 2019 ESC/EAS Guidelines for the management of dyslipidaemias: lipid modification to reduce cardiovascular risk. *Eur Heart J* 2020;**41**:111–188. https://doi.org/10.1093/eurheartj/ehz455
784. Ference BA, Ginsberg HN, Graham I, Ray KK, Packard CJ, Bruckert E, *et al.* Low-density lipoproteins cause atherosclerotic cardiovascular disease. 1. Evidence from genetic, epidemiologic, and clinical studies. A consensus statement from the European Atherosclerosis Society Consensus Panel. *Eur Heart J* 2017;**38**:2459–2472. https://doi.org/10.1093/eurheartj/ehx144
785. Sabatine MS, Giugliano RP, Keech AC, Honarpour N, Wiviott SD, Murphy SA, *et al.* Evolocumab and clinical outcomes in patients with cardiovascular disease. *N Engl J Med* 2017;**376**:1713–1722. https://doi.org/10.1056/NEJMoa1615664
786. Schwartz GG, Steg PG, Szarek M, Bhatt DL, Bittner VA, Diaz R, *et al.* Alirocumab and cardiovascular outcomes after acute coronary syndrome. *N Engl J Med* 2018;**379**:2097–2107. https://doi.org/10.1056/NEJMoa1801174
787. Navarese EP, Kowalewski M, Andreotti F, van Wely M, Camaro C, Kolodziejczak M, *et al.* Meta-analysis of time-related benefits of statin therapy in patients with acute coronary syndrome undergoing percutaneous coronary intervention. *Am J Cardiol* 2014;**113**:1753–1764. https://doi.org/10.1016/j.amjcard.2014.02.034
788. Cannon CP, Blazing MA, Giugliano RP, McCagg A, White JA, Theroux P, *et al.* Ezetimibe added to statin therapy after acute coronary syndromes. *N Engl J Med* 2015;**372**:2387–2397. https://doi.org/10.1056/NEJMoa1410489
789. Koskinas KC, Windecker S, Pedrazzini G, Mueller C, Cook S, Matter CM, *et al.* Evolocumab for early reduction of LDL cholesterol levels in patients with acute coronary syndromes (EVOPACS). *J Am Coll Cardiol* 2019;**74**:2452–2462. https://doi.org/10.1016/j.jacc.2019.08.010
790. Trankle CR, Wohlford G, Buckley LF, Kadariya D, Ravindra K, Markley R, *et al.* Alirocumab in acute myocardial infarction: results from the Virginia Commonwealth University Alirocumab Response Trial (VCU-AlirocRT). *J Cardiovasc Pharmacol* 2019;**74**:266–269. https://doi.org/10.1097/fjc.0000000000000706
791. Iannuzzo G, Gentile M, Bresciani A, Mallardo V, Di Lorenzo A, Merone P, *et al.* Inhibitors of protein convertase subtilisin/kexin 9 (PCSK9) and acute coronary syndrome (ACS): the state-of-the-art. *J Clin Med* 2021;**10**:1510. https://doi.org/10.3390/jcm10071510
792. Räber L, Ueki Y, Otsuka T, Losdat S, Häner JD, Lonborg J, *et al.* Effect of alirocumab added to high-intensity statin therapy on coronary atherosclerosis in patients with acute myocardial infarction: the PACMAN-AMI randomized clinical trial. *JAMA* 2022;**327**:1771–1781. https://doi.org/10.1001/jama.2022.5218
793. Nicholls SJ, Kataoka Y, Nissen SE, Prati F, Windecker S, Puri R, *et al.* Effect of evolocumab on coronary plaque phenotype and burden in statin-treated patients following myocardial infarction. *JACC Cardiovasc Imaging* 2022;**15**:1308–1321. https://doi.org/10.1016/j.jcmg.2022.03.002
794. Schubert J, Lindahl B, Melhus H, Renlund H, Leosdottir M, Yari A, *et al.* Low-density lipoprotein cholesterol reduction and statin intensity in myocardial infarction patients and major adverse outcomes: a Swedish nationwide cohort study. *Eur Heart J* 2021;**42**:243–252. https://doi.org/10.1093/eurheartj/ehaa1011
795. Gencer B, Mach F, Murphy SA, De Ferrari GM, Huber K, Lewis BS, *et al.* Efficacy of evolocumab on cardiovascular outcomes in patients with recent myocardial infarction: a prespecified secondary analysis from the FOURIER trial. *JAMA Cardiol* 2020;**5**:952–957. https://doi.org/10.1001/jamacardio.2020.0882
796. O'Donoghue ML, Giugliano RP, Wiviott SD, Atar D, Keech A, Kuder JF, *et al.* Long-term evolocumab in patients with established atherosclerotic cardiovascular disease. *Circulation* 2022;**146**:1109–1119. https://doi.org/10.1161/circulationaha.122.061620
797. Bhatt DL, Steg PG, Miller M, Brinton EA, Jacobson TA, Ketchum SB, *et al.* Cardiovascular risk reduction with icosapent ethyl for hypertriglyceridemia. *N Engl J Med* 2019;**380**:11–22. https://doi.org/10.1056/NEJMoa1812792
798. Freemantle N, Cleland J, Young P, Mason J, Harrison J. β blockade after myocardial infarction: systematic review and meta regression analysis. *BMJ* 1999;**318**:1730–1737. https://doi.org/10.1136/bmj.318.7200.1730
799. Martínez-Milla J, Raposeiras-Roubín S, Pascual-Figal DA, Ibáñez B. Role of beta-blockers in cardiovascular disease in 2019. *Rev Esp Cardiol (Engl Ed)* 2019;**72**:844–852. https://doi.org/10.1016/j.rec.2019.04.014
800. Dahl Aarvik M, Sandven I, Dondo TB, Gale CP, Ruddox V, Munkhaugen J, *et al.* Effect of oral β-blocker treatment on mortality in contemporary post-myocardial infarction patients: a systematic review and meta-analysis. *Eur Heart J Cardiovasc Pharmacother* 2019;**5**:12–20. https://doi.org/10.1093/ehjcvp/pvy034
801. Dargie HJ. Effect of carvedilol on outcome after myocardial infarction in patients with left-ventricular dysfunction: the CAPRICORN randomised trial. *Lancet* 2001;**357**:1385–1390. https://doi.org/10.1016/s0140-6736(00)04560-8
802. Kim J, Kang D, Park H, Kang M, Park TK, Lee JM, *et al.* Long-term β-blocker therapy and clinical outcomes after acute myocardial infarction in patients without heart failure: nationwide cohort study. *Eur Heart J* 2020;**41**:3521–3529. https://doi.org/10.1093/eurheartj/ehaa376
803. Raposeiras-Roubín S, Abu-Assi E, Redondo-Diéguez A, González-Ferreiro R, López-López A, Bouzas-Cruz N, *et al.* Prognostic benefit of beta-blockers after acute coronary syndrome with preserved systolic function. Still relevant today? *Rev Esp Cardiol (Engl Ed)* 2015;**68**:585–591. https://doi.org/10.1016/j.rec.2014.07.028
804. Dondo TB, Hall M, West RM, Jernberg T, Lindahl B, Bueno H, *et al.* β-blockers and mortality after acute myocardial infarction in patients without heart failure or ventricular dysfunction. *J Am Coll Cardiol* 2017;**69**:2710–2720. https://doi.org/10.1016/j.jacc.2017.03.578
805. Watanabe H, Ozasa N, Morimoto T, Shiomi H, Bingyuan B, Suwa S, *et al.* Long-term use of carvedilol in patients with ST-segment elevation myocardial infarction treated with primary percutaneous coronary intervention. *PLoS One* 2018;**13**:e0199347. https://doi.org/10.1371/journal.pone.0199347
806. Rossello X, Raposeiras-Roubin S, Latini R, Dominguez-Rodriguez A, Barrabés JA, Sánchez PL, *et al.* Rationale and design of the pragmatic clinical trial tREatment with Beta-blockers after myOcardial infarction withOut reduced ejection fracTion (REBOOT). *Eur Heart J Cardiovasc Pharmacother* 2021;**8**:291–301. https://doi.org/10.1093/ehjcvp/pvab060
807. Munkhaugen J, Ruddox V, Halvorsen S, Dammen T, Fagerland MW, Hernæs KH, *et al.* BEtablocker Treatment After acute Myocardial Infarction in revascularized patients without reduced left ventricular ejection fraction (BETAMI): rationale and design of a prospective, randomized, open, blinded end point study. *Am Heart J* 2019;**208**:37–46. https://doi.org/10.1016/j.ahj.2018.10.005
808. Kristensen AMD, Bovin A, Zwisler AD, Cerquira C, Torp-Pedersen C, Bøtker HE, *et al.* Design and rationale of the Danish trial of beta-blocker treatment after myocardial infarction without reduced ejection fraction: study protocol for a randomized controlled trial. *Trials* 2020;**21**:415. https://doi.org/10.1186/s13063-020-4214-6
809. Puymirat E, Riant E, Aissaoui N, Soria A, Ducrocq G, Coste P, *et al.* β blockers and mortality after myocardial infarction in patients without heart failure: multicentre prospective cohort study. *BMJ* 2016;**354**:i4801. https://doi.org/10.1136/bmj.i4801
810. Zeitouni M, Kerneis M, Lattuca B, Guedeney P, Cayla G, Collet JP, *et al.* Do patients need lifelong β-blockers after an uncomplicated myocardial infarction? *Am J Cardiovasc Drugs* 2019;**19**:431–438. https://doi.org/10.1007/s40256-019-00338-4
811. ISIS-4 Collaborative Group. ISIS-4: a randomised factorial trial assessing early oral captopril, oral mononitrate, and intravenous magnesium sulphate in 58,050 patients with suspected acute myocardial infarction. *Lancet* 1995;**345**:669–685. https://doi.org/10.1016/S0140-6736(95)90865-X
812. Held PH, Yusuf S, Furberg CD. Calcium channel blockers in acute myocardial infarction and unstable angina: an overview. *BMJ* 1989;**299**:1187–1192. https://doi.org/10.1136/bmj.299.6709.1187
813. Køber L, Torp-Pedersen C, Carlsen JE, Bagger H, Eliasen P, Lyngborg K, *et al.* A clinical trial of the angiotensin-converting-enzyme inhibitor trandolapril in patients with left ventricular dysfunction after myocardial infarction. *N Engl J Med* 1995;**333**:1670–1676. https://doi.org/10.1056/nejm199512213332503
814. Pfeffer MA, Braunwald E, Moyé LA, Basta L, Brown EJ, Cuddy TE, *et al.* Effect of captopril on mortality and morbidity in patients with left ventricular dysfunction after myocardial infarction–results of the survival and ventricular enlargement trial. *N Engl J Med* 1992;**327**:669–677. https://doi.org/10.1056/nejm199209033271001
815. The AIRE Study Investigators. Effect of ramipril on mortality and morbidity of survivors of acute myocardial infarction with clinical evidence of heart failure. *Lancet* 1993;**342**:821–828. https://doi.org/10.1016/0140-6736(93)92693-N
816. Fox KM. Efficacy of perindopril in reduction of cardiovascular events among patients with stable coronary artery disease: randomised, double-blind, placebo-controlled, multicentre trial (the EUROPA study). *Lancet* 2003;**362**:782–788. https://doi.org/10.1016/s0140-6736(03)14286-9
817. The Heart Outcomes Prevention Evaluation Study Investigators; Yusuf S, Sleight P, Pogue Bosch J, Davies R, *et al.* Effects of an angiotensin-converting-enzyme inhibitor, ramipril, on cardiovascular events in high-risk patients. *N Engl J Med* 2000;**342**:145–153. https://doi.org/10.1056/nejm200001203420301
818. ACE Inhibitor Myocardial Infarction Collaborative Group. Indications for ACE inhibitors in the early treatment of acute myocardial infarction: systematic overview of individual data from 100,000 patients in randomized trials. *Circulation* 1998;**97**:2202–2212. https://doi.org/10.1161/01.cir.97.22.2202
819. Pfeffer MA, McMurray JJ, Velazquez EJ, Rouleau JL, Køber L, Maggioni AP, *et al.* Valsartan, captopril, or both in myocardial infarction complicated by heart failure, left ventricular dysfunction, or both. *N Engl J Med* 2003;**349**:1893–1906. https://doi.org/10.1056/NEJMoa032292
820. Swedberg K, Eneroth P, Kjekshus J, Snapinn S. Effects of enalapril and neuroendocrine activation on prognosis in severe congestive heart failure (follow-up of the CONSENSUS trial). *Am J Cardiol* 1990;**66**:D40–D45. https://doi.org/10.1016/0002-9149(90)90475-g
821. Garg R, Yusuf S. Overview of randomized trials of angiotensin-converting enzyme inhibitors on mortality and morbidity in patients with heart failure. *JAMA* 1995;**273**:1450–1456. https://doi.org/10.1001/jama.1995.03520420066040

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

822. Packer M, Poole-Wilson PA, Armstrong PW, Cleland JGF, Horowitz JD, Massie BM, *et al.* Comparative effects of low and high doses of the angiotensin-converting enzyme inhibitor, lisinopril, on morbidity and mortality in chronic heart failure. *Circulation* 1999; **100**:2312–2318. https://doi.org/10.1161/01.cir.100.23.2312
823. Yusuf S, Pitt B, Davis CE, Hood WB, Cohn JN. Effect of enalapril on survival in patients with reduced left ventricular ejection fractions and congestive heart failure. *N Engl J Med* 1991;**325**:293–302. https://doi.org/10.1056/nejm199108013250501
824. McMurray JJ, Packer M, Desai AS, Gong J, Lefkowitz MP, Rizkala AR, *et al.* Angiotensin-neprilysin inhibition versus enalapril in heart failure. *N Engl J Med* 2014; **371**:993–1004. https://doi.org/10.1056/NEJMoa1409077
825. Pfeffer MA, Claggett B, Lewis EF, Granger CB, Køber L, Maggioni AP, *et al.* Angiotensin receptor-neprilysin inhibition in acute myocardial infarction. *N Engl J Med* 2021;**385**: 1845–1855. https://doi.org/10.1056/NEJMoa2104508
826. Pitt B, Remme W, Zannad F, Neaton J, Martinez F, Roniker B, *et al.* Eplerenone, a selective aldosterone blocker, in patients with left ventricular dysfunction after myocardial infarction. *N Engl J Med* 2003;**348**:1309–1321. https://doi.org/10.1056/NEJMoa030207
827. Montalescot G, Pitt B, Lopez de Sa E, Hamm CW, Flather M, Verheugt F, *et al.* Early eplerenone treatment in patients with acute ST-elevation myocardial infarction without heart failure: the Randomized Double-Blind Reminder Study. *Eur Heart J* 2014;**35**: 2295–2302. https://doi.org/10.1093/eurheartj/ehu164
828. Cowie MR, Fisher M. SGLT2 inhibitors: mechanisms of cardiovascular benefit beyond glycaemic control. *Nat Rev Cardiol* 2020;**17**:761–772. https://doi.org/10.1038/s41569-020-0406-8
829. Neal B, Perkovic V, Mahaffey KW, de Zeeuw D, Fulcher G, Erondu N, *et al.* Canagliflozin and cardiovascular and renal events in type 2 diabetes. *N Engl J Med* 2017;**377**:644–657. https://doi.org/10.1056/NEJMoa1611925
830. Wiviott SD, Raz I, Bonaca MP, Mosenzon O, Kato ET, Cahn A, *et al.* Dapagliflozin and cardiovascular outcomes in type 2 diabetes. *N Engl J Med* 2019;**380**:347–357. https://doi.org/10.1056/NEJMoa1812389
831. Nikolaus M, Massimo F, Katharina S, Dirk M-W, Ramzi AA, Manuel JA, *et al.* 2023 ESC Guidelines for the management of cardiovascular disease in patients with diabetes. *Eur Heart J* 2023; https://doi.org/10.1093/eurheartj/ehad192
832. Anker SD, Butler J, Filippatos G, Ferreira JP, Bocchi E, Böhm M, *et al.* Empagliflozin in heart failure with a preserved ejection fraction. *N Engl J Med* 2021;**385**:1451–1461. https://doi.org/10.1056/NEJMoa2107038
833. Solomon SD, McMurray JJV, Claggett B, de Boer RA, DeMets D, Hernandez AF, *et al.* Dapagliflozin in heart failure with mildly reduced or preserved ejection fraction. *N Engl J Med* 2022;**387**:1089–1098. https://doi.org/10.1056/NEJMoa2206286
834. von Lewinski D, Kolesnik E, Tripolt NJ, Pferschy PN, Benedikt M, Wallner M, *et al.* Empagliflozin in acute myocardial infarction: the EMMY trial. *Eur Heart J* 2022;**43**: 4421–4432. https://doi.org/10.1093/eurheartj/ehac494
835. Harrington J, Udell JA, Jones WS, Anker SD, Bhatt DL, Petrie MC, *et al.* Empagliflozin in patients post myocardial infarction rationale and design of the EMPACT-MI trial. *Am Heart J* 2022;**253**:86–98. https://doi.org/10.1016/j.ahj.2022.05.010
836. Lai KC, Lam SK, Chu KM, Wong BCY, Hui WM, Hu WHC, *et al.* Lansoprazole for the prevention of recurrences of ulcer complications from long-term low-dose aspirin use. *N Engl J Med* 2002;**346**:2033–2038. https://doi.org/10.1056/NEJMoa012877
837. Casado Arroyo R, Polo-Tomas M, Roncalés MP, Scheiman J, Lanas A. Lower GI bleeding is more common than upper among patients on dual antiplatelet therapy: long-term follow-up of a cohort of patients commonly using PPI co-therapy. *Heart* 2012; **98**:718–723. https://doi.org/10.1136/heartjnl-2012-301632
838. Small DS, Farid NA, Payne CD, Weerakkody GJ, Li YG, Brandt JT, *et al.* Effects of the proton pump inhibitor lansoprazole on the pharmacokinetics and pharmacodynamics of prasugrel and clopidogrel. *J Clin Pharmacol* 2008;**48**:475–484. https://doi.org/10.1177/0091270008315310
839. Sibbing D, Morath T, Stegherr J, Braun S, Vogt W, Hadamitzky M, *et al.* Impact of proton pump inhibitors on the antiplatelet effects of clopidogrel. *Thromb Haemost* 2009; **101**:714–719. https://doi.org/10.1160/TH08-12-0808
840. Gilard M, Arnaud B, Cornily JC, Le Gal G, Lacut K, Le Calvez G, *et al.* Influence of omeprazole on the antiplatelet action of clopidogrel associated with aspirin: the randomized, double-blind OCLA (Omeprazole CLopidogrel Aspirin) study. *J Am Coll Cardiol* 2008;**51**:256–260. https://doi.org/10.1016/j.jacc.2007.06.064
841. O'Donoghue ML, Braunwald E, Antman EM, Murphy SA, Bates ER, Rozenman Y, *et al.* Pharmacodynamic effect and clinical efficacy of clopidogrel and prasugrel with or without a proton-pump inhibitor: an analysis of two randomised trials. *Lancet* 2009;**374**: 989–997. https://doi.org/10.1016/s0140-6736(09)61525-7
842. Goodman SG, Clare R, Pieper KS, Nicolau JC, Storey RF, Cantor WJ, *et al.* Association of proton pump inhibitor use on cardiovascular outcomes with clopidogrel and ticagrelor: insights from the platelet inhibition and patient outcomes trial. *Circulation* 2012;**125**:978–986. https://doi.org/10.1161/circulationaha.111.032912
843. Yedlapati SH, Khan SU, Talluri S, Lone AN, Khan MZ, Khan MS, *et al.* Effects of influenza vaccine on mortality and cardiovascular outcomes in patients with cardiovascular disease: a systematic review and meta-analysis. *J Am Heart Assoc* 2021;**10**:e019636. https://doi.org/10.1161/jaha.120.019636
844. Liprandi ÁS, Liprandi MIS, Zaidel EJ, Aisenberg GM, Baranchuk A, Barbosa ECD, *et al.* Influenza vaccination for the prevention of cardiovascular disease in the Americas: consensus document of the Inter-American Society of Cardiology and the World Heart Federation. *Glob Heart* 2021;**16**:55. https://doi.org/10.5334/gh.1069
845. Gurfinkel EP, Leon de la Fuente R, Mendiz O, Mautner B. Flu vaccination in acute coronary syndromes and planned percutaneous coronary interventions (FLUVACS) study. *Eur Heart J* 2004;**25**:25–31. https://doi.org/10.1016/j.ehj.2003.10.018
846. Phrommintikul A, Kuanprasert S, Wongcharoen W, Kanjanavanit R, Chaiwarith R, Sukonthasarn A, *et al.* Influenza vaccination reduces cardiovascular events in patients with acute coronary syndrome. *Eur Heart J* 2011;**32**:1730–1735. https://doi.org/10.1093/eurheartj/ehr004
847. Fröbert O, Götberg M, Erlinge D, Akhtar Z, Christiansen EH, MacIntyre CR, *et al.* Influenza vaccination after myocardial infarction: a randomized, double-blind, placebo-controlled, multicenter trial. *Circulation* 2021;**144**:1476–1484. https://doi.org/10.1161/circulationaha.121.057042
848. Chen Y, Zhang H, Chen Y, Li M, Luo W, Liu Y, *et al.* Colchicine may become a new cornerstone therapy for coronary artery disease: a meta-analysis of randomized controlled trials. *Clin Rheumatol* 2022;**41**:1873–1887. https://doi.org/10.1007/s10067-022-06050-0
849. Razavi E, Ramezani A, Kazemi A, Attar A. Effect of treatment with colchicine after acute coronary syndrome on major cardiovascular events: a systematic review and meta-analysis of clinical trials. *Cardiovasc Ther* 2022;**2022**:8317011. https://doi.org/10.1155/2022/8317011
850. Tardif JC, Kouz S, Waters DD, Bertrand OF, Diaz R, Maggioni AP, *et al.* Efficacy and safety of low-dose colchicine after myocardial infarction. *N Engl J Med* 2019;**381**: 2497–2505. https://doi.org/10.1056/NEJMoa1912388
851. Nidorf SM, Fiolet ATL, Mosterd A, Eikelboom JW, Schut A, Opstal TSJ, *et al.* Colchicine in patients with chronic coronary disease. *N Engl J Med* 2020;**383**:1838–1847. https://doi.org/10.1056/NEJMoa2021372
852. Opstal TSJ, Fiolet ATL, van Broekhoven A, Mosterd A, Eikelboom JW, Nidorf SM, *et al.* Colchicine in patients with chronic coronary disease in relation to prior acute coronary syndrome. *J Am Coll Cardiol* 2021;**78**:859–866. https://doi.org/10.1016/j.jacc.2021.06.037
853. Ji H, Fang L, Yuan L, Zhang Q. Effects of exercise-based cardiac rehabilitation in patients with acute coronary syndrome: a meta-analysis. *Med Sci Monit* 2019;**25**: 5015–5027. https://doi.org/10.12659/msm.917362
854. Candelaria D, Randall S, Ladak L, Gallagher R. Health-related quality of life and exercise-based cardiac rehabilitation in contemporary acute coronary syndrome patients: a systematic review and meta-analysis. *Qual Life Res* 2020;**29**:579–592. https://doi.org/10.1007/s11136-019-02338-y
855. Anthonisen NR, Skeans MA, Wise RA, Manfreda J, Kanner RE, Connett JE, *et al.* The effects of a smoking cessation intervention on 14.5-year mortality: a randomized clinical trial. *Ann Intern Med* 2005;**142**:233–239. https://doi.org/10.7326/0003-4819-142-4-200502150-00005
856. Becerra-Tomás N, Blanco Mejía S, Viguiliouk E, Khan T, Kendall CWC, Kahleova H, *et al.* Mediterranean diet, cardiovascular disease and mortality in diabetes: a systematic review and meta-analysis of prospective cohort studies and randomized clinical trials. *Crit Rev Food Sci Nutr* 2020;**60**:1207–1227. https://doi.org/10.1080/10408398.2019.1565281
857. Liu Y, Lee DC, Li Y, Zhu W, Zhang R, Sui X, *et al.* Associations of resistance exercise with cardiovascular disease morbidity and mortality. *Med Sci Sports Exerc* 2019;**51**: 499–508. https://doi.org/10.1249/mss.0000000000001822
858. Saeidifard F, Medina-Inojosa JR, West CP, Olson TP, Somers VK, Bonikowske AR, *et al.* The association of resistance training with mortality: a systematic review and meta-analysis. *Eur J Prev Cardiol* 2019;**26**:1647–1665. https://doi.org/10.1177/2047487319850718
859. Hartmann-Boyce J, Chepkin SC, Ye W, Bullen C, Lancaster T. Nicotine replacement therapy versus control for smoking cessation. *Cochrane Database Syst Rev* 2018;**5**: CD000146. https://doi.org/10.1002/14651858.CD000146.pub5
860. Howes S, Hartmann-Boyce J, Livingstone-Banks J, Hong B, Lindson N. Antidepressants for smoking cessation. *Cochrane Database Syst Rev* 2020;**4**:CD000031. https://doi.org/10.1002/14651858.CD000031.pub5
861. Cahill K, Lindson-Hawley N, Thomas KH, Fanshawe TR, Lancaster T. Nicotine receptor partial agonists for smoking cessation. *Cochrane Database Syst Rev* 2016;**2016**: CD006103. https://doi.org/10.1002/14651858.CD006103.pub7
862. Lindson N, Chepkin SC, Ye W, Fanshawe TR, Bullen C, Hartmann-Boyce J. Different doses, durations and modes of delivery of nicotine replacement therapy for smoking cessation. *Cochrane Database Syst Rev* 2019;**4**:CD013308. https://doi.org/10.1002/14651858.Cd013308
863. Woolf KJ, Zabad MN, Post JM, McNitt S, Williams GC, Bisognano JD. Effect of nicotine replacement therapy on cardiovascular outcomes after acute coronary syndromes. *Am J Cardiol* 2012;**110**:968–970. https://doi.org/10.1016/j.amjcard.2012.05.028
864. Suissa K, Larivière J, Eisenberg MJ, Eberg M, Gore GC, Grad R, *et al.* Efficacy and safety of smoking cessation interventions in patients with cardiovascular disease: a network meta-analysis of randomized controlled trials. *Circ Cardiovasc Qual Outcomes* 2017;**10**: e002458. https://doi.org/10.1161/circoutcomes.115.002458

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

865. de Lemos JA, Blazing MA, Wiviott SD, Lewis EF, Fox KAA, White HD, *et al.* Early intensive vs a delayed conservative simvastatin strategy in patients with acute coronary syndromes: phase Z of the A to Z trial. *JAMA* 2004;**292**:1307–1316. https://doi.org/10.1001/jama.292.11.1307
866. Ray KK, Cannon CP, McCabe CH, Cairns R, Tonkin AM, Sacks FM, *et al.* Early and late benefits of high-dose atorvastatin in patients with acute coronary syndromes: results from the PROVE IT-TIMI 22 trial. *J Am Coll Cardiol* 2005;**46**:1405–1410. https://doi.org/10.1016/j.jacc.2005.03.077
867. Schwartz GG, Olsson AG, Ezekowitz MD, Ganz P, Oliver MF, Waters D, *et al.* Effects of atorvastatin on early recurrent ischemic events in acute coronary syndromes: the MIRACL study: a randomized controlled trial. *JAMA* 2001;**285**:1711–1718. https://doi.org/10.1001/jama.285.13.1711
868. Baigent C, Blackwell L, Emberson J, Holland LE, Reith C, Bhala N, *et al.* Efficacy and safety of more intensive lowering of LDL cholesterol: a meta-analysis of data from 170,000 participants in 26 randomised trials. *Lancet* 2010;**376**:1670–1681. https://doi.org/10.1016/s0140-6736(10)61350-5
869. Fulcher J, O'Connell R, Voysey M, Emberson J, Blackwell L, Mihaylova B. Efficacy and safety of LDL-lowering therapy among men and women: meta-analysis of individual data from 174,000 participants in 27 randomised trials. *Lancet* 2015;**385**:1397–1405. https://doi.org/10.1016/s0140-6736(14)61368-4
870. CIBIS-II Investigators. The Cardiac Insufficiency Bisoprolol Study II (CIBIS-II): a randomised trial. *Lancet* 1999;**353**:9–13. https://doi.org/10.1016/S0140-6736(98)11181-9
871. Packer M, Coats AJ, Fowler MB, Katus HA, Krum H, Mohacsi P, *et al.* Effect of carvedilol on survival in severe chronic heart failure. *N Engl J Med* 2001;**344**:1651–1658. https://doi.org/10.1056/nejm200105313442201
872. MERIT-HF Study Group. Effect of metoprolol CR/XL in chronic heart failure: Metoprolol CR/XL Randomised Intervention Trial in Congestive Heart Failure (MERIT-HF). *Lancet* 1999;**353**:2001–2007. https://doi.org/10.1016/S0140-6736(99)04440-2
873. Chen ZM, Pan HC, Chen YP, Peto R, Collins R, Jiang LX, *et al.* Early intravenous then oral metoprolol in 45,852 patients with acute myocardial infarction: randomised placebo-controlled trial. *Lancet* 2005;**366**:1622–1632. https://doi.org/10.1016/s0140-6736(05)67661-1
874. Bangalore S, Makani H, Radford M, Thakur K, Toklu B, Katz SD, *et al.* Clinical outcomes with β-blockers for myocardial infarction: a meta-analysis of randomized trials. *Am J Med* 2014;**127**:939–953. https://doi.org/10.1016/j.amjmed.2014.05.032
875. Huang BT, Huang FY, Zuo ZL, Liao YB, Heng Y, Wang PJ, *et al.* Meta-analysis of relation between oral β-blocker therapy and outcomes in patients with acute myocardial infarction who underwent percutaneous coronary intervention. *Am J Cardiol* 2015;**115**:1529–1538. https://doi.org/10.1016/j.amjcard.2015.02.057
876. Goldberger JJ, Bonow RO, Cuffe M, Liu L, Rosenberg Y, Shah PK, *et al.* Effect of beta-blocker dose on survival after acute myocardial infarction. *J Am Coll Cardiol* 2015;**66**:1431–1441. https://doi.org/10.1016/j.jacc.2015.07.047
877. Andersson C, Shilane D, Go AS, Chang TI, Kazi D, Solomon MD, *et al.* β-blocker therapy and cardiac events among patients with newly diagnosed coronary heart disease. *J Am Coll Cardiol* 2014;**64**:247–252. https://doi.org/10.1016/j.jacc.2014.04.042
878. Bangalore S, Steg G, Deedwania P, Crowley K, Eagle KA, Goto S, *et al.* β-blocker use and clinical outcomes in stable outpatients with and without coronary artery disease. *JAMA* 2012;**308**:1340–1349. https://doi.org/10.1001/jama.2012.12559
879. Heart Outcomes Prevention Evaluation Study Investigators. Effects of ramipril on cardiovascular and microvascular outcomes in people with diabetes mellitus: results of the HOPE study and MICRO-HOPE substudy. *Lancet* 2000;**355**:253–259. https://doi.org/10.1016/S0140-6736(99)12323-7
880. Pitt B, Zannad F, Remme WJ, Cody R, Castaigne A, Perez A, *et al.* The effect of spironolactone on morbidity and mortality in patients with severe heart failure. *N Engl J Med* 1999;**341**:709–717. https://doi.org/10.1056/nejm199909023411001
881. Hirpa M, Woreta T, Addis H, Kebede S. What matters to patients? A timely question for value-based care. *PLoS One* 2020;**15**:e0227845. https://doi.org/10.1371/journal.pone.0227845
882. Ebrahimi Z, Patel H, Wijk H, Ekman I, Olaya-Contreras P. A systematic review on implementation of person-centered care interventions for older people in out-of-hospital settings. *Geriatr Nurs* 2021;**42**:213–224. https://doi.org/10.1016/j.gerinurse.2020.08.004
883. Gluyas H. Patient-centred care: improving healthcare outcomes. *Nurs Stand* 2015;**30**:50–57, quiz 59. https://doi.org/10.7748/ns.30.4.50.e10186
884. Kok MM, von Birgelen C. Involving the patient's perspective and preferences concerning coronary angiography and percutaneous coronary intervention. *EuroIntervention* 2020;**15**:1228–1231. https://doi.org/10.4244/eijv15i14a221
885. Astin F, Stephenson J, Probyn J, Holt J, Marshall K, Conway D. Cardiologists' and patients' views about the informed consent process and their understanding of the anticipated treatment benefits of coronary angioplasty: a survey study. *Eur J Cardiovasc Nurs* 2020;**19**:260–268. https://doi.org/10.1177/1474515119879050
886. Flynn D, Knoedler MA, Hess EP, Murad MH, Erwin PJ, Montori VM, *et al.* Engaging patients in health care decisions in the emergency department through shared decision-making: a systematic review. *Acad Emerg Med* 2012;**19**:959–967. https://doi.org/10.1111/j.1553-2712.2012.01414.x
887. Grant EV, Summapund J, Matlock DD, Vaughan Dickson V, Iqbal S, Patel S, *et al.* Patient and cardiologist perspectives on shared decision making in the treatment of older adults hospitalized for acute myocardial infarction. *Med Decis Making* 2020;**40**:279–288. https://doi.org/10.1177/0272989x20912293
888. Shah P, Thornton I, Turrin D, Hipskind JE. Informed consent. In: *StatPearls*. Treasure Island (FL): StatPearls Publishing, Copyright © 2022, StatPearls Publishing LLC, 2022.
889. Prochnow JA, Meiers SJ, Scheckel MM. Improving patient and caregiver new medication education using an innovative teach-back toolkit. *J Nurs Care Qual* 2019;**34**:101–106. https://doi.org/10.1097/ncq.0000000000000342
890. Klingbeil C, Gibson C. The Teach Back Project: a system-wide evidence based practice implementation. *J Pediatr Nurs* 2018;**42**:81–85. https://doi.org/10.1016/j.pedn.2018.06.002
891. Ha Dinh TT, Bonner A, Clark R, Ramsbotham J, Hines S. The effectiveness of the teach-back method on adherence and self-management in health education for people with chronic disease: a systematic review. *JBI Database System Rev Implement Rep* 2016;**14**:210–247. https://doi.org/10.11124/jbisrir-2016-2296
892. Dickert NW, Scicluna VM, Adeoye O, Angiolillo DJ, Blankenship JC, Devireddy CM, *et al.* Emergency consent: patients' and surrogates' perspectives on consent for clinical trials in acute stroke and myocardial infarction. *J Am Heart Assoc* 2019;**8**:e010905. https://doi.org/10.1161/jaha.118.010905
893. Dickert NW, Miller FG. Involving patients in enrolment decisions for acute myocardial infarction trials. *BMJ* 2015;**351**:h3791. https://doi.org/10.1136/bmj.h3791
894. Olsson A, Ring C, Josefsson J, Eriksson A, Rylance R, Fröbert O, *et al.* Patient experience of the informed consent process during acute myocardial infarction: a sub-study of the VALIDATE-SWEDEHEART trial. *Trials* 2020;**21**:246. https://doi.org/10.1186/s13063-020-4147-0
895. El-Haddad C, Hegazi I, Hu W. Understanding patient expectations of health care: a qualitative study. *J Patient Exp* 2020;**7**:1724–1731. https://doi.org/10.1177/2374373520921692
896. Scott JT, Thompson DR. Assessing the information needs of post-myocardial infarction patients: a systematic review. *Patient Educ Couns* 2003;**50**:167–177. https://doi.org/10.1016/s0738-3991(02)00126-x
897. Saczynski JS, McManus DD, Waring ME, Lessard D, Anatchkova MD, Gurwitz JH, *et al.* Change in cognitive function in the month after hospitalization for acute coronary syndromes: findings from TRACE-CORE (Transition, Risks, and Actions in Coronary Events-Center for Outcomes Research and Education). *Circ Cardiovasc Qual Outcomes* 2017;**10**:e001669. https://doi.org/10.1161/circoutcomes.115.001669
898. Goldman JD, Harte FM. Transition of care to prevent recurrence after acute coronary syndrome: the critical role of the primary care provider and pharmacist. *Postgrad Med* 2020;**132**:426–432. https://doi.org/10.1080/00325481.2020.1740512
899. Huriani E. Myocardial infarction patients' learning needs: perceptions of patients, family members and nurses. *Int J Nurs Sci* 2019;**6**:294–299. https://doi.org/10.1016/j.ijnss.2019.05.001
900. Messerli AW, Deutsch C. Implementation of institutional discharge protocols and transition of care following acute coronary syndrome. *Cardiovasc Revasc Med* 2020;**21**:1180–1188. https://doi.org/10.1016/j.carrev.2020.02.013
901. Schiele F, Lemesle G, Angoulvant D, Krempf M, Kownator S, Cheggour S, *et al.* Proposal for a standardized discharge letter after hospital stay for acute myocardial infarction. *Eur Heart J Acute Cardiovasc Care* 2020;**9**:788–801. https://doi.org/10.1177/2048872619844444
902. Murphy B, Le Grande M, Alvarenga M, Worcester M, Jackson A. Anxiety and depression after a cardiac event: prevalence and predictors. *Front Psychol* 2019;**10**:3010. https://doi.org/10.3389/fpsyg.2019.03010
903. Ceccarini M, Manzoni GM, Castelnuovo G. Assessing depression in cardiac patients: what measures should be considered? *Depress Res Treat* 2014;**2014**:148256. https://doi.org/10.1155/2014/148256
904. Moser DK. "The rust of life": impact of anxiety on cardiac patients. *Am J Crit Care* 2007;**16**:361–369. https://doi.org/10.4037/ajcc2007.16.4.361
905. Turgeon RD, Koshman SL, Dong Y, Graham MM. P2Y12 inhibitor adherence trajectories in patients with acute coronary syndrome undergoing percutaneous coronary intervention: prognostic implications. *Eur Heart J* 2022;**43**:2303–2313. https://doi.org/10.1093/eurheartj/ehac116
906. Poitras ME, Maltais ME, Bestard-Denommé L, Stewart M, Fortin M. What are the effective elements in patient-centered and multimorbidity care? A scoping review. *BMC Health Serv Res* 2018;**18**:446. https://doi.org/10.1186/s12913-018-3213-8
907. Hochhalter AK, Song J, Rush J, Sklar L, Stevens A. Making the Most of Your Healthcare intervention for older adults with multiple chronic illnesses. *Patient Educ Couns* 2010;**81**:207–213. https://doi.org/10.1016/j.pec.2010.01.018
908. Hess EP, Knoedler MA, Shah ND, Kline JA, Breslin M, Branda ME, *et al.* The chest pain choice decision aid: a randomized trial. *Circ Cardiovasc Qual Outcomes* 2012;**5**:251–259. https://doi.org/10.1161/circoutcomes.111.964791
909. Hess EP, Hollander JE, Schaffer JT, Kline JA, Torres CA, Diercks DB, *et al.* Shared decision making in patients with low risk chest pain: prospective randomized pragmatic trial. *BMJ* 2016;**355**:i6165. https://doi.org/10.1136/bmj.i6165
910. van Oosterhout REM, de Boer AR, Maas A, Rutten FH, Bots ML, Peters SAE. Sex differences in symptom presentation in acute coronary syndromes: a systematic review

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026

and meta-analysis. *J Am Heart Assoc* 2020;**9**:e014733. https://doi.org/10.1161/jaha.119.014733
911. Hedegaard U, Kjeldsen LJ, Pottegård A, Henriksen JE, Lambrechtsen J, Hangaard J, *et al.* Improving medication adherence in patients with hypertension: a randomized trial. *Am J Med* 2015;**128**:1351–1361. https://doi.org/10.1016/j.amjmed.2015.08.011
912. Bauer LK, Caro MA, Beach SR, Mastromauro CA, Lenihan E, Januzzi JL, *et al.* Effects of depression and anxiety improvement on adherence to medication and health behaviors in recently hospitalized cardiac patients. *Am J Cardiol* 2012;**109**:1266–1271. https://doi.org/10.1016/j.amjcard.2011.12.017
913. Van der Kooy K, van Hout H, Marwijk H, Marten H, Stehouwer C, Beekman A. Depression and the risk for cardiovascular diseases: systematic review and meta analysis. *Int J Geriatr Psychiatry* 2007;**22**:613–626. https://doi.org/10.1002/gps.1723
914. Redfors B, Angerås O, Råmunddal T, Petursson P, Haraldsson I, Dworeck C, *et al.* Trends in gender differences in cardiac care and outcome after acute myocardial infarction in Western Sweden: a report from the Swedish Web System for Enhancement of Evidence-Based Care in Heart Disease Evaluated According to Recommended Therapies (SWEDEHEART). *J Am Heart Assoc* 2015;**4**:e001995. https://doi.org/10.1161/jaha.115.001995
915. Anand SS, Xie CC, Mehta S, Franzosi MG, Joyner C, Chrolavicius S, *et al.* Differences in the management and prognosis of women and men who suffer from acute coronary syndromes. *J Am Coll Cardiol* 2005;**46**:1845–1851. https://doi.org/10.1016/j.jacc.2005.05.091
916. Blomkalns AL, Chen AY, Hochman JS, Peterson ED, Trynosky K, Diercks DB, *et al.* Gender disparities in the diagnosis and treatment of non-ST-segment elevation acute coronary syndromes: large-scale observations from the CRUSADE (Can Rapid Risk Stratification of Unstable Angina Patients Suppress Adverse Outcomes With Early Implementation of the American College of Cardiology/American Heart Association Guidelines) National Quality Improvement Initiative. *J Am Coll Cardiol* 2005;**45**:832–837. https://doi.org/10.1016/j.jacc.2004.11.055
917. Bugiardini R, Yan AT, Yan RT, Fitchett D, Langer A, Manfrini O, *et al.* Factors influencing underutilization of evidence-based therapies in women. *Eur Heart J* 2011;**32**:1337–1344. https://doi.org/10.1093/eurheartj/ehr027
918. Samayoa L, Grace SL, Gravely S, Scott LB, Marzolini S, Colella TJF. Sex differences in cardiac rehabilitation enrollment: a meta-analysis. *Can J Cardiol* 2014;**30**:793–800. https://doi.org/10.1016/j.cjca.2013.11.007
919. Rossello X, Mas-Lladó C, Pocock S, Vicent L, van de Werf F, Chin CT, *et al.* Sex differences in mortality after an acute coronary syndrome increase with lower country wealth and higher income inequality. *Rev Esp Cardiol (Engl Ed)* 2022;**75**:392–400. https://doi.org/10.1016/j.rec.2021.05.006
920. Sardar MR, Badri M, Prince CT, Seltzer J, Kowey PR. Underrepresentation of women, elderly patients, and racial minorities in the randomized trials used for cardiovascular guidelines. *JAMA Internal Medicine* 2014;**174**:1868–1870. https://doi.org/10.1001/jamainternmed.2014.4758
921. Cho L, Vest AR, O'Donoghue ML, Ogunniyi MO, Sarma AA, Denby KJ, *et al.* Increasing participation of women in cardiovascular trials. *J Am College Cardiol* 2021;**78**:737–751. https://doi.org/10.1016/j.jacc.2021.06.022
922. Gong IY, Tan NS, Ali SH, Lebovic G, Mamdani M, Goodman SG, *et al.* Temporal trends of women enrollment in major cardiovascular randomized clinical trials. *Can J Cardiol* 2019;**35**:653–660. https://doi.org/10.1016/j.cjca.2019.01.010
923. Rossello X, Ferreira JP, Caimari F, Lamiral Z, Sharma A, Mehta C, *et al.* Influence of sex, age and race on coronary and heart failure events in patients with diabetes and post-acute coronary syndrome. *Clin Res Cardiol* 2021;**110**:1612–1624. https://doi.org/10.1007/s00392-021-01859-2
924. Mas-Llado C, González-Del-Hoyo M, Siquier-Padilla J, Blaya-Peña L, Coughlan JJ, García de la Villa B, *et al.* Representativeness in randomised clinical trials supporting acute coronary syndrome guidelines. *Eur Heart J Qual Care Clin Outcomes* 2023. https://doi.org/10.1093/ehjqcco/qcad007
925. Aktaa S, Batra G, Wallentin L, Baigent C, Erlinge D, James S, *et al.* European Society of Cardiology methodology for the development of quality indicators for the quantification of cardiovascular care and outcomes. *Eur Heart J Qual Care Clin Outcomes* 2022;**8**:4–13. https://doi.org/10.1093/ehjqcco/qcaa069
926. Minchin M, Roland M, Richardson J, Rowark S, Guthrie B. Quality of care in the United Kingdom after removal of financial incentives. *N Engl J Med* 2018;**379**:948–957. https://doi.org/10.1056/NEJMsa1801495
927. Song Z, Ji Y, Safran DG, Chernew ME. Health care spending, utilization, and quality 8 years into global payment. *N Engl J Med* 2019;**381**:252–263. https://doi.org/10.1056/NEJMsa1813621
928. Arbelo E, Aktaa S, Bollmann A, D'Avila A, Drossart I, Dwight J, *et al.* Quality indicators for the care and outcomes of adults with atrial fibrillation. *Europace* 2021;**23**:494–495. https://doi.org/10.1093/europace/euaa253
929. Schiele F, Aktaa S, Rossello X, Ahrens I, Claeys MJ, Collet JP, *et al.* 2020 Update of the quality indicators for acute myocardial infarction: a position paper of the Association for Acute Cardiovascular Care: the study group for quality indicators from the ACVC and the NSTE-ACS guideline group. *Eur Heart J Acute Cardiovasc Care* 2021;**10**:224–233. https://doi.org/10.1093/ehjacc/zuaa037
930. Aktaa S, Abdin A, Arbelo E, Burri H, Vernooy K, Blomström-Lundqvist C, *et al.* European Society of Cardiology Quality Indicators for the care and outcomes of cardiac pacing: developed by the Working Group for Cardiac Pacing Quality Indicators in collaboration with the European Heart Rhythm Association of the European Society of Cardiology. *Europace* 2022;**24**:165–172. https://doi.org/10.1093/europace/euab193
931. Aktaa S, Gencer B, Arbelo E, Davos CH, Désormais I, Hollander M, *et al.* European Society of Cardiology Quality Indicators for Cardiovascular Disease Prevention: developed by the Working Group for Cardiovascular Disease Prevention Quality Indicators in collaboration with the European Association for Preventive Cardiology of the European Society of Cardiology. *Eur J Prev Cardiol* 2022;**29**:1060–1071. https://doi.org/10.1093/eurjpc/zwab160
932. Schiele F, Gale CP, Simon T, Fox KAA, Bueno H, Lettino M, *et al.* The 2020 ESC-ACVC quality indicators for the management of acute myocardial infarction applied to the FAST-MI registries. *Eur Heart J Acute Cardiovasc Care* 2021;**10**:207–215. https://doi.org/10.1093/ehjacc/zuab010
933. Rossello X, Medina J, Pocock S, Van de Werf F, Chin CT, Danchin N, *et al.* Assessment of quality indicators for acute myocardial infarction management in 28 countries and use of composite quality indicators for benchmarking. *Eur Heart J Acute Cardiovasc Care* 2020;**9**:911–922. https://doi.org/10.1177/2048872620911853
934. Rossello X, Massó-van Roessel A, Perelló-Bordoy A, Mas-Lladó C, Ramis-Barceló MF, Vives-Borrás M, *et al.* Assessment of the ESC quality indicators in patients with acute myocardial infarction: a systematic review. *Eur Heart J Acute Cardiovasc Care* 2021;**10**:878–889. https://doi.org/10.1093/ehjacc/zuab042
935. Batra G, Aktaa S, Wallentin L, Maggioni AP, Wilkinson C, Casadei B, *et al.* Methodology for the development of international clinical data standards for common cardiovascular conditions: European Unified Registries for Heart Care Evaluation and Randomised Trials (EuroHeart). *Eur Heart J Qual Care Clin Outcomes* 2021;**9**:161–168. https://doi.org/10.1093/ehjqcco/qcab052
936. Rossello X, Massó-van Roessel A, Chioncel O, Tavazzi L, Ferrari R, Vahanian A, *et al.* EURObservational Research Programme: a bibliometric assessment of its scientific output. *Eur Heart J Qual Care Clin Outcomes* 2022;**8**:804–811. https://doi.org/10.1093/ehjqcco/qcac041

Downloaded from https://academic.oup.com/eurheartj/article/44/38/3720/7243210 by guest on 04 July 2026